CS 700 Midterm Notes Mon 10/26, open book and notes, no electronics

1. Metrics (Lecture 2)

Good metrics are linear, reliable, repeatable, easy to use, consistent and independent. Reliable, repeatable, consistent and independent are required. Linear and easy to use are nice but not necessary.

MIPS $= \dfrac{n}{T_e \times 10^6}$, MFLOPS $= \dfrac{f}{T_e \times 10^6}$. MIPS fails reliable and consistent (CISC vs RISC, CPI). MFLOPS gives an integer program 0 and a FLOP isn't defined the same everywhere. SPEC normalizes to a basis machine and takes a geometric mean, which is the wrong mean for times.

Example. (1) n is instructions run, f is floating-point ops, T_e is run time in seconds. (2) $n = 4.5 \times 10^9$ in $T_e = 3$ s, and $f = 8 \times 10^8$ in $T_e = 2$ s. (3) MIPS $= 4.5 \times 10^9/(3 \times 10^6) = 1500$, MFLOPS $= 8 \times 10^8/(2 \times 10^6) = 400$. (4) 1500 million instructions and 400 million FLOPs per second. Don't forget the 10^6, and keep T_e in seconds.

Means metrics (clock rate, MIPS, MFLOPS) count what was done. Ends metrics (execution time) count what got accomplished. Execution time is the fundamental metric. Report wall clock and CPU time.

Speedup of system 2 over 1 is $S_{2,1} = \dfrac{T_1}{T_2} = \dfrac{R_2}{R_1}$. Relative change is $\Delta_{2,1} = \dfrac{T_1 - T_2}{T_2} = S_{2,1} - 1$.

Example. (1) T_1 is the old (baseline) time, T_2 the new time, so the old time goes on top. (2) $T_1 = 12$ s, $T_2 = 8$ s. (3) $S_{2,1} = 12/8 = 1.5$ and $\Delta_{2,1} = (12-8)/8 = 4/8 = 0.5$, which matches $S_{2,1} - 1$. (4) The new system is 1.5 times as fast, or 50% faster, even though its time only fell by $4/12 = 33\%$.

Precision $= \dfrac{\text{relevant retrieved}}{\text{retrieved}}$, recall $= \dfrac{\text{relevant retrieved}}{\text{all relevant}}$.

Example (Metrics p.32). (1) Retrieved is everything the search returned, all relevant is found plus missed relevant pages. (2) 30 pages returned, 20 of them relevant, 40 relevant pages missed. (3) Precision $= 20/30$, recall $= 20/(20+40) = 20/60$. (4) 0.667 and 0.333. The search is mostly right about what it returns but finds only a third of what's out there.

2. Summarizing data (Lecture 4)

Center

Mean $\bar{x} = \frac{1}{n}\sum x_i$, median is the middle sorted value (average the two middle ones when n is even), mode is the most frequent value or bin. Use the median when the data is skewed or has outliers, the mode for categorical data, and the mean when the total matters.

Example (Summ p.15). (1) x_i are the measurements, n is how many. (2) 10, 20, 15, 18, 16, then one more run of 200. (3) Mean $= 79/5 = 15.8$, and sorted 10, 15, 16, 18, 20 gives median 16. With 200 added the mean is $279/6 = 46.5$, and with $n = 6$ the median is $(16+18)/2 = 17$. (4) One outlier tripled the mean but moved the median by 1, so the median is the better center for skewed data.

Harmonic $\bar{x}_H = \dfrac{n}{\sum 1/x_i}$ for rates whose numerator (work) is fixed. Geometric $\bar{x}_G = \left(\prod x_i\right)^{1/n}$ for ratios that compound or normalized values. Arithmetic for times, sizes and anything whose sum means something.

Example, harmonic. (1) x_i are rates for jobs that each do the same work. (2) Three 600 MB downloads at 20, 30 and 60 MB/s. (3) $\frac{1}{20} + \frac{1}{30} + \frac{1}{60} = \frac{3+2+1}{60} = \frac{1}{10}$, so $\bar{x}_H = 3/\frac{1}{10} = 30$. Check with totals, the times are 30, 20 and 10 s, so 1800 MB in 60 s is 30 MB/s. (4) 30 MB/s is the true average rate. The arithmetic mean $110/3 = 36.7$ overstates it, and $\bar{x}_G = 36000^{1/3} \approx 33.0$ sits between, as $H \le G \le A$ says.

Example, geometric. (1) x_i are per-step speedup ratios that compound. (2) Three releases with speedups 2, 8 and 4. (3) $2 \times 8 \times 4 = 64$ and $64^{1/3} = 4$. (4) "4 times per release" reproduces the real 64× overall, while the arithmetic mean 4.67 would claim $4.67^3 \approx 102\times$.

The mean of a rate should track total work over total time. If every program does the same work F, then the right average rate is $\dfrac{nF}{\sum t_i}$, which is the harmonic mean of the rates.

Weighted forms, with $\sum w_i = 1$, $\bar{x}_A = \sum w_i x_i$, $\bar{x}_H = 1/\sum \frac{w_i}{x_i}$, $\bar{x}_G = \prod x_i^{w_i}$. Always $\bar{x}_H \le \bar{x}_G \le \bar{x}_A$, equal only when all values are equal.

Example, weighted. (1) w_i is the share of requests of type i, x_i its time. (2) Small, medium and large requests are 70%, 20% and 10% of traffic and take 2, 5 and 20 ms. (3) $\bar{x}_A = 0.7(2) + 0.2(5) + 0.1(20) = 1.4 + 1.0 + 2.0 = 4.4$. (4) The average request takes 4.4 ms. The unweighted $27/3 = 9$ ms pretends large requests are a third of the load.

Spread

Range $= x_{\max} - x_{\min}$. Sample variance $s^2 = \dfrac{\sum (x_i - \bar{x})^2}{n-1} = \dfrac{\sum x_i^2 - n\bar{x}^2}{n-1}$. Standard deviation s. Coefficient of variation COV $= s/\bar{x}$ (unitless). IQR $= Q_3 - Q_1$, SIQR $= \dfrac{Q_3 - Q_1}{2}$. Midhinge $= \dfrac{Q_1 + Q_3}{2}$. Use s with the mean, SIQR or IQR with the median.

Example. (1) $\bar{x}$ is the mean and $n-1$ is the divisor for a sample. (2) 2, 4, 4, 5, 10, so $n = 5$ and $\bar{x} = 25/5 = 5$. (3) Deviations $-3, -1, -1, 0, 5$ sum to 0 (a free check), squares $9, 1, 1, 0, 25$ sum to 36, so $s^2 = 36/4 = 9$. Shortcut check, $\sum x_i^2 = 161$ and $(161 - 5 \cdot 5^2)/4 = 36/4 = 9$. (4) $s = 3$, range $10 - 2 = 8$, COV $= 3/5 = 0.6$, so the typical spread is 60% of the mean.

Quantiles, course convention

Sort $x_{(1)} \le \cdots \le x_{(n)}$ and give $x_{(i)}$ the plotting position $f_i = \dfrac{i - 0.5}{n}$. For a quantile q, find i with $f_i \le q \le f_{i+1}$ and interpolate linearly between $x_{(i)}$ and $x_{(i+1)}$. Histogram bins by Sturges, $k = \lfloor 1 + \log_2 n \rfloor$ on the slide (HW3 rounded instead, so say which you used). Excel's PERCENTILE on the slides interpolates at $1 + (n-1)p$, which gives slightly different quartiles.

Example. (1) $x_{(i)}$ is the i-th smallest value, f_i its plotting position, q the quantile you want. (2) Sorted 2, 3, 5, 6, 8, 9, 12, 20 with $n = 8$, so $f_i = (i - 0.5)/8$ runs $0.0625, 0.1875, 0.3125, \ldots$ in steps of 0.125. (3) $q = 0.25$ sits halfway between $f_2 = 0.1875$ and $f_3 = 0.3125$, so $Q_1 = 3 + 0.5(5 - 3) = 4$. $q = 0.75$ sits halfway between $f_6 = 0.6875$ and $f_7 = 0.8125$, so $Q_3 = 9 + 0.5(12 - 9) = 10.5$. The median is $(6+8)/2 = 7$. (4) IQR $= 10.5 - 4 = 6.5$, SIQR $= 6.5/2 = 3.25$, midhinge $= (4 + 10.5)/2 = 7.25$. The middle half of the data spans 6.5 units.

Example, Sturges. (1) n is the number of values, k the number of bins. (2) $n = 16$, and $n = 1000$. (3) $1 + \log_2 16 = 1 + 4 = 5$. For 1000, $2^{10} = 1024$, so $\log_2 1000 \approx 9.97$ and $1 + 9.97 = 10.97$. (4) 5 bins of width range/5 for $n = 16$. For $n = 1000$ the slide floors to 10, HW3 rounded to 11.

Normal Q-Q plot

Plot $(z_i,\ x_{(i)})$ with $z_i = \Phi^{-1}(f_i)$, or rescale to $m + s\,z_i$ so both axes share units. A straight line means normal. Ends bending away from the line mean heavy tails, and a curve bending one way means skew.

Example. (1) f_i is the plotting position, z_i the standard normal value with area f_i below it, m and s the sample mean and SD. (2) Data 2, 4, 4, 5, 10 with $m = 5$, $s = 3$, so $f_i = 0.1, 0.3, 0.5, 0.7, 0.9$.

(3) In the Φ table, row 1.2, column .08 gives 0.8997, closest to 0.9, so $z_5 = 1.28$. Row 0.5, column .02 gives 0.6985, so $z_4 = 0.52$. By symmetry $z_1 = -1.28$, $z_2 = -0.52$, $z_3 = 0$. Then $5 + 3z_i$ is 1.16, 3.44, 5, 6.56, 8.84. (4) The data sit above the line at both ends (2 vs 1.16, 10 vs 8.84) and below it in the middle, a convex bow, so the data look right-skewed.

3. Confidence intervals (Lecture 6)

Mean

Large n (slides say $n > 30$) $\quad \bar{x} \mp z_{\alpha/2}\dfrac{s}{\sqrt{n}}$.

Small n, normal population $\quad \bar{x} \mp t_{[\alpha/2,\,n-1]}\dfrac{s}{\sqrt{n}}$.

All decks use upper-tail notation, $z_{\alpha/2}$ or $z[\alpha/2]$ is the value with area $\alpha/2$ above it, which the CI deck also calls the $(1-\alpha/2)$-quantile. Same number, e.g. 1.96 for 95%. The p.50 table labels them $z_{0.95}$, $z_{0.975}$. For $n = 15$ use df 14, $t = 2.145$.

Example. (1) $\bar{x}$ is the sample mean, s the sample SD, n the number of runs, $s/\sqrt{n}$ the standard error. (2) 16 runs, $\bar{x} = 50$ ms, $s = 8$ ms, 95%. (3) SE $= 8/\sqrt{16} = 8/4 = 2$. Small n, so use t with $16-1 = 15$ df, and the t table row 15, column .025 gives 2.131. Half-width $2.131 \times 2 = 4.262$. (4) $50 \mp 4.262 = (45.74,\ 54.26)$ ms. With $n = 64$ you'd use z, SE $= 8/8 = 1$, and get $50 \mp 1.96 = (48.04,\ 51.96)$.

Common levels $\quad 90\% \to 1.645$, $95\% \to 1.960$, $99\% \to 2.576$.

Batch means split a correlated series into batches and treat batch means as the sample. A 95% CI means 95% of intervals built this way cover μ. It does not mean μ has a 95% chance of being in this particular interval, and it says nothing about 95% of the data. The width shrinks like $1/\sqrt{n}$, so halving it takes four times as many runs.

How many measurements

For a half-width of e as a fraction of the mean, $n = \left(\dfrac{z_{\alpha/2}\,s}{e\,\bar{x}}\right)^2$, rounded up. Slide example $s = 2.14$, $\bar{x} = 7.94$, 90%, $e = 0.035$ gives $n = 160.5 \to 161$. The slide's "within 7%" is the full width, so the half-width is 3.5%.

Worked out (CI p.44). (1) s is the pilot SD, $\bar{x}$ the pilot mean, e the relative half-width. (2) 90% gives $z_{0.05} = 1.645$, with $s = 2.14$, $\bar{x} = 7.94$, $e = 0.035$. (3) Top $1.645 \times 2.14 = 3.520$, bottom $0.035 \times 7.94 = 0.2779$, ratio $3.520/0.2779 = 12.67$, squared 160.5. (4) Round up to 161 runs to get a 90% CI within $\pm 3.5\%$ of the mean.

Proportion

$p = m/n, \quad p \mp z_{\alpha/2}\sqrt{\dfrac{p(1-p)}{n}}$, valid when $np \geq 10$ (slide) and $n(1-p) \geq 10$ (study guide). Wald interval, poor for small n. Sample size with e as a RELATIVE error (slides) $n = \dfrac{z^2_{\alpha/2}\,p(1-p)}{(e\,p)^2}$.

Slide example $e = 0.005$, $p = 0.1097$ gives 1.25 million 10 ms samples, 3.46 hours. With an absolute half-width e (measurement study guide) drop the p in the denominator. Slide example $m = 658$ of $n = 6000$, so $p = 0.1097$ and the 95% CI is 10.2% to 11.8%.

Example (CI p.50). (1) m is the count with the property, n the total, $p = m/n$ the sample fraction. (2) 650 of 1000 web log entries are gif files, so $p = 0.65$. (3) $np = 650$ and $n(1-p) = 350$ are both over 10. $p(1-p) = 0.65 \times 0.35 = 0.2275$, SE $= \sqrt{0.2275/1000} = \sqrt{2.275} \times 10^{-2} = 0.01508$. Half-widths are $1.645 \times 0.01508 = 0.0248$ and $1.96 \times 0.01508 = 0.0296$. (4) 90% CI (0.625, 0.675) and 95% CI (0.620, 0.680), so about 62% to 68% of entries are gifs.

Worked out, the OS example (CI p.52, p.56). (1) Same symbols, e is the relative half-width. (2) $p = 658/6000 = 0.1097$, 95%, $e = 0.005$. (3) CI half-width $1.96\sqrt{0.1097 \times 0.8903/6000} = 1.96 \times 0.004034 = 0.0079$, giving (0.1018, 0.1176). Sample size $n = 1.96^2 \times 0.1097 \times 0.8903/(0.005 \times 0.1097)^2 = 3.8416 \times 0.8903/(0.1097 \times 0.000025) \approx 1.247 \times 10^6$. (4) About 1.25 million interrupts, and at 10 ms each that's 12,471 s, or 3.46 hours of running.

Variance (normal data only)

$\left(\dfrac{(n-1)s^2}{\chi^2_{1-\alpha/2,\,n-1}},\ \dfrac{(n-1)s^2}{\chi^2_{\alpha/2,\,n-1}}\right)$. The larger χ^2 quantile goes in the left denominator.

Example. (1) s^2 is the sample variance, $n-1$ the df, $\chi^2_{q,\,n-1}$ the value with area q below it. (2) $n = 15$ runs, $s^2 = 4.0$ ms^2, 95%. (3) $(n-1)s^2 = 14 \times 4 = 56$. The χ^2 table row 14 gives 26.12 in column 0.975 and 5.63 in column 0.025. Then $56/26.12 = 2.14$ and $56/5.63 = 9.95$. (4) $\sigma^2 \in (2.14,\ 9.95)$ ms^2, so $\sigma \in (1.46,\ 3.15)$ ms. Fifteen runs pin down a variance much worse than a mean.

4. Hypothesis testing (Lecture 9)

Steps $\quad$ state H_0 and H_1, choose α and one or two sides BEFORE the data, compute the statistic, compare with the critical value or find p, then report the effect with an interval.

$Z = \dfrac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}, \quad t = \dfrac{\bar{x} - \mu_0}{s/\sqrt{n}}$ with $n-1$ df. Two-sided, reject if $|Z| > z[\alpha/2]$. One-sided, reject if $Z > z[\alpha]$. A two-sided test at α rejects exactly when the $(1-\alpha)$ CI excludes μ_0.

Example, Z (HT p.10). (1) $\bar{x}$ is the sample mean, μ_0 the mean under H_0, σ the known SD. (2) $n = 50$ files, $\bar{x} = 12.3$ KB, $\sigma = 0.5$, $\mu_0 = 12.5$, two-sided at $\alpha = 0.05$. (3) $\sigma/\sqrt{n} = 0.5/7.071 = 0.0707$, $Z = (12.3 - 12.5)/0.0707 = -2.83$, and $|Z| > 1.96$. For p, the Φ table row 2.8, column .03 gives 0.9977, so one tail is 0.0023 and $p = 2 \times 0.0023 = 0.0046$. (4) Reject. The 95% CI $12.3 \mp 1.96 \times 0.0707 = (12.16,\ 12.44)$ excludes 12.5 too. The 0.2 KB gap is real, but whether it matters is a separate question.

Example, t (HT p.12). (1) Same, with the sample SD s in place of σ. (2) $n = 5$, $\bar{x} = 12.3$, $s = 0.5$, $\mu_0 = 12.35$. (3) $s/\sqrt{n} = 0.5/2.236 = 0.2236$, $t = -0.05/0.2236 = -0.224$. The t table row 4, column .025 gives 2.776. (4) $|t| < 2.776$, so don't reject ($p = 0.83$). The CI (11.68, 12.92) is so wide that five files couldn't have shown a difference.

p is the probability of a statistic at least this extreme IF H_0 is true. It's not the probability that H_0 is true.

	H_0 true	H_0 false
Reject	Type I (α)	correct (power $1-\beta$)
Don't reject	correct	Type II (β)

Runs needed to detect a difference $\delta \quad n = \left(\dfrac{(z[\alpha/2] + z[\beta])\,\sigma}{\delta}\right)^2$.

At $\alpha = 0.05$ and power 0.80, $(1.96 + 0.84)^2 \approx 7.8$, so $n \approx 7.8\,\sigma^2/\delta^2$.

Example. (1) σ is the SD, δ the smallest true difference you want to catch, $z[\beta]$ sets the power. (2) $\sigma = 10$, $\delta = 4$, $\alpha = 0.05$ two-sided, power 0.80 so $z[0.20] = 0.84$. (3) $(1.96 + 0.84) \times 10/4 = 2.8 \times 2.5 = 7$, and $7^2 = 49$. (4) 49 runs give an 80% chance of rejecting when the true shift is 4. Halving δ to 2 needs $4 \times 49 = 196$.

Not rejecting is not proof of no difference. With m tests at α each, expect about $m\alpha$ false positives, and Bonferroni uses α/m per test. Holm sorts the p values and compares the k-th smallest with $\alpha/(m-k+1)$, stopping at the first failure. Benjamini-Hochberg compares the k-th smallest with $k\alpha/m$ and controls the false discovery rate. Correlated samples count for less, e.g. 400 samples with $n_{\text{eff}} = 100$ halve Z. Statistical significance says nothing about whether the effect matters, so give the effect size and its CI.

Example. (1) $p_{(k)}$ is the k-th smallest p value, m the number of tests. (2) $m = 6$, sorted p values 0.001, 0.008, 0.012, 0.030, 0.041, 0.200, $\alpha = 0.05$. (3) Bonferroni's cutoff $0.05/6 = 0.0083$ passes the first two. Holm's cutoffs $0.05/6, 0.05/5, 0.05/4, 0.05/3$ are 0.0083, 0.010, 0.0125, 0.0167, so it passes three and stops at 0.030. BH's cutoffs $k(0.05)/6$ are 0.0083, 0.0167, 0.025, 0.0333, 0.0417, 0.05, and the largest k that passes is 5 ($0.041 \leq 0.0417$). (4) Bonferroni rejects 2, Holm 3, BH 5. Same data, different answers, so pick the

procedure first.

5. Comparing two alternatives (Lecture 10)

Pick the test

1. Same unit (workload, benchmark) under both? Yes means paired, no means unpaired.
2. Paired with roughly normal differences, paired t on the differences.
3. Paired with outliers, Wilcoxon signed-rank if symmetric, sign test or bootstrap if skewed.
4. Unpaired and roughly normal, Welch's t.
5. Unpaired and skewed or ordinal, Mann-Whitney U or bootstrap.

For paired tests, normality is about the differences, not A or B.

Paired

$d_i = a_i - b_i, \quad \bar{d} \mp t[\alpha/2,\, n-1]\dfrac{s_d}{\sqrt{n}}, \quad n$ is the number of PAIRS.

Pairing removes the shared variation $2\rho\sigma_A\sigma_B$ from the variance of the difference, so it helps whenever $\rho > 0$. For instance with $\sigma_A = \sigma_B = 20$ and $\rho = 0.99$, $\mathrm{Var}(A-B) = 400+400-2(0.99)(20)(20) = 800 - 792 = 8$, not 800.

Example (Comp p.11). (1) d_i is A minus B on workload i, $\bar{d}$ and s_d are the mean and SD of the d_i, n the number of workloads. (2) Six workloads, $\bar{d} = 0.020$, $s_d = 0.0743$, 90%. (3) SE $= 0.0743/\sqrt{6} = 0.0743/2.449 = 0.0303$. The t table row 5, column .05 gives 2.015, so the half-width is $2.015 \times 0.0303 = 0.061$. (4) $0.020 \mp 0.061 = (-0.041,\ 0.081)$. It contains 0, so six workloads can't tell the policies apart, and the width is the real finding.

Unpaired, Welch

$$s = \sqrt{\frac{s_A^2}{n_A} + \frac{s_B^2}{n_B}}, \quad (\bar{x}_A - \bar{x}_B) \mp t[\alpha/2,\, \nu]\, s,$$

$$\nu = \frac{\left(\frac{s_A^2}{n_A} + \frac{s_B^2}{n_B}\right)^2}{\frac{(s_A^2/n_A)^2}{n_A - 1} + \frac{(s_B^2/n_B)^2}{n_B - 1}}$$

Use ν as is or round DOWN (more conservative). It lies between $\min(n_A, n_B) - 1$ and $n_A + n_B - 2$. Pooled t assumes equal variances and uses $n_A + n_B - 2$ df. Welch is the default.

Example. (1) s_A^2, s_B^2 are the sample variances, n_A, n_B the run counts, ν the effective df. (2) A ran 8, 8, 10, 12, 12 ms and B ran 9, 12, 15, 15, 18, 21 ms, so $\bar{x}_A = 10$, $\bar{x}_B = 15$, $s_A^2 = 16/4 = 4$, $s_B^2 = 90/5 = 18$. (3) $s_A^2/n_A = 0.8$ and $s_B^2/n_B = 3.0$, so $s = \sqrt{3.8} = 1.949$. $\nu = 3.8^2/\big(0.8^2/4 + 3^2/5\big) = 14.44/(0.16+1.80) = 7.37$, which is between 4 and 9, and rounds down to 7. The t table row 7, column .025 gives 2.365, so the half-width is $2.365 \times 1.949 = 4.61$. (4) $-5 \mp 4.61 = (-9.61,\ -0.39)$ ms. Zero is outside, so A is faster, by somewhere between 0.4 and 9.6 ms.

Reading results

If the CI for the difference contains 0, you can't say they differ at that level. Overlapping single-system CIs do NOT mean no difference (HW6 Miller example), and checking overlap is not a test. Runs needed paired $n = \big((z[\alpha/2] + z[\beta])\, s_d/\delta\big)^2$, unpaired per group $n = 2\big((z[\alpha/2] + z[\beta])\, s/\delta\big)^2$.

Example (Comp p.25). (1) s_d is the SD of the differences, s the per-system SD, δ the difference to detect. (2) $s_d = 0.0743$, $s = 0.12$, $\delta = 0.05$, $\alpha = 0.05$, power 0.80, so $1.960 + 0.842 = 2.802$. (3) Paired $(2.802 \times 0.0743/0.05)^2 = 4.164^2 = 17.3$. Unpaired $2(2.802 \times 0.12/0.05)^2 = 2 \times 6.725^2 = 90.4$. (4) Round up to 18 pairs against 91 runs per group, so pairing cuts the work about fivefold.

6. Measurement (Lecture 11)

A CI describes sampling error and nothing else. Repetition shrinks random error but never systematic error such as bias, a wrong clock or perturbation. Strategies event counting gives exact totals with tiny overhead but no context, tracing gives full detail and order but the most perturbation and data, sampling sets the overhead you choose but is statistical and misses rare events. Exactness, context and low overhead, you get two. Report the clock and its resolution. Finer observation means more perturbation. Timer rollover after $(2^n - 1)\, T_c$ for an n-bit counter with tick T_c. A single reading is off by up to one tick, so time k repetitions and divide, $(T_k - T_{\text{loop}})/k$. Overhead share of a measurement is about $2c/(T + 2c)$ for read cost c.

Example, rollover. (1) n is the counter's bit width, T_c the time per tick. (2) 16 bits, $T_c = 10$ ns. (3) $(2^{16} - 1) \times 10 = 65{,}535 \times 10 = 655{,}350$ ns. (4) It wraps after about 655 μs, so it can't time anything longer without wrap handling.

Example, loop timing. (1) T_k is the time for k repetitions, T_{loop} the empty loop's time. (2) $k = 10{,}000$, $T_k = 520$ μs, $T_{\text{loop}} = 120$ μs on a 1 μs clock. (3) $(520 - 120)/10{,}000 = 400/10{,}000 = 0.04$ μs. (4) 40 ns per operation, far below the 1 μs tick.

Example, overhead. (1) c is the cost of one clock read, T the event's time. (2) $c = 20$ ns, $T = 40$ ns. (3) $2c/(T+2c) = 40/(40+40) = 0.5$. (4) Half the reading is the timer itself. Timing 1000 operations between one pair of reads drops it to $40/40{,}040 \approx 0.1\%$.

7. Plagiarism and authorship (full research topics in Part 2)

Plagiarism per Luke's TR includes copied text without quotes even when cited, close paraphrase, reused figures, and self-plagiarism of your own published text. Cite ideas, quote words. Authorship goes to people who made an intellectual contribution.

8. More formulas from the handouts and study guides

Metrics

$T_e = \dfrac{n \times \text{CPI}}{f_{\text{clock}}}$ (not on the slides). GFLOPS and TFLOPS divide by 10^9 and 10^{12}. Jain's fairness $J = \dfrac{(\sum x_i)^2}{n \sum x_i^2}$ (HW2 only).

Example, T_e. (1) n is the instruction count, CPI the cycles per instruction, f_{clock} the clock rate. (2) $n = 3 \times 10^9$ on A at 2 GHz with CPI 4 and on B at 1 GHz with CPI 1.5. (3) $T_A = 3 \times 10^9 \times 4/(2 \times 10^9) = 6$ s, $T_B = 3 \times 10^9 \times 1.5/10^9 = 4.5$ s. (4) B wins with half the clock rate, so clock rate alone misleads.

Example, Jain. (1) x_i is flow i's throughput, n the number of flows. (2) Two flows get 90 and 10 Mbps. (3) $(90+10)^2/\big(2(8100+100)\big) = 10000/16400$. (4) $J \approx 0.61$, where 1 is a fair split and $1/n = 0.5$ is one flow taking everything.

Summarizing

Quantile q sits at position $r = nq + 0.5$ in the sorted data, interpolate between $x_{(\lfloor r \rfloor)}$ and the next value. *Example.* (1) n is the sample size, q the quantile, r the position. (2) Sorted 2, 3, 5, 6, 8, 9, 12, 20, $q = 0.9$. (3) $r = 8(0.9) + 0.5 = 7.7$, so go 0.7 of the way from $x_{(7)} = 12$ to $x_{(8)} = 20$, $12 + 0.7(8) = 17.6$. (4) The 90th percentile is 17.6. The same rule gives $r = 2.5$ and $Q_1 = 4$ for $q = 0.25$, matching section 2.

Tukey fences $Q_1 - 1.5\,\text{IQR}$ and $Q_3 + 1.5\,\text{IQR}$, beyond them is an outlier.

Example (Summ p.50). (1) Q_1, Q_3 are the quartiles. (2) The 22 I/O times have $Q_1 = 6.08$, $Q_3 = 8.33$, min 4.26, max 12.74. (3) $\text{IQR} = 8.33 - 6.08 = 2.25$ and $1.5 \times 2.25 = 3.375$, so the fences are $6.08 - 3.375 = 2.705$ and $8.33 + 3.375 = 11.705$. (4) 12.74 is past the upper fence, so it's an outlier, and 4.26 is inside.

Work-weighted harmonic mean uses $w_i = F_i / \sum F_j$.

Example. (1) F_i is job i's work, x_i its rate. (2) Job A does 300 Mops at 100 Mops/s and job B does 100 Mops at 50 Mops/s. (3) $w_A = 300/400 = 0.75$, $w_B = 0.25$, so $0.75/100 + 0.25/50 = 0.0075 + 0.005 = 0.0125$ and $\bar{x}_H = 1/0.0125 = 80$. (4) 80 Mops/s, which matches 400 Mops in $3 + 2 = 5$ s. The unweighted har-

monic mean 66.7 is wrong because the jobs did different work. Inverse CDFs for Q-Q plots, $u = f_i$ exponential with mean a, $-a\ln(1-u)$ (rate $\lambda = 1/a$) Pareto, $k(1-u)^{-1/a}$ geometric, $\lceil \ln(1-u)/\ln(1-p) \rceil$ (Summ p.64 writes $\ln u$, which is the random-variate form) normal, $z \approx 4.91\,[q^{0.14} - (1-q)^{0.14}]$, then $m + s\,z$. Feitelson puts the data on the x-axis, Setia and HW3 on the y-axis, so slope rules read reversed. $\ln 0.9 = -0.1054$, $\ln 0.7 = -0.3567$, $\ln 0.5 = -0.6931$, $\ln 0.3 = -1.2040$, $\ln 0.1 = -2.3026$.

Example, exponential. (1) a is the guessed mean, $u = f_i$ the plotting position. (2) $a = 2$, $n = 5$, so the largest point has $u = (5 - 0.5)/5 = 0.9$. (3) $-2\ln(1-0.9) = -2\ln 0.1 = -2(-2.3026) = 4.605$. (4) The largest value plots against 4.605. Do all five and check the points fall on a line.

Confidence intervals

Batch means with m batches $\bar{b} \mp t[\alpha/2,\, m-1]\, s_b/\sqrt{m}$.

Example. (1) $\bar{b}$ is the mean of the batch means, s_b their SD, m the number of batches. (2) Batch means 10, 14, 12, 16, so $m = 4$, 95%. (3) $\bar{b} = 52/4 = 13$, deviations $-3, 1, -1, 3$, squares sum to 20, $s_b^2 = 20/3 = 6.667$, $s_b = 2.582$, SE $= 2.582/2 = 1.291$. The t table row 3, column .025 gives 3.182, so the half-width is 4.108. (4) 95% CI (8.89, 17.11), wide because 3 df costs a lot.

Sample size scales as $n_2 = n_1(h_1/h_2)^2$ for half-widths h.

Example. (1) h_1 is the half-width you got from n_1 runs, h_2 the one you want. (2) ± 1.4 ms from 25 runs, want ± 0.5. (3) $25(1.4/0.5)^2 = 25 \times 2.8^2 = 25 \times 7.84 = 196$. (4) 196 runs.

Proportion size with ABSOLUTE error e $n = z_{\alpha/2}^2\, p(1-p)/e^2$, use $p = 0.5$ with no pilot (95%, $e = 0.03$ gives 1068).

Worked out. (1) e is the half-width in fraction points. (2) $z = 1.96$, $p = 0.5$, $e = 0.03$. (3) $1.96^2 \times 0.25/0.03^2 = 3.8416 \times 0.25/0.0009 = 0.9604/0.0009 = 1067.1$. (4) Round up to 1068 trials for ± 3 points at 95%. Log transform build the CI on $\ln x$, then (e^L, e^U) is an interval for the median, centred on the geometric mean. Bootstrap percentile 95% CI is the 2.5th and 97.5th percentiles of the B resampled statistics. With B = 10,000 sorted resample means, that's roughly the 251st and 9750th values. Monte Carlo $\bar{Y}(n) = (b-a)\frac{1}{n}\sum g(X_i)$, Var $=$ Var$(Y)/n$.

Example. (1) g is the function to integrate over $[a, b]$, X_i uniform draws on $[a, b]$. (2) $g(x) = x^2$ on $[0, 2]$ with draws 0.5, 1.0, 1.5. (3) g values 0.25, 1, 2.25 average $3.5/3 = 1.167$, times $b - a = 2$ gives 2.333. (4) The estimate is 2.33 against the true $8/3 = 2.67$. More draws shrink the error like $1/\sqrt{n}$.

Hypothesis testing

Two proportions $Z = \dfrac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})(1/n_1 + 1/n_2)}}$, pooled $\hat{p} = \dfrac{m_1 + m_2}{n_1 + n_2}$.

Example. (1) m_i is the success count in group i, n_i its size, $\hat{p}_i = m_i/n_i$. (2) A converts 144 of 800 (0.18), B converts 176 of 800 (0.22). (3) Pooled $\hat{p} = 320/1600 = 0.20$, SE $= \sqrt{0.2 \times 0.8 \times (2/800)} = \sqrt{0.0004} = 0.02$, $Z = 0.04/0.02 = 2.0$. The Φ table row 2.0, column .00 gives 0.9772, so $p = 2 \times 0.0228 = 0.0456$. (4) Reject at 0.05. B converts better.

McNemar (paired yes/no) $Z = \dfrac{b - c}{\sqrt{b + c}}$, only the disagreements b and c count.

Example. (1) b counts items only A got right, c items only B got right. (2) Two classifiers on the same 500 items, $b = 38$, $c = 20$. (3) $Z = 18/\sqrt{58} = 18/7.62 = 2.36$. Row 2.3, column .06 gives 0.9909, so $p = 2 \times 0.0091 = 0.018$. (4) Reject at 0.05. A is better on the items where they disagree.

Paired $t = \bar{d}/(s_d/\sqrt{n})$. Power $\approx P\big[Z > z[\alpha/2] - |\delta|\sqrt{n}/\sigma\big]$.

Example, power. (1) δ is the true shift, σ the SD, n the runs. (2) $\sigma = 10$, $n = 25$, $\delta = 4$, $\alpha = 0.05$ two-sided. (3) $|\delta|\sqrt{n}/\sigma = 4 \times 5/10 = 2$, so power $\approx P[Z > 1.96 - 2] = P[Z > -0.04]$. Row 0.0, column .04 gives 0.5160. (4) Power 0.52, so this experiment misses a real 4-unit effect about half the time.

Family-wise error $1 - (1-\alpha)^m$, e.g. 23% at $m = 5$, 92% at $m = 50$. Simultaneous intervals use $z[\alpha/(2m)]$.

Example. (1) m is the number of tests or intervals. (2) $m = 5$ tests at $\alpha = 0.05$, and $m = 4$ simultaneous 95% intervals. (3) $0.95^2 = 0.9025$, $0.95^4 = 0.8145$, $0.95^5 = 0.7738$, so FWER $= 1 - 0.7738 = 0.226$. For the intervals, $z[0.05/8] = z[0.00625]$ needs $\Phi = 0.99375$, which falls between row 2.4, column .09 (0.9936) and row 2.5, column .00 (0.9938), so $z \approx 2.50$. (4) A 23% chance of at least one false positive, and each of the four intervals uses 2.50 in place of 1.96.

Correlated samples $n_{\text{eff}} = n\dfrac{1-\rho}{1+\rho}$. Pocock cutoffs for peeking 2.18 (2 looks), 2.41 (5 looks).

Example. (1) n is the raw sample count, ρ the lag-1 correlation. (2) 400 consecutive latencies with $\rho = 0.6$ and a naive $Z = 3.0$. (3) $n_{\text{eff}} = 400 \times 0.4/1.6 = 400 \times 0.25 = 100$, so the SE grows by $\sqrt{400/100} = 2$ and $Z = 3.0/2 = 1.5$. (4) Only 100 samples' worth of information, and the result is no longer significant.

Chi-square goodness of fit $\sum (O-E)^2/E$ with $k - m$ df (m includes the 1 for the total plus fitted parameters).

Example. (1) O is the observed count per bin, E the expected count under H_0, k the number of bins. (2) 120 requests over 4 servers came out 36, 24, 33, 27, and H_0 is an even split, so $E = 30$ each. (3) $(36 + 36 + 9 + 9)/30 = 90/30 = 3.0$, with $4 - 1 = 3$ df. The χ^2 table row 3, column 0.95 gives 7.81. (4) $3.0 < 7.81$, so don't reject. The counts fit an even split.

Comparing alternatives

Pooled $s_p^2 = \dfrac{(n_A - 1)s_A^2 + (n_B - 1)s_B^2}{n_A + n_B - 2}$, Cohen's $d = (\bar{x}_A - \bar{x}_B)/s_p$, paired $d_z = \bar{d}/s_d$.

Example. (1) s_A^2, s_B^2 are the sample variances, n_A, n_B the sizes. (2) The Welch data from section 5, $n_A = 5$, $s_A^2 = 4$, $n_B = 6$, $s_B^2 = 18$, $\bar{x}_A - \bar{x}_B = -5$. (3) $s_p^2 = (4 \times 4 + 5 \times 18)/9 = 106/9 = 11.78$, $s_p = 3.432$, $d = -5/3.432 = -1.46$. (4) A's mean sits about 1.5 pooled SDs below B's, a large effect.

Signed-rank, large n $\mu_W = n(n+1)/4$, $\sigma_W^2 = n(n+1)(2n+1)/24$.

Example (Comp p.22). (1) W is the smaller rank sum, n the number of nonzero differences. (2) Six cache workloads, $W^+ = 15$, $W^- = 6$, so $W = 6$. (3) $\mu_W = 6 \times 7/4 = 10.5$, $\sigma_W^2 = 6 \times 7 \times 13/24 = 22.75$, $\sigma_W = 4.77$, $Z = (6 - 10.5)/4.77 = -0.94$. Row 0.9, column .04 gives 0.8264, so $p \approx 2 \times 0.1736 = 0.35$. (4) Not significant, and at $n = 6$ the slide says to report the exact $p = 0.44$.

Sign test $S \sim \text{Bin}(n, \frac{1}{2})$, $z = (S - n/2)/(\sqrt{n}/2)$.

Example. (1) S is the number of positive differences out of n nonzero ones. (2) $n = 20$ pairs, $S = 15$. (3) $z = (15 - 10)/(\sqrt{20}/2) = 5/2.236 = 2.24$. Row 2.2, column .04 gives 0.9875, so $p = 2 \times 0.0125 = 0.025$. (4) Reject at 0.05. One system wins in clearly more than half the pairs.

Mann-Whitney $U_1 = W_1 - n_1(n_1 + 1)/2$, $\mu_U = n_1 n_2/2$, $\sigma_U^2 = n_1 n_2(n_1 + n_2 + 1)/12$, $A_{12} = U_1/(n_1 n_2)$.

Example (Comp p.20). (1) W_1 is group 1's rank sum in the pooled ranking, n_1, n_2 the group sizes. (2) Crash intervals, $n_1 = 15$, $n_2 = 14$, $W_1 = 162$. (3) $U_1 = 162 - 15 \times 16/2 = 162 - 120 = 42$, $\mu_U = 15 \times 14/2 = 105$, $\sigma_U^2 = 15 \times 14 \times 30/12 = 525$, $\sigma_U = 22.91$, $Z = (42 - 105)/22.91 = -2.75$. Row 2.7, column .05 gives 0.9970, so $p = 2 \times 0.003 = 0.006$. (4) Reject at 0.01. $A_{12} = 42/210 = 0.20$, so a System I interval beats a System II one only 20% of the time.

All pairs of c systems $1 - (1-\alpha)^{c(c-1)/2}$ chance of a false positive. For instance $c = 5$ gives $5 \times 4/2 = 10$ pairs and $1 - 0.95^{10} = 1 - 0.599 = 0.40$.

Measurement

Wrapped counter ticks = end + 2^n − start. Instrumentation overhead $(T_{\text{inst}} - T_{\text{base}})/T_{\text{base}}$.

Example. (1) n is the counter width, start and end the two readings. (2) A 16-bit counter with 10 ns ticks reads 65,000 then 1,200, and end < start means it wrapped once. (3) $1200 + 65{,}536 - 65{,}000 = 1736$ ticks, $1736 \times 10 = 17{,}360$ ns. (4) 17.36 μs. Naive subtraction gives a negative time. For overhead, 50.0 s plain and 53.0 s traced gives $3.0/50.0 = 6\%$.

Ratio error $(\sigma_R/R)^2 \approx (\sigma_A/A)^2 + (\sigma_B/B)^2$. Unpaired SE of a difference $\sqrt{SE_A^2 + SE_B^2}$.

Example. (1) $R = A/B$, and σ_A/A, σ_B/B are the relative errors. (2) Baseline 10.0 s and new 8.0 s, each with a 3% relative SE, so $R = 1.25$. (3) $\sqrt{0.03^2 + 0.03^2} = \sqrt{0.0018} = 0.0424$, $\sigma_R = 0.0424 \times 1.25 = 0.053$, $1.96 \times 0.053 = 0.10$. (4) The speedup is 1.25 ± 0.10, so "25% faster" really means 15% to 35%. For a difference, SEs 0.894 and 1.732 give $\sqrt{0.8 + 3.0} = 1.949$, less than their sum 2.626.

Nested runs, p processes with k iterations each $\text{Var}(\bar{x}) = \sigma_p^2/p + \sigma_i^2/(pk)$, cost $c_p p + c_i pk \le B$. Extra processes beat extra iterations.

Example. (1) σ_p is the SD between processes, σ_i between iterations inside one process. (2) $\sigma_p = 6$, $\sigma_i = 3$ ms, budget 36 measurements. (3) 1×36 gives $36/1 + 9/36 = 36.25$, 6×6 gives $6 + 0.25 = 6.25$, 36×1 gives $1 + 0.25 = 1.25$. (4) SEs 6.02, 2.50 and 1.12 ms, so 36 short processes give about a fifth of the SE of one long process.

Student t critical values, upper-tail area

$t[\alpha/2,\, \nu]$ for a two-sided $(1-\alpha)$ CI, e.g. 95% uses the .025 column, 90% uses .05.

df	.10	.05	.025	.01	.005
1	3.078	6.314	12.706	31.821	63.657
2	1.886	2.920	4.303	6.965	9.925
3	1.638	2.353	3.182	4.541	5.841
4	1.533	2.132	2.776	3.747	4.604
5	1.476	2.015	2.571	3.365	4.032
6	1.440	1.943	2.447	3.143	3.707
7	1.415	1.895	2.365	2.998	3.499
8	1.397	1.860	2.306	2.896	3.355
9	1.383	1.833	2.262	2.821	3.250
10	1.372	1.812	2.228	2.764	3.169
11	1.363	1.796	2.201	2.718	3.106
12	1.356	1.782	2.179	2.681	3.055
13	1.350	1.771	2.160	2.650	3.012
14	1.345	1.761	2.145	2.624	2.977
15	1.341	1.753	2.131	2.602	2.947
16	1.337	1.746	2.120	2.583	2.921
17	1.333	1.740	2.110	2.567	2.898
18	1.330	1.734	2.101	2.552	2.878
19	1.328	1.729	2.093	2.539	2.861
20	1.325	1.725	2.086	2.528	2.845
21	1.323	1.721	2.080	2.518	2.831
22	1.321	1.717	2.074	2.508	2.819
23	1.319	1.714	2.069	2.500	2.807
24	1.318	1.711	2.064	2.492	2.797
25	1.316	1.708	2.060	2.485	2.787
26	1.315	1.706	2.056	2.479	2.779
27	1.314	1.703	2.052	2.473	2.771
28	1.313	1.701	2.048	2.467	2.763
29	1.311	1.699	2.045	2.462	2.756
30	1.310	1.697	2.042	2.457	2.750
35	1.306	1.690	2.030	2.438	2.724
40	1.303	1.684	2.021	2.423	2.704
50	1.299	1.676	2.009	2.403	2.678
60	1.296	1.671	2.000	2.390	2.660
80	1.292	1.664	1.990	2.374	2.639
100	1.290	1.660	1.984	2.364	2.626
120	1.289	1.658	1.980	2.358	2.617
∞	1.282	1.645	1.960	2.326	2.576

Square roots

2	1.414	27	5.196	52	7.211	77	8.775
3	1.732	28	5.292	53	7.280	78	8.832
4	2.000	29	5.385	54	7.348	79	8.888
5	2.236	30	5.477	55	7.416	80	8.944
6	2.449	31	5.568	56	7.483	81	9.000
7	2.646	32	5.657	57	7.550	82	9.055
8	2.828	33	5.745	58	7.616	83	9.110
9	3.000	34	5.831	59	7.681	84	9.165
10	3.162	35	5.916	60	7.746	85	9.220
11	3.317	36	6.000	61	7.810	86	9.274
12	3.464	37	6.083	62	7.874	87	9.327
13	3.606	38	6.164	63	7.937	88	9.381
14	3.742	39	6.245	64	8.000	89	9.434
15	3.873	40	6.325	65	8.062	90	9.487
16	4.000	41	6.403	66	8.124	91	9.539
17	4.123	42	6.481	67	8.185	92	9.592
18	4.243	43	6.557	68	8.246	93	9.644
19	4.359	44	6.633	69	8.307	94	9.695
20	4.472	45	6.708	70	8.367	95	9.747
21	4.583	46	6.782	71	8.426	96	9.798
22	4.690	47	6.856	72	8.485	97	9.849
23	4.796	48	6.928	73	8.544	98	9.899
24	4.899	49	7.000	74	8.602	99	9.950
25	5.000	50	7.071	75	8.660	100	10.000
26	5.099	51	7.141	76	8.718		

Standard normal $\Phi(z) = P(Z \le z)$

$P(Z > z) = 1-\Phi(z)$ and $\Phi(-z) = 1-\Phi(z)$. Critical values $z[.05] = 1.645$, $z[.025] = 1.960$, $z[.01] = 2.326$, $z[.005] = 2.576$, $z[.20] = 0.842$.

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990
3.1	.9990	.9991	.9991	.9991	.9992	.9992	.9992	.9992	.9993	.9993
3.2	.9993	.9993	.9994	.9994	.9994	.9994	.9994	.9995	.9995	.9995
3.3	.9995	.9995	.9995	.9996	.9996	.9996	.9996	.9996	.9996	.9997
3.4	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9998

χ^2 quantiles, $P(\chi^2_\nu \le x)$

df	0.005	0.025	0.05	0.95	0.975	0.995
1	0.00	0.00	0.00	3.84	5.02	7.88
2	0.01	0.05	0.10	5.99	7.38	10.60
3	0.07	0.22	0.35	7.81	9.35	12.84
4	0.21	0.48	0.71	9.49	11.14	14.86
5	0.41	0.83	1.15	11.07	12.83	16.75
6	0.68	1.24	1.64	12.59	14.45	18.55
7	0.99	1.69	2.17	14.07	16.01	20.28
8	1.34	2.18	2.73	15.51	17.53	21.95
9	1.73	2.70	3.33	16.92	19.02	23.59
10	2.16	3.25	3.94	18.31	20.48	25.19
11	2.60	3.82	4.57	19.68	21.92	26.76
12	3.07	4.40	5.23	21.03	23.34	28.30
13	3.57	5.01	5.89	22.36	24.74	29.82
14	4.07	5.63	6.57	23.68	26.12	31.32
15	4.60	6.26	7.26	25.00	27.49	32.80
16	5.14	6.91	7.96	26.30	28.85	34.27
17	5.70	7.56	8.67	27.59	30.19	35.72
18	6.26	8.23	9.39	28.87	31.53	37.16
19	6.84	8.91	10.12	30.14	32.85	38.58
20	7.43	9.59	10.85	31.41	34.17	40.00
21	8.03	10.28	11.59	32.67	35.48	41.40
22	8.64	10.98	12.34	33.92	36.78	42.80
23	9.26	11.69	13.09	35.17	38.08	44.18
24	9.89	12.40	13.85	36.42	39.36	45.56
25	10.52	13.12	14.61	37.65	40.65	46.93
26	11.16	13.84	15.38	38.89	41.92	48.29
27	11.81	14.57	16.15	40.11	43.19	49.64
28	12.46	15.31	16.93	41.34	44.46	50.99
29	13.12	16.05	17.71	42.56	45.72	52.34
30	13.79	16.79	18.49	43.77	46.98	53.67
40	20.71	24.43	26.51	55.76	59.34	66.77
50	27.99	32.36	34.76	67.50	71.42	79.49
60	35.53	40.48	43.19	79.08	83.30	91.95

Recipes, decision rules and traps

How to use these pages. Find the question type in Part 1 to pick the method, then follow the numbered recipe in Part 2 and copy the boxed answer line, filling the blanks. Part 3 is for doing the arithmetic without a calculator. Part 4 lists every trap from the guide in one place. Part 5 is the definitions list for true/false questions. Tables of z, t, χ^2 and square roots are elsewhere in the packet. Remember the χ^2 table there is indexed by lower-tail area, so an upper 5% point is the 0.95 column.

Part 1. Decision rules

1.1 Which center, and which mean (Summ p.18, p.25, p.32)

1. Categorical data (instruction types, file types)? Use the **mode**.
2. Outliers present and you want a "typical" value? Use the **median**. If the sum of the values means something (total time, capacity planning), you still want a mean.
3. Otherwise apply the **work/rate test**. Ask what the summary should do if every raw time doubled.
   - It should double, because the values are times, sizes or costs whose sum is physical. Use the **arithmetic** mean.
   - It should halve, because the values are rates (anything "per second", MIPS, MFLOPS, bandwidth, tps). Use the **harmonic** mean. It equals total work over total time when every run did the same work (Summ p.28).
   - Runs did different work? Use the **weighted harmonic** mean with $w_i = F_i / \sum F_j$ (Summ p.33).
   - Effects multiply (successive % gains, per-layer gains, hit ratios across cache levels). Use the **geometric** mean (Summ p.30).
4. Values matter unequally (a request mix of 70/20/10)? Use the **weighted arithmetic** mean with weights summing to 1.
5. Speedups of phases with equal baseline time? The harmonic mean of the speedups is the true overall speedup, not the geometric.
6. Given speedups but the raw times exist? Summarize the times (total time), not the ratios.

Check that $x_H < x_G < x_A$ for positive data that aren't all equal.

1.2 Which dispersion index (Summ p.49)

1. Bounded quantity (hit ratio in $[0, 1]$)? Use the **range**.
2. Unbounded, unimodal and symmetric? Use **COV** (or s). Never COV if the mean is zero, negative or near zero (Summ p.41).
3. Otherwise (skewed, heavy tail, outliers)? Use **percentiles or SIQR**.
4. Claim about the tail or an SLO? Report p95 or p99.
5. Claim about how precisely the mean is known? Report the SE or a CI, and label it.
6. Never use the range to argue stability, since it grows with n.

1.3 z or t (CI p.27, p.31, p.37, HT p.11)

1. σ known? Use z.
2. σ unknown and $n > 30$? Use z with s in place of σ.
3. σ unknown, $n \le 30$, data roughly normal? Use t with $n - 1$ df.
4. Small and clearly skewed? Transform (log), bootstrap, or a rank test (CI p.68-70, HT p.11).
5. Proportions always use z (normal approximation to the binomial).
6. Sample-size formulas use z, since n and hence df are unknown.
7. Batch means use t with $m - 1$ df, m = number of batches.
8. Paired data use t with (pairs -1) df. Welch uses t with ν.

1.4 Comparing two systems (Comp p.5, p.17)

1. Does every measurement of A have a partner of B on the same unit (benchmark, trace, input, user, machine)? Yes means **paired**. Two independent lists, often of different lengths, means **unpaired**. Equal lengths alone don't make data paired.
2. Measuring under A changes the unit for B (cache pollution, learning)? Don't pair.
3. Paired, differences roughly normal. **Paired** t on d_i.
4. Paired, outliers but symmetric differences. **Wilcoxon signed-rank**.
5. Paired, skewed differences. **Sign test** or bootstrap, or log ratios then paired t (Comp p.18).
6. Unpaired, roughly normal. **Welch**, chosen in advance (Comp p.13). Use **pooled** only if the question insists on equal variances (df $n_A + n_B - 2$).
7. Unpaired, skewed or ordinal. **Mann-Whitney U** or bootstrap.
8. Yes/no outcomes. Unpaired gives the two-proportion interval or test. Paired (same test items) gives McNemar or the $\pm 1/0$ scoring.
9. Three or more systems. Omnibus test first (ANOVA, or Friedman if non-normal), then Tukey, Dunnett or a correction (Comp p.28).
10. Claim about p99? Compare p99s (Comp p.18, p.27).

1.5 One-sided or two-sided (HT p.5, p.9)

1. The research question decides, before the data. "Is A faster than B?" is one-sided. "Does the patch change latency?" is two-sided.
2. H_0 stated with no direction (just $\mu = \mu_0$)? Two-sided.
3. The claim always goes in H_1. The equality goes in H_0.

α	two-sided $z[\alpha/2]$	one-sided $z[\alpha]$
0.10	1.645	1.282
0.05	1.960	1.645
0.01	2.576	2.326

For H_1: $\mu < \mu_0$ reject when $Z < -z[\alpha]$. A 90% two-sided interval uses the 0.05 column, a 95% one uses 0.025.

1.6 Which interval for a proportion (CI p.47-49)

1. Compute np and $n(1-p)$. Both at least about 10? Use the Wald interval $p \pm z\sqrt{p(1-p)/n}$ with z, never t.
2. Either below 10, or the interval goes below 0 or above 1? Wald fails. Say Wilson for $n < 40$, Agresti-Coull for $n > 40$ (CI p.49).
3. Two independent proportions? Variances add, $\sqrt{p_1 q_1/n_1 + p_2 q_2/n_2}$.
4. Same items under both (two classifiers, two LLMs on one test set)? Paired. Score $+1/-1/0$ and use the paired interval, or McNemar.

1.7 How many runs

Goal	Formula (round up)
Mean, $\pm e$ relative (CI p.40)	$n = \left(\frac{z_{\alpha/2}\, s}{e\, \bar{x}}\right)^2$
Mean, $\pm e$ absolute	$n = (z_{\alpha/2}\, s/e)^2$
Proportion, relative (CI p.54)	$n = \frac{z^2_{\alpha/2}\, p(1-p)}{(ep)^2}$
Proportion, absolute	$n = z^2_{\alpha/2}\, p(1-p)/e^2$, use $p = 0.5$ if unknown
Test, one sample or paired, power $1-\beta$ (HT p.19, Comp p.25)	$n = \left(\frac{(z[\alpha/2] + z[\beta])\, \sigma}{\delta}\right)^2$, with s_d for paired
Unpaired, per group	$2\times$ the above with raw s
Rescale an interval you have	$n_2 = n_1 (h_1/h_2)^2$

One-sided test, use $z[\alpha]$ for $z[\alpha/2]$. Power 0.80 gives $z[0.20] = 0.842$ (0.84), and 0.90 gives 1.282. At $\alpha = 0.05$ and power 0.80, $(1.96 + 0.84)^2 = 7.84$, so $n \approx 7.8(\sigma/\delta)^2$. Halving δ or the half-width costs $4\times$ the runs. "Within 7%" on CI p.42 is a total width,

so $e = 0.035$. Read whether e is relative or absolute before plugging in.

1.8 Normality fails or data are dependent (CI p.70)

1. Observations correlated (simulation output, consecutive requests)? **Batching**, the only one that handles dependence. Batch in time order, never sorted.
2. Want the median, or the process is multiplicative? **Log transform**, and the back-transformed interval is for the median.
3. Want the mean, independent observations? **Bootstrap** (BCa better than plain percentiles).

1.9 Counting, tracing or sampling (Meas p.7-8)

Need an exact total of a frequent event and no context? Counting. Need context for a rare event? Tracing, ideally triggered by a threshold. No hypothesis yet and need fixed overhead? Sampling. "Exactness, context, overhead, you get two."

Part 2. Recipes, one per problem type

R1. Speedup and relative change (Lilja ch. 2)

1. Name the baseline (system 1) and the new system (system 2).
2. From times, $S_{2,1} = T_1/T_2$ (old over new). From rates, $S_{2,1} = R_2/R_1$.
3. Relative change $\Delta_{2,1} = (T_1 - T_2)/T_2 = S_{2,1} - 1$, as a percent.
4. If asked how much the time fell, that's $(T_1 - T_2)/T_1$, a different number.
5. $S < 1$ is a slowdown and Δ is negative.

Worked example. An old server runs a job in 12 s and the new server runs it in 8 s. How much faster is the new one, and by how much did the time fall?

1. Formula. Speedup $S_{2,1} = T_1/T_2$, where T_1 is the old (baseline) time and T_2 is the new time. Relative change $\Delta_{2,1} = (T_1 - T_2)/T_2$. Time cut $= (T_1 - T_2)/T_1$.

2. Plug in. $T_1 = 12$ s and $T_2 = 8$ s, so $S = 12/8$, $\Delta = (12 - 8)/8$ and the cut is $(12 - 8)/12$.

3. Work it out. $12/8 = 1.5$. Then $12 - 8 = 4$ and $4/8 = 0.5 = 50\%$. Check that $S - 1 = 1.5 - 1 = 0.5$ agrees. The cut is $4/12 = 1/3 = 0.333 = 33.3\%$.

4. Result. The new server is 1.5 times as fast, a 50% relative change, and its time fell by 33.3%. Both describe the same change measured from different baselines.

Answer line. *System 2 is $S_{2,1} =$ ___ times as fast as system 1, a relative change of ___%, while its execution time fell by ___% (baseline is system 1).*

R2. MIPS, MFLOPS, clock rate (Metrics p.16-19)

1. Put T_e in seconds. MIPS $= n/(T_e \times 10^6)$, MFLOPS $= f/(T_e \times 10^6)$.
2. If given CPI and clock, $T_e = n \times \text{CPI}/f_{\text{clock}}$.
3. Compute the real time ratio T_B/T_A and compare it with the MIPS ratio.
4. Explain the failure. MIPS misleads only when instruction counts differ (different ISA or compiler). MFLOPS depends on how operations like $\sqrt{}$ are counted and gives 0 for integer code.

Worked example. Machine A runs a program as 10^9 instructions in 5 s. Machine B's compiler turns the same program into 3×10^8 instructions that run in 4 s. Which is faster, and what do the MIPS ratings say?

1. Formula. MIPS $= n/(T_e \times 10^6)$, where n is the number of instructions executed and T_e is the execution time in seconds. Real speedup of B over A is T_A/T_B.

2. Plug in. A gives $10^9/(5 \times 10^6)$. B gives $3 \times 10^8/(4 \times 10^6)$. Speedup is 5/4.

3. Work it out. For A, $10^9/10^6 = 1000$ and $1000/5 = 200$ MIPS. For B, $3 \times 10^8/10^6 = 300$ and $300/4 = 75$ MIPS. The MIPS ratio is $200/75 = 2.67$ in A's favor. The time ratio is $5/4 = 1.25$ in B's favor.

4. Result. MIPS says A is 2.67 times faster, but B really finishes 1.25 times faster (4 s against 5 s). MIPS gets it backwards because each of B's instructions does more work.

Answer line. *MIPS rates A at ___ and B at ___, but B finishes in ___ s against A's ___ s, so B is faster. MIPS is unreliable here because the instruction counts differ, so an instruction is a different amount of work on each machine (Metrics p.8, p.17).*

R3. Each kind of mean (Summ p.21-23, p.33)

1. Pick the mean with rule 1.1 and say why in one clause.
2. Arithmetic, $\sum x_i/n$.
3. Harmonic, $n/\sum(1/x_i)$. Use a common denominator, e.g. $1/20 + 1/30 + 1/60 = 6/60$.
4. Geometric, $(\prod x_i)^{1/n}$. Convert percents to factors first ($10\% \to 1.10$), take the root, subtract 1 at the end. Take the root by guess and check (Part 3).
5. Weighted arithmetic, $\sum w_i x_i$ with $\sum w_i = 1$ (divide by the total if not).
6. Weighted harmonic, $1/\sum(w_i/x_i)$, no n on top, w_i = share of total work.
7. Check against totals. Harmonic of rates should equal total work over total time.

Worked example. Three 600 MB files copy at 20, 30 and 60 MB/s. What's the average copy rate?

1. Formula. These are rates and every run did the same work (600 MB), so use the harmonic mean $x_H = n/\sum(1/x_i)$, where n is the number of rates and x_i is each rate.

2. Plug in. $x_H = 3/(1/20 + 1/30 + 1/60)$.

3. Work it out. Use the common denominator 60. $1/20 = 3/60$, $1/30 = 2/60$ and $1/60 = 1/60$, so the sum is $6/60 = 0.1$. Then $3/0.1 = 30$. Check with totals. The times are $600/20 = 30$ s, $600/30 = 20$ s and $600/60 = 10$ s, 60 s in all. Total work is $3 \times 600 = 1800$ MB, and $1800/60 = 30$ MB/s, which matches. The arithmetic mean would be $(20 + 30 + 60)/3 = 110/3 = 36.7$.

4. Result. The average rate is 30 MB/s, the one steady rate that copies all three files in the same 60 s. The arithmetic 36.7 overstates it.

Answer line. *These are [times / rates / compounding ratios], so the [arithmetic / harmonic / geometric] mean is appropriate (Summ p.32). The mean is ___ [units]. (Arithmetic would overstate a rate at ___.)*

R4. Variance, SD and COV by hand (Summ p.39-41)

1. Make a table with columns x_i and x_i^2. Sum both.
2. $\bar{x} = \sum x_i/n$.
3. Shortcut $s^2 = \dfrac{\sum x_i^2 - (\sum x_i)^2/n}{n-1} = \dfrac{n\sum x_i^2 - (\sum x_i)^2}{n(n-1)}$.
4. $s = \sqrt{s^2}$ (Part 3). COV $= s/\bar{x}$.
5. Check with deviations if time allows. Deviations must sum to 0.

Worked example. Five run times are 2, 4, 4, 5 and 10 s. Find the mean, variance, SD and COV.

1. Formula. $\bar{x} = \sum x_i/n$ and $s^2 = \left(\sum x_i^2 - (\sum x_i)^2/n\right)/(n-1)$, where x_i is each value and n is the count. Then $s = \sqrt{s^2}$ and COV $= s/\bar{x}$.

2. Plug in. $n = 5$.

x_i	2	4	4	5	10	sum 25
x_i^2	4	16	16	25	100	sum 161

3. Work it out. $\bar{x} = 25/5 = 5$. $(\sum x_i)^2/n = 625/5 = 125$. $s^2 = (161 - 125)/4 = 36/4 = 9$. $s = \sqrt{9} = 3$. COV $= 3/5 = 0.6$. Check with deviations $-3, -1, -1, 0, 5$. They sum to 0 and their squares $9 + 1 + 1 + 0 + 25 = 36$ match the 36 above.

4. Result. Mean 5 s, $s^2 = 9$ s^2, $s = 3$ s, COV 0.6. A typical run is about 3 s from the mean, which is 60% of the mean, so the runs are widely spread.

Answer line. $\bar{x}$ = ___, s^2 = ___ *(divisor $n-1$)*, s = ___ *[units]*, *COV* = ___.

R5. Quartiles and percentiles with $(i-0.5)/n$ (Summ p.59)

1. Sort. Label $x_1 \le \cdots \le x_n$.
2. Position $r = np + 0.5$. Let $j = \lfloor r \rfloor$.
3. $Q(p) = x_j + (r-j)(x_{j+1} - x_j)$. If $r \le 1$ take x_1, if $r \ge n$ take x_n.
4. $Q_1 = Q(0.25)$, median $= Q(0.5)$, $Q_3 = Q(0.75)$.
5. IQR $= Q_3 - Q_1$, SIQR $=$ IQR$/2$, midhinge $= (Q_1 + Q_3)/2$.
6. Tukey fences $Q_1 - 1.5\,\text{IQR}$ and $Q_3 + 1.5\,\text{IQR}$.

Worked example. Eight latencies are 2, 3, 5, 6, 8, 9, 12 and 20 ms, already sorted. Find Q_1, the median, Q_3, the SIQR and the 90th percentile, and say whether 20 is an outlier.

1. Formula. Position $r = np + 0.5$, where n is the count and p is the fraction wanted. With $j = \lfloor r \rfloor$, $Q(p) = x_j + (r-j)(x_{j+1} - x_j)$. Upper fence $= Q_3 + 1.5\,\text{IQR}$.

2. Plug in. $n = 8$. Q_1 has $r = 8(0.25) + 0.5$, the median $r = 8(0.5)+0.5$, Q_3 has $r = 8(0.75)+0.5$ and $Q(0.9)$ has $r = 8(0.9)+0.5$.

3. Work it out. Q_1 has $r = 2 + 0.5 = 2.5$, between $x_2 = 3$ and $x_3 = 5$, so $3 + 0.5(5-3) = 4$. The median has $r = 4.5$, between $x_4 = 6$ and $x_5 = 8$, so $6 + 0.5(2) = 7$. Q_3 has $r = 6.5$, between $x_6 = 9$ and $x_7 = 12$, so $9 + 0.5(3) = 10.5$. IQR $= 10.5 - 4 = 6.5$ and SIQR $= 6.5/2 = 3.25$. $Q(0.9)$ has $r = 7.2 + 0.5 = 7.7$, between $x_7 = 12$ and $x_8 = 20$, so $12 + 0.7(8) = 12 + 5.6 = 17.6$. The fence is $10.5 + 1.5(6.5) = 10.5 + 9.75 = 20.25$.

4. Result. $Q_1 = 4$, median 7, $Q_3 = 10.5$, SIQR 3.25 ms and p90 17.6 ms. The middle half of the latencies spans 6.5 ms. The max 20 is just under the fence 20.25, so Tukey's rule doesn't call it an outlier.

Answer line. *Using plotting positions $(i-0.5)/n$ with linear interpolation, Q_1 = ___, median = ___, Q_3 = ___, so IQR = ___ and SIQR = ___. The fence is ___, so ___ [is / is not] an outlier.*

R6. Histogram with Sturges (Summ p.54)

1. $k = 1 + \log_2 n$. Slide floors it, HW3 rounded. Say which.
2. Width $= (\max - \min)/k$. Bins $[a, a+w)$, last bin closed.
3. Count per bin, then relative frequency $=$ count$/n$.
4. Check counts sum to n and percents to 100.
5. Describe shape (peaks, symmetry, tail). Bin width changes the picture, so confirm with a Q-Q plot.

n	$1 + \log_2 n$	floor	round
16	5.00	5	5
20	5.32	5	5
30	5.91	5	6
50	6.64	6	7
64	7.00	7	7
100	7.64	7	8
1000	10.97	10	11

Worked example. The 12 values on Summ p.55 are −3.0, 0.8, 1.2, 1.5, 2.0, 2.3, 2.4, 3.3, 3.5, 4.0, 4.5 and 5.5. How many bins does Sturges give, and what does the histogram look like?

1. Formula. $k = 1 + \log_2 n$, where k is the number of bins and n is the number of values. Bin width $w = (\max - \min)/k$. Relative frequency $=$ count$/n$.

2. Plug in. $k = 1 + \log_2 12$ and $w = (5.5 - (-3.0))/k$.

3. Work it out. $2^3 = 8$ and $2^4 = 16$, so $\log_2 12$ is between 3 and 4 (it's 3.585, since $12 = 8 \times 1.5$). So $k = 4.585$, floored to 4 as on the slide (rounding gives 5). The range is $5.5 + 3.0 = 8.5$ and $w = 8.5/4 = 2.125$. Step from −3.0 by 2.125 to get edges −3.0, −0.875, 1.25, 3.375, 5.5.

Bin	Values	Count	Rel. freq
$[-3.0, -0.875)$	−3.0	1	$1/12 = 8.3\%$
$[-0.875, 1.25)$	0.8, 1.2	2	$2/12 = 16.7\%$
$[1.25, 3.375)$	1.5, 2.0, 2.3, 2.4, 3.3	5	$5/12 = 41.7\%$
$[3.375, 5.5]$	3.5, 4.0, 4.5, 5.5	4	$4/12 = 33.3\%$

Counts $1 + 2 + 5 + 4 = 12$ and the percents add to 100.

4. Result. Four bins of width 2.125. Most values sit in the top two bins and −3.0 sits alone, so the shape has a long left tail. Bin width changes this picture (Summ p.54 says cell size is critical), so check it with a Q-Q plot.

Answer line. *By Sturges' rule $k = 1 + \log_2$ ___ = ___, [floored / rounded] to ___ bins of width ___. The histogram is [unimodal, roughly symmetric / right-skewed / bimodal].*

R7. Normal Q-Q plot points (Summ p.63, p.69-73)

1. Sort the data. Compute $p_i = (i - 0.5)/n$.
2. Find z_i with $\Phi(z_i) \approx p_i$ from the z table (closest area). For $p_i < 0.5$ use $z_i = -z_{1-p_i}$.
3. Either plot (z_i, x_i), where intercept $\approx$ mean and slope $\approx$ SD (Summ p.70), or rescale to $\bar{x} + s z_i$ and compare with the 45° line (HW3 style).
4. Theory on the horizontal axis, data on the vertical.
5. Read it. Straight line means normal shape. Both ends steeper means longer tails, both ends flatter means shorter tails, a bow means skew (bowed up or convex means right skew) (Summ p.71-73).

n	z_i for $i = 1..n$ (upper half, negate for lower)
4	$\pm 0.32, \pm 1.15$
5	$0, \pm 0.52, \pm 1.28$
6	$\pm 0.21, \pm 0.67, \pm 1.38$
8	$\pm 0.16, \pm 0.49, \pm 0.89, \pm 1.53$
10	$\pm 0.13, \pm 0.39, \pm 0.67, \pm 1.04, \pm 1.645$

Exponential with mean a uses $-a \ln(1 - p_i)$. Slope not 1 means wrong scale, not wrong family.

Worked example. Are the run times 2, 4, 4, 5 and 10 s plausibly normal? From R4, $\bar{x} = 5$ and $s = 3$.

1. Formula. Plotting position $p_i = (i - 0.5)/n$, where i is the rank in sorted order and n is the count. z_i is the z whose table area $\Phi(z_i)$ is closest to p_i. The rescaled theory value is $\bar{x} + s z_i$. Plot the pairs (theory, data) against the line $y = x$.

2. Plug in. $n = 5$, so $p_i = 0.5/5,\ 1.5/5,\ 2.5/5,\ 3.5/5,\ 4.5/5$. The rescaled value is $5 + 3z_i$.

3. Work it out. The positions are 0.1, 0.3, 0.5, 0.7 and 0.9. For 0.9, the z table at row 1.2, column 0.08 gives $\Phi(1.28) = 0.8997$, the closest area, so $z = 1.28$. For 0.7, row 0.5, column 0.02 gives 0.6985, closer than 0.7019 at 0.53, so $z = 0.52$. For 0.5, $z = 0$. By symmetry 0.1 and 0.3 give −1.28 and −0.52. For $i = 1$, $5 + 3(-1.28) = 5 - 3.84 = 1.16$.

i	x_i	p_i	z_i	$5 + 3z_i$	data vs line
1	2	0.1	−1.28	1.16	above
2	4	0.3	−0.52	3.44	above
3	4	0.5	0	5.00	below
4	5	0.7	0.52	6.56	below
5	10	0.9	1.28	8.84	above

4. Result. The low end sits above the line, the middle dips below and the top point is above again. That's a convex bow, so the data look right-skewed. With only $n = 5$ this is weak evidence.

Answer line. *The points (theoretical normal quantile, ordered data) lie [close to a straight line / bowed / with steeper ends], so the data are [consistent with a normal distribution / right-skewed / heavier-tailed than normal]. With n = ___ this is weak evidence.*

R8. CI for the mean, z and t (CI p.31, p.37)

1. From raw data get $\bar{x}$ and s (R4).
2. Choose z or t with rule 1.3. For t, df $= n - 1$ and column $\alpha/2$.
3. SE $= s/\sqrt{n}$ (divide by $\sqrt{n}$, not n).
4. Half-width $h =$ (critical) $\times$ SE.
5. Interval $(\bar{x} - h, \bar{x} + h)$. Check it's centered on $\bar{x}$.

Worked example. Sixteen runs give a mean of 50 ms with $s = 8$ ms. Find a 95% CI for the true mean.

1. Formula. $\bar{x} \pm t[\alpha/2, n-1]\, s/\sqrt{n}$, where $\bar{x}$ is the sample mean, s is the sample SD and n is the number of runs. σ is unknown and $n \le 30$, so use t.

2. Plug in. $50 \pm t[0.025, 15] \times 8/\sqrt{16}$.

3. Work it out. $\sqrt{16} = 4$, so SE $= 8/4 = 2$ ms. The t table at row df 15, column 0.025 gives 2.131. $h = 2.131 \times 2 = 4.262$. Then $50 - 4.262 = 45.738$ and $50 + 4.262 = 54.262$.

4. Result. The 95% CI is $(45.738, 54.262)$ ms. If we repeated the 16-run experiment many times, about 95% of intervals built this way would contain the true mean.

Answer line. *With 95% confidence the mean ___ is between ___ and ___ [units] ($n =$ ___, $t[0.025, ___] =$ ___). The 95% describes the procedure, which captures μ in 95% of repeated experiments (CI p.33).*

R9. CI for a proportion (CI p.48)

1. $p = n_1/n$. Check $np \ge 10$ and $n(1-p) \ge 10$.
2. SE $= \sqrt{p(1-p)/n}$. When n is a perfect square, split as $\sqrt{p(1-p)}/\sqrt{n}$.
3. Half-width $z\times$ SE, with $z = 1.645$, 1.96 or 2.576.
4. Interval $p \pm h$. If it leaves $[0, 1]$ the Wald interval failed (rule 1.6).

Worked example. 80 of 400 sampled requests miss the cache. Give a 95% CI for the miss rate.

1. Formula. $p \pm z[0.025]\sqrt{p(1-p)/n}$, where p is the sample proportion and n is the number sampled.

2. Plug in. $p = 80/400$, and the interval is $p \pm 1.96\sqrt{p(1-p)/400}$.

3. Work it out. $p = 0.2$. Check $np = 80$ and $n(1-p) = 320$, both at least 10. $p(1-p) = 0.2 \times 0.8 = 0.16$. Since 400 is a perfect square, SE $= \sqrt{0.16}/\sqrt{400} = 0.4/20 = 0.02$. The z table area 0.975 sits at row 1.9, column 0.06, so $z = 1.96$. $h = 1.96 \times 0.02 = 0.0392$. Then $0.2 - 0.0392 = 0.1608$ and $0.2 + 0.0392 = 0.2392$.

4. Result. The interval is $(0.1608, 0.2392)$. With 95% confidence between 16.1% and 23.9% of requests miss the cache.

Answer line. *With 95% confidence between ___% and ___% of ___ are ___ ($p =$ ___, $n =$ ___, and np, $n(1-p) \ge 10$ so the normal approximation holds). Compare CI p.53, "95% certain processor spends 10.2-11.8% of its time in OS".*

R10. Sample size for a mean or a proportion (CI p.40-44, p.54-57)

1. Get a pilot $\bar{x}$ and s, or a pilot p (use 0.5 if none).
2. Decide whether e is relative (a fraction of the mean or of p) or absolute (data units or percentage points). Halve a total width.
3. Plug into the 1.7 table with z.
4. Square, then round **up**.
5. Convert to time if asked (samples ÷ rate, runs × minutes per run).

Worked example. (a) A pilot gives $\bar{x} = 20$ ms and $s = 5$ ms. How many runs give a 95% CI within ±2% of the mean?

1. Formula. $n = \big(z\, s/(e\,\bar{x})\big)^2$, where $z = z[0.025]$, s is the pilot SD, e is the relative error as a fraction and $\bar{x}$ is the pilot mean.

2. Plug in. $n = \big(1.96 \times 5/(0.02 \times 20)\big)^2$.

3. Work it out. $1.96 \times 5 = 9.8$ and $0.02 \times 20 = 0.4$. $9.8/0.4 = 24.5$. $24.5^2 = 576 + 24 + 0.25 = 600.25$. Round up to 601.

4. Result. 601 runs. The ±2% is relative, so the half-width is 0.4 ms.

(b) A pilot sample gives $p = 0.2$. How many samples give a 95% CI within ±10% of p (relative)?

1. Formula. $n = z^2 p(1-p)/(ep)^2$, where p is the pilot proportion and e is the relative error.

2. Plug in. $n = 3.8416 \times 0.2 \times 0.8/(0.1 \times 0.2)^2$.

3. Work it out. $0.2 \times 0.8 = 0.16$ and $3.8416 \times 0.16 = 0.614656$. $ep = 0.02$ and $0.02^2 = 0.0004$. $0.614656/0.0004 = 0.614656 \times 2500 = 1536.64$. Round up to 1537. If instead p is unknown and you want ±3 percentage points absolute, use $p = 0.5$. Then $3.8416 \times 0.25 = 0.9604$ and $0.9604/0.0009 = 0.9604 \times 1111.1 = 1067.1$, rounded up to 1068.

4. Result. 1537 samples pin p to 0.2 ± 0.02. The absolute ±3 points version needs 1068.

Answer line. *A pilot of ___ runs gave $s =$ ___ on a mean of ___. For a ±___% interval at ___% confidence we need $n = (___)^2 =$ ___, rounded up to ___ runs.*

R11. CI for a variance (CI p.59-62)

1. Data must be roughly normal. Say so.
2. Numerator $(n-1)s^2$.
3. df $= n - 1$. For 95%, χ^2_L = table column 0.025, χ^2_U = column 0.975.
4. Lower limit $= (n-1)s^2/\chi^2_U$ (big value). Upper limit $= (n-1)s^2/\chi^2_L$.
5. For σ, take square roots of both ends. The interval is not symmetric about s^2.

Worked example. Fifteen runs give $s^2 = 4$ ms^2, and the data look normal. Find 95% CIs for σ^2 and σ.

1. Formula. $\big((n-1)s^2/\chi^2_U,\ (n-1)s^2/\chi^2_L\big)$, where n is the number of runs, s^2 is the sample variance, and χ^2_U and χ^2_L are the table values in columns 0.975 and 0.025 with df $n-1$.

2. Plug in. $(n-1)s^2 = 14 \times 4 = 56$. The χ^2 table at row df 14 gives 26.12 in column 0.975 and 5.63 in column 0.025. So the interval is $(56/26.12,\ 56/5.63)$.

3. Work it out. 56/26.12 is 2 rem 3.76, and $3.76/26.12 \approx 0.144$, so 2.144. $56/5.63 = 9.947$. For σ take roots. $1.4^2 = 1.96$ and $1.5^2 = 2.25$, so $\sqrt{2.144} \approx 1.4 + 0.1(0.184/0.29) = 1.463$, and one Newton step gives 1.464. $\sqrt{9.947}$ is just under $\sqrt{10} = 3.162$, and the bracket $3 + 0.947(0.162) = 3.153$ plus Newton gives 3.154.

4. Result. σ^2 is between 2.14 and 9.95 ms^2 and σ is between 1.46 and 3.15 ms, with 95% confidence. The interval is lopsided around $s^2 = 4$, and it's only valid if the data are normal.

Answer line. *Assuming normal data, with 95% confidence σ^2 is between ___ and ___ [units^2], so σ is between ___ and ___ [units].*

R12. Hypothesis tests (HT p.15 order)

General steps. (1) State H_0 (equality) and H_1 (the claim) and the direction. (2) Choose α. (3) Choose n from power. (4) Choose the test. (5) Compute the statistic once. (6) Report verdict, p, effect with interval, and n.

Test	Statistic
Z, σ known or $n > 30$	$Z = (\bar{x} - \mu_0)/(\sigma/\sqrt{n})$
t, small n, normal	$t = (\bar{x} - \mu_0)/(s/\sqrt{n})$, df $n-1$
Paired	$t = \bar{d}/(s_d/\sqrt{n})$, df $n-1$
Two proportions	$Z = \dfrac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})(1/n_1 + 1/n_2)}}$, $\hat{p}$ pooled
McNemar	$Z = (b-c)/\sqrt{b+c}$, disagreements only
χ^2 goodness of fit	$\chi^2 = \sum(o_i - e_i)^2/e_i$

Chi-square GOF steps. (a) $e_i = n\times$ (claimed probability). (b) Merge neighbouring bins until every $e_i \ge 5$. (c) Sum $(o-e)^2/e$, dividing by expected. (d) df $= k - 1 -$ (parameters estimated from data). (e) Reject if χ^2 exceeds the upper α point (table column $1-\alpha$). Always upper tail.

Equivalence check. Two-sided Z or t rejects exactly when μ_0 is outside the $100(1-\alpha)\%$ interval (HT p.8-9). Compute both and make sure they agree.

Worked example. (a) HW5 Problem 1. Mean training time is claimed to be $\mu_0 = 2.5$ h with $\sigma = 0.8$ h known. 36 runs average 2.74 h. Test H_0: $\mu = 2.5$ at $\alpha = 0.05$.

1. Formula. $Z = (\bar{x} - \mu_0)/(\sigma/\sqrt{n})$, where $\bar{x}$ is the sample mean, μ_0 is the value H_0 claims, σ is the known SD and n is the number of runs. H_0 has no direction, so it's two-sided. Reject if $|Z| > z[0.025]$.

2. Plug in. $Z = (2.74 - 2.5)/(0.8/\sqrt{36})$.

3. Work it out. $\sqrt{36} = 6$, so SE $= 0.8/6 = 0.1333$. $2.74 - 2.5 = 0.24$. $Z = 0.24/0.1333 = 1.80$ (or $0.24 \times 6/0.8 = 1.44/0.8 = 1.80$). The z table area 0.975 at row 1.9, column 0.06 gives $z[0.025] = 1.96$. For p, row 1.8, column 0.00 gives $\Phi(1.80) = 0.9641$, so $p = 2(1 - 0.9641) = 2(0.0359) = 0.0718$. Interval check, $2.74 \pm 1.96(0.1333) = 2.74 \pm 0.2613 = (2.479, 3.001)$, which contains 2.5.

4. Result. $|Z| = 1.80 < 1.96$, so fail to reject H_0 ($p = 0.072$). The data don't show the mean differs from 2.5 h at the 5% level, but that isn't proof it equals 2.5.

(b) 200 requests land on 4 servers as 62, 48, 45 and 45. Is the load balancer splitting evenly? Use $\alpha = 0.05$.

1. Formula. $\chi^2 = \sum(o_i - e_i)^2/e_i$, where o_i is the observed count and $e_i = n\times$ (claimed probability) is the expected count. df $= k - 1$ with k bins.

2. Plug in. $e_i = 200 \times 0.25 = 50$ for each server, all at least 5. $\chi^2 = \big(12^2 + (-2)^2 + (-5)^2 + (-5)^2\big)/50$.

3. Work it out. $144 + 4 + 25 + 25 = 198$ and $198/50 = 3.96$. df $= 4 - 1 = 3$. The upper 5% point is the χ^2 table at row 3, column 0.95, which is 7.81.

4. Result. $3.96 < 7.81$, so fail to reject. The counts are consistent with an even split, which doesn't prove the split is even.

Answer line. *Since $|Z| =$ ___ $>$ $z[\alpha/2] =$ ___, reject H_0 at $\alpha =$ ___ ($p =$ ___). The 95% interval (___,___) excludes μ_0. The difference is ___ [units], and whether that matters is a separate question (HT p.10).*

Answer line. *Since $|t| =$ ___ $< t[\alpha/2,$___$] =$ ___, we fail to reject H_0 (___ $< p <$ ___). This isn't evidence for H_0. With $n =$ ___ we could not have seen a difference of that size (HT p.12, p.14).*

Answer line. *$\chi^2 =$ ___ with ___ df is [below / above] $\chi^2_{0.05} =$ ___, so we [fail to reject / reject] H_0. The counts are [consistent with / not consistent with] the claimed distribution, which doesn't prove it.*

R13. Bracketing a p-value from a table

1. Z. Look up $\Phi(|z|)$. One tail $= 1 - \Phi(|z|)$. Two-sided $p = 2(1 - \Phi(|z|))$.
2. t. Go to the df row. Find the two printed values that $|t|$ sits between. Their column headers are one-tail areas. That gives $a_1 < p_{\text{one}} < a_2$. Double both ends for two-sided.
3. $|t|$ beyond the last column gives $p_{\text{one}} < 0.005$. Below the first column gives $p_{\text{one}} > 0.10$.
4. χ^2. The packet table's headers are lower-tail areas, so the upper-tail p is 1− header. Find the two values bracketing your χ^2.
5. Compare with α. The verdict must match the critical-value verdict.

Worked example. A paired t test on 4 pairs gives $t = 3.0$. What's the two-sided p, and do we reject at $\alpha = 0.05$?

1. Formula. Two-sided $p = 2\,p_{\text{one}}$, where p_{one} is the area above $|t|$ under the t curve with df $=$ pairs -1. Bracket p_{one} by the table columns around $|t|$.

2. Plug in. df $= 4 - 1 = 3$ and $|t| = 3.0$.

3. Work it out. Row 3 of the t table reads 1.638 (0.10), 2.353 (0.05), 3.182 (0.025). 3.0 lies between 2.353 and 3.182, so $0.025 < p_{\text{one}} < 0.05$. Doubling gives $0.05 < p < 0.10$. Same verdict by critical value, $t[0.025, 3] = 3.182$ and $3.0 < 3.182$.

4. Result. $0.05 < p < 0.10$, so fail to reject at 0.05 (software gives exactly 0.058). The data lean toward a difference, but 4 pairs aren't enough to show it at the 5% level.

Answer line. *From the t table with ___ df, ___ $< |t| <$ ___, so ___ $< p <$ ___ (two-sided), [below / above] $\alpha =$ ___.*

R14. Power and runs needed (HT p.19)

1. SE $= \sigma/\sqrt{n}$. Shift in SEs $= |\delta|/\text{SE}$.
2. Two-sided power $\approx \Pr[Z > z[\alpha/2] - |\delta|\sqrt{n}/\sigma] = \Phi(|\delta|\sqrt{n}/\sigma - z[\alpha/2])$. The far tail is negligible.
3. $\beta = 1-$ power.
4. For n, use the 1.7 formula with $z[\beta] = 0.84$ for 80%. Double per group for unpaired. Round up.
5. Check by plugging n back into step 2.

Worked example. Run times have $\sigma = 10$ ms and we plan $n = 25$ runs, two-sided at $\alpha = 0.05$. What's the power to detect a change of $\delta = 4$ ms, and how many runs give 80% power?

1. Formula. Power $\approx \Phi\big(|\delta|\sqrt{n}/\sigma - z[\alpha/2]\big)$ and $n = \big((z[\alpha/2] + z[\beta])\,\sigma/\delta\big)^2$, where δ is the smallest change we care about, σ is the SD and β is the miss rate (1 − power).

2. Plug in. Power $= \Phi(4 \times \sqrt{25}/10 - 1.96)$. For 80% power, $n = \big((1.96 + z[0.20]) \times 10/4\big)^2$.

3. Work it out. $\sqrt{25} = 5$, SE $= 10/5 = 2$, and $4/2 = 2$ SEs of shift. $2 - 1.96 = 0.04$. The z table at row 0.0, column 0.04 gives $\Phi(0.04) = 0.5160$. For $z[0.20]$ find area 0.80, closest at row 0.8, column 0.04 (0.7995), so 0.84. $1.96 + 0.84 = 2.80$, $2.80 \times 10/4 = 7$ and $7^2 = 49$. Check, with $n = 49$ SE $= 10/7 = 1.429$, shift $= 4/1.429 = 2.80$ SEs, and $\Phi(2.80 - 1.96) = \Phi(0.84) = 0.7995 \approx 0.80$.

4. Result. With 25 runs the power is 0.516, so a real 4 ms change gets missed about 48% of the time. 49 runs give 80% power.

Answer line. *With $n =$ ___ the test has power about ___ to detect a difference of ___, so it misses a real effect that size ___% of the time. To detect $\delta =$ ___ with 80% power at $\alpha = 0.05$ we need $n =$ ___, rounded up.*

R15. Multiple comparisons (HT p.26-29)

1. Count tests m. Pairs of k systems $= k(k-1)/2$, times benchmarks, times metrics.
2. FWER $= 1 - (1 - \alpha)^m$ (independent), bound $m\alpha$ for any dependence. Compute 0.95^m by repeated squaring (Part 3).
3. **Bonferroni.** Reject if $p \le \alpha/m$.
4. **Holm.** Sort ascending. Compare $p_{(k)}$ with $\alpha/(m - k + 1)$, starting at α/m. Reject while it passes, stop at the first failure.
5. **BH.** Thresholds $k\alpha/m$. Find the *largest* k that passes, reject $p_{(1)}$ through $p_{(k)}$, even past earlier failures.
6. Simultaneous intervals use $z[\alpha/(2m)]$, e.g. 2.498 at $m = 4$, 2.81 at $m = 10$.

Lay it out as a table with columns k, $p_{(k)}$, α/m, $\alpha/(m - k + 1)$, $k\alpha/m$.

Worked example. Six tests give $p = 0.001$, 0.008, 0.012, 0.030, 0.041 and 0.200. At $\alpha = 0.05$, what's the chance of a false positive with no correction, and how many does each method reject?

1. Formula. FWER $= 1-(1-\alpha)^m$, where m is the number of tests. Bonferroni rejects when $p \le \alpha/m$. Holm compares the k-th smallest $p_{(k)}$ with $\alpha/(m - k + 1)$ and stops at the first failure. BH finds the largest k with $p_{(k)} \le k\alpha/m$.

2. Plug in. $m = 6$, $\alpha = 0.05$. FWER $= 1 - 0.95^6$ and $\alpha/m = 0.05/6$.

3. Work it out. $0.95^2 = 0.9025$, $0.95^4 = 0.9025^2 = 0.8145$ and $0.95^6 = 0.8145 \times 0.9025 = 0.7351$, so FWER $= 0.265$. $0.05/6 = 0.0083$.

k	$p_{(k)}$	Bonf. 0.0083	Holm $\alpha/(m-k+1)$	BH $k\alpha/m$
1	0.001	yes	0.0083 yes	0.0083 yes
2	0.008	yes	0.0100 yes	0.0167 yes
3	0.012	no	0.0125 yes	0.0250 yes
4	0.030	no	0.0167 no, stop	0.0333 yes
5	0.041	no		0.0417 yes
6	0.200	no		0.0500 no

4. Result. Without a correction there's a 26.5% chance of at least one false positive. Bonferroni rejects 2, Holm rejects 3 and BH rejects 5 (the largest passing k is 5).

Answer line. *With $m =$ ___ tests the chance of at least one false positive is $1 - 0.95^{\text{___}} =$ ___. Holm (FWER) rejects ___ hypotheses and BH (FDR) rejects ___. We use [Holm because a single false claim is costly / BH because we are screening].*

R16. Paired CI (Comp p.9)

1. Write the direction, $d_i = A_i - B_i$.
2. $\bar{d} = \sum d_i/n$, n = number of pairs.
3. $s_d^2 = (\sum d_i^2 - n\bar{d}^2)/(n-1)$.
4. SE $= s_d/\sqrt{n}$. Critical $t[\alpha/2, n-1]$.
5. Interval $\bar{d} \pm t$·SE. Does it contain 0?
6. Effect in units and relative to the mean.

Worked example. Five benchmarks each run on A and B. A takes 12, 15, 11, 14, 9 s and B takes 10, 11, 8, 9, 8 s. Find a 95% CI for the difference.

1. Formula. $\bar{d} \pm t[\alpha/2, n-1]\, s_d/\sqrt{n}$, where $d_i = A_i - B_i$, $\bar{d}$ is their mean, s_d is their SD and n is the number of pairs.

2. Plug in. $n = 5$.

d_i	2	4	3	5	1	sum 15
d_i^2	4	16	9	25	1	sum 55

3. Work it out. $\bar{d} = 15/5 = 3$. $s_d^2 = (55 - 5\times 3^2)/4 = (55-45)/4 = 2.5$. SE $= \sqrt{2.5/5} = \sqrt{0.5} = 0.7071$. The t table at row df 4, column 0.025 gives 2.776. $h = 2.776 \times 0.7071 = 1.963$. Then $3 - 1.963 = 1.037$ and $3 + 1.963 = 4.963$.

4. Result. The interval for $A - B$ is $(1.04, 4.96)$ s. Zero is outside, so A is slower by between 1.04 and 4.96 s per benchmark. That's 3 s on B's mean of 9.2 s, about 33% slower.

Answer line. *The interval for $A - B$ is (___, ___). [Zero is outside, so A is slower by between ___ and ___ / It contains 0, so these data do not establish a difference. It is compatible with A being ___ better and B being ___ better (Comp p.11).]*

R17. Welch CI, with ν by hand (Comp p.12)

1. Compute $\bar{x}_A, s_A^2, \bar{x}_B, s_B^2$.
2. Fill the table below. $a = s_A^2/n_A$ and $b = s_B^2/n_B$.
3. $s = \sqrt{a+b}$. This is already the SE of the difference.
4. $\nu = (a+b)^2/\big(a^2/(n_A-1) + b^2/(n_B-1)\big)$.
5. Check $\min(n_A, n_B) - 1 \le \nu \le n_A + n_B - 2$. Round ν down.
6. Interval $(\bar{x}_A - \bar{x}_B) \pm t[\alpha/2, \nu]\, s$.

Worked example. Independent response times are A = 8, 8, 10, 12, 12 ms and B = 9, 12, 15, 15, 18, 21 ms. Find the Welch 95% CI for $\mu_A - \mu_B$.

1. Formula. $(\bar{x}_A - \bar{x}_B) \pm t[\alpha/2, \nu]\, s$, where $s = \sqrt{a+b}$ with $a = s_A^2/n_A$ and $b = s_B^2/n_B$, and $\nu = (a+b)^2/\big(a^2/(n_A-1) + b^2/(n_B-1)\big)$. Here $\bar{x}$ is a group mean, s^2 a group variance and n a group size.

2. Plug in. A sums to 50, so $\bar{x}_A = 50/5 = 10$. Deviations $-2, -2, 0, 2, 2$ square to 16 in total, so $s_A^2 = 16/4 = 4$. B sums to 90, so $\bar{x}_B = 90/6 = 15$. Deviations $-6, -3, 0, 0, 3, 6$ square to $36 + 9 + 0 + 0 + 9 + 36 = 90$, so $s_B^2 = 90/5 = 18$.

3. Work it out. Fill the table. $\sqrt{3.8}$ is row 1 of the square-root example in Part 3.

	A	B	combine
n	5	6	
$\bar{x}$	10	15	diff -5
s^2	4	18	
$a, b = s^2/n$	0.8	3.0	sum 3.8
square a^2, b^2	0.64	9.00	$(a+b)^2 = 14.44$
$\div(n-1)$	0.16	1.80	sum 1.96

$s = \sqrt{3.8} = 1.949$, $\nu = 14.44/1.96 = 7.37 \to 7$
bounds check $4 \le 7.37 \le 9$, $t[0.025, 7] = 2.365$
$-5 \pm 2.365(1.949) = -5 \pm 4.61 = (-9.61, -0.39)$ ms

4. Result. The 95% interval for $\mu_A - \mu_B$ is $(-9.61, -0.39)$ ms. Zero is outside, so A is faster than B by between 0.39 and 9.61 ms.

Answer line. *By Welch, $\bar{x}_A - \bar{x}_B$ = ___, s = ___, ν = ___ (rounded down to ___), and the 95% interval is (___, ___). Zero is [outside, so ___ is faster by between ___ and ___ / inside, so no difference is shown]. Welch was chosen in advance (Comp p.13).*

R18. Pooled t (Comp p.13, study guide)

1. $s_p^2 = \dfrac{(n_A-1)s_A^2 + (n_B-1)s_B^2}{n_A + n_B - 2}$.
2. SE $= s_p\sqrt{1/n_A + 1/n_B}$.
3. df $= n_A + n_B - 2$. Interval $(\bar{x}_A - \bar{x}_B) \pm t$·SE.
4. Note the assumption $\sigma_A = \sigma_B$ and that Welch is the default.

Worked example. Same data as R17 ($n_A = 5$, $s_A^2 = 4$, $n_B = 6$, $s_B^2 = 18$, $\bar{x}_A - \bar{x}_B = -5$ ms). Find the pooled 95% CI.

1. Formula. $s_p^2 = \big((n_A-1)s_A^2 + (n_B-1)s_B^2\big)/(n_A + n_B - 2)$, where s_p^2 is the pooled variance. SE $= s_p\sqrt{1/n_A + 1/n_B}$. Interval $(\bar{x}_A - \bar{x}_B) \pm t[\alpha/2, n_A + n_B - 2]$·SE.

2. Plug in. $s_p^2 = (4\times 4 + 5\times 18)/(5+6-2)$ and SE $= s_p\sqrt{1/5 + 1/6}$, df $= 9$.

3. Work it out. $16 + 90 = 106$ and $106/9 = 11.78$. $\sqrt{11.78}$ is between $\sqrt{11} = 3.317$ and $\sqrt{12} = 3.464$, so $3.317 + 0.78(0.147) = 3.432$. $1/5 + 1/6 = 6/30 + 5/30 = 11/30 = 0.3667$, and $\sqrt{0.3667} = \sqrt{36.67}/10 = 0.6055$. SE $= 3.432 \times 0.6055 = 2.078$. The t table at row df 9, column 0.025 gives 2.262. $h = 2.262 \times 2.078 = 4.70$. Then $-5 \pm 4.70 = (-9.70, -0.30)$.

4. Result. The pooled interval is $(-9.70, -0.30)$ ms, a little wider than Welch's $(-9.61, -0.39)$. B's variance is $18/4 = 4.5$ times A's, above the old rule of about 4, so pooling is doubtful and Welch is the one to report.

Answer line. *Assuming equal variances, s_p = ___ and the pooled interval with ___ df is (___, ___). Here B's variance is ___ times A's, so the equal-variance assumption is doubtful and Welch is preferred.*

R19. Sign test, signed-rank and Mann-Whitney (Comp p.19-23)

Sign test. (1) Drop zero differences, n = remaining. (2) S = number positive. (3) Under H_0, $S \sim \text{Bin}(n, 1/2)$. Exact one tail $= \sum_{k \ge S} \binom{n}{k}/2^n$ (use the upper tail if $S > n/2$). Double for two-sided. (4) Large n, $z = (S - n/2)/(\sqrt{n}/2)$.

Signed-rank. (1) Drop zeros. (2) Rank $|d_i|$ from smallest, ties get the average rank. (3) W^+, W^- = rank sums of positive and negative. Check $W^+ + W^- = n(n+1)/2$. (4) W = min. Small n, exact $p = 2 \times$ (number of the 2^n sign patterns with rank sum $\le W)/2^n$. Above about 20 pairs, $\mu = n(n+1)/4$, $\sigma^2 = n(n+1)(2n+1)/24$. (5) Effect $r = (W^+ - W^-)/(n(n+1)/2)$.

With $n \le 5$ pairs neither test can reach $p < 0.05$, since the minimum is $2/32 = 0.0625$.

Mann-Whitney. (1) Rank all $n_1 + n_2$ together, ties averaged. (2) W_1 = rank sum of sample 1, $U_1 = W_1 - n_1(n_1+1)/2$. (3) $\mu_U = n_1 n_2/2$, $\sigma_U^2 = n_1 n_2(n_1 + n_2 + 1)/12$, $Z = (U_1 - \mu_U)/\sigma_U$ once both $n > 8$. (4) $A_{12} = U_1/(n_1 n_2)$. Check U_1 by counting pairs where sample 1 wins (ties count 0.5).

Worked example. (a) Sign test. A new version is faster on 9 of 10 benchmarks, with no ties. Is it consistently faster, two-sided at $\alpha = 0.05$?

1. Formula. Under H_0 the count of positives $S \sim \text{Bin}(n, 1/2)$, where n is the number of nonzero differences. One tail $= \sum_{k \ge S} \binom{n}{k}/2^n$, doubled for two-sided.

2. Plug in. $n = 10$ and $S = 9$, so one tail $= \big(\binom{10}{9} + \binom{10}{10}\big)/2^{10}$.

3. Work it out. $\binom{10}{9} = 10$ and $\binom{10}{10} = 1$, so 11. $2^{10} = 1024$. One tail $= 11/1024 = 0.0107$. Doubled, $22/1024 = 0.0215$.

4. Result. $p = 0.021 < 0.05$, so reject. The new version wins far more often than coin flips would.

(b) Signed-rank, the six-workload cache data on Comp p.22. The differences $d = A - B$ are $-0.03, -0.05, -0.06, +0.07, +0.09, +0.10$. Is there a difference?

1. Formula. Rank $|d_i|$ from smallest. W^+ and W^- are the rank sums of positive and negative d_i, and $W = \min(W^+, W^-)$. Exact two-sided $p = 2\times$ (sign patterns with rank sum $\le W)/2^n$. Effect $r = (W^+ - W^-)/(n(n+1)/2)$.

2. Plug in. $|d| = 0.03, 0.05, 0.06, 0.07, 0.09, 0.10$ get ranks 1 to 6 with signs $-, -, -, +, +, +$. $n = 6$.

3. Work it out. $W^+ = 4 + 5 + 6 = 15$ and $W^- = 1 + 2 + 3 = 6$. Check $15 + 6 = 21 = 6 \times 7/2$. $W = 6$. Sets of ranks from 1 to

6 summing to 6 or less are {}, {1}, {2}, {3}, {4}, {5}, {6}, {1,2}, {1,3}, {1,4}, {1,5}, {2,3}, {2,4}, {1,2,3}, which is 14. $2^6 = 64$, so $p = 2 \times 14/64 = 28/64 = 0.44$. $r = (15-6)/21 = 9/21 = 0.43$.

4. Result. $p = 0.44$, so fail to reject. Six workloads can't settle whether A and B differ.

(c) Mann-Whitney, the crash data on Comp p.20. System I has $n_1 = 15$, System II has $n_2 = 14$, and the rank sum of System I is $W_1 = 162$. Do the systems differ?

1. Formula. $U_1 = W_1 - n_1(n_1+1)/2$, $\mu_U = n_1 n_2/2$, $\sigma_U^2 = n_1 n_2(n_1+n_2+1)/12$, $Z = (U_1 - \mu_U)/\sigma_U$ and $A_{12} = U_1/(n_1 n_2)$.

2. Plug in. $U_1 = 162 - 15 \times 16/2$, $\mu_U = 15 \times 14/2$, $\sigma_U^2 = 15 \times 14 \times 30/12$.

3. Work it out. $15 \times 16/2 = 120$, so $U_1 = 42$. $\mu_U = 210/2 = 105$. $\sigma_U^2 = 6300/12 = 525$. $\sqrt{525} = 10\sqrt{5.25}$, and $\sqrt{5.25} \approx 2.236 + 0.25(0.213) = 2.289$, Newton 2.291, so $\sigma_U = 22.91$. $Z = (42-105)/22.91 = -63/22.91 = -2.75$. The z table at row 2.7, column 0.05 gives 0.9970, so $p = 2(1-0.9970) = 0.006$. $A_{12} = 42/210 = 0.20$.

4. Result. $p = 0.006$, so reject even at 0.01. $A_{12} = 0.20$ means a random System I gap between crashes beats a random System II gap only 20% of the time.

Answer line. *The [sign / signed-rank / Mann-Whitney] test gives ___ = ___ and p = ___, so we [reject / fail to reject] at α = ___. Effect size ___ = ___. The test was chosen from the design and the shape of the [differences / data], before seeing any p (Comp p.17).*

R20. Timer resolution, overhead, rollover

1. **Elapsed**, $T_e = (\text{stop} - \text{start}) \times T_c$ (Metrics p.23).
2. **Rollover**, $T_{\max} = (2^n - 1)T_c$. If end < start it wrapped once, ticks = end $+2^n-$ start. More than one wrap can't be detected. $2^{16} = 65{,}536$, $2^{32} \approx 4.295 \times 10^9$, $2^{64} \approx 1.845 \times 10^{19}$.
3. **Quantization**, a reading is off by up to one tick, relative error $\le T_c/T_e$. For error under fraction x the timed interval must be at least T_c/x.
4. **Overhead**, two reads of cost c bracketing T give overhead share $2c/(T+2c)$. Want T one to two orders of magnitude above c.
5. **Loop**, per-op $= (T_k - T_{\text{loop}})/k$, resolution T_c/k.
6. **Tick counting** for $T_e < T_c$, $p = m/n$ runs that saw a tick, $T_e \approx pT_c$, interval from R9 times T_c.
7. **Perturbation** $= (T_{\text{inst}} - T_{\text{base}})/T_{\text{base}}$. Not the sum of probe costs (Meas p.17).
8. **Ratio error**, $(\sigma_R/R)^2 \approx (\sigma_A/A)^2 + (\sigma_B/B)^2$.
9. **Levels**, $\text{Var}(\bar{x}) = \sigma_p^2/p + \sigma_i^2/(pk)$. Spend runs on processes.

Worked example. A 16-bit counter ticks every 10 ns. When does it wrap, and how long was an interval that read 65,000 at the start and 1,200 at the end? How big can the quantization error be? Separately, a timer read costs 20 ns and brackets a 40 ns operation. What share is overhead?

1. Formula. $T_{\max} = (2^n - 1)T_c$, where n is the number of bits and T_c the tick. After one wrap, ticks = end $+2^n-$ start. Quantization relative error $\le T_c/T_e$. Overhead share $= 2c/(T+2c)$, where c is the cost of one read and T the true time.

2. Plug in. $T_{\max} = (65{,}536 - 1) \times 10$ ns. Ticks $= 1{,}200 + 65{,}536 - 65{,}000$. Error ≤ 10 ns$/T_e$. Overhead $= 2(20)/(40+2(20))$.

3. Work it out. $65{,}535 \times 10 = 655{,}350$ ns $= 655.35$ µs. $65{,}536 - 65{,}000 = 536$, plus 1,200 gives 1,736 ticks, and $1{,}736 \times 10 = 17{,}360$ ns $= 17.36$ µs. Error $\le 10/17{,}360 = 0.00058 = 0.058\%$. Overhead $= 40/80 = 0.5$. The same wrap formula for a 32-bit counter at 1 µs gives $(2^{32} - 1) \times 1$ µs ≈ 4295 s, and $4295/60 \approx 71.6$ min.

4. Result. The 16-bit counter wraps after about 655 µs. The interval was 17.36 µs, good to within 0.06%. Timing the 40 ns operation on its own is 50% overhead, so time a loop of many operations instead.

Answer line. *The counter wraps after (2— − 1) × ___ = ___, so intervals longer than that can't be timed. One reading can be off by one tick, a relative error of up to ___%. Timer overhead is 2c/(T + 2c) = ___%, so [time it alone / time a loop of k = ___].*

R21. Sampling-profiler proportion (Meas p.7, CI p.48)

1. n = sampling rate × run length.
2. p = samples in f / n. Check $np, n(1-p) \ge 10$.
3. Interval $p \pm z\sqrt{p(1-p)/n}$. Use the $\sqrt{a \times 10^{-k}}$ trick (Part 3).
4. For a target e, decide relative or absolute, then R10. Run length $= n/\text{rate}$.
5. Faster sampling multiplies perturbation, and samples close together are correlated.

Worked example. A profiler samples at 500 Hz over a 20 s run, and 800 samples land in function f. What share of run time does f take, with a 95% CI?

1. Formula. n = rate × run length, p = samples in f/n, and the interval is $p \pm z[0.025]\sqrt{p(1-p)/n}$, where p is the sample proportion and n the number of samples.

2. Plug in. $n = 500 \times 20$, $p = 800/n$, interval $p \pm 1.96\sqrt{p(1-p)/n}$.

3. Work it out. $n = 10{,}000$ and $p = 800/10{,}000 = 0.08$. Check $np = 800$ and $n(1-p) = 9{,}200$, both at least 10. $0.08 \times 0.92 = 0.0736$, and $0.0736/10{,}000 = 7.36 \times 10^{-6}$. $\sqrt{7.36}$ is between $\sqrt{7} = 2.646$ and $\sqrt{8} = 2.828$, so $2.646 + 0.36(0.182) = 2.712$, and Newton gives 2.713. So SE $= 2.713 \times 10^{-3} = 0.002713$. $h = 1.96 \times 0.002713 = 0.00532$. Then $0.08 - 0.00532 = 0.0747$ and $0.08 + 0.00532 = 0.0853$.

4. Result. The interval is $(0.0747, 0.0853)$. With 95% confidence f takes between 7.5% and 8.5% of run time, and the profiler's 8% has a standard error of 0.27 points.

Answer line. *From n = ___ samples, f accounts for between ___% and ___% of run time with 95% confidence. The profiler's "___%" is an estimate with a standard error of ___ points.*

R22. Paper audit checklist

Intervals (CI Wild handout Part III). Is there an interval at all, and for which estimator? What's n and is it justified (pilot, target, confidence)? Is normality assumed, and which response (batching, log, bootstrap) was used? What is the independent unit, or is it pseudo-replication? Name one error the interval leaves out (bias, selection, confounding). Does the conclusion quietly need a credible interval?

Tests (HT Wild handout). What are α and β and which error costs more? Was the direction chosen in advance? Is there an effect size with an interval, or only p? How many tests (the family), and was there a correction? Any peeking or optional stopping? Would an interval have told the reader more than the verdict?

Comparisons (Comp p.26-27). Paired or unpaired, and does the analysis match the design? Test chosen before the data? Welch or pooled? Any conclusion from overlapping bars? Mean compared when the claim is about the tail? Are all comparisons reported?

Measurement (Meas p.30). Which clock at what resolution? Wall, CPU or steady state? How was warm-up chosen? How many repetitions at which level? Perturbation measured or asserted? What is systematic? Is the summary the quantity the claim is about? Open or closed loop for tail claims?

Reporting. Error bars labelled SD, SE or CI with n? Right mean for the data type? Geomean with its baseline and spread? Outlier rule stated with counts? Enough hardware, software and seeds to repeat (Meas p.29)?

Worked example. A paper says "Our prefetcher cuts mean page-load time by 18% ($p < 0.05$). Each configuration ran 5 times on one laptop, error bars shown." Walk the checklist.

Intervals. There's no interval on the 18%, only a p. $n = 5$ has no pilot or target behind it. All runs are on one laptop, so the unit of replication is one machine and the 5 runs are pseudo-replicates. Nothing covers bias like a warm browser cache.

Tests. α is 0.05 by default, with no word on power for $n = 5$. We aren't told if the direction was fixed first, how many pages or metrics were tested, or whether there was a correction.

Comparisons. The same pages load under both settings, so the design is paired. Check that the test was paired too, and that nobody

read significance off overlapping bars.

Measurement. Which clock, and was there a warm-up? Was the prefetcher's own overhead measured or just assumed small?

Reporting. The error bars don't say SD, SE or CI. An 18% figure on means hides the tail, which matters for page loads.

Result. The claim is weak. It rests on 5 runs on one machine with an unlabeled spread. I'd ask for a paired CI on the per-page difference across several machines, with n chosen from a pilot.

Answer line. *The paper reports ___ from ___ [units] with n = ___. The unit of replication is ___, so [the interval is too narrow / n is right]. The interval doesn't cover ___, a systematic error repetition can't fix. I'd ask for ___.*

Part 3. Arithmetic without a calculator

Square roots, bracket then one Newton step

1. Move the decimal point in **pairs** of digits until the number N lies between 1 and 100. Count the pairs.
2. Bracket N between two table entries and interpolate linearly to get x_0. Below 2, use known squares ($1.1^2 = 1.21$, $1.2^2 = 1.44$, $1.3^2 = 1.69$, $1.4^2 = 1.96$).
3. One Newton step $x_1 = (x_0 + N/x_0)/2$. This is good to about 4 digits.
4. Shift the decimal back by one place per pair moved.
5. Check $x_1^2 \approx N$.

Target	Bracket and x_0	Newton x_1
$\sqrt{3.8}$	$1.732 + 0.8(0.268) = 1.946$	1.949
$\sqrt{54.08}$	$7.348 + 0.08(0.068) = 7.353$	7.354
$\sqrt{0.000528}$	$5.28 \to 2.236 + 0.28(0.213) = 2.296$	$2.298 \to 0.02298$
$\sqrt{219.16}$	$2.1916 \to 1.414 + 0.19(0.318) = 1.475$	$1.480 \to 14.80$

Row 1 in full is $3.8/1.946 = 1.953$, then $(1.946 + 1.953)/2 = 1.949$. In the third row 0.000528 moved two pairs to 5.28, so the root moves back two places. In the fourth, 219.16 moved one pair the other way to 2.1916, so multiply the root by 10. Shortcuts are $\sqrt{a/b} = \sqrt{a}/\sqrt{b}$ when b is a perfect square ($\sqrt{0.16/400} = 0.4/20$), and $\sqrt{n}$ for common n, $\sqrt{5} = 2.236$, $\sqrt{6} = 2.449$, $\sqrt{8} = 2.828$, $\sqrt{10} = 3.162$, $\sqrt{50} = 7.071$.

Other roots (geometric means)

Guess and check by powering. Cube root of 0.36 is between 0.71 ($0.71^3 = 0.3579$) and 0.72 (0.3732), interpolate $0.71 + 0.01 \times (0.0021/0.0153) = 0.7114$. A square root of a product is the usual route for $n = 2$, and a fourth root is two square roots.

Dividing

1. Scale both to whole numbers first, $0.24/0.1333 = 2400/1333 \approx 1.80$.
2. Dividing by a small decimal is multiplying by its reciprocal, $\div 0.0004 = \times 2500$, $\div 0.0009 = \times 1111.1$, $\div 0.000025 = \times 40{,}000$.
3. Handy reciprocals $1/1.96 = 0.5102$, $1/3 = 0.3333$, $1/6 = 0.1667$, $1/7 = 0.1429$, $1/9 = 0.1111$, $1/12 = 0.0833$.
4. Estimate by whole multiples, then the remainder, 56/26.119 is 2 rem 3.762, and $3.762/26.119 \approx 0.144$, so 2.144.
5. Always multiply back to check.

Squares and powers

$(a + b)^2 = a^2 + 2ab + b^2$, so $24.5^2 = 576 + 24 + 0.25 = 600.25$. Memorize $1.96^2 = 3.8416$, $1.645^2 = 2.706$, $2.576^2 = 6.636$, $(1.96{+}0.84)^2 = 7.84$, $2.802^2 = 7.851$. Powers by repeated squaring, $0.95^2 = 0.9025$, $0.95^4 = 0.8145$, $0.95^5 = 0.7738$, $0.95^6 = 0.7351$, $0.95^{10} = 0.5987$, $0.95^{20} = 0.3585$. $\log_2$ from $2^{10} = 1024$, so $\log_2 1000$ is just under 10.

Sanity checks before you write the answer

- Deviations from the mean sum to 0. Batch means average to the grand mean.
- s is positive and has the data's units. Variance is in squared units.
- A mean CI is centered on $\bar{x}$. A variance CI is lopsided, lower end from the big χ^2.
- A proportion interval stays in $[0, 1]$, otherwise Wald failed.
- $x_H < x_G < x_A$. A weighted mean lies between the smallest and largest value.
- Welch ν lies between $\min(n_A, n_B) - 1$ and $n_A + n_B - 2$.
- $W^+ + W^- = n(n+1)/2$. $U_1 + U_2 = n_1 n_2$.
- Histogram counts sum to n, relative frequencies to 1.
- Test verdict matches whether μ_0 (or 0) is in the interval.
- Sample sizes are rounded up and are whole numbers.
- A t critical value is bigger than the matching z. Bigger confidence means wider.
- Probabilities, power and p are in $[0, 1]$. Percent vs fraction is stated.
- Units are converted (ms to s, minutes to seconds) before dividing.

Part 4. Easy to get wrong, master list

Intro and metrics

- The course focus is Feitelson's third definition, rigorous evaluation, not building systems.
- A hypothesis needs a metric and a direction or it can't be refuted.
- The CS loop on Intro p.11 starts from an idea, not an observation.
- Normalize counts to a rate before comparing runs of different lengths, and use one time unit.
- Rates averaged across runs get the harmonic mean.
- Linear and easy to use are the two non-required properties.
- Repeatable is same number twice. Reliable is ranking systems like their true performance.
- Consistent means same definition across machines, not same value across runs.
- "Easy to use" (p.10) and "easy to measure" (p.15-25) are the same property.
- Keep the 10^6 in MIPS and MFLOPS and put T_e in seconds.
- MIPS misleads only when instruction counts differ. Say why it fails.
- Metrics p.14 has a typo. The belief is "2 GHz is always better than 1 GHz".
- Clock rate is "reliable ×(ish)", often right within one family.
- MFLOPS fails independent and consistent. MIPS keeps independent.
- SPEC normalized value is reference over measured, bigger is better.
- Execution time is only ≈ repeatable, so repeat and report mean and variance.
- SPEC is means-based by the study guide, though closer to ends than MIPS.
- Wall times vary but CPU times don't means outside interference. Report both.
- Speedup from times is old over new, T_1/T_2.
- A 33% cut in time is a 50% relative change in performance.
- Relative change isn't symmetric, +25% one way is −20% the other.
- Speedup below 1 is a slowdown with negative relative change.
- Recall divides by all relevant items (found plus missed).
- High precision with low recall is easy to produce.
- Objective QoS (PSNR, jitter) and subjective (MOS) capture different things.
- An ends-based metric can still fail required properties (QoE not consistent).
- Extra criteria need a reason, not a bare list.

Summarizing data

- Sort before the median, and average the two middles for even n.
- The median isn't always better. The mean matches totals.
- A sample mean is an estimate, not the true mean.
- Don't average rates arithmetically.
- Convert percents to factors before a geometric mean.
- Plain harmonic mean equals total work over time only with equal work per run.
- Equal baseline time phases, harmonic mean of speedups gives the true speedup.
- Geomeans against different baselines can't be compared, and geomeans over successes only hide failures.
- Weights must sum to 1. Weighted harmonic is $1/\sum(w_i/x_i)$, no n.
- Divide by $n - 1$, not n.
- Don't report variance beside the mean. Use s.
- COV is meaningless for means near zero or negative.

- Variance is very sensitive to outliers (14.2 to 5666.3 by adding 200).
- Slides p.43-44 use Excel's $1 + (n-1)p$. Use $(i-0.5)/n$ on the exam.
- SIQR is half the IQR.
- Midhinge below the median hints at left skew in the middle half.
- Sturges is floored on the slide, rounded in HW3. Say which.
- Histograms show relative frequency, count over n.
- Scatter plots are for paired variables, not for comparing distributions.
- Don't trust one histogram's shape. Check with a Q-Q plot.
- Two-sample Q-Q uses the smaller n.
- Slides put theory on x and data on y. Feitelson flips them, so his slope rules invert.
- A straight line at any slope means same shape.
- Q-Q plots ignore order, so they can't show independence.
- The slide's geometric inverse with $\ln u$ reverses a Q-Q plot. Use $\ln(1-u)$.
- Exponential mean a vs rate $\lambda = 1/a$.
- Small samples wiggle off the line even when the model is right.
- Label error bars SD, SE or CI and give n.
- Don't give a bare geomean speedup without spread.
- Range grows with n, so it can't show stability.

Confidence intervals

- Narrow means precise, not accurate.
- Resolution is the tool, precision is the scatter.
- Quantization is random error (CI p.9).
- Randomization doesn't remove a bias shared by all conditions.
- s describes single values, $s/\sqrt{n}$ describes the mean.
- The CLT needs independence. More consecutive samples don't fix dependence.
- CI p.29 writes $N(\mu, \sigma/\sqrt{n})$ with an SD, p.36 writes $N(\mu, \sigma^2)$ with a variance.
- df is $n-1$. Use the $\alpha/2$ column for two-sided.
- Divide s by $\sqrt{n}$, not n.
- t assumes normal data. Comment on skew at small n.
- A population file's variance divides by N, a sample's by $n-1$.
- Never say "95% probability μ is in this interval".
- More data narrows around $\bar{x}$, which converges to μ only without bias.
- Overlapping intervals imply nothing. Non-overlap implies significance.
- Coverage is a long-run rate, one sample can't check it.
- $10\times$ runs is $\sqrt{10} = 3.16\times$ narrower.
- Proportion intervals use z, need $np \geq 10$, and have n inside the root.
- Compare proportions with an interval on the difference.
- Relative e (slides) vs absolute e (study guide).
- "Within 7%" is a total width, so $e = 0.035$.
- Round sample sizes up (160.5 to 161).
- Sample size uses z, and remember to square.
- Round a low pilot p toward 0.5.
- "We ran it three times" isn't a sample size.
- Variance CI puts the big χ^2 under the lower limit, uses $n-1$ df and $(n-1)s^2$.
- Variance CI isn't $s^2 \pm$ something, and needs normal data.
- Batching widens the interval (fewer df).
- Never sort before batching.
- The bootstrap still needs independent draws and must resample with replacement.
- A back-transformed log interval is for the median.
- A narrow interval in a biased setup isn't trustworthy.
- Count units, not measurements.
- Intervals do apply to randomized algorithms (seed distribution).

Hypothesis testing

- The claim goes in H_1, the equality in H_0.
- Say "fail to reject", never "accept H_0".
- One-sided 0.05 is 1.645, two-sided 1.96, two-sided 0.01 is 2.576, one-sided 0.01 is 2.326.
- For H_1: $\mu < \mu_0$ reject in the lower tail.
- Choosing one-sided after seeing the data doubles the real α.
- Divide by $\sigma/\sqrt{n}$, not σ.
- Small n with s needs $t[\alpha/2, n-1]$.
- Unpaired tests on paired data throw away power.
- A rank test answers a different null.
- Two-proportion SE uses the pooled $\hat{p}$.
- McNemar uses only disagreements.
- Double the tail for two-sided p, not for one-sided.
- p is not the probability H_0 is true or that it's a fluke.
- Small p isn't a big effect. Report the value, effect and interval.
- $\beta \neq 1 - \alpha$. Type I is the false positive, Type II the miss.
- $z[\beta]$ for 80% power is 0.84, not 1.96 or 1.282.
- Round n up.
- Low α with no power is a useless test (Bouthillier, 90% misses).
- "Highly significant" isn't "large".
- Not significant isn't equivalent. Look at the interval width.
- Don't lean on Cohen's labels for systems data.
- More samples from one run sharpen a wrong answer.
- Correlation invalidates the test, not just widens it.
- No test fixes interference.
- Positive ρ shrinks $n_{\text{eff}} = n(1-\rho)/(1+\rho)$.
- $m\alpha$ is only a bound. Use $1-(1-\alpha)^m$.
- Holm divides by $m-k+1$ and stops at the first failure.
- BH continues past failures to the largest passing k.
- Reporting only significant cells hides a family of tests.
- "$p = 0.07$, run a few more" is peeking.
- Design problems (interference, copied baselines, maximization bias) aren't fixed by a better test.
- Type III is a correct rejection in the wrong direction.
- A fixed percentage alarm sets a different α and β per test.
- χ^2 divides by expected, df $k-1-$params, merge bins below 5.
- The χ^2 handout quotes 16.919 (0.05) for $\alpha = 0.01$. The 0.01 value is 21.666.

Comparing alternatives

- Equal sample sizes don't make data paired.
- n counts units. Iterations in one run are one unit.
- The estimate is the same under both designs, only the SE changes.
- Pairing doesn't create ρ, it uses it.
- Paired isn't always more powerful ($\rho \approx 0$, small n).
- Pairing is blocking for two systems.
- Paired df is pairs -1 (HW6 P1 has 8 df, not 16).
- State the subtraction direction.
- "Contains 0" means "not shown", not "equal".
- Don't pair unpaired data just because rows line up.
- Square s^2/n in the ν denominator, and use $n-1$ there but n inside s.
- A 90% interval uses $t[0.05, \nu]$.
- Missing df row? Go to the next smaller df.
- Welch s is already divided by n.
- Pre-testing variances then choosing is worse than always Welch.
- Pooled weights are $n-1$, divisor $n_A + n_B - 2$.
- Unequal n is fine for Welch, but ask why.
- "Student's t assuming equal variances" with unequal groups is a red flag.
- Heavy overlap is no evidence, not equality.
- Separate intervals on paired data are even more misleading.
- Same test set percentages are paired.
- Signed-rank is paired, rank-sum (Mann-Whitney) unpaired.
- Mann-Whitney doesn't test medians.
- Signed-rank assumes symmetric differences.
- Switching to a rank test after t missed 0.05 is p-hacking.
- Unpaired run count has the factor 2 and raw s.
- Wild handout slide numbers are from an older deck. Cite Comp p.24-25.
- Interleaving doesn't fix an instance slow for the whole run.
- "Only eight benchmarks" is usually about generality.

Measurement

- Profiler percentages are estimates with an SE.
- Sampling overhead depends on rate, not workload.
- Don't sample for rare events. Trace them.
- Write $2^n - 1$ in the rollover formula.
- Don't divide an invariant cycle count by rated MHz.
- CPU time isn't automatically "correct".
- "Discarded the first 10" is a choice to justify.
- Repetition doesn't improve resolution. Lengthen the interval.
- Bracketing costs $2c$, not c.
- Four significant figures from three runs confuses accuracy with precision.
- Subtract the empty loop, and remember loops run hot.
- Tick counting fails if the event start is synchronized with the clock.
- Perturbation isn't additive.
- Overhead can hit one configuration more than the other.
- Derived ratios are noisier than their inputs.
- 1000 runs makes a result more precise, not more accurate.
- A on Monday and B on Wednesday confounds system and time.
- Iterations in one process are pseudo-replication.
- Report the minimum only with n.
- A mean interval can't support a p99 claim.
- Don't drop outliers quietly or after seeing the effect.
- Closed-loop harnesses can't show overload tails.
- With-and-without catches perturbation, not bias.
- Time-of-day variation is random only if runs are spread over it.
- The median is justified by one-sided noise, still say so.

Research process and ethics

- Academic advisor isn't the research advisor (form within 24 credits).
- PRCP is Aydin's term, not official.
- Co-advising names both, one is official. The deck is Aydin's talk.
- 701 need not be publishable but must match 3 credits.
- 701 co-supervisors allowed, one tenure-line CS member registered.
- 701 supervisor isn't automatically the advisor. No deferring 701.
- CS 700 term project is individual. Linking to 701 isn't graded.
- Proposal moved to Nov 2. 701 form due Oct 31.
- Report is 6-8 pages in ACM or IEEE template.
- Correct, important and novel results still aren't guaranteed acceptance.
- Reviewers are always anonymous. Double-blind hides authors too.
- Not every paper gets a rebuttal.
- The h-index is a count of papers.
- Honor Code for coursework, Policy 4007 for research and theses.
- Honest error isn't misconduct.
- Falsification includes omission.
- The conference-to-journal exception covers your own paper, cited and attached.
- Dual submission is banned even if you plan to withdraw one.
- A citation doesn't make copied words OK (Luke Ex. 2, 3).
- Not plagiarism isn't allowed. A cited figure without permission is copyright violation.
- Equations count as text. Captions are text.
- Matching bibliography entries are fine.
- Luke ranks a marked quote (Ex. 7) above an own-words rewrite (Ex. 6).
- Permission doesn't cover large pieces of text.
- AI disclosure doesn't shift responsibility.
- IEEE recommends disclosing grammar help, ACM exempts it.
- AI disclosure goes in acknowledgments, never the author list.
- Midterm is 35%, homework 40%. Resubmit within two weeks.

Part 5. Definitions glossary

Measurement error and estimation

Accuracy. How close the mean of the measurements is to the true value. Set by systematic error (CI p.17).

Precision. How repeatable the measurements are. Set by random error and quantified by statistics (CI p.17-18).

Resolution. Smallest increment between measured values, a property of the tool (CI p.17).

Systematic error. An experimental mistake giving constant or slowly varying bias, e.g. clock drift, cache not cleared. Repetition can't fix it (CI p.7, Meas p.20).

Random error. Unpredictable, unbiased, from tool limits, the observer or random processes (CI p.8).

Quantization error. Same event read as 13 or 14 ticks depending on phase. Random (CI p.9).

Statistic. A quantity computed from a sample, an estimate of a population parameter (CI p.4).

Point vs interval estimate. A single best guess vs an interval with a stated confidence (CI p.22-23).

Unbiased, efficient, consistent. Average over samples equals the parameter, precise, and closer as n grows (CI p.23).

Standard error. SD of the sample mean, $\sigma/\sqrt{n}$ (CI p.29).

Central limit theorem. Independent values from one population give $\bar{x} \approx N(\mu, \sigma/\sqrt{n})$ for large n (CI p.29).

Confidence level. About $100(1-\alpha)$ of 100 intervals contain μ. A property of the method, not one interval (CI p.33).

Wald interval. The basic $p \pm z\sqrt{p(1-p)/n}$. Problematic near 0 or 1 (CI p.49).

Degrees of freedom. $n-1$, because one is spent estimating the mean (HT p.11).

Batching, transformation, bootstrap. The three responses when normality or independence fails (CI p.68-70).

Testing

Null and alternative. H_0 holds the equality and is kept until the data force you off it. Failing to reject isn't evidence for it (HT p.4).

One vs two-sided. All of α in one tail vs split. Decided by the question in advance (HT p.5).

Type I error. Rejecting a true H_0, a false positive, probability α (HT p.6).

Type II error. Failing to reject a false H_0, a missed effect, probability β (HT p.6).

Confidence. $1-\alpha$, not crying wolf when there's no wolf (HT p.6).

Power. $1-\beta$, seeing the effect when it's really there. Bought with sample size (HT p.6-7).

p-value. Pr[a statistic at least this extreme | H_0]. Reject if $p \leq \alpha$ (HT p.13). Not Pr[H_0], not a size, not a sharp line (HT p.14).

Statistical vs practical significance. Same 0.3 ms effect is "highly significant" at $n = 10^6$ (HT p.17).

Effect size. Raw difference, relative change, Cohen's d, A_{12} (HT p.18, Comp p.24).

A_{12}. Probability a random value from A beats one from B, 0.5 means no difference (HT p.18, Comp p.19).

Experimental unit. The thing independently replicated (run, machine, benchmark, user). n counts units (HT p.20, Comp p.4).

Pseudo-replication. Treating non-independent measurements (iterations in one process) as replicates, giving too narrow an interval (Meas p.25, study guides).

Autocorrelation. Lag-1 ρ. Positive ρ inflates Type I error (HT p.21).

Interference. Treated units change control outcomes, so the test measures the wrong quantity (HT p.22).

Peeking. Testing after every batch and stopping at $p < \alpha$. Inflates Type I error (HT p.23-24).

FWER. Probability of at least one false positive among m tests (HT p.27).

FDR. Expected fraction of rejections that are false. Controlled by BH (HT p.28).

HARKing. Hypothesizing after the results are known (HT p.30).

Comparing alternatives

Paired design. Every unit measured under both systems, analyse the n differences (Comp p.5).

Unpaired design. Each unit under one system, no correspondence (Comp p.5).

Why pairing works. $\mathrm{Var}(A-B) = \sigma_A^2 + \sigma_B^2 - 2\rho\sigma_A\sigma_B$ (Comp p.6).

Welch-Satterthwaite ν. Effective df for unequal variances, usually fractional (Comp p.12).

Pooled t. Assumes equal variances, df $n_A + n_B - 2$ (Comp p.13).

Overlap rule. Non-overlap implies significant, overlap implies nothing (Comp p.15).

Mann-Whitney U. Unpaired rank test. Null is $P(X > Y) = 1/2$, not equal medians (Comp p.19).

Wilcoxon signed-rank. Paired rank test. Robust to outliers, not skew (Comp p.21).

Sign test. Paired, counts positive differences, $S \sim \mathrm{Bin}(n, 1/2)$. Least powerful (Comp p.23).

Measurement

Event. A defined change in system state. What counts is a decision (Meas p.6).

Counting, tracing, sampling. Increment per event, record each event with state, inspect state at fixed intervals (Meas p.7).

Wall, CPU, steady-state time. Elapsed real time, time holding a core, time after caches and JIT settle (Meas p.11).

CLOCK_MONOTONIC. Elapsed-time clock that never runs backwards. Avoid gettimeofday() (Meas p.12).

Invariant cycle counter. Ticks at a fixed rate, so it already measures elapsed time (Meas p.13).

Warm-up. Early iterations unrepresentative. How N was chosen must be stated (Meas p.14).

Perturbation. Instrumentation changes the system it measures. Not additive (Meas p.15-17).

Indirect measurement. Measure something visible and derive the rest, e.g. load from a calibrated loop (Meas p.18).

Measurement bias. Innocuous setup (link order, environment size) shifts results repeatably (Meas p.21).

Randomization. Turns an unknown nuisance factor from systematic into random (Meas p.22).

Levels of repetition. Iterations in runs in processes in rebuilds, each with its own variance (Meas p.25).

Open vs closed loop. Arrivals independent of replies vs fixed clients waiting. Closed can't overload (Meas p.28).

Metrics and summaries

Rate metric. Events over the time interval, i.e. throughput (Metrics p.5).

Linear. Metric doubles when performance doubles. Nice, not required (Metrics p.7, p.13).

Reliable. Metric A > B implies performance A > B. Required (Metrics p.8).

Repeatable. Same value every run, deterministic. Required (Metrics p.9).

Easy to use. Easy to measure, so measured correctly. Nice, not required (Metrics p.10).

Consistent. Units and definition same across systems. Required (Metrics p.11).

Independent. Not bent by commercial pressure. Required (Metrics p.12).

Means vs ends metrics. Counts what was done useful or not, vs only real progress. Execution time is ends-based (Metrics p.27-28).

Execution time. The reference metric, smallest time is highest performance (Metrics p.22).

MIPS, MFLOPS, SPEC. Instruction rate, FP-operation rate, geomean of reference over measured times (Metrics p.16-21).

Precision and recall. Relevant retrieved over retrieved, and over all relevant (Metrics p.32).

Mean, median, mode. Sum meaningful, outliers present, categorical (Summ p.18).

COV. $s/\bar{x}$, dimensionless (Summ p.41).

SIQR. $(Q_3 - Q_1)/2$, preferred to s for skewed data (Summ p.45).

Quantile. $Q((i - 0.5)/n) = x_i$, interpolate between (Summ p.59).

Q-Q plot. Equal shape means linearly related quantiles (Summ p.61).

Research process

Scientific method. Testable, refutable hypothesis, design, conduct, test statistically (Intro p.9).

Feitelson's three meanings. Building systems, experimentation as feedback, rigorous evaluation (Intro p.10).

Academic vs research advisor. Assigned at admission vs declared by paperwork within 24 credits (Advisor p.3).

PRCP. Informal research phase of a semester or more before advising, Aydin's term (Advisor p.8-9).

Peer review. Experts review independently before publication, editors or chairs decide (Pub p.3).

Acceptance rate. Share accepted. First-tier 10-25%, sometimes under 10% (Pub p.13).

Rebuttal. Early reviews let authors clarify. Not for every paper (Pub p.17).

Camera-ready. Revised final copy of an accepted paper (Pub p.18).

h-index. Largest h with h papers each cited at least h times (Pub p.6).

Double-blind. Reviewers don't know authors either. Non-compliant papers rejected right away (Pub p.21).

Dual submission. Same or mostly same paper to several venues at once. Not allowed (Pub p.20, Ethics p.14).

Self-plagiarism. Reusing your own published text. Rephrase, except conference to journal (Ethics p.12).

Conference to journal. About 30% more material for IEEE, cite and attach the conference version (Ethics p.14).

Research misconduct. Fabrication, falsification or plagiarism in research. Excludes honest error (Ethics p.18).

Fabrication. Making up data or results and recording or reporting them (Ethics p.20).

Falsification. Manipulating materials, equipment or processes, or changing or omitting data (Ethics p.20).

Plagiarism. Using another's ideas, processes, results or words without credit (Ethics p.20).

Policy 4007 and ORIA. GMU research-misconduct policy, run by ORIA. Honor Code covers coursework (Ethics p.15-16).

Copyright violation. Legal offense, e.g. a cited figure without permission. Separate from plagiarism (Luke TR Ex. 14).

AI use. Allowed with detailed disclosure and full author responsibility. AI can't be an author (Ethics p.22-26).

How the worked examples read. Step (1) is the formula with every symbol named, (2) plugs in, (3) works it out, (4) gives the result and what it means.

Metrics (Lecture 2)

What a metric is made of. A performance metric is a count (how many times an event happens), a duration (length of a time interval), a size (of some parameter), or a value derived from those (Metrics p.4, p.29).

Hard to measure areas. QoS, programmer productivity and security are the slide's examples of things with no obvious metric. In security and software engineering, inventing the metric is itself a research topic (Metrics p.2).

What a good metric does. It allows accurate and detailed comparisons, leads to correct conclusions, is well understood by everyone, and has a quantitative basis. Its job is to keep you from erroneous conclusions (Metrics p.6).

Slide example for each property. The dB scale for sound is the nonlinear example (Metrics p.7). Time-sharing changing the measured execution time is the repeatability example (p.9). For easy to use, a metric that's hard to measure is less likely to be measured correctly, and "a wrong value is worse than a bad metric" (p.10). Independence is about money riding on results, which pressures vendors to optimize for a metric or to influence its definition (p.12).

Clock rate caveats (Metrics p.14-15). Is the gain even proportional? Architectures differ in operations per cycle, in CPI, and in branch penalty from pipeline depth. The processor may not be the bottleneck at all, since memory and I/O delays can dominate. Verdict, faster clock $\neq$ better performance, but it's "a good first-order metric".

MIPS and MFLOPS framing. MIPS is like distance traveled per unit time, with instructions as the distance, which only works if every instruction is the same amount of work (Metrics p.16). Nickname "Meaningless Indicator of Performance" (p.17). MFLOPS tries a better unit of distance, one floating-point operation (p.18). Transcendental ops and roots leave too much freedom in what counts as one FLOP (p.19).

SPEC details. SPEC is the System Performance Evaluation Cooperative. Manufacturers pick "representative" programs, the times are normalized to a standard basis machine, and the SPECmark is the geometric mean of the normalized values (Metrics p.20). Problems on p.21 are that the rating doesn't match execution times of non-SPEC programs, and it's subject to tinkering because a committee picks the suite and vendors use targeted compiler optimizations.

System noise in wall-clock time. Background OS tasks, virtual-to-physical page mapping, random cache mapping and replacement, variable system load (Metrics p.24).

Full scorecard (Metrics p.25). A blank cell means the property fails.

	Clock	MIPS	MFLOPS	SPEC	Time
Linear					✓
Reliable					✓
Repeatable	✓	✓	✓	✓	≈ ✓
Easy to measure	✓	✓	✓	$\frac{1}{2}$ ✓	✓
Consistent	✓			✓	✓
Independent	✓	✓			✓

The slide's clock column (p.15, p.25) leaves Reliable blank. The study guide writes it as "×(ish)".

Other metrics (Metrics p.26). Response time is elapsed time from request to response. Throughput is jobs or operations completed per unit time (video frames per second). Bandwidth is bits per second. Ad hoc metrics are defined for one specific need.

Means vs ends detail. Means metrics count nop instructions and multiplies by zero as work, which is why they come out unreliable (Metrics p.27). The p.28 spectrum runs from ends to means as execution time, SPEC, MFLOPS, MIPS, clock rate.

Metrics in other fields (Metrics p.31-32). Fields without measurement get called "soft". Fault tolerance uses MTTF (mean time to failure) and MTTR (mean time to repair). Evolutionary computation compares techniques by how often each finds the optimum, or gets within ε of it, which is harder with multiple objectives. Software engineering has code metrics (lines of code), design metrics, test metrics and process metrics (people, time, schedule). Security has the strength of one-way hash functions and intrusion detection.

Study guide calculations (Metrics study guide).

MIPS (Q15).

(1) MIPS $= n/(T_e \times 10^6)$, where n is the number of instructions executed and T_e is the execution time in seconds.
(2) $n = 4.5 \times 10^9$ and $T_e = 3$ s, so MIPS $= 4.5 \times 10^9/(3 \times 10^6)$.
(3) Split it up. $4.5/3 = 1.5$ and $10^9/10^6 = 10^3$, so 1.5×10^3.
(4) 1500 MIPS. The CPU gets through 1.5 billion instructions a second, which says nothing about how much work each one does.

MFLOPS (Q16).

(1) MFLOPS $= f/(T_e \times 10^6)$, where f is the number of floating-point operations and T_e the execution time in seconds.
(2) $f = 8 \times 10^8$, $T_e = 2$ s, so $8 \times 10^8/(2 \times 10^6)$.
(3) $8/2 = 4$ and $10^8/10^6 = 10^2$, so 4×10^2.
(4) 400 MFLOPS. An integer-only program would score 0 here even while doing useful work.

Precision and recall (Q17).

(1) Precision = (relevant items retrieved) / (total items retrieved). Recall = (relevant items retrieved) / (relevant items that exist in the whole collection).
(2) 35 relevant out of 50 retrieved, and 60 relevant in total. Precision = 35/50, recall = 35/60.
(3) $35/50 = 70/100 = 0.70$. $35/60 = 7/12$, and $7 \div 12 = 0.583$.
(4) Precision 0.70 and recall 0.583. So 70% of what came back was useful, and the search found 58% of the useful pages.

MIPS comparison (Q18).

(1) MIPS $= n/(T_e \times 10^6)$ as above. For the same workload, how much faster A is $= T_B/T_A$.
(2) A runs $n = 2 \times 10^8$ in 0.5 s, B runs $n = 1.5 \times 10^8$ in 0.6 s.
(3) A is $200/0.5 = 400$ MIPS. B is $150/0.6 = 250$ MIPS. MIPS ratio $400/250 = 1.6$. Real time ratio $0.6/0.5 = 1.2$. The gap comes from instruction count, since A needs $2.0/1.5 = 1.33$ times as many instructions for the same job, and $1.6/1.33 = 1.2$.
(4) A really is faster, but by 1.2×, not the 1.6× MIPS claims. Each of A's instructions does less of the job, so counting instructions flatters A. Compare execution times.

Clock rate claim (Q19), made-up numbers. Claiming 4.5 GHz is 50% faster than 3.0 GHz fails because the clock ratio says nothing without CPI, instruction counts and whether the CPU is the bottleneck.

(1) CPU time $= IC \times \text{CPI}/f$, where IC is the instruction count, CPI the average cycles per instruction and f the clock rate in Hz.
(2) Same program, $IC = 10^9$. Old chip $f = 3 \times 10^9$ with CPI 1.2. New chip $f = 4.5 \times 10^9$ with CPI 2 (deeper pipeline, bigger branch penalty).
(3) Old $= 10^9 \times 1.2/(3 \times 10^9) = 1.2/3 = 0.4$ s. New $= 10^9 \times 2/(4.5 \times$

$10^9) = 2/4.5 = 0.444$ s.

(4) The "50% faster" chip is $0.444/0.4 = 1.11$, so 11% slower on this program.

Vendor-only compiler flags for a benchmark run hurt independence most directly, the SPEC tinkering problem.

Summarizing data (Lecture 4)

Three properties of numerical data (Summ p.2). Central tendency (arithmetic, geometric, harmonic mean, median, mode), variability (range, IQR, variance, SD, COV, mean absolute deviation) and distribution (its type).

Why one number is a problem. Performance is multidimensional (CPU, I/O and network time and their interactions), and systems are specialized, so one system can show a wide range of times across benchmarks (Summ p.4-5). You'll be pressured to give a mean anyway, so learn to pick the right one and to spot bad ones in other people's results (p.7). The summary slide says avoid means if possible because they lose information (p.32).

Mean as expected value (Summ p.9-11). $E[X] = \sum_{i=1}^{n} x_i p_i$, the "first moment" of X, where x_i is a measured value and $p_i = \Pr(\text{we measure } x_i)$. With no other information take $p_i = 1/n$, which gives the arithmetic mean. The true mean can't be known, since it would take infinitely many measurements.

(1) $E[X] = \sum x_i p_i$, value times its probability, summed.

(2) Values 2, 4, 9. First with $p_i = \frac{1}{3}$ each, then with 9 showing up half the time and 2 and 4 a quarter each.

(3) Equal weights give $(2 + 4 + 9)/3 = 15/3 = 5$. Unequal weights give $0.25(2) + 0.25(4) + 0.5(9) = 0.5 + 1 + 4.5 = 6$.

(4) 5 is the plain arithmetic mean. Putting more probability on 9 pulls the expected value up to 6.

Outlier example (Summ p.12, p.15, study guide A1). The mean weights every value equally, so outliers drag it.

(1) Mean $\bar{x} = \frac{1}{n}\sum x_i$, with n the count. Median is the middle value of the sorted data, or the average of the two middle values when n is even.

(2) Data 10, 20, 15, 18, 16, so $n = 5$. Then add 200, so $n = 6$.

(3) Sum = 79, mean $79/5 = 15.8$. Sorted 10, 15, 16, 18, 20, middle is the 3rd value, 16. With 200 the sum is 279, mean $279/6 = 46.5$. Sorted 10, 15, 16, 18, 20, 200, middle two are 16 and 18, median $\frac{1}{2}(16 + 18) = 17$.

(4) One outlier moved the mean by 30.7 and the median by 1. The mean of 46.5 is bigger than five of the six values, so it isn't typical of anything.

Same data, other indices (study guide C1).

(1) Range = max − min. Sample variance $s^2 = \frac{1}{n-1}\sum(x_i - \bar{x})^2$, SD $s = \sqrt{s^2}$, COV $= s/\bar{x}$, MAD $= \frac{1}{n}\sum|x_i - \bar{x}|$.

(2) $\bar{x} = 15.8$, so the deviations $x_i - \bar{x}$ are -5.8, 4.2, -0.8, 2.2, 0.2.

(3) Range $20-10 = 10$. Squares $33.64+17.64+0.64+4.84+0.04 = 56.8$, so $s^2 = 56.8/4 = 14.2$ and $s = \sqrt{14.2} = 3.77$ (check $3.77^2 = 14.21$). COV $= 3.77/15.8 = 0.239$. Absolute deviations sum to $5.8+4.2+0.8+2.2+0.2 = 13.2$, MAD $= 13.2/5 = 2.64$. With 200, $\bar{x} = 46.5$ and deviations -36.5, -26.5, -31.5, -28.5, -30.5, 153.5. Range $200 - 10 = 190$. Squares sum to 28331.5, $s^2 = 28331.5/5 = 5666.3$, $s = 75.27$. COV $= 75.27/46.5 = 1.619$. Absolute deviations sum to 307, MAD $= 307/6 = 51.17$.

(4) Before, range 10, $s = 3.77$, COV 0.239, MAD 2.64. After, range 190, $s = 75.27$, COV 1.619, MAD 51.17. Squaring makes s the most outlier-sensitive, and the spread is now bigger than the mean itself.

Mode. It may not exist, and it may not be unique, e.g. a bimodal distribution (Summ p.17).

The p.19 list, answered (study guide A2). Message sizes, arithmetic (total bytes is meaningful, also give the median since sizes are skewed). Cache hits, arithmetic. Execution time, arithmetic. MFLOPS, MIPS and bandwidth, harmonic. Speedup, geometric by the lecture rule, but summarize the raw times if you have them. Cost, arithmetic.

A mean is still a metric. It has to meet the same six good-metric properties, and execution time is still the best measure (Summ p.24).

Harmonic mean worked example (Summ p.29, study guide B1).

(1) $M_H = n/\sum_{i=1}^{n}\frac{1}{x_i}$, where x_i is the rate (MFLOPS) of run i and n the number of runs. The true overall rate is total work over total time, $\sum F_i/\sum T_i$.

(2) $n = 5$, rates 405, 367, 405, 419, 388. Work F_i (in 10^9 FLOPs) 130, 160, 115, 252, 187 and times T_i 321, 436, 284, 601, 482 s.

(3) Reciprocals $0.002469 + 0.002725 + 0.002469 + 0.002387 + 0.002577 = 0.012627$, so $M_H = 5/0.012627 = 396.0$. Total work $130 + 160 + 115 + 252 + 187 = 844 \times 10^9$, total time $321+436+284+601+482 = 2124$ s, rate 844,000/2124 = 397.4 MFLOPS. For comparison the arithmetic mean is $1984/5 = 396.8$.

(4) $M_H = 396.0$ against the true 397.4. They differ because the runs did different work (115 to 252×10^9 FLOPs). Only the work-weighted harmonic mean $\sum F_i/\sum(F_i/x_i)$ gives 397.4 exactly.

Geometric mean worked example (Summ p.31). Ten tests, each with an improvement factor for the OS, middleware and application layers.

(1) $M_G = \left(\prod_{i=1}^{n} x_i\right)^{1/n}$, where x_i are the improvement factors and n how many there are.

(2) Test 1 has 1.18, 1.23, 1.10, so $n = 3$.

(3) $1.18 \times 1.23 = 1.4514$, times 1.10 is 1.5965. Cube root by bracketing, $1.16^3 = 1.561$ and $1.17^3 = 1.602$, so about 1.17. The overall 1.20 is the geometric mean of the ten per-test values (same as the geometric mean of all 30 factors), which is 1.195. The ten values are so close together that their arithmetic mean, $11.96/10 = 1.196$, is nearly the same.

(4) Test 1 averages 1.17 per layer and the whole table about 1.20, so a typical layer got about 20% faster.

More study guide mean examples.

Two phases (B2).

(1) Speedup = old time / new time. AM $= \frac{1}{n}\sum S_i$, GM $= (\prod S_i)^{1/n}$, HM $= n/\sum(1/S_i)$.

(2) Two 100 s phases, phase 1 cut to 1 s. $S_1 = 100/1 = 100$, $S_2 = 100/100 = 1$.

(3) AM $= (100 + 1)/2 = 50.5$. GM $= \sqrt{100 \times 1} = 10$. HM $= 2/(1/100 + 1/1) = 2/1.01 = 1.98$. True speedup is total old over total new, $200/101 = 1.98$.

(4) Only the harmonic mean is right, because the phases had equal baseline time. AM overstates by 25×, GM by 5×.

Releases (B3).

(1) GM of the improvement factors, $(\prod x_i)^{1/n}$.

(2) 10%, 25%, 5% are factors 1.10, 1.25, 1.05, $n = 3$.

(3) $1.10 \times 1.25 = 1.375$, times 1.05 is 1.44375. $1.13^3 = 1.443$, so the cube root is 1.130.

(4) 13.0% per release. Three releases at 13% each compound to the same 44% total.

Hit ratios (B4).

(1) GM $= (\prod h_i)^{1/n}$ with h_i the hit ratio at level i.

(2) 0.90, 0.80, 0.50, $n = 3$.

(3) $0.9 \times 0.8 = 0.72$, times 0.5 is 0.36. $0.71^3 = 0.358$, so $0.36^{1/3} = 0.711$.

(4) 0.711. The product 0.36 is the compounded pass-through fraction, and the GM keeps that product.

Weighted mean (B5).

(1) Weighted mean $= \sum w_i x_i$, where w_i is the relative frequency

of task i (weights sum to 1).

(2) Times 2, 5, 10 s with $w = 0.5, 0.3, 0.2$.

(3) $0.5(2) + 0.3(5) + 0.2(10) = 1 + 1.5 + 2 = 4.5$. Unweighted $17/3 = 5.67$.

(4) Report 4.5 s and disclose the weights, since the reader can't recover them from the summary.

Arithmetic mean of rates (B6). The arithmetic mean of rates F/T_i is $\frac{F}{n}\sum 1/T_i$, which isn't a function of $\sum T_i$, so it fails "inversely proportional to total time" and always overstates.

(1) True rate $= \sum F_i / \sum T_i$. AM of rates $= \frac{1}{n}\sum F/T_i$, HM of rates $= n/\sum(T_i/F)$.

(2) Two programs, $F = 100$ operations each, times 1 s and 9 s, so rates 100 and 11.11.

(3) True rate $200/10 = 20$. AM $= (100 + 11.11)/2 = 55.6$. HM $= 2/(1/100 + 1/11.11) = 2/(0.01 + 0.09) = 2/0.1 = 20$.

(4) AM claims 55.6, almost 3× the real 20. The HM gets it exactly because the work was equal.

Extra dispersion indices. The p.36 list includes the 10th and 90th percentiles. Maximum distance from the mean is $\max|x_i - \bar{x}|$. Neither it nor the range uses all the information efficiently (Summ p.38). Tiny example with 10, 20, 15, 18, 16 and $\bar{x} = 15.8$. The distances are 5.8, 4.2, 0.8, 2.2, 0.2, so the max distance is 5.8 (from the 10). Only that one value matters.

Variance facts (Summ p.39). Variance is the second moment. The $\sum x_i^2$ form can be computed "on the fly" in one pass through the data. It has $n - 1$ degrees of freedom. *In plain words.* The deviations are measured from $\bar{x}$, which came from the same data, so one degree of freedom is used up and dividing by n would come out too small (study guide C3).

(1) One-pass form $s^2 = \dfrac{n\sum x_i^2 - \left(\sum x_i\right)^2}{n(n-1)}$, where you only keep the running count n, the running sum and the running sum of squares.

(2) Data 10, 20, 15, 18, 16. $n = 5$, $\sum x_i = 79$, $\sum x_i^2 = 100 + 400 + 225 + 324 + 256 = 1305$.

(3) $5 \times 1305 = 6525$. $79^2 = 6241$. Top $6525 - 6241 = 284$. Bottom $5 \times 4 = 20$. $284/20 = 14.2$.

(4) $s^2 = 14.2$, same as the two-pass answer in C1, with no need to store the data.

Quartile definition (Summ p.42). Q_1 is the value with 25% of observations below it, Q_2 50%, Q_3 75%. Percentiles split the data into hundredths.

Quartile worked example (Summ p.43-44). 24 sorted values from 1.05 to 144.91.

(1) Course rule $p = (i - 0.5)/n$, so the p quantile sits at position $i = np + 0.5$ in the sorted list, with linear interpolation between neighbours. Excel's PERCENTILE (the slide) uses position $(n-1)p + 1$ instead.

(2) $n = 24$. Needed sorted values $x_{(6)} = 1.28$, $x_{(7)} = 1.34$, $x_{(12)} = 2.15$, $x_{(13)} = 2.21$, $x_{(18)} = 2.83$, $x_{(19)} = 3.51$, $x_{(20)} = 3.77$.

(3) Course rule. Q_1 at $24(0.25) + 0.5 = 6.5$, halfway from 1.28 to 1.34, so 1.31. Median at 12.5, $(2.15 + 2.21)/2 = 2.18$. Q_3 at 18.5, $(2.83 + 3.51)/2 = 3.17$. p_{80} at $19.2 + 0.5 = 19.7$, $3.51 + 0.7(3.77 - 3.51) = 3.51 + 0.182 = 3.692$.
Excel. Q_1 at $23(0.25) + 1 = 6.75$, $1.28 + 0.75(0.06) = 1.325$. Q_3 at 18.25, $2.83 + 0.25(0.68) = 3.00$. p_{80} at 19.4, $3.51 + 0.4(0.26) = 3.614$. Midhinge $(1.325 + 3.00)/2 = 2.16$.

(4) Slide (Excel) $Q_1 = 1.32$, $Q_2 = 2.18$, $Q_3 = 3.00$, midhinge 2.16, $p_{80} = 3.613$. Course rule $Q_1 = 1.31$, median 2.18, $Q_3 = 3.17$, $p_{80} = 3.692$. Both say half the values are under 2.18 even though the mean is 9.51.

IQR. Not affected by extreme values (Summ p.45).

Coefficient of skewness (Summ p.46). Positive means a long right tail, negative a long left tail. A coefficient of -1.8 means left skew, so report the median and SIQR (study guide D4).

(1) Skewness $= \dfrac{1}{ns^3}\sum_{i=1}^{n}(x_i - \bar{x})^3$, with n the count, $\bar{x}$ the mean and s the sample SD.

(2) The 24 quartile values. Too long by hand, so use the summaries, $n = 24$, $\bar{x} = 9.51$, $s = 29.50$ (from the 24 values), and the slide's column of cubes, which sums to about 2,484,600. The slide's column header reads $(X - X_i)^3$ but the numbers are $(X_i - \bar{X})^3$, e.g. $(1.05 - 9.51)^3 = -8.46^3 = -606$.

(3) $s^3 = 29.499^3 = 25{,}671$, and $ns^3 = 24 \times 25{,}671 = 616{,}100$. Then $2{,}484{,}600/616{,}100 = 4.033$.

(4) 4.033, a strong right skew. Almost all of it is from 144.91, whose cube alone is 2,482,007.

Mean absolute deviation (Summ p.47).

(1) MAD $= \dfrac{1}{n}\sum_{i=1}^{n}|x_i - \bar{x}|$, the average distance from the mean.

(2) Same 24 values, $\bar{x} = 9.51$. Each distance is on the slide, e.g. $|1.05 - 9.51| = 8.46$ and $|144.91 - 9.51| = 135.39$.

(3) The 24 distances sum to 315.90, so MAD $= 315.90/24 = 13.16$.

(4) MAD 13.16. A typical value is 13 away from the mean, larger than the mean itself, because the mean sits far from the pile of small values.

Mean, median, mode order (Summ p.48). Right-skewed, mode < median < mean. Left-skewed, mean < median < mode. Symmetric, all equal.

Box-and-whisker plot (Summ p.50, study guide D1). Draws the five-number summary min, Q_1, median, Q_3, max. The box holds the middle 50% of the data. Slide example, 22 I/O times, gives 4.26, 6.08, 7.35, 8.33, 12.74.

(1) Quartile position $i = np + 0.5$ (course rule). IQR $= Q_3 - Q_1$, SIQR $=$ IQR/2, midhinge $= (Q_1 + Q_3)/2$, upper fence $= Q_3 + 1.5\,\text{IQR}$, lower fence $= Q_1 - 1.5\,\text{IQR}$.

(2) $n = 22$. Q_1 at $22(0.25) + 0.5 = 6$, the 6th sorted value 6.08. Median at 11.5, $(7.24 + 7.46)/2 = 7.35$. Q_3 at 17, the 17th value 8.33. Min 4.26, max 12.74.

(3) IQR $= 8.33 - 6.08 = 2.25$. SIQR $= 1.125$. Midhinge $= 14.41/2 = 7.205$. $1.5 \times 2.25 = 3.375$, so upper fence $8.33 + 3.375 = 11.705$ and lower fence $6.08 - 3.375 = 2.705$.

(4) $12.74 > 11.705$, so the max is an outlier, while nothing is below 2.705. The slide's quartiles match the course rule exactly here.

Same mean, still different (study guide F3). Two systems with equal means can differ in spread (SD, COV), in the tail (p90, max, box plot) or in shape (histogram, Q-Q plot).

Bare geomean in a paper (study guide F2). Ask for the per-benchmark raw times, a dispersion measure (range, COV, box plot), and the rule for which benchmarks were included or dropped.

Why know the distribution (Summ p.52). You need it whenever the mean and variability feed a simulation or an analytical model. Methods are visual (histograms, Q-Q plots) or goodness-of-fit tests (chi-square, Kolmogorov-Smirnov).

Comparing two data sets (Summ p.53). Histograms are the simple visual check. A scatter plot is for the relationship between two variables. A Q-Q plot is for comparing two distributions.

Histogram example (Summ p.55-56). 12 values (−3.0, 0.8, 1.2, 1.5, 2.0, 2.3, 2.4, 3.3, 3.5, 4.0, 4.5, 5.5) in width-1 bins with right-closed edges $(1 < x \le 2)$ give counts 1, 1, 3, 2, 3, 1, 1. Halving the width to 0.5 changes the shape and creates empty bins.

(1) Count each value in the bin $a < x \le b$. Sturges number of bins $k = 1 + \log_2 n$, with n the count.

(2) Bins ≤ 0, $(0, 1]$, $(1, 2]$, $(2, 3]$, $(3, 4]$, $(4, 5]$, > 5, and $n = 12$.

(3) -3.0 goes in ≤ 0. 0.8 in $(0, 1]$. 1.2, 1.5, 2.0 in $(1, 2]$, where 2.0 stays in this bin because the right edge is closed. 2.3, 2.4 in $(2, 3]$. 3.3, 3.5, 4.0 in $(3, 4]$. 4.5 in $(4, 5]$. 5.5 in > 5. For Sturges, $2^3 = 8$ and $2^4 = 16$, so $\log_2 12$ is between 3 and 4, it's 3.585, and $k = 4.58$.

(4) Counts 1, 1, 3, 2, 3, 1, 1. Sturges says 4 bins (slide floors) or 5

(HW3 rounds), fewer than the 7 used.

Scatter plot (Summ p.57-58). Plots one data set against another to see a relationship, e.g. CPU time against I/O time per job.

Quantile plot (Summ p.59-60). Plot $Q(p)$ against p. With $n = 10$ the p values are $p_i = (i - 0.5)/10$, so $i = 1$ gives $0.5/10 = 0.05$, $i = 2$ gives 0.15, up to $i = 10$ giving $9.5/10 = 0.95$. Why $(i-0.5)/n$ and not i/n. It centres each point in its $1/n$ slice. With i/n the largest value gets $p = 1$, whose theoretical quantile is infinite for an unbounded distribution (study guide D5).

Two-sample Q-Q example (Summ p.62). $n = 15$, fitted line $y = 1.0957x + 0.4712$ with $R^2 = 0.9416$. Reasonably linear, so similar shapes.

Exponential Q-Q example (Summ p.65-68). 1000 values suspected exponential. Fitted line $y = 0.9642x + 0.016$, $R^2 = 0.9988$, which confirms it.

CDFs behind the inverses (Summ p.64). Exponential $F(x) = 1 - e^{-x/a}$, Pareto $F(x) = 1 - x^{-a}$, geometric $F(x) = 1 - (1-p)^x$. The slide's inverses for a uniform u are $-a\ln(1-u)$, $1/(1-u)^{1/a}$ and $\lceil \ln u / \ln(1-p) \rceil$.

(1) Set $F(x) = u$ and solve for x.
(2) Exponential with $a = 2$ at $u = 0.5$. Pareto with $a = 2$ at $u = 0.75$.
(3) Exponential $x = -2\ln(0.5) = 2(0.693) = 1.386$. Pareto $x = 1/(0.25)^{1/2} = 1/0.5 = 2$.
(4) The median of Exp($a = 2$) is 1.386, and 75% of Pareto($a = 2$) values are below 2. These are the theoretical quantiles you put on a Q-Q plot.

Slope and intercept (study guide E1, E3). Same shape at twice the scale and shifted by 5 gives a line with slope 2 and intercept 5. So a data-set-1 quantile of 3 lines up with $2(3) + 5 = 11$ in data set 2.

(1) Plotting position $p_i = (i - 0.5)/n$. Exponential quantile $F^{-1}(p) = -\ln(1-p)/\lambda$, with λ the rate.
(2) Data 0.4, 1.1, 1.9, 3.2, 6.8, $n = 5$, $\lambda = 0.5$, so $p = 0.1, 0.3, 0.5, 0.7, 0.9$.
(3) $-\ln 0.9 = 0.105$, $-\ln 0.7 = 0.357$, $-\ln 0.5 = 0.693$, $-\ln 0.3 = 1.204$, $-\ln 0.1 = 2.303$. Divide by 0.5 (double them) to get 0.211, 0.713, 1.386, 2.408, 4.605. Least squares on the pairs (quantile, data) gives slope 1.45, intercept -0.02, $R^2 = 0.995$.
(4) Straight, so exponential is fine. The slope says the data are stretched 1.45 times, so $\lambda \approx 0.5/1.45 = 0.345$.

Means handout, who uses which mean. SPEC CPU2017 ratio is reference time over system time, and its run rules mandate the geomean, which is why architecture papers use it. Graph500 uses the harmonic mean of TEPS over 64 search keys (a rate), and the same paper uses the arithmetic mean for repeated timings. The performance-portability metric Ψ is the harmonic mean of relative performance across platforms, so the worst platform dominates. F1 is the harmonic mean of precision and recall. A geomean speedup below 1 means an average slowdown.

(1) $F_1 = \dfrac{2}{1/P + 1/R} = \dfrac{2PR}{P+R}$, with P precision and R recall.
(2) The Q17 search, $P = 0.70$, $R = 7/12 = 0.583$.
(3) Top $2(0.7)(7/12) = 49/60$. Bottom $0.7 + 7/12 = 77/60$. Ratio $49/77 = 7/11 = 0.636$.
(4) $F_1 = 0.636$, pulled toward the weaker recall, as a harmonic mean does.

Means handout, the argument. Fleming and Wallace (1986) say use the geomean for normalized results since only its ranking doesn't depend on the reference machine. Smith (1988) replies that it's consistent but tracks no physical quantity, so summarize total time and use the harmonic mean for rates. John (2004) uses weighted arithmetic or harmonic means with weights chosen so the summary equals the real aggregate. Mashey (2004) says ratios are roughly lognormal, so the geomean is their natural centre. Iqbal and John (2010) say report dispersion (COV) with any mean and pull out outlier benchmarks. Hoefler and Belli (2015) rule 3 is arithmetic for costs and harmonic for rates, and rule 4 is avoid summarizing ratios, summarize the underlying costs or rates, and use the geomean only if those aren't available. Eeckhout (2024) says abandon the geomean speedup. Equal-time speedup (ETS) is the harmonic mean of per-benchmark speedups, weighting each benchmark by its baseline time. Equal-work speedup (EWS) is the ratio of each system's harmonic-mean IPC, weighting by work. Say which one you used.

(1) ETS $= n/\sum(1/S_i)$, with S_i the speedup on benchmark i, valid when baseline times are equal.
(2) Two benchmarks, 10 s each on the baseline, speedups 2 and 4.
(3) ETS $= 2/(1/2 + 1/4) = 2/0.75 = 2.67$. Check with times, new times $10/2 = 5$ and $10/4 = 2.5$, total $20/7.5 = 2.67$.
(4) 2.67, which is the real ratio of total times. The arithmetic mean of 3 would overstate it.

Q-Q plots handout. Q-Q plots get used to check normality before a parametric method, to check model residuals, to check a fitted point-process model (time-rescaled residuals against exponential), and to check calibration of a test (observed p-values against uniform, where a lift only at the far right is the real signal). Systems and networking papers mostly use CDF or log-log CCDF overlays plus a K-S statistic instead. The correlation of the plotted points works as a scalar fit score, $r = 1$ is perfect. K-S and Q-Q verdicts can disagree at small n. A Z-score transform only shifts and rescales, so it can't change shape and its Q-Q plot looks like the raw one. Normal residuals don't prove a model is accurate, since a biased model can have Gaussian residuals. A P-P plot resolves the centre and squashes the tails, a Q-Q plot does the opposite, so a p99 question needs Q-Q. Simultaneous confidence bands answer "how far off the line is too far".

Reporting variability handout. If the claim is about a typical run, report spread of the data (SD, IQR, percentiles). If it's about how precisely a summary is known, report a CI for that summary. SD fails on skewed or heavy-tailed data, and mean − SD can go below physically possible values. SE shrinks like $\sqrt{n}$ while the data's spread doesn't. For example, with $s = 10$ ms, SE $= s/\sqrt{n}$ is $10/\sqrt{4} = 5$ at $n = 4$ and $10/\sqrt{100} = 1$ at $n = 100$, while the SD stays about 10 either way. A p99.9 needs thousands of observations. Median absolute deviation is the robust typical deviation. The interquartile mean ignores both tails, good for few runs with a wild outlier. Always give the number of runs or seeds. Hoefler and Belli rule 5 says state whether values are deterministic and give CIs for nondeterministic data, rule 6 says don't assume normality without checking, rule 7 says compare with non-overlapping CIs or ANOVA, and rule 8 says ask whether mean or median is even useful, since worst-case latency needs percentiles. Example papers. Georges et al. (Java) show violin plots. Arcuri and Briand use many runs, nonparametric tests and Vargha-Delaney $\hat{A}_{12}$. A timing side-channel paper pairs ECDFs and violins with K-S distance and Cliff's delta.

Confidence intervals (Lecture 6)

Inference picture (CI p.5, p.21). Take a sample from the population, compute statistics, infer the population's parameters. The lecture assumes random errors are normally distributed.

Error types, extra details. Systematic errors are controlled through the skill of the experimenter (CI p.7). Random errors typically can't be controlled, so you characterize them statistically (p.8).

Error model with n sources (CI p.14-15). Each source adds $\pm E$. With n sources the possible measurements run from $x - nE$ to $x + nE$ in steps of $2E$. $\Pr(x_i)$ is the number of paths to x_i, a binomial, which becomes Gaussian as $n \to \infty$. Tiny example with

$n = 2$ sources. The four equally likely paths give $x-2E$ (one path), x (two paths, $+E-E$ and $-E+E$) and $x+2E$ (one path), so probabilities $\frac{1}{4}, \frac{1}{2}, \frac{1}{4}$. Already peaked in the middle.

Quantifying the three (CI p.18). True accuracy is hard to know and is judged relative to a predefined standard (the definition of a second). Resolution depends on the tools. Precision is quantified with statistics.

Sample mean is unbiased (CI p.24). $E[\bar{X}] = \dfrac{E\left[\sum X_i\right]}{n} = \dfrac{\sum E[X_i]}{n} = \dfrac{n\mu}{n} = \mu$.

Sample size demo (CI p.25-26). Population mean 0.2083, variance 0.0440. Three samples of $n = 15$ (1.7% of the population) have means 0.1718, 0.2467, 0.3744. Three samples of $n = 87$ (10%) have means 0.2239, 0.2203, 0.2178.

(1) The "efficiency" row is |sample − population| divided by the population value, a relative error.
(2) Worst $n = 15$ sample, 0.3744. Average of the $n = 15$ means, and average of the $n = 87$ means.
(3) $|0.3744 - 0.2083|/0.2083 = 0.1661/0.2083 = 0.80$. Average of the three $n = 15$ means $(0.1718 + 0.2467 + 0.3744)/3 = 0.7929/3 = 0.2643$, error $0.0560/0.2083 = 0.269$. Average of the $n = 87$ means $(0.2239 + 0.2203 + 0.2178)/3 = 0.2206$, error $0.0123/0.2083 = 0.059$. Variance at $n = 87$ is $|0.0459 - 0.0440|/0.0440 = 0.043$.
(4) Small samples are off by up to 80% (average 26.9% off), while $n = 87$ gets within 5.9% for the mean and 4.3% for the variance.

Interval endpoints (CI p.30). $c_1 = \mu - z_{\alpha/2}\,\sigma/\sqrt{n}$ and $c_2 = \mu + z_{\alpha/2}\,\sigma/\sqrt{n}$, and $z_{1-\alpha/2} = -z_{\alpha/2}$.

Higher confidence, wider interval (CI p.32). With the $n = 87$ samples the full widths are 0.0750 at 90%, 0.0894 at 95%, 0.1175 at 99%, and every interval covered μ.

(1) Interval $\bar{x} \mp z_{\alpha/2}\, s/\sqrt{n}$, where $\bar{x}$ is the sample mean, s the sample SD, n the sample size and $z_{\alpha/2}$ the normal value with $\alpha/2$ in the upper tail. Full width is $2z_{\alpha/2}s/\sqrt{n}$.
(2) Sample 1, $n = 87$, $\bar{x} = 0.2239$, $s^2 = 0.0452688$. From the z table, cumulative 0.95 gives 1.645 (90%), 0.975 gives 1.96 (95%), 0.995 gives 2.576 (99%).
(3) $s = \sqrt{0.0452688} = 0.2128$ and $\sqrt{87} = 9.327$, so $s/\sqrt{n} = 0.02281$. 95% half-width $1.96 \times 0.02281 = 0.0447$, interval (0.1792, 0.2686), width 0.0894. 90% half-width $1.645 \times 0.02281 = 0.0375$, interval (0.1864, 0.2614), width 0.0750. 99% half-width $2.576 \times 0.02281 = 0.0588$, width 0.1175.
(4) Going from 95% to 99% widens the interval by $0.1175/0.0894 = 1.31$, which is just $2.576/1.96$. All three contain $\mu = 0.2083$.

t distribution (CI p.35). $t(\nu) \sim \dfrac{N(0,1)}{\sqrt{\chi^2(\nu)/\nu}}$, where ν is the degrees of freedom and $\chi^2(\nu)$ is the sum of squares of ν unit normal variates. Its pdf looks like $N(0,1)$, and for $\nu > 30$ you can use $N(0,1)$ instead.

Why t (CI p.36). For $N(\mu, \sigma^2)$ data, $(\bar{X} - \mu)/(\sigma/\sqrt{n})$ is $N(0,1)$ and $(n-1)s^2/\sigma^2$ is χ^2 with $n-1$ df. Dividing them gives $(\bar{X} - \mu)/\sqrt{s^2/n}$, a t with $n-1$ df. By $n = 30$, t is within about 5% of z, e.g. $t[0.025, 29] = 2.045$ vs 1.96, and $2.045/1.96 = 1.043$ (CI study guide).

Excel functions on the slides. Half-width = CONFIDENCE(α, s, n) (CI p.32). TINV$(0.05, 14) = 2.145$, the two-sided t for $n = 15$ (p.38). NORMSINV$(1 - \alpha/2)$ gives z (p.50). CHIINV$(0.975, 99) = 73.36$ is the lower χ^2 critical value and CHIINV$(0.025, 99) = 128.42$ the upper (p.61).

t interval with $n = 15$ (CI p.38).

(1) $\bar{x} \mp t_{[\alpha/2,\, n-1]}\, s/\sqrt{n}$, with $t_{[\alpha/2,\, n-1]}$ the t value with $\alpha/2$ in the upper tail and $n-1$ degrees of freedom.
(2) Sample 1, $n = 15$, $\bar{x} = 0.1718$, $s^2 = 0.0180$. t table row df = 14, column 0.025 (95% two-sided) gives 2.145.
(3) $s = \sqrt{0.0180} = 0.1342$ and $\sqrt{15} = 3.873$, so $s/\sqrt{n} = 0.03464$. Half-width $2.145 \times 0.03464 = 0.0743$. $0.1718 - 0.0743 = 0.0975$ and $0.1718 + 0.0743 = 0.2461$.
(4) (0.0975, 0.246), which contains 0.2083. The slide prints 0.2462 from unrounded inputs.

Sample-size derivation (CI p.40). Set the interval $(1 \mp e)\bar{x}$ equal to $\bar{x} \mp z_{\alpha/2}s/\sqrt{n}$. Then $z_{\alpha/2}s/\sqrt{n} = \bar{x}e$, so $n = (z_{\alpha/2}s/(\bar{x}e))^2$.
Example (CI study guide Q12).

(1) $n = \left(\dfrac{z_{\alpha/2}\, s}{\bar{x}\, e}\right)^2$, with e the relative error as a fraction, s and $\bar{x}$ from a pilot.
(2) Pilot of 10 runs, $\bar{x} = 20$ ms, $s = 5$ ms, want ±2% so $e = 0.02$, and $z = 1.96$.
(3) Top $1.96 \times 5 = 9.8$. Bottom $20 \times 0.02 = 0.4$. $9.8/0.4 = 24.5$, and $24.5^2 = 600.25$. Always round up.
(4) 601 runs. The 10-run pilot only supports about ±16%, since $1.96(5)/(20\sqrt{10}) = 0.155$.

Proportion SE (CI p.47). Over all samples of size n drawn with replacement, p is roughly normal with mean π (the true proportion) and SE $\sigma_p = \sqrt{\pi(1-\pi)/n}$. The normal is approximating the binomial, hence $n\pi \geq 10$. Tiny example with $\pi = 0.5$ and $n = 100$. $\sigma_p = \sqrt{0.5 \times 0.5/100} = \sqrt{0.0025} = 0.05$, and $n\pi = 50 \geq 10$, so the normal is fine. Sample proportions mostly land within 0.5 ± 0.10.

Web log example (CI p.50).

(1) $p \mp z_{\alpha/2}\sqrt{p(1-p)/n}$, with p the sample proportion and n the sample size. Needs $np \geq 10$.
(2) 650 of 1000 entries are gif, so $p = 0.65$, $n = 1000$, and $np = 650 > 10$. z table gives 1.645 for 90% (cumulative 0.95) and 1.96 for 95% (cumulative 0.975).
(3) $0.65 \times 0.35 = 0.2275$, divided by 1000 is 0.0002275, square root 0.01508 (since $0.015^2 = 0.000225$). 90% half-width $1.645 \times 0.01508 = 0.0248$. 95% half-width $1.96 \times 0.01508 = 0.0296$.
(4) 90% is (0.625, 0.675) and 95% is (0.620, 0.680). So 62 to 68% of files are gifs, at 95%.

OS-time interval, exact (CI p.53).

(1) Same proportion interval, $p \mp z_{\alpha/2}\sqrt{p(1-p)/n}$.
(2) $p = 0.1097$, $n = 6000$, 95% so $z = 1.96$. Check $np = 658 \geq 10$.
(3) $0.1097 \times 0.8903 = 0.09767$, divided by 6000 is 1.628×10^{-5}, square root 0.004035. Half-width $1.96 \times 0.004035 = 0.00791$.
(4) (0.1018, 0.1176), so the processor spends 10.2 to 11.8% of its time in the OS, with 95% confidence.

Variance CI example (CI p.61). $n = 100$ from $N(3, 2)$, so $\sigma = 2$ and $\sigma^2 = 4$.

(1) $\left(\dfrac{(n-1)s^2}{\chi^2_{[\alpha/2,\, n-1]}},\ \dfrac{(n-1)s^2}{\chi^2_{[1-\alpha/2,\, n-1]}}\right)$, where s^2 is the sample variance, $n-1$ the degrees of freedom, and $\chi^2_{[q,\,\nu]}$ the chi-square value with q in the upper tail. The big χ^2 value gives the lower limit.
(2) $n = 100$, $s^2 = 4.71435$, df = 99. The slide's Excel values are $\chi^2_{[0.025,\,99]} = 128.422$ and $\chi^2_{[0.975,\,99]} = 73.361$. A printed table usually jumps from row 90 to row 100, and row 100 gives 129.56 and 74.22, close enough by hand.
(3) $(n-1)s^2 = 99 \times 4.71435 = 466.72$. Lower $466.72/128.422 = 3.634$. Upper $466.72/73.361 = 6.362$.
(4) (3.634, 6.362), which contains the true $\sigma^2 = 4$. Not symmetric around $s^2 = 4.71$, because χ^2 is skewed.

Fifteen runs (CI study guide Q6).

(1) Same formula, then take square roots of both ends to get an interval for σ.
(2) $n = 15$, $s^2 = 4.0$ ms^2, df = 14. Table row df = 14, column 0.025 gives 26.119 and column 0.975 gives 5.629.
(3) $(n-1)s^2 = 14 \times 4 = 56$. $56/26.119 = 2.14$ and $56/5.629 = 9.95$. Square roots $\sqrt{2.14} = 1.46$ and $\sqrt{9.95} = 3.15$.
(4) Variance (2.14, 9.95) ms^2 and SD (1.46, 3.15) ms. Fifteen runs pin down a mean much better than a variance.

CLT "large number" (CI p.64). Usually taken as about 6 or 7 or more.

Transforms and batching, extra (CI p.68). The transform can be log, square root or Box-Cox, picked from the data, and log isn't automatically right. Batches are groups of four or more. After batching you no longer have an interval for individual values. Use at least $m \geq 4$ batches (CI study guide).

Bootstrap, extra (CI p.69). B = 10,000 resamples take seconds. Plain percentile intervals under-cover at small n, meaning the real coverage is below the promised level. BCa (bias-corrected and accelerated) is SciPy's default. It works for medians, quantiles, ratios, correlations and differences of medians. BCa shifts the percentiles using a bias correction (share of resamples below the observed estimate) and an acceleration from the jackknife (CI study guide). *Which order statistics (study guide Q11).*

(1) Sort the B resampled statistics. The 95% percentile interval cuts $\alpha/2 = 2.5\%$ off each end, so it uses positions about $0.025B$ and $0.975B$.
(2) B = 10,000, then $B = 40$.
(3) $0.025 \times 10{,}000 = 250$, so the 251st and the 9750th values give 95%. For 90%, $0.05 \times 10{,}000 = 500$, so the 501st and the 9500th. With $B = 40$ each value is $1/40 = 2.5\%$ of the resamples.
(4) With $B = 40$ the tails can only move in 2.5% steps and the endpoints are very noisy, so use a large B and report it.

Which response, extra (CI p.70). The bootstrap is the modern default and gives the narrowest interval at equal coverage. Also ask whether the mean is even the summary you want for strongly skewed data.

Summary slide wording trap (CI p.74). It writes the confidence level as "Pr(population parameter within computed interval)". For the exam use the p.33 reading, a property of the procedure.

CI study guide worked results.

Mean interval at three levels (Q1, Q3).

(1) $\bar{x} \mp t_{[\alpha/2,\, n-1]}\, s/\sqrt{n}$.
(2) $n = 10$, $\bar{x} = 7.2$, $s = 0.8$, df = 9. t table row df = 9 gives 1.833 (column 0.05, 90%), 2.262 (column 0.025, 95%), 3.250 (column 0.005, 99%).
(3) $\sqrt{10} = 3.162$, so SE $= 0.8/3.162 = 0.253$. Half-widths $1.833(0.253) = 0.464$, $2.262(0.253) = 0.572$, $3.250(0.253) = 0.822$.
(4) 90% (6.74, 7.66), 95% (6.63, 7.77), 99% (6.38, 8.02). The narrower one is bought by accepting that 1 in 10 such intervals misses μ.

Same numbers at $n = 100$ (Q2).

(1) $\bar{x} \mp z_{\alpha/2}\, s/\sqrt{n}$, since $n > 30$ lets z replace t.
(2) $n = 100$, $\bar{x} = 7.2$, $s = 0.8$, $z = 1.96$.
(3) SE $= 0.8/10 = 0.08$. Half-width $1.96 \times 0.08 = 0.157$. Shrink factor $0.572/0.157 = 3.6$, which is $\sqrt{10} = 3.16$ from n times $2.262/1.96 = 1.15$ from dropping t.
(4) (7.04, 7.36). Not 10× narrower because width goes like $1/\sqrt{n}$. Half-width $\propto 1/\sqrt{n}$, so one more decimal place costs 100× the data.

Accuracy of one model (Q4).

(1) $p \mp z_{\alpha/2}\sqrt{p(1-p)/n}$, valid when np and $n(1-p)$ are both at least 10.
(2) 348 of 400 correct, so $p = 0.87$, $n = 400$, $z = 1.96$. Check $np = 348$ and $n(1-p) = 52$.
(3) $0.87 \times 0.13 = 0.1131$, divided by 400 is 0.00028275, square root 0.01682. Half-width $1.96 \times 0.01682 = 0.0330$.
(4) (0.837, 0.903), so true accuracy is 83.7 to 90.3%.

Difference against a 0.89 model (Q5).

(1) $(p_2 - p_1) \mp z_{\alpha/2}\sqrt{p_1(1-p_1)/n_1 + p_2(1-p_2)/n_2}$.
(2) $p_1 = 0.87$, $p_2 = 0.89$, both $n = 400$.
(3) $0.89 \times 0.11 = 0.0979$, divided by 400 is 0.00024475. Add Q4's 0.00028275 to get 0.0005275, square root 0.02297. Half-width $1.96 \times 0.02297 = 0.045$.
(4) $0.02 \pm 0.045 = (-0.025, 0.065)$. It contains 0, so the models can't be told apart.

99 of 100 (Q7). Likewise 98% on 200 questions has $n(1-p) = 4$, Wald territory fails (CI in practice handout, pitfall 4).

(1) Wald $p \mp z\sqrt{p(1-p)/n}$. Wilson centre $\dfrac{p + z^2/(2n)}{1 + z^2/n}$ and half-width $\dfrac{z\sqrt{p(1-p)/n + z^2/(4n^2)}}{1 + z^2/n}$.
(2) $p = 0.99$, $n = 100$, $z = 1.96$ so $z^2 = 3.8416$.
(3) Wald SE $= \sqrt{0.0099/100} = 0.00995$, half-width 0.0195. Wilson $z^2/(2n) = 0.0192$ and $1 + z^2/n = 1.0384$, centre $(0.99 + 0.0192)/1.0384 = 0.9719$. Inside the root $0.000099 + 3.8416/40{,}000 = 0.000195$, root 0.01397, half-width $1.96(0.01397)/1.0384 = 0.0264$.
(4) Wald gives (0.9705, 1.0095), past 1, because $n(1-p) = 1$. Wilson gives (0.946, 0.998), lopsided and inside $[0, 1]$.

Paired allocator differences (Q8).

(1) $\bar{d} \mp t_{[\alpha/2,\, n-1]}\, s_d/\sqrt{n}$, where d_i is new minus baseline on benchmark i, $\bar{d}$ their mean and s_d their SD.
(2) $d = -0.8, -1.2, 0.3, -0.5, -1.5, -0.2, -0.9$, so $n = 7$, df = 6. t table row 6, column 0.025 gives 2.447.
(3) Sum -4.8, $\bar{d} = -4.8/7 = -0.686$. One-pass, $\sum d^2 = 5.52$ and $n\bar{d}^2 = 7(0.4702) = 3.291$, so $\sum(d-\bar{d})^2 = 2.229$, $s_d^2 = 2.229/6 = 0.3714$, $s_d = 0.609$. $\sqrt{7} = 2.646$, SE $= 0.609/2.646 = 0.230$. Half-width $2.447 \times 0.230 = 0.564$.
(4) $(-1.25, -0.12)$, which excludes 0, so the new allocator is significantly faster, by somewhere between 0.12 and 1.25 s.

Data set S, ordinary and batch means (Q9).

(1) Ordinary $\bar{x} \mp t_{[0.025,\, n-1]}\, s/\sqrt{n}$. Batch means $\bar{x} \mp t_{[0.025,\, m-1]}\, s_b/\sqrt{m}$, with m batches and s_b the SD of the batch means.
(2) 16 latencies, $\bar{x} = 18.394$, $s = 14.501$, t row 15 gives 2.131. Batches of four, $m = 4$, t row 3 gives 3.182.
(3) Ordinary SE $= 14.501/4 = 3.625$, half-width $2.131 \times 3.625 = 7.725$. Batch means $79.7/4 = 19.925$, $53.8/4 = 13.45$, $100.2/4 = 25.05$, $60.6/4 = 15.15$. Deviations from 18.394 are $1.531, -4.944, 6.656, -3.244$, squares sum to 81.61, $s_b^2 = 81.61/3 = 27.20$, $s_b = 5.216$. SE $= 5.216/2 = 2.608$, half-width $3.182 \times 2.608 = 8.30$.
(4) Ordinary (10.67, 26.12). Batch means (10.10, 26.69), wider (the guide prints 10.09 after rounding). The loss is all in the multiplier, 3.182 against 2.131.

Data set S in logs (Q10).

(1) Interval on logs $\overline{\ln x} \mp t_{[0.025,\, n-1]}\, s_{\ln}/\sqrt{n}$, then $e^{(\cdot)}$ of both ends.
(2) $\overline{\ln x} = 2.7392$, $s_{\ln} = 0.5335$, $n = 16$, $t = 2.131$.
(3) SE $= 0.5335/4 = 0.1334$, half-width $2.131 \times 0.1334 = 0.2843$. Log interval (2.4549, 3.0235). Exponentiate to (11.65, 20.56). Centre $e^{2.7392} = 15.5$.
(4) (11.65, 20.56) ms is an interval for the median (the geometric mean 15.5), not the mean 18.4. It's multiplicatively symmetric, $15.5/11.65 = 20.56/15.5 = 1.33 = e^{0.2843}$.

Trials for a failure rate (Q13, Q14).

(1) $n = z_{\alpha/2}^2\, p(1-p)/e^2$, with p the guessed proportion and e the absolute half-width.
(2) $z^2 = 1.96^2 = 3.8416$, $e = 0.03$ so $e^2 = 0.0009$. Pilot $p = 0.05$, or $p = 0.5$ with no prior.
(3) $0.05 \times 0.95 = 0.0475$, $3.8416 \times 0.0475 = 0.1825$, $0.1825/0.0009 = 202.8$. With $p = 0.5$, $3.8416 \times 0.25 = 0.9604$, $0.9604/0.0009 = 1067.1$.
(4) 203 trials with the pilot and 1068 without, rounding up. But ±3 points around 5% is a 60% relative error, so for rare events

specify relative error.

Narrowing error bars (Q15).

(1) Half-width $h \propto 1/\sqrt{n}$, so $n_2 = n_1(h_1/h_2)^2$.
(2) $n_1 = 25$, $h_1 = 1.4$ ms, $h_2 = 0.5$ ms, 4 min per run.
(3) $1.4/0.5 = 2.8$, $2.8^2 = 7.84$, $25 \times 7.84 = 196$. Time $196 \times 4 = 784$ min, $784/60 = 13.1$ h.
(4) 196 runs, about 13 h, against 100 min for the first 25. A 2.8× better precision costs 7.84× the time.

Error bars at $n = 5$ (Q21).

(1) SE $= s/\sqrt{n}$. 95% CI half-width $= t_{[0.025,\, n-1]}\, s/\sqrt{n}$. SD is just s.
(2) $n = 5$, df $= 4$, t row 4 column 0.025 gives 2.776. Then $n = 10$, $t = 2.262$.
(3) $\sqrt{5} = 2.236$, so SE $= s/2.236 = 0.45s$ and CI $= 2.776s/2.236 = 1.24s$. At $n = 10$, CI $= 2.262s/3.162 = 0.72s$.
(4) At $n = 5$ it's SE < SD < CI. At $n = 10$ the CI half-width ($0.72s$) is below the SD. The guide's ranking sentence has CI and SD swapped.

CI study guide design answers. Simulation waiting times for 100,000 consecutive customers, use batching after deleting warm-up. Speedups on 12 benchmarks, use the log transform. 40 independent skewed job times when the claim is the mean, use the bootstrap with BCa (Q18). 200 iterations inside one JVM are pseudo-replication. Bootstrap over JVM invocations instead, or delete warm-up and use batch means (Q19). Speed tests users run when their connection feels slow are a selection (collider) bias. A bigger sample only narrows the interval around the wrong value (Q23). Reporting the best of 3 runs is biased (the max grows with n) and not reproducible. Show all runs, or use interquartile mean and bootstrap intervals (Q25). The bootstrap still assumes independent draws and can't add information the sample lacks (Q26).

CI in practice handout, paper one-liners. Agarwal et al. (NeurIPS 2021, deep RL with 3 to 5 runs) use stratified bootstrap CIs, resampling within tasks, on the interquartile mean, plus performance profiles. Miller (2024, LLM evals) treats benchmark questions as a sample, adds clustered SEs for related questions and power for planning. Card et al. (EMNLP 2020) compute minimum detectable effects, the sample-size formula run backwards. Arcuri and Briand, ICSE 2011 "Practical Guide" and STVR 2014 "Hitchhiker's Guide", want many runs, effect sizes and nonparametric tests for randomized algorithms. Georges et al. (OOPSLA 2007) build t CIs for Java start-up and steady state, treat JIT warm-up as systematic, and use interval overlap as an informal test. Mytkowicz et al. (ASPLOS 2009) show link order and environment size shift results enough to flip conclusions, and none of 133 surveyed papers handled it. Stabilizer (ASPLOS 2013) re-randomizes code, stack and heap layout so the CLT applies. SMARTS (ISCA 2003) samples simulation to get CPI within ±3% at 99.7% confidence from under 50 million instructions, with a pilot for σ. Sakai (SIGIR Forum 2014) wants effect sizes and CIs in IR, Fuhr (2017) lists ten mistakes, Sakai (2020) warns against making them mandatory. Dragicevic (2016) says SD, SE and 95% CI bars differ by about $\sqrt{n}$ and then about 2, and uses logs for ratios so 2× and $\frac{1}{2}$ are symmetric. Kohavi et al. (KDD 2013, Bing A/B tests) show that with millions of users every tiny effect is significant. Furia et al. (TSE 2021) use Bayesian credible intervals. Schloegel et al. (S&P 2024) check 150 fuzzing papers from 2018 to 2023 against Klees et al., who used 30 trials, and find the guidance widely ignored. Salamatian (HotNets 2025) brings causal inference to Internet measurement, uses synthetic control and placebo p-values, and finds the about 7 ms IXP drop not significant. Henning et al. (FSE 2025) ran 2366 cloud runs and found daily and weekly throughput patterns, so time of day is a factor. Raasveldt et al. (DBTest 2018) report medians with quantile CIs, and config choices alone gave M beats P, P beats S, S beats M. Pawlikowski et al. (2002) found most simulation papers justified neither run length nor uncertainty, and prescribe warm-up deletion plus batch means.

Hypothesis testing (Lecture 9)

Error trade-off. At a fixed n, lowering α raises β. The only way to lower both is more data. $\alpha = 0.05$ is a convention, a false-positive rate you agreed to, not a law of nature (HT p.6).

One-sided power. A one-sided test has a smaller critical value (1.645 vs 1.96 at 0.05), so it has more power, but only in the direction you named first. It can't reject at all in the other direction (HT p.9).

Example. Suppose $Z = 1.80$ and H_1 said "larger". One-sided, $1.80 > 1.645$, so reject ($p = \Pr[Z \geq 1.80] = 0.036$ from the normal table). Two-sided, $1.80 < 1.96$, so don't reject ($p = 2 \times 0.036 = 0.072$). Same data, and only the one-sided test sees it. If Z had been -3, the one-sided "larger" test couldn't reject at all.

Worked example, σ known (HT p.10). $n = 50$ files, $\bar{x} = 12.3$ KB, $\sigma = 0.5$ KB, H_0: $\mu = 12.5$, two-sided, $\alpha = 0.05$.

Formula. $Z = \dfrac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$, where $\bar{x}$ is the sample mean, μ_0 the mean that H_0 claims, σ the known population SD and n the number of files. The CI is $\bar{x} \pm z[\alpha/2]\,\sigma/\sqrt{n}$.

Plug in. $Z = \dfrac{12.3 - 12.5}{0.5/\sqrt{50}}$ and CI $12.3 \pm 1.96 \times 0.5/\sqrt{50}$.

Work it out. $\sqrt{50} = 7.071$, so the standard error is $0.5/7.071 = 0.0707$. Top is $12.3 - 12.5 = -0.2$. $Z = -0.2/0.0707 = -2.83$. Critical value from the z table at $\alpha/2 = 0.025$ in the upper tail is 1.96, and $|-2.83| > 1.96$. For p, the normal table at $z = 2.83$ gives $\Phi = 0.9977$, so one tail is 0.0023 and $p = 2 \times 0.0023 = 0.005$. Margin is $1.96 \times 0.0707 = 0.139$, so the CI is $12.3 - 0.139 = 12.16$ to $12.3 + 0.139 = 12.44$.

Result. Reject H_0, $p = 0.005$, 95% CI (12.16, 12.44), which excludes 12.5. The true mean size really is below 12.5 KB, but only by about 0.2 KB, and whether that matters is a separate question.

Worked example, σ unknown (HT p.12). $n = 5$, $\bar{x} = 12.3$ KB, $s = 0.5$ KB, sizes assumed normal, H_0: $\mu = 12.35$, two-sided, $\alpha = 0.05$.

Formula. $t = \dfrac{\bar{x} - \mu_0}{s/\sqrt{n}}$ with $n - 1$ df, where s is the sample SD (it replaces the unknown σ). CI $\bar{x} \pm t[\alpha/2;\, n-1]\, s/\sqrt{n}$.

Plug in. $t = \dfrac{12.3 - 12.35}{0.5/\sqrt{5}}$, df $= 5 - 1 = 4$, CI $12.3 \pm t[0.025; 4] \times 0.5/\sqrt{5}$.

Work it out. $\sqrt{5} = 2.236$, so the SE is $0.5/2.236 = 0.2236$. Top is -0.05, so $t = -0.05/0.2236 = -0.224$. In the t table go to row 4 df, column 0.025 (two-sided 95%), which reads 2.776. 0.224 is far inside ± 2.776. It's even below the 0.10 column (1.533), so p is well above 0.20. Software gives $p = 0.83$. Margin $2.776 \times 0.2236 = 0.621$, so the CI is $12.3 - 0.621 = 11.68$ to $12.3 + 0.621 = 12.92$.

Result. Don't reject, $p = 0.83$, 95% CI (11.68, 12.92). It contains 12.35 and almost everything else. In plain words, with five files we couldn't have seen a difference.

Two intervals, same verdict (HT p.16). $(-100, 100)$ and $(-1, 1)$ both contain 0, so both tests say "don't reject". The first means "we have no idea", the second means "any effect is tiny". Only the interval tells them apart, so report the interval (Jain's argument).

Same effect, three sample sizes (HT p.17). Effect 0.3 ms, $\sigma = 50$ ms, $Z = \text{effect}/(\sigma/\sqrt{n})$.

n	Z	p	verdict
1,000	0.19	0.85	not significant
100,000	1.90	0.058	not significant
1,000,000	6.00	2×10^{-9}	"highly significant"

One row by hand. Formula. $Z = \text{effect}/(\sigma/\sqrt{n})$, the effect divided by its standard error.

Plug in. For $n = 1{,}000{,}000$, $Z = 0.3/(50/\sqrt{1{,}000{,}000})$.

Work it out. $\sqrt{1{,}000{,}000} = 1000$, so the SE is $50/1000 = 0.05$ and $Z = 0.3/0.05 = 6.00$. For $n = 1000$, $\sqrt{1000} = 31.6$, SE $= 50/31.6 = 1.58$, $Z = 0.3/1.58 = 0.19$, and the normal table at 0.19 gives $\Phi = 0.5753$, so $p = 2(1 - 0.5753) = 0.85$. For $n = 100{,}000$, $\sqrt{100{,}000} = 316$, SE $= 0.158$, $Z = 1.90$, $\Phi(1.90) = 0.9713$, $p = 2 \times 0.0287 = 0.057$ (slide rounds to 0.058).

Result. Same 0.3 ms, and only n moved Z from 0.19 to 6. Page load 250.0 to 249.7 ms is a 0.12% change no user notices. With too little data the reverse happens, an important effect is "not significant". Since Z grows like $\sqrt{n}$, four times the data doubles Z for the same effect (HT study guide).

Effect size families (HT p.18). Raw difference in original units (systems and networking). Relative change or speedup, usually a geometric mean across benchmarks (architecture, compilers). Cohen's $d = (\bar{x}_1 - \bar{x}_2)/s$, unit-free, where $\bar{x}_1, \bar{x}_2$ are the group means and s the SD (HCI, CS education, SE meta-analyses). Probability of superiority, which is the same quantity as ROC AUC in ML (fuzzing, test generation). Cohen's small, medium, large are 0.2, 0.5, 0.8 and come from behavioural science, so with low-noise systems data $d = 5$ can still be a negligible speedup.

Example. Formula. $d = (\bar{x}_1 - \bar{x}_2)/s$.

Plug in. Study times 52 and 50 min with $s = 4$ min gives $d = (52 - 50)/4$. A kernel at 10.1 vs 10.0 ms with $s = 0.02$ ms gives $d = (10.1 - 10.0)/0.02$.

Work it out. $2/4 = 0.5$ and $0.1/0.02 = 5$.

Result. The first is "medium" by Cohen. The second is "huge" by Cohen, yet it's a 1% speedup that may not matter to anyone. Decide the smallest effect that matters before the experiment (it sets n), and report the effect with an interval after.

Power choices (HT p.19). Power 0.80 at $\alpha = 0.05$ is Cohen's convention. $\beta = 0.20$ against $\alpha = 0.05$ treats a missed effect as a quarter as serious as a false alarm. δ is a judgement and σ comes from a pilot or published measurements.

Formula. $n = \big((z[\alpha/2] + z[\beta])\,\sigma/\delta\big)^2$, where $z[\alpha/2]$ is the two-sided critical value, $z[\beta]$ the upper-tail z for the miss rate $\beta = 1 - \text{power}$, σ the run-to-run SD and δ the smallest difference worth detecting. Since σ/δ is the same whatever power you pick, the cost of extra power is just the ratio of the $(z[\alpha/2] + z[\beta])^2$ factors.

Plug in. From the z table, $z[0.20] = 0.842$, $z[0.10] = 1.282$, $z[0.05] = 1.645$, with $z[0.025] = 1.96$ each time.

Work it out. Power 0.80 gives $(1.96 + 0.842)^2 = 2.802^2 = 7.85$. Power 0.90 gives $3.242^2 = 10.51$, and $10.51/7.85 = 1.34$. Power 0.95 gives $3.605^2 = 13.0$, and $13.0/7.85 = 1.66$.

Result. Power 0.90 needs 34% more runs and 0.95 needs 66% more.

Small example. $\sigma = 10$ ms, $\delta = 5$ ms, power 0.80. $n = 7.84 \times (10/5)^2 = 7.84 \times 4 = 31.4$, round up to 32 runs. Halving δ to 2.5 ms gives $7.84 \times 16 = 125.4$, so 126 runs, four times as many. Power computed after the fact from your own p tells you nothing. "We ran it five times" is not a sample size, it's the absence of one. An underpowered study doesn't just miss effects, the ones it does find are exaggerated.

Where dependence hides (HT p.20). Within one run (consecutive latencies share caches, queues, GC, background load). Within one source (many requests from one user, session or connection). Within one host (many runs on one machine or one boot). The classic mistake is 10,000 requests from one run of each system analysed as $n = 10{,}000$. It's really one replicate per system.

Autocorrelation formulas (HT p.21). Lag-1 $\rho = \text{Corr}(x_t, x_{t+1}) = \text{Cov}(x_t, x_{t+1})/\sigma^2$, between -1 and $+1$, where x_t

is the measurement at time t. $\mathrm{Var}(\bar{x}) \approx \frac{\sigma^2}{n} \cdot \frac{1+\rho}{1-\rho}$. Positive ρ (drift, warm-up, caches) gives $n_{\text{eff}} < n$. Negative ρ (high and low alternate, e.g. GC every other run) gives $n_{\text{eff}} > n$. At $\rho = 0.5$, $n_{\text{eff}} = n/3$, so the naive $|Z|$ is too large by $\sqrt{3} = 1.73$.

Formula. $n_{\text{eff}} \approx n(1-\rho)/(1+\rho)$, the number of independent runs your n correlated runs are worth.

Plug in. $n = 30$, $\rho = 0.5$, so $n_{\text{eff}} = 30(1-0.5)/(1+0.5)$.

Work it out. $0.5/1.5 = 1/3$, so $n_{\text{eff}} = 30/3 = 10$. The true SE is $\sigma/\sqrt{10}$ instead of $\sigma/\sqrt{30}$, which is $\sqrt{30/10} = \sqrt{3} = 1.73$ times bigger. So a naive $Z = 2.5$ is really $2.5/1.73 = 1.44$.

Result. The naive test rejects ($2.5 > 1.96$) but the honest one doesn't ($1.44 < 1.96$). Thirty correlated runs carry the information of ten. Actual Type I rate at nominal 0.05 is 5%, 11%, 26%, 51%, 65% at $\rho = 0$, 0.2, 0.5, 0.8, 0.9. In plain words, correlation doesn't just widen the interval, it makes the test lie.

Designing for independence (HT p.22). Analyse one summary per run (mean, p99) and test across runs. Replicate what varies (runs, reboots, machines, days). Interleave the order of the alternatives. Block or pair (same benchmark, machine, time). Plot measurements in time order and check lag-1 correlation. With interference (shared link or cluster) randomise whole clusters. Netflix bitrate capping (Spang et al., IMC 2021) showed about 5% lower throughput when tested on a fraction of traffic but about 12% higher when most traffic was capped, so the sign flipped. The study guide adds switchback designs (switch a whole link or region between treatment and control over time) and paired-link designs.

Peeking cost, simulated A/A test (HT p.24). Two-sided Z, stop the first time $|Z| > 1.96$. Actual Type I rate is 5% with 1 look, about 8% with 2 (HT study guide), 14% with 5, 25% with 20, 37% with 100. Keep looking long enough and you'll always reject.

Stopping rules that keep α (HT p.25). Fixed horizon, pick n from power and look once. Stopping for a deadline or budget is fine. Group sequential, plan K looks in advance with a stricter boundary each time. Pocock uses the same boundary every look. O'Brien-Fleming is very strict early and comes back to about 1.96 at the final look. Always-valid p-values (sequential probability ratio tests) allow any stopping rule and are used on A/B platforms. Stopping the moment a result is significant also inflates the effect size you report, so state the stopping rule either way.

The family (HT p.26). The family is the set of tests whose conclusions are read together, and you must define it before seeing p-values. 5 schedulers × 8 workloads × 3 metrics = 240 tests.

Formula. Expected false positives $= m\alpha$, with m tests each at level α and every H_0 true.

Plug in. 240×0.05.

Result. About 12 "findings" even when nothing is real. Picking out the significant ones afterwards doesn't keep Pr[Type I] at α.

More FWER values (HT p.27).

Formula. $\mathrm{FWER} = 1 - (1-\alpha)^m$, the chance at least one of m independent true-null tests at level α rejects. It's 1 minus the chance all m come out clean.

Plug in. $1 - 0.95^{10}$, $1 - 0.95^{20}$, $1 - 0.95^{100}$.

Work it out. Square by hand. $0.95^2 = 0.9025$, $0.95^4 = 0.9025^2 = 0.8145$, $0.95^5 = 0.8145 \times 0.95 = 0.7738$, $0.95^{10} = 0.7738^2 = 0.5987$, $0.95^{20} = 0.5987^2 = 0.3585$. For 100, $0.95^{40} = 0.3585^2 = 0.1285$, $0.95^{80} = 0.1285^2 = 0.0165$, $0.95^{100} = 0.0165 \times 0.3585 = 0.0059$.

Result. $1 - 0.5987 = 40\%$ at $m = 10$, $1 - 0.3585 = 64\%$ at $m = 20$, $1 - 0.0059 = 99\%$ at $m = 100$. Twenty tests is a coin flip on whether the table holds a false claim.

Holm beats Bonferroni (HT p.28). Holm has the same FWER guarantee and always rejects at least as many, so there's no reason to use plain Bonferroni once you have the p-values. Bonferroni's guarantee holds under any dependence.

Worked through HT p.29. Sorted p-values 0.001, 0.008, 0.012, 0.030, 0.041, 0.200, so $m = 6$, $\alpha = 0.05$.

Formulas. Bonferroni rejects when $p \le \alpha/m$. Holm compares the k-th smallest $p_{(k)}$ with $\alpha/(m-k+1)$ and stops at the first failure. BH finds the largest k with $p_{(k)} \le k\alpha/m$ and rejects the first k.

Plug in and work it out. Bonferroni cut $0.05/6 = 0.0083$. 0.001 and 0.008 pass, 0.012 fails, so 2. Holm cuts are $0.05/6 = 0.0083$, $0.05/5 = 0.010$, $0.05/4 = 0.0125$, $0.05/3 = 0.0167$. Then $0.001 \le 0.0083$, $0.008 \le 0.010$, $0.012 \le 0.0125$, but $0.030 > 0.0167$ stops it, so 3. BH cuts are $k \times 0.0083$, which is 0.0083, 0.0167, 0.025, 0.0333, 0.0417, 0.05. The largest k that passes is $k = 5$ ($0.041 \le 0.0417$), and $0.200 > 0.05$, so 5.

Result. No correction 5 (everything except 0.200), Bonferroni 2, Holm 3, BH 5. Same data, different number of discoveries.

Common misreading of α (HT study guide Q15). α is Pr[reject | H_0 true], the rate among true nulls. It is not the share of significant published results that are false. That share also depends on power and on how many tested effects are real, and can be far above 5%. Showing two systems are equivalent needs an equivalence test or a narrow interval around 0, not a large p.

McNemar condition (HT study guide). The $Z = (b-c)/\sqrt{b+c}$ normal approximation wants $b + c \gtrsim 25$ disagreements. Here b is the number of test items only classifier A got right and c the number only B got right. Items both got right or both got wrong are ignored.

Example. Plug in. $b = 40$, $c = 24$, so $Z = (40-24)/\sqrt{40+24}$.

Work it out. $b + c = 64 \ge 25$, so the approximation is OK. $\sqrt{64} = 8$ and $Z = 16/8 = 2.00$. The normal table gives $\Pr[Z \ge 2.00] = 0.0228$, so $p = 2 \times 0.0228 = 0.046$.

Result. Reject at 0.05, A is more accurate, but only just.

CS communities that hit multiple testing (HT p.30). FuzzBench (ESEC/FSE 2021) uses per-benchmark Mann-Whitney tests plus Friedman/Nemenyi critical-difference diagrams across benchmarks. Otero et al. (ECIR 2025) found unadjusted pairwise tests of many IR systems badly inflate Type I error and recommend BH. Zhao et al. (SIGMOD 2017) treat every chart in an interactive exploration tool as an implicit test, 100 tests at 0.05 give about 13 "discoveries", roughly 40% bogus. Cockburn et al. (CACM 2020) cover p-hacking and HARKing in empirical CS. A results table with one row per benchmark and a column of asterisks is a family of tests.

Effect size reporting is rare. Kampenes et al. (2007) found only 29% of 92 controlled SE experiments reported any effect size (HT study guide reading list). A_{12} comes from Vargha and Delaney (2000) (HT p.33).

Comparing two alternatives (Lecture 10)

Three things to settle first (Comp p.4). The response (mean latency, p99, hit ratio, energy per request), the experimental unit, and the design (can both systems run on the same unit). You can change your test after the fact, but you can't change your design after the fact.

Wrong vs wasteful (Comp p.5). Don't pair when measuring one system destroys or changes the unit, or the two samples come from different populations. A paired analysis on unpaired data is wrong. An unpaired analysis on paired data is merely wasteful. Pairing is free power.

Cost of pairing (Comp p.6). You get $n - 1$ df instead of $n_A + n_B - 2$. With 8 benchmarks that's 7 df instead of 14, so the 95% critical t goes from 2.145 (row 14) to 2.365 (row 7), a small price. Worth it whenever $\rho > 0$. Benchmarks, traces and users differ from each other far more than systems do, so on a paired unit that difference cancels in $A_i - B_i$. Unpaired means different

units, so $\rho = 0$ and that variation hides the effect.

Eight-benchmark demo, full numbers (Comp p.7-8). B is slower on every benchmark by 0.5 to 4.2 s, but the benchmarks range from 7 to 64 s.

	Paired t	Welch
estimate	+1.83 s	+1.83 s
noise used	$s_d = 1.41$ s	between-benchmark SD ≈ 21.5 s
t	3.67 (df 7)	0.16 (df ≈ 13.9)
p	0.008	0.87
95% CI	(+0.65, +3.00)	(−22.0, +25.7)

Paired, by hand. Formula. $t = \bar{d}/(s_d/\sqrt{n})$ with $n-1$ df, where $d_i = B_i - A_i$, $\bar{d}$ is their mean and s_d their SD. CI $\bar{d} \pm t[0.025; n-1]\, s_d/\sqrt{n}$.

Plug in. $d = 0.8, 2.2, 0.5, 4.2, 1.5, 1.3, 3.6, 0.5$ and $n = 8$.

Work it out. $\sum d = 14.6$, so $\bar{d} = 14.6/8 = 1.825$. $\sum d^2 = 40.52$ and $n\bar{d}^2 = 8 \times 1.825^2 = 26.645$, so $\sum(d-\bar{d})^2 = 40.52 - 26.645 = 13.875$. Then $s_d^2 = 13.875/7 = 1.982$ and $s_d = 1.408$. SE $= 1.408/\sqrt{8} = 1.408/2.828 = 0.498$. $t = 1.825/0.498 = 3.67$. The t table row 7, column 0.025 gives 2.365, and the 0.005 column gives 3.499, so 3.67 beats both and $p < 0.01$ (software 0.008). Margin $2.365 \times 0.498 = 1.18$, CI (0.65, 3.00).

Welch, summary numbers plugged in. Formula. $t = (\bar{x}_B - \bar{x}_A)/s$ with $s = \sqrt{s_A^2/n_A + s_B^2/n_B}$.

Plug in. $\bar{x}_A = 29.79$, $\bar{x}_B = 31.61$, $s_A = 21.51$, $s_B = 22.90$, $n = 8$ each.

Work it out. $21.51^2/8 = 57.8$ and $22.90^2/8 = 65.6$, so $s = \sqrt{123.4} = 11.11$. $t = 1.825/11.11 = 0.16$. With $\nu \approx 13.9$, $t[0.025; 13.9] \approx 2.146$, margin $2.146 \times 11.11 = 23.8$, CI (−22.0, +25.7).

Result. Paired width is $3.00 - 0.65 = 2.35$ s and Welch width is $25.7 + 22.0 = 47.7$ s, about 20 times wider. Same estimate, a 20× wider interval and the opposite conclusion. No amount of extra runs makes the unpaired design as efficient here.

In software (Comp p.10). `scipy.stats.ttest_rel(a,b)` paired t, `ttest_ind(a,b,equal_var=False)` Welch, `wilcoxon(a,b)` signed-rank, `mannwhitneyu(a,b)` Mann-Whitney. None returns a CI by default. `stats.bootstrap` gives an interval for any statistic, like a difference of medians or p99s. In R, `t.test(a,b,paired=TRUE)` and `wilcox.test()` print the interval.

Paired cache example, the test (Comp p.11). Hit ratios on six workloads, $d = A - B = +0.07, +0.09, -0.05, -0.06, +0.10, -0.03$.

Formula. $t = \bar{d}/(s_d/\sqrt{n})$ with $n-1$ df, where $\bar{d}$ is the mean difference and s_d the SD of the differences. 90% CI $\bar{d} \pm t[0.05; n-1]\, s_d/\sqrt{n}$.

Plug in. $n = 6$, $\sum d = 0.12$.

Work it out. $\bar{d} = 0.12/6 = 0.020$. Deviations from 0.020 are $0.05, 0.07, -0.07, -0.08, 0.08, -0.05$. Squares are $0.0025, 0.0049, 0.0049, 0.0064, 0.0064, 0.0025$, summing to 0.0276. $s_d^2 = 0.0276/5 = 0.00552$, $s_d = 0.0743$. SE $= 0.0743/\sqrt{6} = 0.0743/2.449 = 0.0303$. $t = 0.020/0.0303 = 0.66$ with 5 df. In the t table row 5, even the 0.10 column is 1.476, so one tail is above 0.10 and two-sided $p > 0.20$ (software 0.54). For the CI, row 5 column 0.05 gives 2.015. Margin $2.015 \times 0.0303 = 0.061$, so $0.020 \pm 0.061 = (-0.041, +0.081)$.

Result. Don't reject, $p = 0.54$. The interval fits B being 4 points better and A being 8 points better. That's "six workloads told us very little", not "equivalent".

Pooled vs Welch (Comp p.13). The old rule was to pool if one sample variance is under about 4× the other. Current advice is Welch by default since it's barely less powerful when variances are equal (Delacre, Lakens and Leys 2017, Comp p.30). On the Comp p.14 data pooled gives $p = 0.0501$ and Welch $p = 0.0404$, opposite verdicts at 0.05.

Data. $n_A = 7$, $\bar{x}_A = 0.2657$, $s_A = 0.0934$, $n_B = 9$, $\bar{x}_B = 0.4011$, $s_B = 0.1447$, and $\bar{x}_A - \bar{x}_B = -0.1354$.

Formulas. Pooled $s_p^2 = \big((n_A-1)s_A^2 + (n_B-1)s_B^2\big)/(n_A+n_B-2)$ is the shared variance, and $t = (\bar{x}_A - \bar{x}_B)/\big(s_p\sqrt{1/n_A + 1/n_B}\big)$ with $n_A + n_B - 2$ df. Welch $t = (\bar{x}_A - \bar{x}_B)/\sqrt{s_A^2/n_A + s_B^2/n_B}$ with ν df (formula below).

Work it out, pooled. $s_A^2 = 0.00872$, $s_B^2 = 0.02094$. $s_p^2 = (6 \times 0.00872 + 8 \times 0.02094)/14 = (0.0523 + 0.1675)/14 = 0.01570$, $s_p = 0.1253$. $\sqrt{1/7 + 1/9} = \sqrt{0.2540} = 0.504$, so SE $= 0.1253 \times 0.504 = 0.0632$. $t = -0.1354/0.0632 = -2.14$ with 14 df. Row 14 column 0.025 is 2.145, so 2.14 just misses.

Work it out, Welch. $0.00872/7 = 0.001246$ and $0.02094/9 = 0.002326$, sum 0.003573, $s = \sqrt{0.003573} = 0.0598$. $t = -0.1354/0.0598 = -2.26$. With $\nu = 13.6$ the critical value is about 2.15 (between row 13's 2.160 and row 14's 2.145), so 2.26 clears it.

Result. Pooled $p = 0.0501$ (don't reject), Welch $p = 0.0404$ (reject). The variance ratio is $0.02094/0.00872 = 2.4$, under 4, so the old rule would have said pool. Pick Welch before you look.

Unpaired example, the interval (Comp p.14).

Formula. $(\bar{x}_A - \bar{x}_B) \pm t[\alpha/2; \nu]\, s$, with $s = \sqrt{s_A^2/n_A + s_B^2/n_B}$, and the Welch df is

$$\nu = \frac{(s_A^2/n_A + s_B^2/n_B)^2}{(s_A^2/n_A)^2/(n_A-1) + (s_B^2/n_B)^2/(n_B-1)}.$$

Plug in. $s_A^2/n_A = 0.001246$, $s_B^2/n_B = 0.002326$, sum 0.003573, $s = 0.0598$, difference −0.1354, 90% so $\alpha/2 = 0.05$.

Work it out. Top of ν is $0.003573^2 = 1.276 \times 10^{-5}$. Bottom is $0.001246^2/6 + 0.002326^2/8 = 2.59\times10^{-7} + 6.77\times10^{-7} = 9.35\times10^{-7}$. $\nu = 12.76/0.935 = 13.6$. The t table column 0.05 reads 1.771 at row 13 and 1.761 at row 14, so 13.6 gives about 1.765. Margin $1.765 \times 0.0598 = 0.1055$. Ends $-0.1354 - 0.1055 = -0.241$ and $-0.1354 + 0.1055 = -0.030$ (slide prints −0.029 from unrounded values).

Result. 90% CI (−0.241, −0.029). It means B's hit ratio is higher by between 3 and 24 points. That's a wide range, so say so. Unequal n is fine for Welch, but ask why A has only 7 workloads.

Overlap by eye (Comp p.15-16). Three cases. No overlap means they differ at that level. A little overlap means possibly still significant, so do the test. Each mean inside the other's interval means no evidence of a difference (not evidence of equality). For 95% intervals, an overlap of up to about half a margin of error is still significant at 0.05. Comp p.16 example, 90% intervals from $\bar{x} \pm t[0.05; n-1]\, s/\sqrt{n}$. For A, $0.0934/\sqrt{7} = 0.0353$, row 6 gives 1.943, margin 0.0686, so $0.2657 \pm 0.0686 = (0.197, 0.334)$. For B, $0.1447/3 = 0.0482$, row 8 gives 1.860, margin 0.0897, so $0.4011 \pm 0.0897 = (0.311, 0.491)$. They overlap from 0.311 to 0.334, neither mean is inside the other interval (case 2), and the difference interval (−0.241, −0.029) excludes 0. Eyeballing would have said "no difference". In plain words, the SE of a difference is smaller than the sum of the two half-widths (Comp study guide). Here the half-widths add to $0.0686 + 0.0897 = 0.158$, which is more than the gap 0.135, but the margin on the difference is only 0.1055.

When normality fails, extra points (Comp p.18). Analysing $\log x$ turns ratios into differences and makes skewed positive data closer to normal. t tests are fairly robust with equal-sized samples, and the deeper question is whether the mean is your quantity at all.

Mann-Whitney worked example (Comp p.20). Time between crashes in weeks, System I $n_1 = 15$, System II $n_2 = 14$.

Formula. Rank all $n_1 + n_2$ values together from smallest. W_1 is the sum of System I's ranks. $U_1 = W_1 - n_1(n_1+1)/2$ counts how many (I, II) pairs have the System I value bigger. Under H_0, $\mu_U = n_1 n_2/2$ and $\sigma_U = \sqrt{n_1 n_2 (n_1+n_2+1)/12}$, and $Z = (U_1 - \mu_U)/\sigma_U$. $A_{12} = U_1/(n_1 n_2)$.

Plug in. Pooled ranks of System I's 15 values (0.12 up to 1.36) are 1, 2, 3, 4, 6, 7, 9, 10, 11, 12, 14, 15, 19, 20, 29. No ties.

Work it out. $W_1 = 162$. $n_1(n_1+1)/2 = 15 \times 16/2 = 120$, so $U_1 = 162 - 120 = 42$. $\mu_U = 15 \times 14/2 = 105$. $\sigma_U = \sqrt{15 \cdot 14 \cdot 30/12} = \sqrt{6300/12} = \sqrt{525} = 22.9$. $Z = (42 - 105)/22.9 = -63/22.9 = -2.75$. The normal table at 2.75 gives $\Phi = 0.9970$, one tail 0.0030, so $p = 2 \times 0.0030 = 0.006$. $A_{12} = 42/(15 \times 14) = 42/210 = 0.20$.

Result. $p = 0.006$, reject at $\alpha = 0.01$. $A_{12} = 0.20$, so a random System I interval beats a random System II interval only 20% of the time. Medians 0.43 vs 0.885 weeks, so System II runs about twice as long between crashes. Mann-Whitney tests stochastic dominance, not medians (Comp p.19).

Signed-rank worked example (Comp p.22). Same six cache differences.

Formula. Rank $|d_i|$ from smallest. W^+ and W^- are the rank sums of the positive and negative d_i, $W = \min(W^+, W^-)$. Exact $p = 2 \times$ (patterns with W this small)$/2^n$. Normal version $\mu = n(n+1)/4$, $\sigma = \sqrt{n(n+1)(2n+1)/24}$, $Z = (W - \mu)/\sigma$. Effect $r = (W^+ - W^-)/(n(n+1)/2)$.

Plug in. Sorted $|d|$ are 0.03, 0.05, 0.06 (all negative) then 0.07, 0.09, 0.10 (all positive), ranks 1 to 6.

Work it out. $W^+ = 4 + 5 + 6 = 15$, $W^- = 1 + 2 + 3 = 6$, check $15 + 6 = 21 = 6 \times 7/2$. $W = 6$. For the exact count, list the sets of ranks that could be negative with sum ≤ 6. Empty is 1, a single rank is 6 sets, pairs $\{1,2\}, \{1,3\}, \{1,4\}, \{1,5\}, \{2,3\}, \{2,4\}$ are 6, and the triple $\{1,2,3\}$ is 1. That's 14 of $2^6 = 64$, so $p = 2 \times 14/64 = 28/64 = 0.44$. Normal version $\mu = 6 \times 7/4 = 10.5$, $\sigma = \sqrt{6 \cdot 7 \cdot 13/24} = \sqrt{22.75} = 4.77$, $Z = (6 - 10.5)/4.77 = -0.94$, $\Phi(0.94) = 0.8264$, $p = 2 \times 0.1736 = 0.35$. $r = (15 - 6)/21 = 0.43$.

Result. Exact $p = 0.44$, don't reject. The normal $p = 0.35$ is too rough at $n = 6$, report the exact one. Hodges-Lehmann shift estimate $+0.02$. At $n = 6$ the smallest possible two-sided p is $2/64 = 0.031$ (Comp study guide).

Sign test on the same data (Comp p.23).

Formula. S is the number of positive d_i, and under H_0 $S \sim \text{Binomial}(n, \frac{1}{2})$, so $\Pr[S = k] = \binom{n}{k}/2^n$.

Plug in. $S = 3$ of $n = 6$.

Work it out. $\binom{6}{k}$ for $k = 0..6$ is 1, 6, 15, 20, 15, 6, 1. $\Pr[S \geq 3] = (20 + 15 + 6 + 1)/64 = 42/64 = 0.66$, and doubling gives more than 1, so it caps at 1.

Result. $p = 1.00$, a 3 to 3 split is exactly what H_0 predicts. Compare signed-rank 0.44 and paired t 0.54. In Python it's `scipy.stats.binomtest(S, n, 0.5)`. Safe under skew but least powerful, prefer $\log(A/B)$ or a bootstrap when you can.

Speedup form (Comp p.24). The demo as a ratio uses the two means, $\bar{x}_A = 238.3/8 = 29.79$ s and $\bar{x}_B = 252.9/8 = 31.61$ s, so $31.61/29.79 = 1.06\times$ slower. The raw difference with its interval is usually the best choice. A ratio helps when the units mean little to the reader.

Runs needed, slide numbers (Comp p.25).

Formula. Paired $n = \big((z[\alpha/2] + z[\beta])\, s_d/\delta\big)^2$, unpaired n per group $= 2\big((z[\alpha/2] + z[\beta])\, s/\delta\big)^2$, where s_d is the SD of the differences, s the SD within one system and δ the smallest difference worth detecting.

Plug in. $\alpha = 0.05$, power 0.80, so $1.96 + 0.84 = 2.80$. Paired $s_d = 0.074$, unpaired $s = 0.12$, $\delta = 0.05$.

Work it out. Paired $0.074/0.05 = 1.48$, $2.80 \times 1.48 = 4.144$, squared 17.2, round up to 18 pairs. Unpaired $0.12/0.05 = 2.4$, $2.80 \times 2.4 = 6.72$, squared 45.2, times 2 is 90.3, round up to 91 per group.

Result. 18 pairs vs 91 per group. Pairing cut the runs by a factor of five. The six-workload study had roughly a one-in-five chance of detecting a 0.05 difference. "We ran it six times" is not a sample size.

What to report (Comp p.26). The design (paired or not, the unit, how many, how they were assigned), the test and why plus the assumption check, t or Z with df and the exact p (not "$p < 0.05$"), the difference with a CI in the response's units, and every comparison, including non-significant ones. If you measured three systems and report one pair, say so.

Three alternatives (Comp p.28). $c = 4$ gives $c(c-1)/2 = 4 \times 3/2 = 6$ pairs. $0.95^6 = 0.95^4 \times 0.95^2 = 0.8145 \times 0.9025 = 0.735$, so $1 - 0.735 \approx 0.26$ chance of at least one false positive. The paired design generalises to blocking.

Names (Comp p.30). Mann-Whitney U and the Wilcoxon rank-sum test are the same test. The signed-rank test is not.

Measurement (Lecture 11)

The lecture's question (Meas p.3). Which errors in this experiment shrink when I repeat it, and which don't? Repetition is cheap and feels like rigour, but it only buys you one of the two. Reported improvements in systems work are often around 10%, inside the range a careless setup can manufacture (Meas p.2, Mytkowicz et al. via Kalibera and Jones).

Three design questions (Meas p.6). How often does the event occur? How much context do you need when it does? How much disturbance can you tolerate? Pick the strategy from those answers. Most bad setups are a tool chosen by habit.

Strategy table rows not elsewhere (Meas p.8). Overhead grows with event rate for counting and tracing, but with sampling rate for sampling. Rare events are counted, captured, or likely missed. Each fails when, respectively, events are frequent, overhead runs out, or you need exactness.

Tools map to the three (Meas p.9). Counting is `perf stat`, hardware counters, app counters. Tracing is ftrace, eBPF/bpftrace, OpenTelemetry spans. Sampling is `perf record`, async-profiler, py-spy, flame graphs. Ask which of the three a tool is and its cost model follows.

Clocks (Meas p.12). `clock_gettime( CLOCK_PROCESS_CPUTIME_ID)` gives CPU time for this process (MONOTONIC gives elapsed). Both are served by the OS without a system call, so they're cheap enough to ignore. `gettimeofday()` is obsolescent and not monotonic, NTP can move it backwards. Report the clock and its resolution like a microscope's magnification.

Cycle counter, two more expired assumptions (Meas p.13). Out-of-order execution reorders the counter read, so it needs serialising. The counter isn't guaranteed synchronised across cores. An instrument's error model has an expiry date.

Warm-up (Meas p.14). The assumption under "discard the first N" is that a steady state exists, and small, well-studied benchmarks often never reach one (Barrett et al., OOPSLA 2017). Plot iterations before summarising. If the series has structure, its mean means nothing.

Clock quantisation example (Meas study guide). A 180 ns event timed on a 100 ns clock reads 100 or 200 ns.

Perturbation details (Meas p.16-17). Instrumentation uses the cycles, cache lines, memory bandwidth, I/O and log storage it is measuring. A traced concurrency bug can vanish because you traced it. Two probes together can cost more than the sum (evicted working set) or less (the second probe's data is already in cache). If you can't run the uninstrumented system, say you can't bound the perturbation instead of calling it small.

Indirect measurement example (Meas p.18). To find the load on a machine you don't control, time a known program, calibrate it when the machine is idle, then invert to recover the competing load. Same shape elsewhere, energy from power draw, contention from latency, capacity from response time.

Measurement bias numbers (Meas p.21). Of 133 papers from ASPLOS, PACT, PLDI and CGO, 0 adequately accounted for measurement bias (Mytkowicz et al., ASPLOS 2009). The effect can be large enough to reverse a paper's conclusion, and every affected measurement is perfectly repeatable.

Randomising, concretely (Meas p.22). Re-randomise code, stack and heap layout between runs (Curtsinger and Berger, STABILIZER, ASPLOS 2013). Randomise the order of conditions instead of all of A then all of B. Spread repetitions across hours and days. Standard design elsewhere, it reached systems measurement about thirty years late.

Cloud machines (Meas p.23). Shared infrastructure brings noisy neighbours, migration, throttling and hardware heterogeneity. Henning et al. (FSE Companion 2025, 2366 benchmark runs over three months on AWS) found detectable variation by hour of day and day of week. Your system at 3 a.m. and the baseline at 2 p.m. gives narrow, repeatable, meaningless results.

Variation reporting numbers (Meas p.25). Kalibera and Jones (ISMM 2013) found 71 surveyed papers gave no measure of variation at all. van der Kouwe et al. (Benchmarking Crimes, 2018) found 96% of 50 security papers had preventable flaws.

Which summary (Meas p.26). Mean is pulled by outliers you know are artifacts, assumes a symmetric error model, and is rarely right for latency. Median and quantiles ignore a handful of slow outliers and assume nothing about the distribution, but need different interval machinery. Fastest runs are the least polluted since interference only slows a run, but the minimum drifts lower the more runs you do.

Effect vs artifacts (Meas p.27). Ask whether your effect is larger than the artifacts you didn't control. If that's unknown, it's the finding to report.

What to record (Meas p.29). Hardware (CPU model and count, governor, turbo, SMT state). Software (OS and kernel, VM or shared host, compiler, flags, library versions). Experiment (workload and how generated, random seeds, warm-up policy and how it was chosen, repetitions at each level). Analysis (which summary and why). Publish the scripts and raw results. A measurement nobody can repeat is an anecdote.

Advisors, CS 701, publishing and ethics

These four lectures have no math and no homework. Expect true/false, "is this acceptable and why", classify-the-misconduct and one h-index calculation. Every fact below has a cite like (CS701 p.4), which is the PDF page of that deck. Luke's plagiarism tech report GMU-CS-TR-2013-8 is cited as (Luke TR p.N). A myth-vs-truth table for all four decks is at the end.

0. Why these lectures are in CS 700 (Intro deck)

The big idea

CS 700 used to be only the "systems" course on experimental methods and statistics. In 2018 the PhD program was restructured to cut time to completion, so the course now also teaches how research works and helps match you with a faculty member for CS 701 (Intro p.2, p.3). That's why advisor, publishing and ethics lectures sit next to confidence intervals.

Facts

- CS 700 is the required "systems" counterpart to the required theory courses CS 583 and CS 600. Its subject is experimental methods and quantitative techniques in CS (Intro p.2).
- Course topics are an overview of the research process (meeting CS faculty, matching for CS 701) plus the "old CS 700" of measurement, simulation and statistical methods (Intro p.3).
- PhD-process topics are what a PhD is and why get one, finding an advisor and working well with one, picking a dissertation topic, finding a job (academia vs industry), research ethics and integrity, how and when to use AI, and tools like LaTeX, plotting and scripting. Reading papers, reviewing papers and giving talks are left to the advanced seminar courses, CS 800 (Intro p.4).
- Early weeks use the second half of class to discuss one of these topics, with pointers to CRA workshop talks. Faculty research talks run through the semester to help you find a 701 match. Guest lectures at the end cover experimental techniques in areas like software engineering and ML (Intro p.5).
- Scientific method has four steps. Identify a problem and form a hypothesis that is testable and refutable, design an experiment, conduct it, then do hypothesis testing with statistics (Intro p.9). *Example.* "Cache policy A gives a lower mean miss rate than LRU on these benchmarks" is a fine hypothesis, since 30 runs could prove it false. "A is a more elegant design" isn't, since no experiment can refute it.
- Feitelson (2006) gives three meanings of experimental CS. Building systems (hardware or software) as the counterpart to theory, experimentation as a feedback step in the engineering loop, and evaluating computer systems with the rigorous methods of the natural sciences. The third one is the focus of this class. Feitelson also argues there's a place for just observing the real world, like analyzing measured network traffic (Intro p.10). *Example.* Building a new file system is meaning one. Timing it, finding a slow path and redesigning is meaning two. Running 30 timed trials against ext4 and comparing CIs is meaning three, the one this class teaches.
- The figure on Intro p.11 compares the scientific method loop (observation, hypothesis or model, concrete prediction, experimental test) with the system design loop (idea, system design, concrete implementation, experimental evaluation).
- Course goals include understanding the trade-offs between simulation, measurement and analytical modeling, rigorously comparing systems in the presence of noise (often "quality" of output matters more than speed, like face recognition), and deciding if a change has a statistically significant impact (Intro p.13). Also using statistics to cut the number of simulations and designing experiments to get the most information for the effort (Intro p.14), drawing conclusions from noisy data (Intro p.15) and aggregating large data for a big-picture view (Intro p.16).
- Grading is 40% homework, 35% midterm, 25% term project. All work is individual under the Mason Honor Code. Lost homework points can be won back by resubmitting within two weeks. The term project should ideally use experimental methods tied to your 701 research (Intro p.8).
- Office hours Wednesday 1.30 to 2.30 PM in Nguyen 5305, GTA Mohammadreza Noei, Piazza for assignment discussion, material on Canvas (Intro p.8).
- Textbook is Lilja, "Measuring Computer Performance". Jain is the alternative and somewhat dated (Intro p.7).
- LaTeX is required for all homework submissions (Intro p.26).

Likely exam questions

1. **Q.** What are the three definitions of experimental CS, and which one is this course about?
   A. Building systems, experimentation as feedback in the engineering loop, and evaluating systems with the rigorous methods of the natural sciences. This course is about the third one (Intro p.10).
2. **Q.** You get 90 on homework, 80 on the midterm, 85 on the project. What's your grade?
   A. Rule. Grade is the weighted mean $0.40\,H + 0.35\,M + 0.25\,P$ (weights from Intro p.8). Plug in $H = 90$, $M = 80$, $P = 85$. Step 1, homework $0.40 \times 90 = 36$. Step 2, midterm $0.35 \times 80 = 28$. Step 3, project $0.25 \times 85 = 21.25$. Step 4, add $36 + 28 + 21.25 = 85.25$. Result 85.25. Check, it must land between 80 and 90 since the weights sum to 1.
3. **Q.** Why was CS 700 changed in 2018?
   A. The PhD program was restructured to reduce time to completion, so the course now also gives a broader introduction to the research process (Intro p.2).
4. **Q.** What must a hypothesis be in the scientific method?
   A. Testable and refutable (Intro p.9).

1. Selecting your PhD advisor (Advisor deck)

The big idea

Picking your research advisor is the biggest decision of your PhD. You can't rush it. Most professors want to work with you informally for a semester or more first, so both sides can see if it's a good fit. Start early, do your homework on their papers, and don't open with "will you be my advisor?".

Facts

- The talk is based on a presentation by Prof. Hakan Aydin, director of the CS PhD program (Advisor p.2).
- Every PhD student gets an **academic** advisor at admission, usually someone with compatible interests. The actual **research** advisor is declared by separate paperwork within the first **24 credits**, which is a hard deadline (Advisor p.3).
- It's the most important decision of the PhD. It may decide your success in the program, how long the PhD takes and your long-term career path. Choosing the advisor and choosing the research area go hand in hand (Advisor p.4).
- "Easy" case. You were already in close contact with a faculty member during admission, so they're probably your academic advisor and maybe already your GRA boss. You know each other. You still go through an initial research phase before officially declaring, and the checklist still helps (Advisor p.5).
- Common case. You have a broad idea of your interests but should stay open-minded. Find faculty in those or close areas, read their recent projects and papers, and contact them after doing that homework (Advisor p.6).
- Three misconceptions are on Advisor p.7. Delaying the search to the last semester because the deadline is 24 credits. Thinking

faculty X will agree just because you want X. Thinking faculty will answer "can you be my advisor?" right away or within a few weeks. All three are wrong.

- **PRCP** means Preliminary Research Co-Operation Phase. It's an unofficial early research involvement before a professor agrees to advise you, unless they already know you from prior publications or class work. It may take one semester or more and lets both sides judge fit (Advisor p.8).
- PRCP is a term Prof. Aydin made up for the talk. It's not an official procedure, but nearly all faculty agree an initial phase matters (Advisor p.9).
- PRCP parts are weekly or biweekly meetings, discussing old and new papers, discussing ideas, group research meetings and working on a real (maybe limited) project. CS 701 in spring, mandatory for PhD students, is a great formal PRCP (Advisor p.10).
- Fall timing. Use Fall to plan your first research involvement, attend research talks (including the CS 700 ones), request meetings with faculty, explain your background and interests, ask how to get involved. Don't ask "can you be my PhD advisor?" this early (Advisor p.11).
- Faculty side. Dedicated students with good backgrounds and work ethic are the most important asset for research-active faculty. They still may decline because they're already overcommitted with advisees (advising takes a lot of time) or they saw a mismatch in the PRCP (Advisor p.12).
- Positive signs faculty look for (the more the better) are good background in the area, willingness to learn by reading and discussing papers, being independent and open to brainstorming, proposing new good ideas, and being able to join their actual research right away even in a limited way (Advisor p.13).
- Student side. During the PRCP look for answers to key questions before committing. Don't expect all answers after one meeting or a few weeks (Advisor p.14).
- You'll work closely with this person for the rest of your PhD. Ask if their research approach and style match you, and if not, whether you can adapt (Advisor p.15).
- "Making progress in research" is vague. Peer-reviewed research is the ultimate objective. Ask what submission or publication numbers and pace they expect and if you're comfortable with it, but don't underestimate how productive you'll become (Advisor p.16).
- Ask how they interact with students (regular meetings, as needed, guidance by a postdoc), whether you want a hands-on or hands-off advisor (each has pros and cons), and funding odds if becoming a GRA next year matters to you (Advisor p.17).
- Ask how they pick students' research projects and dissertation topics, and how much the student gets a say (Advisor p.18).
- Co-advising. Two faculty can co-advise, but only one is declared the official advisor. The thesis names both. Their areas are usually related but complementary. Ask if they're open to it and have done it before. It has pros and cons (Advisor p.19).
- Changing advisor is technically possible and sometimes unavoidable, but it's a last resort. That's why the informal early phase is so important (Advisor p.20).
- Wrap-up. Most faculty take their time before accepting new students. Think carefully yourself, do your homework and show you're committed once you start talking with someone (Advisor p.21).
- More advice is in the CRA-W Grad Cohort workshop talks online ("Resources from Past Years"). The department may sponsor you to attend (Advisor p.22).

Likely exam questions

1. **Q.** True or false. The academic advisor assigned at admission is your official research advisor.
   A. False. The research advisor is declared with separate paperwork within the first 24 credits (Advisor p.3).
2. **Q.** True or false. Since the deadline is 24 credits, I can wait until my last semester to look for an advisor.
   A. False. It's the first misconception on Advisor p.7. The informal PRCP alone can take a semester or more (Advisor p.8), so waiting makes you miss the deadline.
3. **Q.** What is the PRCP and is it an official GMU procedure?
   A. It's an informal phase of working together, one semester or more, before a professor agrees to advise you, so both sides can check fit (Advisor p.8). It's not official. Prof. Aydin made up the name (Advisor p.9).
4. **Q.** Give two reasons a professor may say no to a good student.
   A. They're already over-committed with PhD advisees, since advising takes a lot of time, or they noticed a mismatch during the PRCP (Advisor p.12).
5. **Q.** A student emails a professor in week 2 asking "Can you be my PhD advisor?" and waits for a reply before contacting anyone else. What's wrong?
   A. Asking that this early is what Advisor p.11 says to avoid. Instead read their papers and ask for a meeting about getting involved (Advisor p.6, p.11). Expecting a quick reply is a misconception (Advisor p.7), and faculty take their time (Advisor p.21).
6. **Q.** With co-advising, how many official advisors are there?
   A. One. Only one is declared official, but the thesis names both (Advisor p.19).

2. CS 701 and the CS 700 term project (CS701 deck)

The big idea

CS 701 is a required spring course where each first-year PhD student does a semester-long research project with one faculty supervisor. There are no classes. You and your supervisor sign a form that works like a contract with the department, then you do real hands-on research, write a report and give a short talk. The form is due Oct 31, so you need a supervisor now.

Facts about CS 701

- Full name is Research Experience in Computer Science. Mandatory for all first-year PhD CS students. Offered **only in Spring**. Must be passed with **B or better**. Prerequisite is **B or better in CS 700** (CS701 p.2).
- Each student pairs with a CS faculty supervisor for a semester-long project. The point is to interact with faculty and get research experience (CS701 p.3).
- At the end the student submits a **report to the supervisor**, every 701 student gives a short public talk of about **10 minutes** to the whole department, and the **supervisor submits the grade** to the registrar (CS701 p.3).
- Registration. Find a CS faculty member willing to supervise. Fill out the CS 701 form with the project details (description and deliverables). Student and supervisor sign it. Submit it to `csphd@gmu.edu` and `aydin@cs.gmu.edu`. The department must approve it and may ask for revisions (CS701 p.4).
- The form is a **3-way contract** among the student, the supervisor and the CS department (CS701 p.4).
- After approval the department gives you a **CRN** (course reference number), and you use it to register on **PatriotWeb**. The form, revisions and getting the CRN all take time (CS701 p.5).
- Who can supervise. Any **tenure-track or tenured** CS faculty, meaning assistant, associate or full professors. Not people whose directory title says "Teaching" or "Instructor" (CS701 p.6).
- You can have a **co-supervisor**, who co-signs the form. CS-affiliated faculty can co-supervise. Even with a co-supervisor, the registrar records the course under one CS faculty member (CS701 p.6).
- Email the form to `csphd@gmu.edu` with the subject line "[CS701 form]" and cc `aydin@cs.gmu.edu`. Submitting on time is critical.

Deadline Oct 31, 2026 (CS701 p.7).

- **No regular class meetings.** Student and supervisor meet periodically or as needed. Any required meeting pattern (weekly, biweekly) goes on the form. Periodic meetings help you stay in touch and discuss progress and unclear points (CS701 p.8).
- The project doesn't have to be groundbreaking or immediately publishable. But the effort must match a **3-credit semester-long course** for first-year PhD students. It's much more than a regular course's term project. Don't expect to finish by "sprinting" in the last few weeks. Timely planning and progress in spring is critical (CS701 p.9).
- Literature review or paper reading is likely needed first and is an essential part of many projects. But the project **can't be only** a literature review or survey (CS701 p.10).
- A **hands-on** (implementation) part is required beyond the literature review. Software, simulations or experiments. For a Theoretical CS project, working on a proof for a non-trivial open problem or a non-trivial novel complexity analysis is also OK (CS701 p.11). *Example.* A theory student who proves a new bound for a non-trivial open problem, with no code at all, meets the hands-on rule. Reading ten papers on that problem and summarizing them doesn't.
- Implementing a set of techniques from possibly several papers and evaluating them thoroughly, possibly with new performance metrics, is fine. Proposing new techniques or extending existing ones is also reasonable. Discuss details with the supervisor (CS701 p.12).
- Form field "Description of the research activity". Must be **more than a title**, about one paragraph (attach sheets if needed), and must state the hands-on part (CS701 p.13).
- Form field "Deliverables and other requirements". A **project report is the default and mandatory** deliverable. List other requirements like software or hardware artefacts and periodic meetings. The report goes to the supervisor at the end of spring. Ask the supervisor about format and length (CS701 p.14).
- A conference or journal submission completed by the end of spring **can't** be a deliverable. "A draft paper to be revised and submitted for publication in the future" **is** OK, and the report can then be that draft paper (CS701 p.15).
- A successful 701 can be the first step to an advisor relationship, but that's not the primary intent. Many students end up with a different area and advisor than their 701 (CS701 p.18).
- Suggested steps (CS701 p.27). (1) Identify a few CS faculty in your preferred areas. (2) Contact them and request meetings about possible projects. (3) Collect information from the ones who are available. (4) Decide, fill out the form, get the faculty signature and submit it to `csphd@gmu.edu` and `aydin@cs.gmu.edu`. If nobody in the first group is available, pick a second group (maybe a different area) and repeat.

The 701 misconceptions, myth then truth

- *Myth.* My 701 supervisor automatically becomes my PhD advisor. *Truth.* No. Declaring the PhD advisor is a separate process with a separate form within the first 24 credits (CS701 p.16).
- *Myth.* My dissertation will be in the area of my 701 project. *Truth.* Not necessarily. You can do 701 in area X and the dissertation in area Y (CS701 p.17). *Example.* You do 701 on network measurement with Prof. A, then in year two declare Prof. B in ML as your advisor. Allowed, and many students end up this way (CS701 p.18).
- *Myth.* My first-choice professor will agree. *Truth.* Not necessarily, they may be unavailable for various reasons (CS701 p.19).
- *Myth.* After I contact a professor, we'll meet and decide quickly. *Truth.* No guarantee (CS701 p.20).
- *Myth.* I should wait for the first professor's answer before contacting a second. *Truth.* You shouldn't. Contact several faculty and learn about different projects, to save time (CS701 p.21).
- *Myth.* I can work with only one faculty member. *Truth.* You can have co-supervisors, but only one is the official primary supervisor, and that one must be tenured or tenure-track CS faculty. The registrar records the primary. The co-supervisor can be CS-affiliated faculty, is named next to the primary on the form and co-signs it. You can acknowledge the co-supervisor in the report and talk (CS701 p.22).
- *Myth.* I can wait until late October to search. *Truth.* That would be a big mistake (CS701 p.23).
- *Myth.* If I'm a GRA, my sponsor will be my 701 supervisor. *Truth.* That's most natural but not required (CS701 p.24).
- *Myth.* If I can't find a project in my favorite area, I'll take 701 next year. *Truth.* Not possible, it's a catalog rule (CS701 p.25).
- *Myth.* If I can't find a supervisor, the department will assign one. *Truth.* No. A CS faculty member has to agree to supervise (CS701 p.26).

Facts about the CS 700 term project

- **Individual** project. Pick an experimental CS problem with quantitative analysis of a computer system, technique, algorithm or method. Comparing several techniques quantitatively is also fine. Do the experiments or analysis and write a report (CS701 p.28).
- Must include **experiments and/or simulations**, and you must apply class statistics like confidence intervals and regression (CS701 p.29).
- Ideally pick a topic in your research area, even better tied to your 701, but that's **not required and not graded** (CS701 p.29).
- **Not acceptable** are a literature survey of quantitative or statistical techniques, and applying statistics to polls or surveys (CS701 p.29).
- **1-page proposal** as a PDF on Canvas. It briefly covers the problem or algorithms, the experimental techniques and the methodology. The slide says due midnight Sunday Oct 25, but it was moved to **Nov 2** (CS701 p.30).
- **Status update by Nov 15**, reporting any changes or unexpected problems (CS701 p.30).
- **Dec 7** class is for **5-minute** presentations covering the problem, the techniques you implemented or simulated, and preliminary results (CS701 p.31).
- Final report is a **6-8 page** technical report in an **ACM or IEEE** conference template, submitted on Canvas **by Dec 7** (CS701 p.32).

Likely exam questions

1. **Q.** When is CS 701 offered, what grade is needed, and what's the prerequisite?
   A. Spring only, B or better, and B or better in CS 700 (CS701 p.2).
2. **Q.** A draft form lists an "Assistant Professor, Teaching" as supervisor, the description "Federated learning", the work "read and summarize 15 papers", and the deliverable "paper submitted to INFOCOM by May". Fix every problem.
   A. The supervisor must be tenured or tenure-track, and "Teaching" titles don't qualify (CS701 p.6). That person could only co-supervise and co-sign next to a tenure-line CS primary (CS701 p.22). The description is just a title and must be a paragraph naming the hands-on part (CS701 p.13). A literature review alone isn't allowed (CS701 p.10), so add implementation, simulation or experiments (CS701 p.11). A conference submission can't be a deliverable, but "a draft paper to be revised and submitted in the future" can (CS701 p.15). The mandatory report must be listed (CS701 p.14).
3. **Q.** Who gets the CS 701 form, and what happens after approval?
   A. Email it to csphd@gmu.edu with "[CS701 form]" in the subject and cc aydin@cs.gmu.edu (CS701 p.7). The department

approves it or asks for a revision (CS701 p.4), then gives you a CRN to register on PatriotWeb (CS701 p.5).

4. **Q.** True or false. The 701 project must be publishable, so it can be small.
   A. False both ways. It doesn't need to be groundbreaking or publishable, but the effort must match a full 3-credit course, much more than a term project (CS701 p.9).
5. **Q.** Which is acceptable as a CS 700 term project? (a) CIs on a survey of 40 classmates' IDE preference. (b) A review of how papers use regression. (c) Timing two hash tables 30 runs each at four load factors, compared with CIs and regression.
   A. Only (c). (a) applies statistics to a poll, and (b) is a literature survey of statistical techniques, both banned (CS701 p.29). (c) is an individual quantitative comparison with experiments and class methods (CS701 p.28-29).
6. **Q.** Give the term project deadlines.
   A. Proposal Nov 2 (slide says Oct 25), status update Nov 15, 5-minute talk Dec 7, 6-8 page ACM or IEEE report Dec 7 (CS701 p.30-32).

3. Publication practices in CS (Pub deck)

The big idea

A PhD means doing original research good enough to pass peer review, where other experts judge your paper before it's published. In CS the main venues are conferences, and they're very competitive. Good results aren't enough, because there are a fixed number of slots and writing and organization count too. Rejection is normal, so you fix the paper and resubmit. Never send the same paper to two places at once.

Facts

- A PhD needs original research ending in a dissertation you write and defend. The research must be worthy of publication in peer-reviewed journals and/or conference proceedings (Pub p.2).
- **Peer review** means experts ("peers") review articles before they're officially published or presented. The aim is to keep science honest by filtering out invalid or poor-quality work. Reviewers evaluate independently. Journal editors or conference chairs make the final decision, sometimes after discussion (Pub p.3).
- Why publish. It shows other experts, not just you, your advisor or your group, approve your work. It makes the work widely accessible. Your publication record is your main track record. The slide flags quality vs quantity (Pub p.4).
- Long-term impact is often correlated with the number of **citations**. Important papers can get hundreds or thousands. Citation counts are used to judge researchers and even venues (Pub p.5).
- **h-index** is the largest h such that an author (or journal) has at least h papers each cited at least h times. Slide example, citations 10, 9, 8, 6, 3, 1 (already sorted). Rank 1, $10 \geq 1$ yes. Rank 2, $9 \geq 2$ yes. Rank 3, $8 \geq 3$ yes. Rank 4, $6 \geq 4$ yes. Rank 5, $3 \geq 5$ no. So $h = 4$, since four papers have 4 or more but there aren't five with 5 or more. PhD students' h-index usually isn't high even at the end, but it grows as you publish more good papers (Pub p.6).
- How to compute h. Sort citations from high to low. Go down the list. h is the last rank i where the count is still $\geq i$. The answer is a number of papers, not a citation count.
- Most PhD students start publishing peer-reviewed papers during or after their **3rd year**. It takes time to build background, find a problem, solve it, learn to write a scholarly paper and beat other researchers (Pub p.7).
- Knowing the process early helps you appreciate the competition and build reading, evaluating, writing and presenting skills alongside your research (Pub p.8).
- Venue types are conferences, journals and workshops (Pub p.10).
- Conferences solicit papers on a topic. The **call for papers (CFP)** gives dates, topics, formatting rules and the organizing committee. Authors submit by a deadline and wait for accept or reject. The decision includes comments from **anonymous reviewers** who are on the **program committee (PC)** (Pub p.11).
- Format per the CFP, meaning page limits, font size limits and IEEE or ACM templates. Deadlines might be extended, less likely for first-tier conferences, so never rely on it (Pub p.12).
- **Acceptance rate** is the percent of papers accepted, so rate = accepted ÷ submitted ×100%. For example 60 accepted out of 400 submitted is $60/400 = 0.15$, which is 15%, inside the first-tier range. Lower rate generally means more prestige. First-tier conferences are usually **10-25%**, sometimes under 10%. Your advisor can tell you how competitive venues are, since picking the right venue isn't trivial (Pub p.13).
- Decision process. The PC chair assigns papers to PC members (professors or industry researchers). Each writes an independent review scoring **writing quality, organization, technical contribution and novelty**, with a final recommendation of Strong Accept, Accept, Neutral, Reject or Strong Reject. Note the impact of non-technical factors (Pub p.14).
- The PC meets to discuss all papers and decide, which can take all day or several days. Each conference has a **fixed number of slots** (about 30 to 1000+). Often not all above-threshold papers fit, so some quite reasonable papers get rejected. The PC in a sense "orders" the good papers (Pub p.15). *Example.* Forty papers clear the bar but there are 30 slots. Your paper with Accept, Accept, Neutral lands at rank 35 and is rejected. That's not an error or bias, it's the misconception on Pub p.9.
- Big conferences have **Area Chairs (ACs)** appointed by the PC chair, each overseeing reviews in a sub-area (Pub p.16).
- **Rebuttal**. Increasingly, reviews are released before the final decision so authors can clear up misunderstandings. Not every paper gets this, many are directly accepted or rejected. Author replies are considered in the decision (Pub p.17).
- Accepted papers are resubmitted as a **camera-ready copy** by another deadline, with reviewer comments included when possible. Authors present them at the conference, which is a chance to network and travel (Pub p.18).
- On rejection, don't take it personally or treat it like failing an exam. Use the comments to improve, and talk with your advisor and co-authors about revising and resubmitting to another venue or the same conference's next edition (Pub p.19).
- **Dual submission** is sending the same or mostly the same paper to several venues at once, usually to raise acceptance odds, or trying to get a second paper from the same material. Not allowed. It's often caught and authors get severe sanctions (Pub p.20).
- **Anonymity**. Authors never know the reviewers. In **double-blind** review, increasingly common, reviewers don't know the authors either. The CFP then asks authors to leave off names and affiliations and avoid indirect hints. A non-compliant paper is rejected right away (Pub p.21). The slide asks why it's spreading. The usual answer is so a famous name or school can't sway the reviewers and the paper is judged on content. *Example.* "In our earlier work [7] we built X" or a link to github.com/yourname is an indirect hint, and the paper can be rejected unread. Writing "Smith et al. [7] built X" in the third person and hiding the link fixes it (Pub p.21, the fix is the usual practice).

The 8 publication misconceptions (Pub p.9), all false

1. A publication is basically a graduate-level project report.
2. If the results are correct, my paper will be accepted.
3. If the results are correct, important and novel, it will be accepted.

4. If my advisor says "it's good", it will be accepted.
5. If my paper is rejected, the conference made an error or is biased.
6. My paper was rejected, so all that time and effort was wasted.
7. Only publishing matters, the venue isn't important.
8. The number of publications matters, not quality.

Why they're false. Slots are fixed so good papers get rejected (Pub p.15), non-technical things like writing and organization are scored (Pub p.14), rejection comments help you improve and resubmit (Pub p.19), venue prestige differs (Pub p.13), and quality counts over quantity (Pub p.4).

Likely exam questions

1. **Q.** Find the h-index for citations 5, 15, 2, 6, 11, 0, 6, 1.
   A. Rule. h is the largest h with at least h papers cited at least h times (Pub p.6). Step 1, sort high to low to get 15, 11, 6, 6, 5, 2, 1, 0. Step 2, compare each count with its rank. Rank 1, $15 \geq 1$ yes. Rank 2, $11 \geq 2$ yes. Rank 3, $6 \geq 3$ yes. Rank 4, $6 \geq 4$ yes. Rank 5, $5 \geq 5$ yes. Rank 6, $2 \geq 6$ no, so stop. Step 3, the last rank that passed is 5. Result $h = 5$. Five papers have at least 5 citations, but there aren't six with at least 6.
2. **Q.** An author has 3 papers with 500 citations each. What's h?
   A. Rule as above (Pub p.6). Step 1, sorted list is 500, 500, 500. Step 2, rank 1, $500 \geq 1$ yes. Rank 2, $500 \geq 2$ yes. Rank 3, $500 \geq 3$ yes. There's no rank 4 paper, so h can't be 4. Result $h = 3$. Huge citation counts can't push h past the number of papers.
3. **Q.** True or false. If my results are correct, important and novel, a good conference will accept it.
   A. False. It's a listed misconception (Pub p.9). There's a fixed number of slots so reasonable papers get rejected (Pub p.15), and writing quality and organization are scored too (Pub p.14).
4. **Q.** Name the four review criteria and the five recommendations.
   A. Writing quality, organization, technical contribution and novelty. Strong Accept, Accept, Neutral, Reject, Strong Reject (Pub p.14).
5. **Q.** What must authors do for double-blind review, and what if they don't?
   A. Leave names and affiliations off and avoid indirect hints about who they are. A non-compliant paper is rejected right away (Pub p.21).
6. **Q.** A student submits the same paper to two conferences to improve the odds, planning to withdraw one. Is that OK?
   A. No. That's dual submission, which isn't allowed, is often caught and gets severe sanctions (Pub p.20, Ethics p.14).

4. Research integrity and ethics (Ethics deck, plus Luke TR)

The big idea

You must do research honestly. The three big sins are making up data (fabrication), changing or dropping data (falsification) and using others' words or ideas without credit (plagiarism). Coursework falls under the Honor Code, but research and your dissertation fall under GMU Policy 4007. Citing doesn't make copied words OK, you need quotes or your own words. AI is usually allowed if you disclose it and take full responsibility.

Facts on rules and who governs what

- The PhD goal is original research and contributions to your field, but done ethically and with the highest integrity (Ethics p.2).
- Coursework and academic activities are governed by the **GMU Honor Code**. Research activities, including dissertation research, are governed by **GMU Policy 4007** (Misconduct in Research and Scholarship) (Ethics p.3).
- IEEE, ACM and IFIP each have a Code of Ethics and Professional Conduct for their submissions and publications (Ethics p.4). Much of the deck is adapted from the IEEE Author Ethics Guidelines (Ethics p.5).
- IEEE keeps a list of authors penalized for ethics violations. They can't publish for extended periods, and IEEE informs their institutions, which usually add their own sanctions (Ethics p.6).
- Policy 4007 misconduct is monitored and investigated by GMU's **Office of Research Integrity and Assurance (ORIA)** (Ethics p.15).
- Policy 4007 covers all full-time and part-time employees and anyone affiliated with GMU, graduate students doing research or scholarship producing reports, conference papers, publications or creative works that list GMU affiliation, undergraduates in sponsored research, and anyone at GMU who teaches, researches or does scholarship (Ethics p.16).
- For graduate students, academic misconduct goes **solely through the Honor Code**, except (1) research activities, whether sponsored or not, and (2) master's theses and doctoral dissertations. Those two go through Policy 4007 (Ethics p.16).
- All research misconduct allegations are handled under 4007. The university aims for an impartial, unbiased process. Investigators are chosen for relevant expertise and screened beforehand for bias or personal, professional or financial conflicts of interest. Everyone must keep it confidential and cooperate (Ethics p.17).
- **Research misconduct** means fabrication, falsification or plagiarism in proposing, performing or reviewing research, or in reporting results. Fabricating or falsifying credentials on funding applications and manuscripts can count. It **does not include honest error or differences of opinion** (Ethics p.18).
- 4007 sets rules to receive and review allegations, runs inquiries and investigations (committees, hearings) and communicates results (Ethics p.19).
- GMU offers in-person **Responsible Conduct in Research (RCR)** training through ORIA (Ethics p.21).

FFP definitions (Policy 4007, Ethics p.20)

- **Fabrication**. Making up data or results and recording or reporting them. Slide short form is "inventing or making up data or results" (Ethics p.13).
- **Falsification**. Manipulating research materials, equipment or processes, or changing or omitting data or results, so the research isn't accurately represented. Slide short form adds manipulating equipment or software (Ethics p.13).
- **Plagiarism**. Taking another person's ideas, processes, results or words without appropriate credit.
- Even if you can't explain some trends right away, don't omit or manipulate them. Another group will likely have a similar implementation and get the real results (Ethics p.13).
- *Example.* You rerun the simulator with 20 random seeds and report only the seed where your scheme wins. That's falsification, since the process and results were manipulated so the research isn't accurately shown (Ethics p.13, p.20).
- Quick test. Never measured it? Fabrication. Measured it but changed, hid or tweaked it? Falsification. Someone else's words or ideas without credit? Plagiarism.

Facts on plagiarism and citing

- IEEE defines plagiarism as using another researcher's ideas, results or words without explicitly acknowledging the original author and source. Plagiarism in any form is unacceptable and a serious breach of professional conduct with possibly severe ethical and legal consequences (Ethics p.7).
- Always cite. For a **direct quotation**, put the verbatim text in quotation marks and cite the source (Ethics p.8).
- Usually better is to **paraphrase or summarize** with an in-text citation, for ideas, processes, arguments or conclusions (Ethics p.9).
- Copying the sentence with no quotes and no citation is plagiarism (Ethics p.10).
- Data, results, information, graphics and tables also need a cita-

tion to the original whenever you refer to, adapt or reuse them (Ethics p.11).

- **Self-plagiarism**. The same rules apply to your own published work. Rephrase instead of cut and paste. Exception is extending your own conference paper to a journal submission (Ethics p.12). *Example.* Pasting the background section of your own 2025 workshop paper into a new conference paper is self-plagiarism, so rephrase it and cite the old paper. Turning a conference paper into its journal version is the allowed case.
- **Multiple or concurrent submissions**. A submission must not be under review or already published elsewhere, and that includes papers with substantial overlap. Check with organizers and your advisor if in doubt (Ethics p.14). *Example.* While your conference paper is under review you send a workshop paper with the same experiments and a new title. Not allowed, since substantial overlap counts too (Ethics p.14, Pub p.20).
- **Conference to journal**. The journal version needs substantially more material, typically **30% more for IEEE**, and must cite and include as an attachment the conference version, explaining the differences (Ethics p.14).

Luke's plagiarism report

- Plagiarism is the **second worst** thing in research. The worst is faking your experiments. Plagiarists are often fired, publicly denounced by coauthors, and PhD students get disowned by advisors and thrown out. Avoid even the appearance of plagiarism (Luke TR p.1).
- You can plagiarize someone's **words** or their **descriptive idea**. Light rewording isn't enough unless the idea is well-known and general (Luke TR p.1).
- It harms two groups, the source who loses credit and readers misled into thinking you came up with it. Plagiarism is an ethical offense, copyright violation is a separate legal one you can be sued for. It's not allowed anywhere in a paper, including abstract, intro, conclusion, tutorial material and especially related work (Luke TR p.1).
- Pasted text is plagiarism (Ex. 1). Pasted text plus a citation is still plagiarism (Ex. 2). Pasted text naming the authors and citing is still plagiarism, since the words are still stolen (Ex. 3) (Luke TR p.2).
- Using less of their text, more than a phrase, is still open to accusation. Under a phrase is relatively safe (Ex. 4). Light rewording with the same structure plagiarizes the idea. Never reword "to avoid plagiarizing", write in your own words from the start (Ex. 5). For instance the source says "the assumption of symmetry makes possible update rules which are much more efficient" and you write "when you assume a symmetric model, you can come up with update rules which are more efficient [1]". Same structure, still plagiarism. Rewriting in your own words without looking at the source is much better, but a quote is better still (Ex. 6) (Luke TR p.2).
- An offset block quote with citation is fine, but don't quote huge chunks without reason (Ex. 7). An inline quote in quotation marks is OK, but offset larger ones (Ex. 8) (Luke TR p.3).
- An equation copied without attribution is plagiarism, since math is dense text (Ex. 9). "As Luke and Panait have shown [1], the update rule is…" is dangerously close (Ex. 10). "The update rule (from Luke and Panait [1]) is" is acceptable. Standard notation like the Bellman equation needn't be cited, but when in doubt cite (Ex. 11). More than a few equations needs permission from the author and maybe the publisher (Ex. 12) (Luke TR p.3).
- A copied figure with no citation is plagiarism (Ex. 13). A copied figure with citation isn't plagiarism but is **copyright violation**, because figures are the author's creation and take more effort (Ex. 14). Use a figure only with permission, stated in the caption, however small (Ex. 15). Copying the original caption is plagiarism, treat captions like body text (Ex. 16) (Luke TR p.4).
- An identical bibliography entry is fine since formats are rigid (Ex. 17). A small algorithm clearly marked as from the source is OK, though rewriting it is better (Ex. 18). A large algorithm (a third of a page or more) is like a figure and needs author and publisher permission, and still looks like padding (Ex. 19) (Luke TR p.5).

Facts on AI use

- In general AI tools are OK in submissions if the use is acknowledged in detail and authors take full responsibility, including for any resulting plagiarism or falsification. Check the venue's current rules, which keep changing (Ethics p.22).
- **IEEE**. Any AI-generated content (text, figures, images, code) must be disclosed in the **acknowledgments**. Name the AI system, identify the sections that used it and briefly explain how much. Editing and grammar help is generally outside the policy, but disclosure is still **recommended** (Ethics p.23). *Example.* ChatGPT drafted your related work and drew Fig. 3 in an IEEE paper. The acknowledgments must name ChatGPT, name the related work section and Fig. 3, and say briefly how much it generated.
- **ACM**. Generative AI like ChatGPT **can't be listed as an author**. Using it to create content is permitted but must be fully disclosed, for example in the Acknowledgements. When unsure, disclose. Basic word processing that suggests text, spelling and grammar checks, and translation tools are **exempt and need not be disclosed**. The policy will be updated as the line with tools like MS-Word or Grammarly blurs (Ethics p.24).
- **Elsevier**. AI only to improve readability and language, not to replace key tasks like producing scientific insights or drawing conclusions. Disclose it in the manuscript and a statement appears in the published work. AI can't be an author or co-author or be cited as an author (Ethics p.25). *Example.* For an Elsevier journal, AI smoothing your English is fine if you disclose it. AI writing your conclusions section isn't allowed even with disclosure, since drawing conclusions is a key authoring task.
- **ICML 2024 CFP**. LLMs are allowed as general writing aids. Authors take full responsibility, including for LLM content that's plagiarism or fabrication. LLMs can't be authors (Ethics p.26).
- Always check the latest CFP or journal website. If it says nothing, the organization's general rules (IEEE, ACM) apply. Ask the organizers and your advisor if unsure (Ethics p.27).

Likely exam questions

1. **Q.** Classify each. (a) Two of 30 timing runs are slow and unexplained, so you delete them. (b) A simulator crashed, so you fill the table cell with an interpolated value and don't say so. (c) You copy a sentence from a paper without quotes or citation. (d) You report a wrong mean because of a spreadsheet typo you didn't notice.
   A. (a) Falsification, since omitting data misrepresents the research (Ethics p.20, p.13). (b) Fabrication, a result that was never measured is reported (Ethics p.20). (c) Plagiarism (Ethics p.10, p.20). (d) Not misconduct, it's honest error (Ethics p.18).
2. **Q.** A grad student is accused of misconduct on a homework and, separately, in their dissertation. Which rules apply?
   A. The homework goes through the Honor Code. The dissertation goes through Policy 4007, investigated by ORIA (Ethics p.16, p.15).
3. **Q.** You copy a paragraph word for word, write "As Luke and Panait have pointed out" and add [1]. Plagiarism?
   A. Yes. The idea is credited but the words are passed off as yours (Luke TR p.2, Ex. 3). Use quotation marks or a block quote (Luke TR p.3, Ex. 7-8, Ethics p.8), or paraphrase with a citation (Ethics p.9).

4. **Q.** You put another paper's figure in yours with "from [1]" in the caption. Plagiarism? OK?
   A. Not plagiarism but not OK. It's copyright violation (Luke TR p.4, Ex. 14). Get author and maybe publisher permission and write "reproduced with permission from [1]" in the caption (Ex. 15).
5. **Q.** You turn your 10-page conference paper into an IEEE journal paper. What's required?
   A. Rule. The journal version needs typically 30% more material for IEEE (Ethics p.14), so new length is old length $\times 1.30$. Plug in 10 pages. Step 1, $10 \times 1.30 = 13$ pages' worth. Step 2, new material is $13 - 10 = 3$ pages. Result, about 3 pages of genuinely new content. Also cite and attach the conference version and explain the differences (Ethics p.14). Reusing your own conference text is the allowed self-plagiarism exception (Ethics p.12).
6. **Q.** Can ChatGPT be a co-author on an ACM paper? Must you disclose a grammar checker?
   A. No, AI can't be an author, and content it creates must be fully disclosed (Ethics p.24). A grammar checker is exempt under ACM. Under IEEE it's outside the policy but disclosure is recommended (Ethics p.23).

Myth vs truth across all four lectures

Myth (false)	Truth	Cite
My academic advisor from admission is my research advisor	Research advisor is declared separately within 24 credits	Advisor p.3
The 24-credit deadline lets me search in my last semester	Too late, PRCP alone can take a semester or more	Advisor p.7-8
If I want to work with X, X will agree	Not guaranteed, they may be over-committed or see a mismatch	Advisor p.7, p.12
Faculty answer "be my advisor?" within weeks	Most faculty take their time	Advisor p.7, p.21
PRCP is an official GMU procedure	Aydin's own term, not official	Advisor p.9
Co-advised students have two official advisors	One official advisor, thesis names both	Advisor p.19
Changing advisor is a normal step	Possible but a last resort	Advisor p.20
My 701 supervisor becomes my PhD advisor	Separate form within 24 credits	CS701 p.16
My dissertation will be in my 701 area	Can do 701 in X and dissertation in Y	CS701 p.17
My first-choice professor will agree	They may be unavailable	CS701 p.19
A meeting and a quick decision will follow	No guarantee	CS701 p.20
Wait for one professor before asking another	Contact several at once	CS701 p.21
Only one faculty member can work with me	Co-supervisors allowed, one official tenure-line CS primary	CS701 p.22
I can start searching in late October	Big mistake, form due Oct 31	CS701 p.23, p.7
My GRA sponsor must be my 701 supervisor	Natural but not required	CS701 p.24
I can take 701 next year instead	Not possible, catalog rule	CS701 p.25
The department will assign a supervisor	Faculty must agree to supervise	CS701 p.26
701 can be just a literature review	Needs a hands-on part	CS701 p.10-11
A conference submission can be a 701 deliverable	No, but a draft paper for later submission can	CS701 p.15
701 needn't be publishable so it can be small	Effort of a 3-credit course, no sprinting	CS701 p.9
A publication is like a grad project report	False, peer reviewed in a competition	Pub p.9, p.3
Correct results get accepted	False, fixed slots and writing is scored	Pub p.9, p.14-15
Correct, important and novel gets accepted	Still not guaranteed	Pub p.9, p.15
My advisor says it's good, so it'll be accepted	False	Pub p.9
Rejection means the conference erred or is biased	Reasonable papers are rejected for lack of slots	Pub p.9, p.15
Rejection wastes all the effort	Use comments to improve and resubmit	Pub p.9, p.19
Venue doesn't matter	Lower acceptance rate means more prestige	Pub p.9, p.13
Number of papers matters, not quality	Quality vs quantity	Pub p.9, p.4
Every paper gets a rebuttal	Many are directly accepted or rejected	Pub p.17
Dual submission is fine if I withdraw one later	Not allowed, severe sanctions	Pub p.20
Citing a source makes pasted text OK	Still plagiarism without quotes	Luke TR p.2, Ethics p.8
A cited figure is fine	Copyright violation, need permission	Luke TR p.4
Light rewording avoids plagiarism	Same structure still plagiarizes the idea	Luke TR p.2
Reusing my own published text is fine	Self-plagiarism, except conf to journal	Ethics p.12
Dropping unexplained data is OK	That's falsification	Ethics p.13, p.20
An honest error is research misconduct	Not misconduct	Ethics p.18
Dissertation misconduct goes to the Honor Code	Policy 4007	Ethics p.16
AI can be a co-author if disclosed	Never an author (ACM, Elsevier, ICML)	Ethics p.24-26
Disclosing AI shifts responsibility to the tool	Authors take full responsibility	Ethics p.22

Intro deck

- The "old CS 700" part covers measurement tools, simulation and design of experiments, statistics for designing experiments, comparing alternatives and characterizing measured data, plus simple analytical modeling. Most examples come from performance measurement of systems and networks, but the methods apply to all of CS (Intro p.6).
- Specialized readings are Cohen "Empirical Techniques in AI", Crovella et al. "Internet Measurement" and Feitelson "Workload Modeling for Computer Systems Performance Evaluation" (Intro p.7). The distribution fitting reading is chapter 4 of that Feitelson book.
- Schedule order is intro, metrics, summarizing data, measurement, simulation, comparing alternatives, linear regression, design of experiments, characterizing measured data, analytical modeling (Intro p.12).
- More course goals. Give an intuitive conceptual background for standard statistical tools and let you apply them correctly in new situations (Intro p.15). Get new insights from complex measurement and simulation results, like how a new feature affects the whole system (Intro p.16).
- Measurement agenda lists fundamental strategies, interval timers, cycle counters, program profiling, tracing, indirect measurement, and measuring network delay and bandwidth (Intro p.18). Simulation covers types of simulation, random number generation, and verification and validation (Intro p.19).
- Statistical interpretation covers arithmetic, harmonic and geometric means, sources of measurement error, CIs and comparing alternatives (Intro p.20). Design of experiments covers one and two factor ANOVA, m-factor, fractional factorial and Plackett and Burman designs (Intro p.21, the slide spells it "Berman").
- *Example for the three means.* Course rule is arithmetic for times, harmonic for rates, geometric for ratios. Times 2 s and 4 s average to $(2+4)/2 = 3$ s. Rates of 100 and 50 MB/s, each over the same amount of data, average to $2/(\frac{1}{100} + \frac{1}{50}) = 2/0.03 = 66.7$ MB/s, not 75. Speedups of 2 and 0.5 average to $\sqrt{2 \times 0.5} = 1$, so no net speedup.
- Characterizing measured data means workload characterization and fitting data to known distributions with Q-Q plots, chi-square and K-S tests (Intro p.22). Analytical modeling means single queues, networks of queues and operational analysis with Little's Law (Intro p.23).
- First readings were Feitelson "Experimental Computer Science, the need for a cultural change" and Lilja chapter 1. HW0 is a probability refresher you solve but don't submit (Intro p.24).
- HW1 asks for a LaTeX document with a photo in the upper right corner, submitted on Canvas as a zip of all the source files (Intro p.26).

Advisor deck

Nothing missing. Every slide is already in the advisor section.

CS701 deck

Nothing missing. Every slide, including the term project pages, is already in the CS 701 section.

Pub deck

- The h-index was proposed as an extra metric on top of the total number of publications and citations (Pub p.6). *Plain words.* It's meant to reward many well-cited papers, not one hit or many ignored ones.
- Publishing is a testament to the quality and originality of your work (Pub p.4).

Ethics deck

- Elsevier's reason AI can't be an author is that authorship carries responsibilities and tasks only humans can take on (Ethics p.25).
- ACM's sample disclosure goes in the Acknowledgements, e.g. "ChatGPT was utilized to generate sections of this Work, including text, tables, graphs, code, data, citations" (Ethics p.24).

Luke TR

- A very large quote without good cause makes reviewers wonder why you lean on someone else's text and can bring a copyright violation charge (Luke TR p.3, Ex. 7). *Example.* One offset sentence from Luke and Panait with [1] is fine. Their whole related work section offset and cited isn't, since there's no good cause for that much.
- Offset long quotes because at first glance an inline one can be mistaken for your own words (Luke TR p.3, Ex. 8).
- A moderate equation needs the same care as several sentences. For a few equations write "This and the next four equations are from Luke and Panait [1]" (Luke TR p.3, Ex. 9, 11).
- Many equations without permission look like padding your paper with the source's ideas. Permission covers equations, not large pieces of text (Luke TR p.3, Ex. 12). *Example.* You need eight equations of someone's derivation. Ask the author (and the publisher if it holds the copyright), then write "The next several equations are cited, with permission, from Luke and Panait [1]".
- Figures differ from text for two reasons. A figure is set off from the text so it looks free-standing with no citation, and it's the author's creation with more effort in it (Luke TR p.4, Ex. 14).
- Publisher permission is needed too if the publisher holds the copyright. You may also mention the permission in the body text (Luke TR p.4, Ex. 15). *Example caption.* "Figure 4. The State Transitions (reproduced with permission from Luke and Panait [1])."
- For a very small algorithm, getting the author's permission is wise but not required (Luke TR p.5, Ex. 18). *Example.* A short SELECT procedure headed "From Luke [2]" is OK, though rewriting it is better. A third-page procedure counts as a figure and needs author and publisher permission (Luke TR p.5, Ex. 19).

Feitelson ch. 4 (distribution fitting)

Cites are PDF pages of distribution-fitting.pdf. Book page is PDF page plus 137. The notes only had his Q-Q axis flip, the exponential and Pareto inverses and the chi-square recipe, so most of this is new.

Big picture

- You never get a perfect fit. Samples are random, and real data (like Unix runtimes) was never drawn from a neat formula. "Close enough" means the model gives reliable evaluation results, which we settle for judging with moments or shape (Feitelson p.2).
- Two approaches. Pick a known distribution type and fit its parameters (e.g. Poisson arrivals give exponential interarrivals, whose θ is the mean), or skip known types and build a distribution with the right shape, or use the empirical one (e.g. parallel job sizes with peaks at powers of two) (Feitelson p.3).
- The standard three steps are (1) select a candidate distribution, (2) estimate its parameters from the data, (3) check goodness of fit (Feitelson p.3). Step 3 is needed because the best parameters for a wrong family are still wrong (Feitelson p.4).

Step 1, which distribution (Feitelson p.4-5)

- Best reason is knowing the true one. Poisson arrivals mean exponential interarrival times. A sum of many finite-variance parts is normal. A product of many parts is lognormal.

- No idea? Fit many families, run goodness of fit on each, keep the best score. This is what fitting packages do.
- Sometimes the exact shape doesn't matter. M/G/1 mean wait $W = \dfrac{\lambda E[S^2]}{2(1-\lambda E[S])}$ only needs two moments of service time. Then use hyper-exponential if CV > 1 and Erlang if CV < 1.

  Worked example, M/G/1 wait. **Formula.** $W = \lambda E[S^2]/(2(1-\lambda E[S]))$, where W is the mean wait in the queue, λ the arrival rate, $E[S]$ the mean service time and $E[S^2]$ the second moment of service time, which equals $(1+\mathrm{CV}^2)E[S]^2$. **Plug in.** $\lambda = 0.5$ jobs/s and $E[S] = 1$ s, once with exponential service ($\mathrm{CV}^2 = 1$) and once with $\mathrm{CV}^2 = 3$. **Work.** $E[S^2] = 2 \times 1 = 2$ in the first case, so $W = 0.5 \times 2/(2 \times 0.5) = 1/1 = 1$ s. In the second $E[S^2] = 4$, so $W = 0.5 \times 4/1 = 2$ s. **Result.** Same mean service, but the more variable one doubles the wait, which is why only the first two moments need to be right.
- Weakest reason is convenience, like the exponential's memoryless property. Never use exponential (or assume Poisson arrivals) when the CV is clearly above 1 (Feitelson p.5).

Step 2, estimating parameters

- Method of moments. Write the moments as functions of the parameters and invert. You need as many moments as parameters (Feitelson p.5, p.10). Gamma has $\bar{X} = \alpha\beta$ and $\mathrm{Var} = \alpha\beta^2$, so $\hat{\alpha} = \bar{X}^2/\mathrm{Var}$ and $\hat{\beta} = \mathrm{Var}/\bar{X}$ (Feitelson p.5).

  Worked example, gamma by moments. **Formula.** $\hat{\alpha} = \bar{X}^2/s^2$ and $\hat{\beta} = s^2/\bar{X}$, where $\bar{X}$ is the sample mean, s^2 the sample variance (course $n-1$ version), $\hat{\alpha}$ the shape and $\hat{\beta}$ the scale. **Plug in.** Data 2, 2, 4, 6, 8, 10, 10, so $n = 7$. **Work.** Mean $= 42/7 = 6$. Deviations are $-4, -4, -2, 0, 2, 4, 4$, squares add to $16 + 16 + 4 + 0 + 4 + 16 + 16 = 72$, so $s^2 = 72/6 = 12$. Then $\hat{\alpha} = 6^2/12 = 36/12 = 3$ and $\hat{\beta} = 12/6 = 2$. Check, $\alpha\beta = 3 \times 2 = 6$ and $\alpha\beta^2 = 3 \times 4 = 12$. **Result.** Gamma with shape 3 and scale 2, the gamma whose mean and variance equal the data's.
- Maximum likelihood. $L(\theta) = \prod f(x_i \mid \theta)$, assuming independent samples. Steps (1) write L, (2) take the log, (3) differentiate in θ, (4) set to zero, (5) solve for θ, (6) check the second derivative is negative (Feitelson p.6-7). *Plain words.* Pick the parameter that makes the data you saw most probable. Logs turn the product into a sum.
- For the exponential, MLE gives $\hat{\theta}$ = sample mean (Feitelson p.7). No closed form? Maximize numerically (Feitelson p.7).

  Worked example, exponential MLE. **Formula.** $f(x) = \frac{1}{\theta}e^{-x/\theta}$, so $\ln L(\theta) = -n\ln\theta - \frac{1}{\theta}\sum x_i$, where θ is the mean, n the number of samples and $\sum x_i$ their total. **Plug in.** Data 2, 3, 7, so $n = 3$ and $\sum x_i = 12$, giving $\ln L = -3\ln\theta - 12/\theta$. **Work.** Differentiate, $-3/\theta + 12/\theta^2$. Set to 0 and multiply by θ^2, $-3\theta + 12 = 0$, so $\theta = 4$. Second derivative $3/\theta^2 - 24/\theta^3$ at 4 is $3/16 - 24/64 = 0.1875 - 0.375 = -0.1875 < 0$, so it's a maximum. Spot check, $\ln L$ is -7.30, -7.16, -7.23 at $\theta = 3, 4, 5$. **Result.** $\hat{\theta} = 4$, which is just the mean 12/3.
- Exponential family (exponential, hyper-exponential, Erlang, normal, lognormal, gamma) has the moments as sufficient statistics, so moment formulas give the MLE (Feitelson p.8).
- Lognormal. $\hat{\mu} = \frac{1}{n}\sum \ln x_i$ and $\hat{\sigma} = \sqrt{\frac{1}{n-1}\sum(\ln x_i - \hat{\mu})^2}$. Here moments and MLE agree (Feitelson p.10).

  Worked example, lognormal. **Formula.** As above, where μ and σ are the mean and SD of $\ln x$, not of x. **Plug in.** Data 2.72, 7.39, 20.09, which are e^1, e^2, e^3. **Work.** Logs are 1, 2, 3. $\hat{\mu} = 6/3 = 2$. Squared deviations $1 + 0 + 1 = 2$, divided by $n - 1 = 2$ gives 1, so $\hat{\sigma} = 1$. **Result.** Lognormal with $\mu = 2$, $\sigma = 1$. Its median is $e^2 = 7.39$, the middle data value.
- Pareto. $\hat{k} = \min x_i$, then $\hat{a} = 1 \Big/ \frac{1}{n}\sum \ln(x_i/\hat{k})$ (Feitelson p.10).

  Worked example, Pareto. **Formula.** $\hat{k} = \min x_i$ and $\hat{a} = 1/\left[\frac{1}{n}\sum \ln(x_i/\hat{k})\right]$, where k is where the distribution starts and a is its shape (smaller a means a heavier tail). **Plug in.** Data 2, 4, 8, and $\ln 2 = 0.693$. **Work.** $\hat{k} = 2$. Ratios $x_i/\hat{k}$ are 1, 2, 4, with logs 0, 0.693, 1.386 ($\ln 4 = 2\ln 2$). Sum 2.079, mean $2.079/3 = 0.693$. $\hat{a} = 1/0.693 = 1.443$. **Result.** Pareto starting at 2 with shape 1.44.
- Erlang with k stages has $\mathrm{CV} = 1/\sqrt{k}$, so take $k = \lfloor 1/\mathrm{CV}^2 \rfloor$ and $1/\lambda = \mathrm{mean}/k$ (Feitelson p.9).

  Worked example, Erlang. **Formula.** $\mathrm{CV} = s/\bar{X}$, $k = \lfloor 1/\mathrm{CV}^2 \rfloor$, $1/\lambda = \bar{X}/k$, where k is the number of stages and $1/\lambda$ the mean time of one stage. **Plug in.** Mean 10, SD 6. **Work.** CV $= 6/10 = 0.6$, $\mathrm{CV}^2 = 0.36$, $1/0.36 = 2.78$, floor gives $k = 2$. Then $1/\lambda = 10/2 = 5$, so $\lambda = 0.2$. **Result.** Two stages of mean 5 each. Its CV is $1/\sqrt{2} = 0.707$, the closest Erlang to 0.6 by Allen's rule.
- Two-stage hyper-exponential has 3 parameters, so two moments leave a free choice. One recipe is $p = \frac{1}{2}\big(1 - \sqrt{(\mathrm{CV}^2 - 1)/(\mathrm{CV}^2 + 1)}\big)$, $\lambda_1 = 2p/\mu$, $\lambda_2 = 2(1-p)/\mu$, which needs CV ≥ 1 (Feitelson p.8-9).

  Worked example, hyper-exponential. **Formula.** $\mathrm{CV}^2 = (\mu_2' - \mu^2)/\mu^2$, then p, λ_1, λ_2 as above, where μ is the mean, μ_2' the second moment, p the chance of using stage 1 and λ_1, λ_2 the stage rates. **Plug in.** $\mu = 1$, $\mu_2' = 4$. **Work.** $\mathrm{CV}^2 = (4-1)/1 = 3$. $(3-1)/(3+1) = 0.5$, $\sqrt{0.5} = 0.7071$, $1 - 0.7071 = 0.2929$, and half of that is $p = 0.1464$. $\lambda_1 = 2 \times 0.1464/1 = 0.293$ and $\lambda_2 = 2 \times 0.8536/1 = 1.707$. Check mean $p/\lambda_1 + (1-p)/\lambda_2 = 0.5 + 0.5 = 1$, and second moment $2p/\lambda_1^2 + 2(1-p)/\lambda_2^2 = 3.414 + 0.586 = 4$. **Result.** A rare slow stage (14.6% of jobs, mean $1/0.293 = 3.41$) and a common fast one (mean 0.586). Each stage carries half the mean (Feitelson p.9).
- Weibull and gamma MLE need hard equations, so gamma is usually fitted by moments (Feitelson p.9-10).

Outliers and shape

- High moments are ruled by a few huge samples. For 1,1,1,1,2,2,2,3,3,4,5,15 the fifth moment is 63,675.8 and 99.4% of the sum comes from the 15. Changing 15 to 16 raises it 37.8% (Feitelson p.11).

  Worked example, fifth moment. **Formula.** $\mu_5' = \frac{1}{n}\sum x_i^5$, the average of each value to the fifth power. **Plug in.** The 12 values above. **Work.** Four 1s give 4, three 2s give $3 \times 32 = 96$, two 3s give $2 \times 243 = 486$, then $4^5 = 1024$, $5^5 = 3125$, $15^5 = 759{,}375$. Sum 764,110, divided by 12 is 63,675.8. The 15 alone is $759{,}375/764{,}110 = 99.4\%$. With 16, $16^5 = 1{,}048{,}576$, the sum becomes 1,053,311 and $\mu_5' = 87{,}775.9$, which is $87{,}775.9/63{,}675.8 = 1.378$, up 37.8%. **Result.** One sample sets the moment, so a fit built on it rests on that one value.
- Dropping just the top 4 of 184,612 Unix processes cut the mean 4.5%, the second moment 70% and the CV 44%, while the median barely moved (Feitelson p.12). This is real, since another time period would show different tail samples.
- MLE is less sensitive to outliers than moments (every sample gets equal weight) and usually less biased, so prefer MLE when you can (Feitelson p.12).
- Matching moments can still give the wrong shape. Jann's 3-moment hyper-Erlang fits came out bimodal, and Lazowska (1977) found moment-matched hyper-exponentials mispredicted a queue, with different ones giving different answers. Matching percentiles may be better (Feitelson p.12-13).

Mixtures and the EM algorithm

- A mixture has $f(x) = \sum p_i f_i(x)$, each sample from exactly one component. Hyper-exponential is a mixture of exponentials (Feitelson p.13-14).
- Two reasons to use one. Hand-tailor a shape (extend a tail, or Gaussians for many modes), or model a multiclass workload

(Feitelson p.14).

- EM repeats an E-step (split each observation among components in proportion $f_1(x)/p$ and $f_2(x)/p$ with $p = f_1(x) + f_2(x)$) and an M-step (MLE fit of each component to its share) until the likelihood converges (Feitelson p.15-16). It can stop at a local maximum, so rerun from another start and compare (Feitelson p.17). *E-step example.* At $x = 0.25$ say $f_1 = 0.3$ and $f_2 = 0.1$. Then $p = 0.4$, so $0.3/0.4 = 0.75$ of that sample goes to component 1 and 0.25 to component 2.
- Example. The inferno runtimes were Pareto background plus a normal bump at 0.2 to 0.3 s. EM split them in two iterations, and the bump was runaway `ps` calls from an OS course exercise bug (Feitelson p.14, p.17).

Using the data's own shape

- The empirical distribution is always available and best when no formula fits, like modal data (Feitelson p.18). To sample from it, build the cumulative table $c(x_i)$, draw u uniform on $[0, 1]$, and take the smallest x_i with $c(x_i) \geq u$. Interpolate between values for continuous data (Feitelson p.19).

  Worked example, sampling the empirical distribution. **Formula.** Discrete, $c(x_i) = \#\{x_j \leq x_i\}/n$. Continuous, $c(x_i) = \#\{x_j < x_i\}/(n-1)$ and $x = x_i + \frac{u - c(x_i)}{c(x_{i+1}) - c(x_i)}(x_{i+1} - x_i)$, where u is the uniform draw and x_i, x_{i+1} the values whose c brackets it. **Plug in.** Discrete data 1, 1, 2, 4 with $u = 0.6$. Continuous data 1, 2, 4, 8 with $u = 0.5$. **Work.** Discrete, $c(1) = 2/4 = 0.5$, $c(2) = 0.75$, $c(4) = 1$, and the smallest $c \geq 0.6$ is at 2. Continuous, $c = 0, \frac{1}{3}, \frac{2}{3}, 1$, so $u = 0.5$ sits between 2 and 4 and $x = 2 + \frac{0.5 - 0.333}{0.333} \times 2 = 2 + 0.5 \times 2 = 3$. **Result.** Draws 2 and 3. Repeating gives values with the data's own shape.
- Censored data (sessions still running, killed processes) is handled with Kaplan-Meier, $\bar{F}_n(x) = \prod_{X_i < x}(1 - d_i/n_i)$. It assumes random censoring and can't rebuild the tail past the largest sample (Feitelson p.20-21).

  Worked example, Kaplan-Meier. **Formula.** $\bar{F}_n(x) = \prod_{X_i < x}(1 - d_i/n_i)$, where $\bar{F}_n(x)$ is the chance of lasting past x, d_i the number of real (uncensored) samples equal to X_i, and n_i the number of samples, real or censored, that are $\geq X_i$. **Plug in.** Session lengths 2, 3, 4+, 6, where 4+ was still running. Find $\bar{F}_n(5)$. **Work.** At 2, $1 - 1/4 = 0.75$. At 3, $1 - 1/3 = 0.667$. At 4+, $d = 0$ so the factor is 1. Product $0.75 \times 0.667 \times 1 = 0.5$. **Result.** $\bar{F}_n(5) = 0.5$. Dropping the censored session would give 1/3 (only the 6 of 2, 3, 6), which is biased low.
- Empirical distributions overfit. Check by cross-validation, comparing the first half of the data with the second half because workloads change over time (Feitelson p.21).
- Modal examples are parallel job sizes at powers of two, packet sizes at 1514 bytes (1500 Ethernet MTU plus 14 header bytes), and file sizes in multiples of blocks (Feitelson p.21-23). A mixture of Gaussians, fit by EM, models modes with spread (Feitelson p.24).
- Tail samples are few but huge and can dominate system behavior, so the tail must be modeled well (Feitelson p.24-25). Feldmann and Whitt fit a hyper-exponential phase by phase starting from the tail end. Four or five phases fit the 1994 Unix tail. It may not match the moments and depends on the chosen c_i and q, so check it visually (Feitelson p.25-27). LLCD plots show survival probability on log-log axes (Feitelson p.27).

Goodness of fit, the four methods (Feitelson p.28-29) Q-Q plots (graphical), Kolmogorov-Smirnov, Anderson-Darling, and χ^2. All compare the data with the model and ask if they're the same distribution.

4.5.1 Q-Q plots

- If the distributions match, the plot is a straight line with slope 1 (Feitelson p.29). Example, the 0.4 point of the KTH runtimes is 171 s in the data and 297 s in a 3-stage hyper-exponential model, plotted at (171, 297). The plot was close to the diagonal except for jobs under 10 s, while a plain exponential model didn't give a line (Feitelson p.29-30).
- Steps (Feitelson p.30-31). (1) Pick samples, all of them or a subsample, e.g. every 10th of 1000 sorted values. (2) Give $X_{(i)}$ the percentile $p_i = (i - 0.5)/n$, because i/n is lopsided, giving the last sample 1 and no sample 0. (3) Plot $\left(X_{(i)}, F^{-1}(p_i)\right)$. For the exponential, Table 4.2 gives $F^{-1}(p) = -\theta \ln(1-p)$ (Feitelson p.31).

  Worked example, exponential Q-Q. **Formula.** $p_i = (i - 0.5)/n$ and $y_i = -\hat{\theta} \ln(1 - p_i)$, where $X_{(i)}$ is the i-th smallest value and y_i the model value at the same percentile. **Plug in.** Data 1, 2, 4, 9, so $n = 4$ and $\hat{\theta} = 16/4 = 4$ (the MLE). **Work.**

i	$X_{(i)}$	p_i	$\ln(1 - p_i)$	y_i
1	1	0.125	−0.134	0.53
2	2	0.375	−0.470	1.88
3	4	0.625	−0.981	3.92
4	9	0.875	−2.079	8.32

  Result. Points (1, 0.53), (2, 1.88), (4, 3.92), (9, 8.32) sit just under the slope-1 line, so the exponential is a fair fit for these four values.
- Log-form Q-Q coordinates, x then y (Feitelson p.31). Weibull $\ln X_{(i)}$ vs $\frac{1}{\alpha}\ln[-\ln(1-p_i)] + \ln\beta$. Lognormal $\ln X_{(i)}$ vs $\text{sgn}(p_i - 0.5)(1.238t + 0.0324t^2)$ with $t = \sqrt{-\ln[4p_i(1-p_i)]}$. Pareto $\ln X_{(i)}$ vs $\ln k - \frac{1}{a}\ln(1-p_i)$.

  Worked example, Pareto row. **Formula.** $x_i = \ln X_{(i)}$, $y_i = \ln k - \frac{1}{a}\ln(1 - p_i)$, with k, a the fitted Pareto start and shape. **Plug in.** Data 2, 4, 8 with $\hat{k} = 2$, $\hat{a} = 1.443$ from above, so $1/a = 0.693$ and $\ln k = 0.693$. **Work.** $p_i = \frac{1}{6}, \frac{1}{2}, \frac{5}{6}$, $\ln(1 - p_i) = -0.182, -0.693, -1.792$. $y_i = 0.693 + 0.693 \times (0.182,\ 0.693,\ 1.792) = 0.82,\ 1.17,\ 1.94$. $x_i = 0.69,\ 1.39,\ 2.08$. **Result.** Points (0.69, 0.82), (1.39, 1.17), (2.08, 1.94) hug the diagonal, worst in the middle.

  Worked example, Weibull row, one point. **Formula.** $y = \frac{1}{\alpha}\ln[-\ln(1-p_i)] + \ln\beta$, where α is the shape and β the scale. **Plug in.** $\alpha = 2$, $\beta = 3$, $p_i = 0.5$. **Work.** $\ln[-\ln 0.5] = \ln 0.693 = -0.367$, halved is -0.183, plus $\ln 3 = 1.099$ gives $y = 0.915$. **Result.** The median sample should have $\ln X \approx 0.915$, which is $X \approx 2.50$.

  Worked example, lognormal row, one point. **Formula.** $y = \text{sgn}(p_i - 0.5)(1.238t + 0.0324t^2)$ with $t = \sqrt{-\ln[4p_i(1-p_i)]}$, where sgn is $+1$ above the median and -1 below, t is a helper and y approximates the standard normal quantile. **Plug in.** $p_i = 0.875$. **Work.** $4 \times 0.875 \times 0.125 = 0.4375$, $-\ln 0.4375 = 0.827$, $t = \sqrt{0.827} = 0.909$. $y = +1 \times (1.238 \times 0.909 + 0.0324 \times 0.827) = 1.126 + 0.027 = 1.152$. **Result.** $y = 1.152$, close to the exact $z = 1.150$. A straight line of $\ln X$ vs y means lognormal fits, crossing $y = 0$ at $\ln X = \hat{\mu}$.
- If F^{-1} has no closed form (e.g. hyper-exponential), generate many model samples and compare the two empirical distributions (Feitelson p.31).
- Reading his plots, with data on x and model on y. Straight but slope above 1 means right shape, wrong scale, so shrink the model scale (slope below 1, the reverse). Tapering off to the right means the data's tail is longer, so use a heavier-tailed model. Shooting straight up means the model's tail is too long (Feitelson p.32). *Example.* If every model value is about twice the data value (straight, slope 2), the shape is right but the scale is double, so halve it, e.g. exponential θ from 8 to 4.
- A P-P plot pairs the two percentiles at each value. It shows vertical CDF gaps and is more sensitive in the body, while Q-Q shows horizontal gaps and is more sensitive in the tails. Workload work prefers Q-Q because tails matter (Feitelson p.32).
- With heavy tails one Q-Q plot is all tail. In the 1994 Unix data 34% of runtimes are 0.005 s and 17% are 0.02 s, but the max is

1573.50 s, so every other point sits at the origin. Dropping the extremes hides the tail, and a log scale is the good compromise showing both body and tail (Feitelson p.32-33).

Kolmogorov-Smirnov (Feitelson p.33-34)

- Sort the data, form the step ECDF F_n (jumps of $1/n$), and take the largest gap $D_n = \max_i \max\big(\big|\frac{i}{n} - F(X_{(i)})\big|,\ \big|F(X_{(i)}) - \frac{i-1}{n}\big|\big)$. Large D_n rejects the fit, and the cutoff depends on n and α from a table. *Plain words.* Check each data point's gap to both the top and bottom of its step.

- *Worked example, K-S.* **Formula.** D_n as above, where F is the model CDF, $X_{(i)}$ the i-th smallest value, i/n the top of its ECDF step and $(i-1)/n$ the bottom. **Plug in.** Data 0.1, 0.4, 0.7, $n = 3$, model Uniform(0,1) so $F(x) = x$. **Work.** $i = 1$, $\frac{1}{3} - 0.1 = 0.233$ and $0.1 - 0 = 0.1$. $i = 2$, $\frac{2}{3} - 0.4 = 0.267$ and $0.4 - \frac{1}{3} = 0.067$. $i = 3$, $1 - 0.7 = 0.3$ and $0.7 - \frac{2}{3} = 0.033$. Largest is 0.3. **Result.** $D_3 = 0.3$. A standard K-S table (not in the reading) gives 0.708 for $n = 3$ at $\alpha = 0.05$, so $0.3 < 0.708$ and we don't reject Uniform.
- Caveats. Parameters must be known in advance, not fitted from the data, or the general table makes it conservative. Bootstrapping fixes this, by drawing n samples from the model about 1000 times and seeing where the real D_n falls. With huge workload datasets the test always rejects, so test a random subsample of about 500. It's insensitive to tails, since both CDFs are near 1 there (Feitelson p.34).

Anderson-Darling (Feitelson p.35) A K-S variant that stresses the tails. It takes a weighted mean of squared CDF gaps with weight $f(x)/[F(x)(1-F(x))]$, large in both tails. Computed as $A^2 = -n - \frac{1}{n}\sum(2i-1)\big[\ln F(X_{(i)}) + \ln(1 - F(X_{(n-i+1)}))\big]$ and compared with a table.

Worked example, Anderson-Darling. **Formula.** As above, where n is the sample size, F the model CDF and term i pairs the i-th smallest value with the i-th largest. **Plug in.** Same data 0.1, 0.4, 0.7 against Uniform(0,1). **Work.** $i = 1$, $1 \times (\ln 0.1 + \ln 0.3) = -2.303 - 1.204 = -3.507$. $i = 2$, $3 \times (\ln 0.4 + \ln 0.6) = 3 \times (-0.916 - 0.511) = -4.281$. $i = 3$, $5 \times (\ln 0.7 + \ln 0.9) = 5 \times (-0.357 - 0.105) = -2.310$. Sum -10.098. $A^2 = -3 - \frac{1}{3}(-10.098) = -3 + 3.366 = 0.366$. **Result.** $A^2 = 0.37$, far below the usual 5% cutoff of about 2.5 for a fully known model (standard table, not in the reading), so no reason to reject.

χ^2 in Feitelson (Feitelson p.35) Used when $F(x)$ has no simple formula. Draw as many model samples as data points and get E_i from them, or use $E_i = n(F(b_{i+1}) - F(b_i))$ if F is known. Bootstrap also works. Since each range needs at least 5, the tail ends up in a few wide ranges, so χ^2 is poor for heavy-tailed data.

Worked example, χ^2 with known F. **Formula.** $E_i = n(F(b_{i+1}) - F(b_i))$ and $\chi^2 = \sum(O_i - E_i)^2/E_i$, where b_i is where range i starts, O_i the observed count and E_i the expected count. **Plug in.** $n = 50$, model exponential with $\theta = 2$ fixed in advance, so $F(x) = 1 - e^{-x/2}$. Ranges $[0,1)$, $[1,2)$, $[2,4)$, $[4,\infty)$ with $O = 22, 10, 12, 6$. Use $e^{-0.5} = 0.6065$, $e^{-1} = 0.3679$, $e^{-2} = 0.1353$. **Work.** $F(1) = 0.3935$, $F(2) = 0.6321$, $F(4) = 0.8647$. Range probabilities 0.3935, 0.2386, 0.2326, 0.1353, times 50 gives $E = 19.67, 11.93, 11.63, 6.77$, all at least 5. $O - E = 2.33, -1.93, 0.37, -0.77$. Terms $5.43/19.67 = 0.276$, $3.72/11.93 = 0.312$, $0.14/11.63 = 0.012$, $0.59/6.77 = 0.088$. Sum 0.69. **Result.** With the course's $k - 1 = 3$ df the 5% cutoff is 7.815, and $0.69 < 7.815$, so the exponential isn't rejected.

Software (Feitelson p.36) Packages test around 40 distributions and suggest the best, but you still must check the graphs (e.g. Q-Q plots), and they can't handle mixtures or the hyper-exponential.

Every homework problem from HW0 and HW2 to HW6, in the form the exam is likely to copy. Each one starts with **In plain words**, which says what the question really wants. Then every calculation goes in the same four beats. *Formula* gives the formula in symbols with each symbol named. *Plug in* puts the real numbers in. *Work it out* does the arithmetic one step at a time. *Result* gives the number. The final answer is marked **Answer**. Every number below was recomputed in Python from the data files. Where the homework needed a computer there's a small version you can do with a pencil. Places where the submitted answer drifts from the slides or the source are marked **Check.** When an interval is built from rounded pieces the last digit can be 1 off from the listed answer, which came from unrounded values. Either is fine on the exam.

Table values used in this section

$z[0.05]$	1.645	$t[0.05, 15]$	1.753
$z[0.025]$	1.960	$t[0.025, 15]$	2.131
$z[0.01]$	2.326	$t[0.005, 15]$	2.947
$z[0.005]$	2.576	$t[\cdot, 3]$ at .05/.025/.005	2.353/3.182/5.841
$\Phi(0.33)$	0.6293	$t[0.025, 8]$	2.306
$\Phi(0.5)$	0.6915	$t[0.05, 8]$	1.860
$\Phi(0.83)$	0.7967	$t[0.05, 6]$	1.943
$\Phi(0.9)$	0.8159	$t[0.05, 37]$	1.687
$\Phi(1.3)$	0.9032	$t[0.025, 2]$	4.303
$\Phi(1.8)$	0.9641	$\Phi(0.21)$	0.5832
$\Phi(2.0)$	0.9772	$\Phi(0.67)$	0.7486
$\Phi(2.5)$	0.9938	$\Phi(1.38)$	0.9162
$t[0.05, 5]$	2.015	$t[0.025, 4]$	2.776
$t[0.05, 24]$	1.711	$t[0.10, 37]$	1.304
$z[0.20]$	0.84	$t[0.025, 7]$	2.365
$\Phi(2.4)$	0.9918	$t[0.025, 5]$	2.571

$z[a]$ is the upper-tail value with area a above it, so $z[0.025] = 1.960$ (CI p.30). $\Phi(z)$ is the area below z, which is the chance a standard normal lands below z. Handy roots are $\sqrt{50} = 7.071$, $\sqrt{12} = 3.464$, $\sqrt{3} = 1.732$, $\sqrt{6} = 2.449$, $\sqrt{5} = 2.236$.

Which homework to open for which exam question

If the exam asks you to	Model
compute a probability, mean or variance from a table	HW0 Q5 to Q9
judge a metric against the six criteria	HW2
pick and compute a mean (times, rates, ratios)	HW3 Q1
find quartiles, a histogram, a Q-Q plot	HW3 Q2
build a z or t CI and compare widths	HW4 P1
fix a CI for skewed data	HW4 P2
run a one-sample test, p-value, threshold	HW5 P1
set up H_0 and H_1 for a claim	HW5 P2
compare two systems, paired or Welch	HW6 P1, P2
audit a described experiment	HW4 to HW6 P3

Answer key at a glance (his submitted answers, all recomputed)

HW0 Q4	1/16, 15/16, 3/8
HW0 Q5	discrete, $\mu = 2.84$, Var 5.134, SD 2.27, COV 0.80
HW0 Q6	0.75, 0.425, $4/17 = 0.235$, $17/23 = 0.739$
HW0 Q7	mean 50, SD 23.09, 0.125, 0.375, 0.125
HW0 Q8	0.0873, 0.1841, 0.8159
HW0 Q9	0.321, 0.203, 0.309
HW3 Q1	GM 1.4294 (42.94%), AM 203.33 ms, HM 2485.66 tps
HW3 Q2	means 2.0142, 0.2083, s^2 1.0457, 0.0440, Q-Q r 0.9993, 0.904
HW4 P1	$\mu = 2.0142$, $\sigma^2 = 1.0447$, all six CIs contain μ, 95% widths 0.52 vs 1.21
HW4 P2	$\mu = 0.2083$, $\sigma^2 = 0.0440$, batched widest, bootstrap narrowest, 99% batched dips to -0.158
HW5 P1	$Z = 1.80$, fail to reject, $p = 0.0718$, threshold 2.843 h
HW5 P2	$H_1: \mu < 740$, $Z = -2.50$, reject, $p = 0.0062$
HW6 P1	paired, $\bar{d} = 0.611$, CI $(-10.77, 11.99)$, not shown
HW6 P2	Welch, $\nu = 37.89$, CI $(-1.540, 0.781)$, not shown

HW0. Probability refresher (ungraded, solved here)

HW0 Q1 and Q2. Independent events, and $P[A$ and $B]$

Question. *1. Explain what is meant by "independent events" in Probability theory. 2. If A and B are independent events, what is $P[A$ and $B]$?*

In plain words. Two events are independent when one happening tells you nothing about the other, like two separate coin flips. Then the chance that both happen is just the two chances multiplied. This is the rule that lets us treat repeated runs as separate pieces of evidence later in the course.

1. Q1. Independent means knowing B happened doesn't change the chance of A. In symbols $P[A \mid B] = P[A]$, where $P[A \mid B]$ is the chance of A once you already know B happened.
2. Q2. *Formula.* Multiplication rule $P[A \cap B] = P[A \mid B]\, P[B]$, where $A \cap B$ means A and B both happen.
3. *Plug in.* Independence lets you swap $P[A \mid B]$ for $P[A]$.
4. *Result.* $P[A \cap B] = P[A]\, P[B]$.
5. Example. Two fair coins. $P[\text{both heads}] = P[H_1]\, P[H_2] = 0.5 \times 0.5 = 0.25$.

Answer. Q1, independent means $P[A \mid B] = P[A]$, so one event carries no information about the other. Q2, $P[A \cap B] = P[A]\, P[B]$.

Usual mistake. Mixing up independent with mutually exclusive. Exclusive events with nonzero probability are never independent, since $P[A \cap B] = 0$. In the course, independence is what makes $\sigma/\sqrt{n}$ the right standard error (HT p.20, CI p.65).

HW0 Q3. $E[kX + mY]$ for constants k, m

Question. *If the expected values of two random variables X and Y are $E[X]$ and $E[Y]$, respectively, what is the expected value of $k \cdot X + m \cdot Y$ (i.e., $E[k \cdot X + m \cdot Y]$), if k and m are constants?*

In plain words. Expected value is the long-run average. Averages are linear, so scaling a variable scales its average, and the average of a sum is the sum of the averages. You don't need independence for this, only for the variance version.

1. *Formula.* $E[X] = \sum_x x\, P[X = x]$, each value times its chance, added up. It's a weighted sum (Summ p.10).
2. *Plug in.* $E[kX + mY] = \sum (kx + my)\, P[x, y]$ over all pairs of values.
3. *Work it out.* Split the sum into $k \sum x\, P[x, y] + m \sum y\, P[x, y]$. Each piece collapses to $E[X]$ and $E[Y]$.
4. *Result.* $E[kX + mY] = k\, E[X] + m\, E[Y]$.
5. Example. $E[X] = 3$, $E[Y] = 5$, $k = 2$, $m = -1$ gives $2(3) + (-1)(5) = 6 - 5 = 1$.

Answer. $k\,E[X] + m\,E[Y]$, with or without independence.

Usual mistake. Doing the same for variance. $\text{Var}(X-Y) = \sigma_X^2 + \sigma_Y^2 - 2\rho\sigma_X\sigma_Y$ needs the correlation, and that term is exactly what pairing buys (Comp p.6). With $k=1$, $m=-1$ you get $E[A-B] = E[A] - E[B]$, which is why $\bar{d} = \bar{x}_A - \bar{x}_B$ (Comp p.8).

HW0 Q4. Fair coin tossed 4 times. P(all heads), P(at least one tail), P(exactly 2H 2T)

Question. *A fair coin is tossed 4 times. (i) What is the probability that all four tosses will be "heads"? (ii) What is the probability that at least one toss will give "tail"? (iii) What is the probability that there will be exactly 2 heads and 2 tails? Show your work (i.e., explain how you obtained your answers).*

In plain words. With a fair coin every sequence of 4 tosses is equally likely. So each probability is the number of sequences you want divided by the total number of sequences. For "at least one" it's easier to count the opposite, no tails at all, and subtract from 1.

1. *Formula.* $P[\text{event}] = \dfrac{\text{sequences in the event}}{\text{all sequences}}$. All sequences $= 2^n$, where n is the number of tosses.
2. *Plug in.* $2^4 = 2\times2\times2\times2 = 16$ sequences.
3. All heads. Only HHHH, so 1 sequence. $P = 1/16 = 0.0625$.
4. At least one tail. *Formula.* $P[\text{at least one}] = 1 - P[\text{none}]$. *Plug in.* $1 - 1/16$. *Work it out.* $16/16 - 1/16 = 15/16$. *Result.* 0.9375.
5. Exactly 2 heads. *Formula.* count $= \binom{n}{k} = \dfrac{n!}{k!\,(n-k)!}$, the number of ways to choose which k of the n tosses are heads. *Plug in.* $\binom{4}{2} = \dfrac{4!}{2!\,2!}$. *Work it out.* $\dfrac{24}{2\times2} = 6$ sequences (HHTT, HTHT, HTTH, THHT, THTH, TTHH). *Result.* $6/16 = 3/8 = 0.375$.

Answer. $1/16 = 0.0625$, $15/16 = 0.9375$, $3/8 = 0.375$.

Usual mistake. Adding probabilities instead of using the complement for "at least one". The same coin is the error model on CI p.12. Two $\pm E$ sources give $x-2E, x, x+2E$ with $1/4, 1/2, 1/4$. With 3 sources, $P[x+E] = \binom{3}{2}/8 = 3/8$, and many sources turn the binomial into a normal (CI p.15).

HW0 Q5. Requests per ms with $P_0, P_1, P_2, P_4, P_5, P_6, P_7$ = .18, .22, .14, .18, .12, .10, .06. Type? Why is $P_7 = 0.09$ invalid? Mean, variance, SD, COV

Question. *On a certain server, the number of client requests per millisecond varies between 0 and 7, with the following probabilities (P_i is the probability that there are i requests per millisecond). $P_0 = 0.18$, $P_1 = 0.22$, $P_2 = 0.14$, $P_4 = 0.18$, $P_5 = 0.12$, $P_6 = 0.10$, $P_7 = 0.06$. (i) What type of probability distribution is this (discrete or continuous)? (ii) If P_7 was 0.09, this would not be a valid probability distribution. Why? (iii) Let X denote the random variable that shows the number of client requests per millisecond (i.e., $P_i = P(X=i)$). Find the mean, variance, standard deviation, coefficient of variation for X. Show your work.*

In plain words. You get a table of values and their chances. Discrete means it only takes separate values you can count, like 0, 1, 2 requests. The mean is each value times its chance, added up. Variance says how spread out it is, the SD is its square root, and COV is the SD divided by the mean, so it's the spread relative to the size.

1. Whole-number values only, so it's discrete.
2. Probabilities must add to exactly 1. Listed values give $.18+.22+.14+.18+.12+.10+.06 = 1.00$, so $P_3 = 0$. With $P_7 = 0.09$ the sum is $1.00 - 0.06 + 0.09 = 1.03 > 1$, so it's invalid.
3. Mean. *Formula.* $\mu = E[X] = \sum x\,p(x)$, where x is a number of requests and $p(x)$ its probability. *Plug in.* $0(.18) + 1(.22) + 2(.14) + 4(.18) + 5(.12) + 6(.10) + 7(.06)$. *Work it out.* $0 + 0.22 + 0.28 + 0.72 + 0.60 + 0.60 + 0.42$. *Result.* $\mu = 2.84$.
4. Average of squares. *Formula.* $E[X^2] = \sum x^2\,p(x)$. *Plug in.* $0(.18) + 1(.22) + 4(.14) + 16(.18) + 25(.12) + 36(.10) + 49(.06)$. *Work it out.* $0 + 0.22 + 0.56 + 2.88 + 3.00 + 3.60 + 2.94$. *Result.* 13.20.
5. Variance. *Formula.* $\text{Var} = E[X^2] - \mu^2$, the average of the squares minus the square of the average. *Plug in.* $13.20 - 2.84^2$. *Work it out.* $2.84^2 = 8.0656$, so $13.20 - 8.0656$. *Result.* 5.1344.
6. SD. *Formula.* $\sigma = \sqrt{\text{Var}}$. *Plug in.* $\sqrt{5.1344}$. *Result.* 2.266.
7. COV. *Formula.* $\text{COV} = \sigma/\mu$ (Summ p.41). *Plug in.* $2.266/2.84$. *Result.* 0.798.

Answer. Discrete, mean 2.84, variance 5.134, SD 2.27, COV 0.80.

Usual mistake. Dividing by $n-1$ here. These are known probabilities of a population, so there's no $n-1$. Also forgetting the missing P_3 and putting P_4's value at $x=3$.

Small version. $X \in \{0,1,2\}$ with $.25, .5, .25$. $E[X] = 0(.25) + 1(.5) + 2(.25) = 0 + 0.5 + 0.5 = 1$. $E[X^2] = 0(.25) + 1(.5) + 4(.25) = 0 + 0.5 + 1 = 1.5$. $\text{Var} = 1.5 - 1^2 = 0.5$, $\text{SD} = \sqrt{0.5} = 0.707$, $\text{COV} = 0.707/1 = 0.707$. COV is meaningless if the mean is 0 or negative (Summ p.41).

HW0 Q6. Cohort with Spring 10 (6 BS, 4 MS) and Fall 30 (17 BS, 13 MS). P(Fall), P(MS), P(Spring|MS), P(Fall|BS)

Question. *The 2019 PhD CS cohort has the following student profiles. Spring 2019 Start Term, 10 students started. 6 have only BS degree, 4 have MS degrees. Fall 2019 Start Term, 30 students started. 17 have only BS degree, 13 have MS degrees. If a random student is selected from the 2019 cohort, (i) What is the probability that the student started in Fall 2019? (ii) What is the probability that the student has MS degree? (iii) If we randomly select a student and the student has MS degree, what is the probability that the student started in Spring'19? (iv) If we randomly select a student and the student has only BS degree, what is the probability that the student started in Fall'19?*

In plain words. This is counting from a table. A plain probability divides by all 40 students. A conditional one, written $P[A \mid B]$ and read "A given B", only looks at the students in group B, because you already know the student is in B.

1. Totals. Students $= 10 + 30 = 40$. MS $= 4 + 13 = 17$. BS-only $= 6 + 17 = 23$.
2. P[Fall]. *Formula.* $P[A] = \dfrac{\text{students in } A}{\text{all students}}$. *Plug in.* 30/40. *Result.* 0.75.
3. P[MS]. *Plug in.* 17/40. *Result.* 0.425.
4. P[Spring | MS]. *Formula.* $P[A \mid B] = \dfrac{P[A \text{ and } B]}{P[B]}$, the chance of A once you know B happened. Here A is started in Spring and B is holds an MS. *Plug in.* P[Spring and MS] = 4/40 (4 of the 40 are Spring MS) and P[MS] = 17/40, so $\dfrac{4/40}{17/40}$. *Work it out.* The /40 cancels, leaving 4/17. *Result.* 0.235. Shortcut, just count inside the MS group, 4 of 17.
5. P[Fall | BS]. *Plug in.* P[Fall and BS] = 17/40, P[BS] = 23/40, so $\dfrac{17/40}{23/40}$. *Work it out.* 17/23. *Result.* 0.739.

Answer. 0.75, 0.425, 0.235, 0.739.

Usual mistake. Dividing by the wrong total, for example 4/10,

which is $P[\text{MS} \mid \text{Spring}]$. The same direction mix-up is the p-value trap, since $P[\text{data} \mid H_0] \neq P[H_0 \mid \text{data}]$ (HT p.13 to 14).

HW0 Q7. $X \sim U(10, 90)$**. Mean, SD,** $P[X < 20]$**,** $P[X > 60]$**,** $P[20 < X < 30]$

Question. *Assume a random variable has uniform distribution between 10 and 90. What is the mean? What is the standard deviation? What is the probability that the random variable will take a value (i) less than 20? (ii) more than 60? (iii) between 20 and 30?*

In plain words. Uniform means every value between 10 and 90 is equally likely, so the density is a flat line. Any probability is just the length of the stretch you care about divided by the full length, 80. The mean is the middle, and the SD has a fixed formula.

1. Mean. *Formula.* $\mu = (a+b)/2$, where $a = 10$ is the low end and $b = 90$ the high end. *Plug in.* $(10 + 90)/2$. *Work it out.* $100/2$. *Result.* 50.
2. Variance. *Formula.* $\sigma^2 = (b-a)^2/12$. *Plug in.* $(90-10)^2/12$. *Work it out.* $80^2 = 6400$, then $6400/12$. *Result.* 533.3.
3. SD. *Formula.* $\sigma = \sqrt{\sigma^2} = (b-a)/\sqrt{12}$. *Plug in.* $80/\sqrt{12}$. *Work it out.* $80/3.464$. *Result.* 23.09.
4. Probabilities. *Formula.* $P[c < X < d] = \dfrac{d-c}{b-a}$, length of the stretch over 80.
5. $P[X < 20]$. *Plug in.* $(20-10)/80$. *Work it out.* $10/80$. *Result.* 0.125.
6. $P[X > 60]$. *Plug in.* $(90-60)/80$. *Work it out.* $30/80$. *Result.* 0.375.
7. $P[20 < X < 30]$. *Plug in.* $(30-20)/80$. *Work it out.* $10/80$. *Result.* 0.125.

Answer. Mean 50, SD 23.09, then 0.125, 0.375, 0.125.

Usual mistake. Using $(b-a)/12$ or forgetting the square. Rounding error of a timer is uniform on $\pm\Delta$, so its SD is $2\Delta/\sqrt{12}$ (quantization, CI p.9 to 10).

HW0 Q8. $Z \sim N(0, 1)$**.** $P[0.9 < Z < 1.3]$**,** $P[Z > 0.9]$**,** $P[Z > -0.9]$

Question. *Assume a random variable has standard normal distribution $N(0, 1)$. What is the probability that this random variable will take on a value (i) between 0.9 and 1.3, (ii) greater than 0.9, (iii) greater than -0.9? Explain how you obtained your results.*

In plain words. Z is the standard bell curve with mean 0 and SD 1. The table gives $\Phi(z)$, the area to the left of z, which is the chance of landing below z. Every question is a left area, a right area (1 minus the left area), or the gap between two left areas.

1. Between. *Formula.* $P[a < Z < b] = \Phi(b) - \Phi(a)$. *Plug in.* $\Phi(1.3) - \Phi(0.9)$. *Work it out.* $0.9032 - 0.8159$. *Result.* 0.0873.
2. Above. *Formula.* $P[Z > a] = 1 - \Phi(a)$. *Plug in.* $1 - \Phi(0.9)$. *Work it out.* $1 - 0.8159$. *Result.* 0.1841.
3. Above a negative. *Formula.* Symmetry gives $\Phi(-a) = 1 - \Phi(a)$. *Plug in.* $P[Z > -0.9] = 1 - \Phi(-0.9) = 1 - (1 - \Phi(0.9))$. *Work it out.* That's just $\Phi(0.9)$. *Result.* 0.8159.

Answer. 0.0873, 0.1841, 0.8159.

Usual mistake. Answering $P[Z > -0.9]$ with 0.1841. Sketch the curve and shade before subtracting.

HW0 Q9. Highs $\sim N(85, 6^2)$**.** $P[83 < X < 88]$**,** $P[X > 90]$**,** $P[X < 82]$ **with z-scores**

Question. *Over the course of several years, in a certain city, it is observed that the highest daily temperature in August, in Fahrenheit, is normally distributed with mean = 85 and standard deviation = 6. What is the probability that a randomly selected August day's highest temperature is (i) between 83 and 88, (ii) greater than 90, (iii) smaller than 82? Solve problem 9 using z scores. Show your work.*

In plain words. Turn each temperature into a z-score, which counts how many SDs it sits away from the mean. Then it's a standard normal question and you use the same table as Q8.

1. *Formula.* $z = (x - \mu)/\sigma$, where x is the temperature, $\mu = 85$ the mean and $\sigma = 6$ the SD.
2. Between 83 and 88. *Plug in.* $z_1 = (83 - 85)/6 = -2/6 = -0.33$ and $z_2 = (88 - 85)/6 = 3/6 = 0.5$. *Work it out.* $\Phi(0.5) - \Phi(-0.33) = 0.6915 - (1 - 0.6293) = 0.6915 - 0.3707$. *Result.* 0.3208.
3. Above 90. *Plug in.* $z = (90 - 85)/6 = 5/6 = 0.83$. *Work it out.* $1 - \Phi(0.83) = 1 - 0.7967$. *Result.* 0.2033.
4. Below 82. *Plug in.* $z = (82 - 85)/6 = -3/6 = -0.5$. *Work it out.* $\Phi(-0.5) = 1 - \Phi(0.5) = 1 - 0.6915$. *Result.* 0.3085.

Answer. About 0.321, 0.203, 0.309 (unrounded z gives 0.322 and 0.202).

Backwards version. Where do the hottest 5% of days start? *Formula.* $x = \mu + z\sigma$. *Plug in.* $85 + z[0.05] \times 6 = 85 + 1.645 \times 6$. *Work it out.* $1.645 \times 6 = 9.87$, so $85 + 9.87$. *Result.* 94.9.

HW2. Beyond performance metrics (report)

HW2. Pick metrics from your area, evaluate them on the lecture's criteria, propose extra criteria (Metrics p.33)

Question. *HW 2, Beyond Performance Metrics. (i) Identify one or more metrics commonly used in your area of research. If you haven't yet decided on your research area, pick one of the areas you might be interested in. Look at one or more seminal research articles in your area and see what metrics were used. (ii) Evaluate the metric(s) based on the criteria discussed in today's lecture. What additional criteria might be useful in evaluating metrics in your research area? (iii) Submit a short typed report (at most 4 pages), using LaTeX.*

In plain words. Pick the numbers people in your research area use to say "my system is better", then grade each one against the six properties a good metric should have. The key idea is that a metric can be easy to measure and still mislead, like MIPS. It's graded on argument, so every yes or no needs a concrete example of how the metric fails.

1. List the six criteria (Metrics p.7 to 12). Linear (2x metric means 2x performance). Reliable (A scores higher means A really is better). Repeatable (same experiment, same value). Easy to use (easy to measure correctly). Consistent (same units and definition on every system). Independent (vendors can't bend it).
2. Say which are required. Reliable, repeatable, consistent and independent are required, while linear and easy to use are nice to have (Metrics p.13).
3. Classify means versus ends. Means-based counts what was done, useful or not. Ends-based counts what got accomplished (Metrics p.27 to 28).
4. For each metric, give a verdict per criterion with one concrete failure.
5. Propose extra criteria your area needs.

His answer. Area is low-latency video streaming. Metrics were average throughput and the QoE score (Yin et al. 2015, chunk

quality minus a switch penalty minus a stall penalty, as a weighted sum).

Criterion	Throughput	QoE score
Linear	Yes	No
Reliable	No	No
Repeatable	Partly	Partly
Easy to use	Yes	Yes
Consistent	Partly	No
Independent	No	No

- Throughput isn't reliable because a sender can inflate it by filling a deep queue (bufferbloat), the same failure as MIPS (Metrics p.17). It isn't independent because ISPs advertise it. Repeatability is partial because Pantheon found algorithm rankings change with the path. It's a means metric used for ends claims, since 8 Mbps with two stalls beats 5 Mbps with none.
- QoE isn't consistent because each paper picks its own weights (Yin's stall weight 3000, Pensieve's three variants), the MFLOPS problem. It isn't linear since it adds unlike units. It isn't reliable because the linear form was never validated against subjective tests (Peroni and Gorinsky 2024). It isn't independent because authors set the weights they're judged under.
- Extra criteria. Externality, meaning what a flow does to others on the link, which Jain's index hides. Report the distribution and the tail, not just the mean. State the conditions the metric was checked over.

By hand, Jain's index. *Formula.* $J = \dfrac{(\sum x_i)^2}{n \sum x_i^2}$, where x_i is flow i's throughput and n the number of flows. It's 1 when all flows get the same and $1/n$ when one flow gets everything. *Plug in.* Flows of 90 and 10 Mbps, $n = 2$, so $\dfrac{(90+10)^2}{2(90^2+10^2)}$. *Work it out.* $\dfrac{100^2}{2(8100+100)} = \dfrac{10000}{16400}$. *Result.* 0.61.

By hand, precision and recall (Metrics p.32). *Formula.* Precision $=$ $\dfrac{\text{relevant retrieved}}{\text{all retrieved}}$ and recall $=$ $\dfrac{\text{relevant retrieved}}{\text{all relevant}}$. *Plug in.* 30 retrieved, 20 of them relevant, 40 relevant ones missed. Precision $= 20/30$, recall $= 20/(20+40) = 20/60$. *Result.* Precision 0.67, recall 0.33.

Check. The repeatability paragraph says Puffer's 95% CI on stall ratio was "around 10 to 17 percent of the mean" after "1.75 years" per scheme. Those numbers are from the arXiv preprint. The NSDI 2020 version he cites (and quotes in HW4) says ±13% to ±21% with at least 2.5 years per scheme. The argument stands, but the two homeworks quote different versions.

Usual mistake. Saying a metric "fails" without naming the failure. On the exam, MIPS is repeatable, easy and independent but not linear, reliable or consistent (Metrics p.17).

How to write it on the exam.

1. Name the metric and what it counts, then say means or ends.
2. Go down the six criteria in order, one line each, with a yes, no or partly and a concrete reason.
3. Flag which failures matter most, the required four (Metrics p.13).
4. Add one criterion your area needs and say why the six miss it.

Exam variant, MIPS. *Formula.* MIPS $= \dfrac{n}{T_e \times 10^6}$ (Metrics p.16), where n is the number of instructions run and T_e the run time in seconds. *Plug in.* A CISC machine runs 50M instructions in 1 s, so $\dfrac{50 \times 10^6}{1 \times 10^6} = 50$. A RISC machine runs 100M simpler instructions for the same program in 0.8 s, so $\dfrac{100 \times 10^6}{0.8 \times 10^6} = 125$. *Work it out.* MIPS ratio $125/50 = 2.5$. Real speedup is old time over new time, $1/0.8 = 1.25$. *Result.* MIPS ranks RISC 2.5x better, but it's really 1.25x faster. So MIPS isn't linear, and with different instruction sets it isn't consistent either. Execution time is the ends-based alternative (Metrics p.27 to 28).

HW3. Choosing a mean, quartiles, histograms, Q-Q plots

HW3 Q1a. Five stacked features improve quality 10, 50, 70, 85, 15%. Mean improvement per feature, and which mean?

Question. *Q1 (3 pts). For each of the following sub-questions, i) indicate what type of mean (arithmetic, geometric, or harmonic) should be used, with explanation, ii) Compute the mean value based on your choice. a. A researcher considers adding five new features to her optimization project to improve the quality of the solution. Each feature is implemented on top of the previous one (i.e., Feature 2 is implemented on top of Feature 1, Feature 3 is implemented on top of Feature 2, and so on.) She observes the following quality improvements after adding each feature. Feature 1, 10%. Feature 2, 50%. Feature 3, 70%. Feature 4, 85%. Feature 5, 15%. What is the mean improvement per feature? What type of mean did you use and why?*

In plain words. The mean you pick depends on how the numbers combine. Here each feature's gain stacks on top of the previous one, so the gains multiply instead of adding. The geometric mean is the one number that, multiplied by itself 5 times, gives the same total gain as the real five.

1. Each feature builds on the last, so effects multiply. Use the geometric mean (Summ p.30, p.32).
2. *Formula.* GM $= (x_1 x_2 \cdots x_n)^{1/n}$, where $x_i = 1 + \text{gain}_i$ is the ratio new over old and $n = 5$ features (Summ p.23).
3. *Plug in.* Ratios 1.10, 1.50, 1.70, 1.85, 1.15, so GM $= (1.10 \times 1.50 \times 1.70 \times 1.85 \times 1.15)^{1/5}$.
4. *Work it out.* $1.10 \times 1.50 = 1.65$, $\times 1.70 = 2.805$, $\times 1.85 = 5.18925$, $\times 1.15 = 5.9676$. Fifth root by hand with logs, $\ln 5.9676 = 1.786$, $1.786/5 = 0.357$, $e^{0.357} = 1.4294$.
5. *Result.* GM $= 1.4294$. Subtract 1 to get back to a percentage, $0.4294 = 42.94\%$.

Answer. Geometric mean, because the gains compound. 1.4294, so 42.94% per feature. Matches his main.tex.

Usual mistake. Taking the GM of the percentages (10, 50, ...) instead of the ratios, or forgetting to subtract 1 at the end.

By hand. Gains of 100, 300, 700% are ratios 2, 4, 8. *Plug in.* GM $= (2 \times 4 \times 8)^{1/3}$. *Work it out.* $2 \times 4 \times 8 = 64$ and $\sqrt[3]{64} = 4$. *Result.* $4 - 1 = 3$, so 300% per version. The AM of the ratios, $(2+4+8)/3 = 4.67$, would compound to $4.67^3 \approx 102$, not 64.

HW3 Q1b. Compiler times 150, 200, 120, 300, 270, 180 ms. Mean, and which?

Question. *b. The running time of a new compiler is measured with source programs of comparable size. In 6 experiments, the running times are measured as 150 ms, 200 ms, 120 ms, 300 ms, 270 ms, and 180 ms. What is the mean execution time? What type of mean did you use and why?*

In plain words. These are running times, and times add up into a total. The arithmetic mean, the usual sum divided by count, is the one whose total matches the real total time, so it's the right one.

1. These are times, so the mean should be proportional to total time. Arithmetic mean (Summ p.25, p.27).
2. *Formula.* $\bar{x} = \dfrac{1}{n} \sum x_i$, where x_i are the times and $n = 6$ runs.

3. *Plug in.* $\dfrac{150+200+120+300+270+180}{6}$.
4. *Work it out.* $150+200=350$, $+120=470$, $+300=770$, $+270=1040$, $+180=1220$. Then $1220/6$.
5. *Result.* 203.33 ms.

Answer. Arithmetic mean, because times add up to a total. 203.33 ms.

Usual mistake. Using the harmonic mean because it's "performance". Times take the AM.

Weighted version (Summ p.33). *Formula.* $\bar{x}_w = \sum w_i x_i$ with weights w_i adding to 1. *Plug in.* Times 2, 5, 10 s with weights .5, .3, .2 give $.5(2)+.3(5)+.2(10)$. *Work it out.* $1+1.5+2$. *Result.* 4.5 s.

HW3 Q1c. DB at 2500, 3500, 3700, 1800, 2000 tps. Mean, and which?

Question. *c. A new database system is implemented and its performance is tested. In five different experiments, the following performance (speed) values are obtained. 2500 transactions per second, 3500 transactions per second, 3700 transactions per second, 1800 transactions per second, 2000 transactions per second. What is the mean performance (in terms of transactions per second)? What type of mean did you use and why?*

In plain words. These are rates, things per second, and you can't just average rates. If every run did the same amount of work, the honest average rate is total work over total time, and that works out to the harmonic mean, n divided by the sum of the $1/x_i$.

1. Rates have time in the denominator, so the mean should be inversely proportional to total time. Harmonic (Summ p.25, p.28).
2. *Formula.* $x_H = \dfrac{n}{\sum 1/x_i}$ (Summ p.22), where x_i are the rates and $n=5$ runs.
3. *Plug in.* $x_H = \dfrac{5}{1/2500+1/3500+1/3700+1/1800+1/2000}$.
4. *Work it out.* The reciprocals are $0.000400+0.000286+0.000270+0.000556+0.000500=0.0020115$. Then $5/0.0020115$.
5. *Result.* $x_H = 2485.66$ tps.

Answer. Harmonic mean, because these are rates with equal work per run. 2485.66 tps.

Usual mistake. The AM of rates, $(2500+3500+3700+1800+2000)/5 = 13500/5 = 2700$, overstates. The HM is right only if every run did the same work F (Summ p.28).

By hand. Rates 2, 3, 6. *Plug in.* $x_H = \dfrac{3}{1/2+1/3+1/6}$. *Work it out.* $1/2+1/3+1/6 = 3/6+2/6+1/6 = 6/6 = 1$, so $3/1$. *Result.* $x_H = 3$. The AM would say $11/3 = 3.67$.

Exam variant, unequal work. Program 1 does 100 Mops at 50 Mops/s, program 2 does 300 Mops at 100 Mops/s. *Formula.* True rate $= \dfrac{\text{total work}}{\text{total time}}$, with time = work over rate. *Plug in.* Times $100/50 = 2$ s and $300/100 = 3$ s, so rate $= \dfrac{100+300}{2+3}$. *Work it out.* $400/5$. *Result.* 80 Mops/s. The plain HM gives $2/(1/50+1/100) = 2/0.03 = 66.7$, which is wrong because it assumes equal work per run (Summ p.26, p.28).

HW3 Q2a,b. For normal.txt ($n=1000$) and not-normal.txt ($n=870$), mean, variance, median, Q1, Q2, Q3

Question. *Q2. Consider the two data sets, normal.txt and not-normal.txt. For each data set, do the following exercises. a. (0.5 pts) Find the mean and variance of the sample data. b. (0.5 pts) Find the median, first, second, and third quartile of the sample data. Note, you can use a program such as Microsoft Excel/OpenOffice Calc/Matlab/Mathematica/etc. for doing these exercises, or you can do them without the help of any program.* [The files hold 1000 and 870 values.]

In plain words. Summarize each data set by its center (mean and median) and its spread (variance and quartiles). Quartiles are the values that cut the sorted data into four equal parts. The course has a fixed recipe for where each sorted value sits, $(i-0.5)/n$, and you interpolate between neighbours when a quartile falls between two values.

1. Mean. *Formula.* $\bar{x} = \frac{1}{n}\sum x_i$, where x_i are the data values and n the count. *Plug in.* normal has $\sum x = 2014.15$, so $2014.15/1000$. not-normal has $\sum x = 181.25$, so $181.25/870$. *Result.* 2.0142 and 0.2083.
2. Variance. *Formula.* Sample variance with $n-1$, one-pass form $s^2 = \dfrac{\sum x_i^2 - (\sum x_i)^2/n}{n-1}$ (Summ p.39). *Plug in.* normal has $\sum x^2 = 5101.51$, so $\dfrac{5101.51 - 2014.15^2/1000}{999}$. *Work it out.* $2014.15^2/1000 = 4056.82$, $5101.51 - 4056.82 = 1044.69$, then $1044.69/999$. *Result.* 1.0457. not-normal, $\dfrac{76.01 - 181.25^2/870}{869} = \dfrac{76.01-37.76}{869} = \dfrac{38.25}{869} = 0.0440$.
3. Quartiles. *Formula.* Sort the data. The i-th smallest value $x_{(i)}$ sits at fraction $f_i = (i-0.5)/n$. Solving for the position of quantile p gives $i = pn + 0.5$ (Summ p.59). If i isn't a whole number, go that far between the two neighbours.
4. normal ($n=1000$). Q_1 at $i = 0.25(1000)+0.5 = 250.5$, halfway between $x_{(250)} = 1.2646$ and $x_{(251)} = 1.2714$, so 1.2680. Median at $i = 500.5$, halfway between 2.0320 and 2.0324, so 2.0322. Q_3 at $i = 750.5$, halfway between 2.6905 and 2.6948, so 2.6927.
5. not-normal ($n=870$). Q_1 at $i = 0.25(870)+0.5 = 218$, exactly $x_{(218)} = 0.0587$. Median at $i = 435.5$, halfway between 0.1426 and 0.1432, so 0.1429. Q_3 at $i = 653$, exactly $x_{(653)} = 0.2865$.

Answer.

	normal	not-normal
Mean	2.0142	0.2083
Variance ($n-1$)	1.0457	0.0440
Min	−1.6554	0.0007
Q_1	1.2680	0.0587
Median $= Q_2$	2.0322	0.1429
Q_3	2.6927	0.2865
Max	5.2128	1.2957

not-normal is right-skewed. The mean 0.208 sits above the median 0.143, and $Q_3 - Q_2 = 0.2865 - 0.1429 = 0.144$ against $Q_2 - Q_1 = 0.1429 - 0.0587 = 0.084$ (Summ p.48). For skewed data report the SIQR. *Formula.* SIQR $= (Q_3 - Q_1)/2$, half the width of the middle 50%. *Plug in.* $(0.2865 - 0.0587)/2$. *Work it out.* 0.2278/2. *Result.* 0.114 (Summ p.45, p.49).

By hand. Data 7, 1, 11, 5, 9, 3. Sorted 1, 3, 5, 7, 9, 11.

1. Mean. $\bar{x} = (1+3+5+7+9+11)/6 = 36/6 = 6$.
2. Variance. $\sum x^2 = 1+9+25+49+81+121 = 286$. $(\sum x)^2/n = 36^2/6 = 1296/6 = 216$. $s^2 = (286-216)/5 = 70/5 = 14$, so $s = \sqrt{14} = 3.742$.
3. Quartiles. The f_i are .083, .25, .417, .583, .75, .917. Q_1 at $i = 0.25(6)+0.5 = 2$, so $x_{(2)} = 3$. Q_3 at $i = 0.75(6)+0.5 = 5$, so $x_{(5)} = 9$. Median at $i = 3.5$, halfway between 5 and 7, so 6. With $n=10$, Q_1 at $i=3$ and Q_3 at $i=8$ exactly.

Exam variant with interpolation. Data 2, 4, 4, 5, 7, 9, 10, 12 ($n = 8$). The f_i are .0625, .1875, .3125, .4375, .5625, .6875, .8125, .9375. Q_1 at $i = 0.25(8) + 0.5 = 2.5$, between $x_{(2)} = 4$ and $x_{(3)} = 4$, so 4. Median at $i = 4.5$, halfway between 5 and 7, so 6. Q_3 at $i = 6.5$, halfway between $x_{(6)} = 9$ and $x_{(7)} = 10$, so 9.5. SIQR $= (9.5 - 4)/2 = 5.5/2 = 2.75$.

Usual mistake. Dividing by n for a sample variance, or taking Q_1 as $x_{(n/4)}$ without the $(i - 0.5)/n$ positions.

HW3 Q2c. Histogram of each

Question. *c. (1 pts) Plot the sample data using a histogram.* [For each of normal.txt and not-normal.txt.]

In plain words. A histogram chops the range into equal-width bins and counts how many values land in each, so you can see the shape of the data. The only real decision is how many bins, and Sturges' rule gives a default from n.

1. Equal-width bins over the range, plot relative frequency (Summ p.54).
2. Number of bins. *Formula.* Sturges, $k = \lfloor 1 + \log_2 n \rfloor$ (Summ p.54), where k is the bin count and n the number of values. *Plug in.* $1 + \log_2 1000 = 1 + 9.97 = 10.97$ and $1 + \log_2 870 = 1 + 9.76 = 10.76$. *Work it out.* Round down. *Result.* $k = 10$ for both.
3. Bin width. *Formula.* width $= (\max - \min)/k$. *Plug in.* normal $(5.2128 - (-1.6554))/10$, not-normal $(1.2957 - 0.0007)/10$. *Work it out.* 6.868/10 and 1.295/10. *Result.* 0.687 and 0.130.
4. Counts with 10 bins. normal gives 1, 6, 54, 128, 214, 258, 217, 83, 33, 6. not-normal gives 402, 228, 94, 69, 33, 20, 12, 6, 2, 4.

Answer. normal.txt is a symmetric bell with its peak in the bin around 1.8 to 2.5, close to the mean 2.01. not-normal.txt piles up in the first bin near 0 and trails off in a long right tail out to about 1.3. So one looks normal and the other is right-skewed.

Check. His main.tex says 11 bins for both because analyze.py used round$(1 + \log_2 n)$, which rounds 10.97 and 10.76 up. The slide floors, giving 10. No conclusion changes, but use the floor on the exam.

By hand. Six points 1, 3, 5, 7, 9, 11. $k = \lfloor 1 + \log_2 6 \rfloor = \lfloor 1 + 2.58 \rfloor = 3$. Width $= (11 - 1)/3 = 3.33$. Bins $[1, 4.33)$, $[4.33, 7.67)$, $[7.67, 11]$ hold 2, 2, 2. Bin width changes the shape (Summ p.55 to 56), so judge shape with the Q-Q plot.

HW3 Q2d. Quantile plot

Question. *d. (2 pts) Draw the quantile plot for the sample data.* [For each of normal.txt and not-normal.txt.]

In plain words. A quantile plot is just the sorted data plotted against where each point sits, its fraction f_i. You can read any percentile straight off it, and the shape shows the tails. A long flat stretch then a sharp climb means a long right tail.

1. Plot $(f_i, x_{(i)})$ with $f_i = (i - 0.5)/n$, where $x_{(i)}$ is the i-th smallest value (Summ p.59).
2. normal.txt gives an S shape. not-normal.txt stays flat, then shoots up near $f = 1$, the long right tail.

Answer. The S shape for normal.txt and the flat-then-steep curve for not-normal.txt in step 2.

By hand. In the Summ p.60 table ($n = 10$), $Q(0.80)$ sits at $i = 0.80(10) + 0.5 = 8.5$, halfway between $x_{(8)} = 44.5$ and $x_{(9)} = 50.3$. *Work it out.* $(44.5 + 50.3)/2 = 94.8/2$. *Result.* 47.4.

HW3 Q2e. Compare with a normal using a Q-Q plot

Question. *e. (3 pts) Compare the distribution of the sample data against the theoretical normal distribution using a Q-Q plot.* [For each of normal.txt and not-normal.txt.]

In plain words. A Q-Q plot lines up your sorted data against what a perfect normal with the same mean and SD would give at the same positions. If the data are normal, the points fall on a straight line. Bends at the ends tell you which way the data aren't normal.

1. For each f_i find z_i with $\Phi(z_i) = f_i$, where z_i is the standard normal value with area f_i below it. Use the table or $z_i = 4.91[f_i^{0.14} - (1 - f_i)^{0.14}]$ (Summ p.69).
2. *Formula.* Rescale $x_i = m + sz_i$, where m is the sample mean and s the sample SD, so normal data falls on the 45° line. Unscaled, the intercept is the mean and the slope the SD (Summ p.70).
3. Plot theoretical against observed (Summ p.63) and read the shape (Summ p.71 to 73).

Answer. normal.txt follows the line with correlation 0.9993, so it's consistent with normal. not-normal.txt sits above the line at both ends and below in the middle, correlation 0.904. The normal predicts about -0.47 at the low end but the data never go below 0, and the top values are far too large. Right-skewed, not normal.

By hand. Data 1, 3, 5, 7, 9, 11 with $m = 6$, $s = 3.742$.

1. f_i for $i = 1, 2, 3$ are $0.5/6 = .083$, $1.5/6 = .25$, $2.5/6 = .417$. From the table $z = -1.38, -0.67, -0.21$, and the top three are the same with a plus sign.
2. *Plug in.* $x = 6 + 3.742z$. *Work it out.* $6 - 3.742(1.38) = 6 - 5.16 = 0.84$, $6 - 3.742(0.67) = 6 - 2.51 = 3.49$, $6 - 3.742(0.21) = 6 - 0.79 = 5.21$, then by symmetry 6.79, 8.51, 11.16.
3. *Result.* Pairs (0.84, 1), (3.49, 3) and so on are all within 0.5 of the line, so normal.

Skewed version 1, 1, 1, 2, 3, 10. $m = 18/6 = 3$. Deviations $-2, -2, -2, -1, 0, 7$, squares sum to $4 + 4 + 4 + 1 + 0 + 49 = 62$, $s^2 = 62/5 = 12.4$, $s = 3.52$. Theoretical $3 - 3.52(1.38) = -1.86$, $3 - 3.52(0.67) = 0.64$, $3 - 3.52(0.21) = 2.26$, then 3.74, 5.36, 7.86. Ends above the line and middle below is the right-skew pattern.

Usual mistake. Plotting f_i instead of z_i on the x-axis, or reading "longer tails" (Summ p.71) when the shape is asymmetric (Summ p.73).

HW4. Confidence intervals

HW4 1.1 and 2.1. The file is the population. Population mean and variance

Question. *State which tool you used (Excel, Python, R) for each part. Problem 1, Normally distributed data. Use the population data in the file normal.txt. This data is normally distributed. 1. Find the population mean and variance. Problem 2, Data that is not normally distributed. Repeat parts 1 to 3 of Problem 1 for the population data in the file not-normal.txt. This data is not normally distributed.*

In plain words. The homework pretends the whole file is the entire population, so you know the true answer. Later you check whether your sample-based intervals actually caught it. For a whole population you divide by N, the number of values, not $N - 1$.

1. Mean. *Formula.* $\mu = \frac{1}{N}\sum x_i$, where N is the population size. *Plug in.* 2014.15/1000 and 181.25/870. *Result.* 2.0142 and 0.2083.
2. Variance. *Formula.* $\sigma^2 = \dfrac{\sum x_i^2 - (\sum x_i)^2/N}{N}$. Divide by N, not $N - 1$. The $n - 1$ is for estimating from a sample (Summ p.39).

Plug in. normal, 1044.69/1000. not-normal, 38.25/870. *Result.* 1.0447 and 0.0440.

3. Shortcut from HW3's value. $\sigma^2 = s^2 \times (N-1)/N = 1.0457 \times 999/1000 = 1.0447$.

Answer. normal has $\mu = 2.0142$, $\sigma^2 = 1.0447$. not-normal has $\mu = 0.2083$, $\sigma^2 = 0.0440$ (the same values as CI p.32).

HW4 1.2. Sample of 50 from normal.txt. 90, 95, 99% CIs. Do they contain μ?

Question. *2. Generate a random sample of size 50, that is, randomly select 50 numbers from the population. Find 90%, 95% and 99% confidence intervals for the mean. Is the population mean within the intervals?*

In plain words. A confidence interval is a range built from one sample that should catch the true mean most of the time. It's the sample mean plus or minus a margin. The margin is a critical value (bigger for more confidence) times the standard error $s/\sqrt{n}$, which is how much the sample mean would wobble if you redrew the sample.

1. $n > 30$, so use z (CI p.27, p.31). *Formula.* $\bar{x} \pm z[\alpha/2]\, s/\sqrt{n}$, where $\bar{x}$ is the sample mean, s the sample SD, n the sample size, and $z[\alpha/2]$ the normal value with area $\alpha/2$ above it. α is 1 minus the confidence, so 0.10, 0.05, 0.01.
2. From the sample, $\bar{x} = 2.2093$ and $s = 0.9443$.
3. SE. *Formula.* $\text{SE} = s/\sqrt{n}$. *Plug in.* $0.9443/\sqrt{50}$. *Work it out.* 0.9443/7.071. *Result.* 0.1335.
4. Margins. *Formula.* margin $= z[\alpha/2] \times \text{SE}$. *Plug in.* 1.645(0.1335), 1.960(0.1335), 2.576(0.1335). *Result.* 0.2196, 0.2617, 0.3439.
5. CIs. *Plug in.* 2.2093 ± 0.2196, 2.2093 ± 0.2617, 2.2093 ± 0.3439. *Result.* below.

Answer. 90% (1.9897, 2.4290), 95% (1.9476, 2.4710), 99% (1.8653, 2.5533). All contain $\mu = 2.0142$.

By hand. $n = 100$, $\bar{x} = 10$, $s = 2$. $\text{SE} = 2/\sqrt{100} = 2/10 = 0.2$. Margin $1.96 \times 0.2 = 0.392$. CI $10 \pm 0.392 = (9.61, 10.39)$.

Sample size for $\pm 5\%$ at 95% (CI p.44). *Formula.* $n = \left(\dfrac{z\, s}{r\, \bar{x}}\right)^2$, where $r = 0.05$ is the allowed error as a fraction of the mean. *Plug in.* $\left(\dfrac{1.96 \times 2}{0.05 \times 10}\right)^2$. *Work it out.* $3.92/0.5 = 7.84$, $7.84^2 = 61.5$. *Result.* Round up to 62 runs.

HW4 1.3. Sample of 16. Same three CIs

Question. *3. Generate a random sample of size 16. Find 90%, 95% and 99% confidence intervals for the mean. Is the population mean within the intervals?*

In plain words. Same thing with only 16 points. With a small sample s itself is shaky, so you use the t distribution, which has fatter tails and gives a bigger critical value to pay for that. Its shape depends on the degrees of freedom (df), which is $n - 1$.

1. Small n and a normal population. *Formula.* $\bar{x} \pm t[\alpha/2, n-1]\, s/\sqrt{n}$, where $t[\alpha/2, n-1]$ is the t value with area $\alpha/2$ above it and $n-1$ df (CI p.37). Here df $= 16 - 1 = 15$.
2. From the sample, $\bar{x} = 2.2911$, $s = 1.1332$.
3. SE. *Plug in.* $1.1332/\sqrt{16}$. *Work it out.* 1.1332/4. *Result.* 0.2833.
4. Margins. *Plug in.* 1.753(0.2833), 2.131(0.2833), 2.947(0.2833). *Result.* 0.4966, 0.6037, 0.8349.
5. CIs. *Plug in.* 2.2911 ± 0.4966, 2.2911 ± 0.6037, 2.2911 ± 0.8349.

Answer. 90% (1.7944, 2.7877), 95% (1.6872, 2.8949), 99% (1.4562, 3.1259). All contain $\mu = 2.0142$.

By hand. $n = 9$, $\bar{x} = 20$, $s = 6$. $\text{SE} = 6/\sqrt{9} = 6/3 = 2$. df $= 8$, $t[0.025, 8] = 2.306$. Margin $2.306 \times 2 = 4.612$. CI $20 \pm 4.612 = (15.39, 24.61)$. If σ is given, use z even for small n (CI p.27).

Exam variant from raw values. Data 8, 10, 12, 14.

1. Mean. $(8 + 10 + 12 + 14)/4 = 44/4 = 11$.
2. SD. *Formula.* $s^2 = \dfrac{\sum (x_i - \bar{x})^2}{n-1}$. *Plug in.* Deviations $-3, -1, 1, 3$, squares $9, 1, 1, 9$. *Work it out.* $20/3 = 6.67$, $s = \sqrt{6.67} = 2.582$.
3. SE. $2.582/\sqrt{4} = 2.582/2 = 1.291$.
4. CI. df $= 3$, $t[0.025, 3] = 3.182$. Margin $3.182 \times 1.291 = 4.11$. *Result.* $11 \pm 4.11 = (6.89, 15.11)$.

HW4 1.4. Compare widths from 1.2 and 1.3, explain with sample size and distribution

Question. *4. Compare the widths of the intervals from parts 2 and 3. Explain the difference in terms of both the sample size and the distribution you used for the critical value.*

In plain words. Why is the $n = 16$ interval so much wider? Width is critical value times $s/\sqrt{n}$, so three things can change it. Fewer points shrink $\sqrt{n}$, the t value is bigger than z, and the sample SD came out different just by luck.

1. *Formula.* Width $= 2 \times \text{crit} \times s/\sqrt{n}$.
2. *Plug in.* $n = 50$, $2 \times 1.960 \times 0.1335$. $n = 16$, $2 \times 2.131 \times 0.2833$. *Result.* 0.5235 and 1.2077 (unrounded).
3. Ratio. $1.2077/0.5235 = 2.31$.
4. Split the ratio into factors. Sample size $\sqrt{50/16} = \sqrt{3.125} = 1.77$ (CI p.39). Critical value $2.131/1.960 = 1.09$, from t having heavier tails (CI p.35). Luck in s, $1.1332/0.9443 = 1.20$.
5. Check, $1.77 \times 1.09 \times 1.20 = 2.31$.

Answer. The 95% interval at $n = 16$ is 2.31 times as wide as at $n = 50$. About 1.77x comes from the smaller sample, 1.09x from using t with 15 df instead of z, and 1.20x from the second sample happening to have a bigger SD.

Usual mistake. Blaming only n. The question asks for the critical value too. Halving a width takes 4 times the data.

HW4 2.2 and 2.3. Repeat with not-normal.txt

Question. *Problem 2. Repeat parts 1 to 3 of Problem 1 for the population data in the file not-normal.txt. This data is not normally distributed.* [So 2.2 is a sample of 50 and 2.3 a sample of 16, each with 90%, 95% and 99% CIs for the mean, and is the population mean within the intervals?]

In plain words. Same intervals, but on skewed data. The point is to see whether the normal-based recipes still work when the data aren't normal. At $n = 50$ the central limit theorem (CLT), which says averages of many values look normal even when the values don't, saves you. At $n = 16$ it's shakier.

1. $n = 50$. The data aren't normal but the CLT makes $\bar{x}$ close to normal, so z is fine (CI p.29, p.67). $\bar{x} = 0.2168$, $s = 0.2278$. $\text{SE} = 0.2278/\sqrt{50} = 0.2278/7.071 = 0.0322$. Margins 1.645(0.0322) = 0.0530, 1.960(0.0322) = 0.0631, 2.576(0.0322) = 0.0829. CIs are 0.2168 plus or minus each.
2. $n = 16$, t with 15 df. $\bar{x} = 0.2406$, $s = 0.2040$. $\text{SE} = 0.2040/\sqrt{16} = 0.2040/4 = 0.0510$. Margins 1.753(0.0510) = 0.0894, 2.131(0.0510) = 0.1087, 2.947(0.0510) = 0.1503. CIs are 0.2406 plus or minus each.

Answer.

Level	$n = 50$ (z)	$n = 16$ (t_{15})
90%	(0.1638, 0.2698)	(0.1512, 0.3300)
95%	(0.1537, 0.2800)	(0.1319, 0.3493)
99%	(0.1339, 0.2998)	(0.0903, 0.3909)

All six contain $\mu = 0.2083$.

Usual mistake. Trusting the t interval at $n = 16$ on skewed data. The CI p.67 example, $n = 16$ from a skewed population, covers only about 91% at nominal 95% and runs too narrow.

HW4 2a. Batching. 16 values in draw order into 4 batches of 4, CI from the batch means with t on 3 df

Question. *Then, for the smaller sample only (size 16), compute the 90%, 95% and 99% intervals two further ways. These correspond to the first and third responses on the lecture slides for when the normality assumption fails. 2a. Batching. Group your 16 observations into 4 batches of 4 and compute the mean of each batch. You now have 4 numbers. Compute the confidence intervals from those 4 batch means, using the t distribution with 3 degrees of freedom. Batch the observations in the order you drew them. Do not sort the sample first, sorting would group the small values together and the large values together, and the batch means would no longer be four draws from the same distribution.*

In plain words. Batching is a trick to make data look more normal. Averaging groups of 4 values gives batch means that are closer to normal than the raw values, the CLT done by hand. You then treat the 4 batch means as your whole sample, which leaves only $4 - 1 = 3$ degrees of freedom.

1. Average groups of 4 or more to push toward normal (CI p.68). Keep draw order, since sorting would make each batch a different slice.
2. Batch means 0.4212, 0.2368, 0.0904, 0.2139. *Formula.* $\bar{x} = \frac{1}{b}\sum \bar{x}_j$, where $\bar{x}_j$ is batch j's mean and $b = 4$ batches. *Plug in.* 0.9623/4. *Result.* 0.2406, same as the raw mean.
3. Variance of the batch means. *Formula.* $s^2 = \dfrac{\sum \bar{x}_j^2 - (\sum \bar{x}_j)^2/b}{b-1}$. *Plug in.* $\sum \bar{x}_j^2 = 0.2874$ and $(\sum \bar{x}_j)^2/4 = 0.9623^2/4 = 0.2315$. *Work it out.* $(0.2874 - 0.2315)/3 = 0.0559/3 = 0.0186$, $s = \sqrt{0.0186} = 0.1365$. *Result.* $\text{SE} = s/\sqrt{b} = 0.1365/2 = 0.0683$.
4. Margins with t on 3 df. $2.353(0.0683) = 0.161$, $3.182(0.0683) = 0.217$, $5.841(0.0683) = 0.399$. CIs are 0.2406 plus or minus each.

Answer. 90% (0.0799, 0.4012), 95% (0.0233, 0.4578), 99% (−0.1581, 0.6393). All contain μ.

By hand. 12 values into 3 batches with means 2, 4, 6. Mean $12/3 = 4$. $s^2 = ((2-4)^2 + (4-4)^2 + (6-4)^2)/2 = 8/2 = 4$, $s = 2$. $\text{SE} = 2/\sqrt{3} = 2/1.732 = 1.155$. df = 2, $t[0.025, 2] = 4.303$, margin $4.303 \times 1.155 = 4.97$. 95% CI $= 4 \pm 4.97 = (-0.97, 8.97)$. Batching needs lots of data.

HW4 2b. Bootstrap with $B = 10000$, seed 12345

Question. *2b. Bootstrap. Resample your 16 observations with replacement, recompute the mean, and repeat many times. The percentiles of those resampled means give the interval.* [The given NumPy script draws the 16 from not-normal.txt without replacement using seed 12345, makes $B = 10000$ resamples of size 16 with replacement, and prints the $a = (1 - \text{level})/2$ and $1 - a$ quantiles of the resampled means for levels 0.90, 0.95, 0.99.] *If you are working in Excel or R instead, you may use any equivalent implementation, but report B, your seed, and the tool you used.*

In plain words. The bootstrap pretends your 16 values are the whole population. You draw new samples of 16 from them with replacement, many times, and watch how much the mean jumps around. The middle 95% of those means is your 95% interval, with no normal assumption needed.

1. Treat the sample as the population. Draw 16 with replacement, take the mean, repeat B times, where $B = 10000$ is the number of resamples (CI p.69).
2. Sort the means. 95% runs from the 2.5th to the 97.5th percentile, which is about the 250th and the 9750th of 10000 sorted means. 90% runs from the 5th to the 95th percentile, 99% from the 0.5th to the 99.5th.

Answer. 90% (0.1618, 0.3242), 95% (0.1493, 0.3407), 99% (0.1230, 0.3705). All contain 0.2083.

By hand. Sample $\{2, 4, 9\}$. Resamples $\{2, 2, 9\}$ mean $13/3 = 4.33$, $\{4, 9, 9\}$ mean $22/3 = 7.33$, $\{2, 4, 4\}$ mean $10/3 = 3.33$, $\{9, 2, 4\}$ mean $15/3 = 5$. Repeat thousands of times, sort, read the percentiles.

Usual mistake. Resampling without replacement gives the same mean every time. If the 16 values missed the tail, every resample misses it too (CI p.69 to 70).

HW4 2c. Compare raw, batched and bootstrap at $n = 16$

Question. *2c. Compare.*

1. *Put the three sets of intervals for the sample of 16, raw, batched, and bootstrapped, into a single table, at all three confidence levels. State for each whether it contains the population mean.*
2. *Which method gives the widest intervals, and which the narrowest? Was that what you expected before computing them?*
3. *The batched interval is meant to correct for the data not being normal. Explain why the correction makes the interval wider rather than narrower, and say what you gave up to obtain it.*
4. *Look at your 99% batched interval. Does it contain any values that are impossible for this quantity? What does that tell you about the method at this sample size?*
5. *Your sample either did or did not contain the population mean. Explain why that single outcome does not, on its own, tell you whether a method's coverage is correct.*

In plain words. This lines up the three methods side by side and asks you to explain the pattern. Batching is widest because 3 df make the t value big. Bootstrap is narrowest because it can only see the spread inside the 16 values you already have.

Width = upper end minus lower end. For example raw 95% is $0.3493 - 0.1319 = 0.217$.

	Raw t	Batched	Bootstrap
90% width	0.179	0.321	0.162
95% width	0.217	0.435	0.191
99% width	0.301	0.797	0.248

Steps 1 to 5 answer parts 1 to 5 of 2c.

1. All nine contain 0.2083.
2. Batching is widest and bootstrap narrowest at every level. That's what we expected, since the slides predict it (CI p.68 says about 50% wider, CI p.70 calls bootstrap the narrowest). Here batching was about twice as wide ($0.435/0.217 = 2.0$) because the batch means were spread out.
3. Batching widens because df fall from 15 to 3, so t goes from 2.131 to 3.182, and here the SE also rose from 0.0510 to 0.0683. The cost is precision and any statement about individual values.
4. The 99% batched interval reaches −0.158, impossible for positive data. Too few df at high confidence.

5. Coverage is a long-run rate over repeated samples (CI p.33). One hit is consistent with 95% and with 91%, so you'd have to repeat the draw and count.

Which fix when. Correlated observations, batch. Median or a multiplicative process, transform. Mean with independent data, bootstrap (CI p.70).

The third fix by hand, transform. HW4 skipped it, but it's the other response on CI p.68. Data 1, 10, 100.

1. Take $\log_{10}$ to get 0, 1, 2. Mean $3/3 = 1$. $s^2 = ((0-1)^2 + 0 + (2-1)^2)/2 = 2/2 = 1$, $s = 1$.
2. SE $= 1/\sqrt{3} = 0.577$. With $t[0.025, 2] = 4.303$ the margin is $4.303 \times 0.577 = 2.48$.
3. Log-scale CI $1 \pm 2.48 = (-1.48, 3.48)$.
4. Back-transform with 10^x. $10^{-1.48} = 0.033$ and $10^{3.48} \approx 3050$, centred on $10^1 = 10$.

That's an interval for the median (here the geometric mean), not the mean, which is $111/3 = 37$ (CI p.68 trap).

HW4 Problem 3. Confidence intervals in the wild, 8 questions

Question. *Problem 3, A paper from the wild. Work through the homework exercise in Part III of the companion handout Confidence Intervals in the Wild. Read only the experimental setup and results sections of the paper you choose, and answer all eight questions there. Two or three sentences per question is enough. Give the full citation at the top. Choosing the paper is part of the exercise. Pick the entry from Part I of the handout that sits closest to your own research area, or any peer-reviewed paper from that area, ideally one you would cite anyway, that reports experimental results with confidence intervals, error bars, or any other statement of uncertainty. Papers that report results with no statement of uncertainty are also fair game, and often more revealing.* [Defaults if no area fits are entry 4, Georges et al., Statistically Rigorous Java Performance Evaluation, OOPSLA 2007, or entry 13, Henning et al., When Should I Run My Application Benchmark?, FSE Companion 2025. The eight handout questions are printed in italics at the start of each answer under "His paper" below.]

In plain words. You pick a real paper and audit how it reports uncertainty, using 8 fixed questions. The point is to spot when an error bar promises more than it should, usually because the observations weren't independent or the interval ignores systematic error. Two or three sentences per question is enough.

Checklist.

1. Paper and why.
2. Locate the first interval or error bar. Finding none is a finding.
3. Estimator. Mean, proportion, ratio, quantile, GM? Which lecture formula fits?
4. n. Justified, or just asserted?
5. Assumption. What does the interval assume? If skewed with small n, which of batch, transform, bootstrap (CI p.68)?
6. Independence. What's the unit, and are the units really independent (CI p.66)?
7. What the interval excludes. Systematic error, setup, population (CI p.17).
8. Interpretation. Do they claim "95% probability the true value is inside", which needs a credible interval?

His paper. Yan et al., Puffer, NSDI 2020.

1. *Say what you picked and why. Give the full citation and one or two sentences on how the paper relates to your own work.* Video streaming, already cited in HW2, and no Part I entry fits.
2. *Locate the interval. Quote the sentence or identify the figure where an interval or error bar first appears. If you cannot find one, say so, that is a finding, and the rest of the questions still apply to whatever the paper reports in its place.* Sec. 3.4, "95% CI on stall ratio between ±13% and ±21% of the mean". Fig. 9 error bars are 95% CIs.
3. *Identify the estimator. Mean, proportion, variance, quantile, geometric mean, or something else? Which formula from lecture applies, if any?* Stall ratio is a ratio, so bootstrap. Average SSIM is a duration-weighted mean with a z interval.
4. *Find n. How many measurements? Is the choice justified, or asserted?* 637,189 streams from 54,612 IPs, about 2.5 to 2.8 stream-years per scheme. That's whatever 8 months gave, not planned. The "2 stream-years to separate a 15% difference" is unexplained.
5. *Check the assumption. What distributional assumption does their interval require, and do the authors address it? If the response is plainly not normal and n is small, which of the three responses on slide 68 did they use, and if none, which would you have used?* Stalls are very skewed (4% of streams stalled). Bootstrap fits, but B and the interval type aren't stated.
6. *Check the independence. What is the unit being treated as an independent draw, a run, a request, a benchmark, a user, a seed? Are consecutive units plausibly independent, or is this an entry-15 situation dressed up as an entry-1 situation?* Randomized by session but the bootstrap resamples streams, which share a connection. The intervals are likely too narrow. Resample sessions.
7. *Check what the interval excludes. Name one source of error in their setup that the interval does not cover (pitfall 3).* One Stanford server, self-selected viewers, no Safari. Says nothing about other users or servers.
8. *Check the interpretation. Do the authors state a conclusion that requires a credible interval while computing a confidence interval?* No credible-interval claim. But they compare schemes by error-bar overlap, which isn't a test.

Exam scenario. "Mean RPC latency was 4.2 ms ± 0.1 ms (95% CI) from 10,000 requests in one run on one machine, so there is a 95% chance the true mean is between 4.1 and 4.3." Walk the checklist.

1. The claim is a single mean latency with a 95% CI, plus a probability reading of that CI. There's no paper to pick, so say what's being claimed.
2. The interval is 4.2 ± 0.1 ms, labelled 95%. Working backwards, $0.1 = 1.96\, s/\sqrt{10000}$ gives $s = 0.1 \times 100/1.96 = 5.1$ ms, so the per-request SD is about 5 ms, bigger than the mean. That fits skewed latency.
3. Estimator is a mean, z formula since n is large (CI p.31).
4. $n = 10{,}000$ is asserted, and it counts requests, not runs.
5. Latency is skewed, but for a mean at this n the CLT is fine (CI p.29).
6. The real problem. Consecutive requests share caches, queues and load, so they aren't independent and the interval is too narrow (CI p.66). Batch the requests or do several runs with one mean per run (CI p.70, HT p.22).
7. One machine, one run. Systematic error and the setup aren't in the interval (CI p.17).
8. "95% chance the true mean is inside" is a credible-interval claim. A CI only says the procedure covers μ in 95% of repeats (CI p.33). Honest headline. "In one run on one machine, mean RPC latency was 4.2 ms. The ±0.1 ms treats 10,000 back-to-back requests as independent, so the real uncertainty is larger, and we'd need several independent runs to give an honest inter-

val. Nothing here covers other machines."

HW5. Hypothesis testing

HW5 1.1. $\sigma = 0.8$ h known, $n = 36$, $\bar{x} = 2.74$. Test $H_0 : \mu = 2.5$ at $\alpha = 0.05$

Question. *Problem 1, Testing a mean with σ known. For a certain class of a new neural network architecture, we would like to test the hypothesis that the mean training time μ is equal to $\mu_0 = 2.5$ hours. The standard deviation is known to be 0.8 hours. 1. (7 points) Suppose we run experiments with 36 such systems and observe that the mean training time in our experiments is 2.74 hours. Test the null hypothesis $H_0 : \mu = 2.5$ hours at the level of significance $\alpha = 0.05$. Do you reject H_0 or fail to reject it?*

In plain words. A hypothesis test asks whether the data would be surprising if the boring claim H_0 (the mean is 2.5 h) were true. You measure how far the sample mean is from 2.5 in units of standard error, and that's Z. If $|Z|$ goes past 1.96, that would happen less than 5% of the time under H_0, so you reject it.

1. $H_0 : \mu = 2.5$, $H_1 : \mu \neq 2.5$. No direction is named, so two-sided (HT p.5).
2. *Formula.* σ is known, so $Z = \dfrac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$ (HT p.8), where $\bar{x} = 2.74$ is the sample mean, $\mu_0 = 2.5$ the value in H_0, $\sigma = 0.8$ the known SD and $n = 36$ the number of systems.
3. *Plug in.* $Z = \dfrac{2.74 - 2.5}{0.8/\sqrt{36}}$.
4. *Work it out.* SE $= 0.8/6 = 0.1333$. Top $= 0.24$. $Z = 0.24/0.1333$.
5. *Result.* $Z = 1.80$.
6. Decision rule. Two-sided at 0.05, reject if $|Z| > z[0.025] = 1.960$. Since $1.80 < 1.96$, fail to reject.
7. Interval check (HT p.9). $2.74 \pm 1.96(0.1333) = 2.74 \pm 0.261 = (2.479, 3.001)$ contains 2.5, same verdict.

Answer. $Z = 1.80$, fail to reject H_0. Matches main.tex.

Small version. $n = 16$, $\sigma = 0.8$, $\bar{x} = 2.9$. SE $= 0.8/\sqrt{16} = 0.8/4 = 0.2$. $Z = (2.9 - 2.5)/0.2 = 0.4/0.2 = 2.0 > 1.96$, reject. The HT p.10 example gives $Z = -2.83$ and rejects.

Usual mistake. Writing "accept H_0". Failing to reject isn't evidence for H_0 (HT p.4).

HW5 1.2. p-value and conclusion

Question. *2. (5 points) For the same experiments, find the p-value. Using the p-value you computed, restate the conclusion you reached in part 1.*

In plain words. The p-value is the chance of getting a result at least this far from 2.5 if H_0 were true. A small p means the data would be surprising under H_0. You reject when $p < \alpha$, and that always gives the same verdict as the critical-value rule.

1. *Formula.* $p = P[\text{statistic this extreme} \mid H_0]$ (HT p.13). Two-sided, $p = 2\,(1 - \Phi(|Z|))$.
2. *Plug in.* $p = 2\,(1 - \Phi(1.80))$.
3. *Work it out.* $1 - 0.9641 = 0.0359$, then 2×0.0359.
4. *Result.* $p = 0.0718$.

Answer. $p = 0.072 > 0.05$, so fail to reject H_0. Same conclusion as part 1, the data don't show the mean differs from 2.5 h.

Usual mistake. Forgetting to double for two-sided, or saying "7% chance H_0 is true" (HT p.14).

HW5 1.3. $\alpha = 0.01$, $n = 36$. Smallest $\bar{x} > 2.5$ that rejects

Question. *3. (8 points) Suppose we reduce the level of significance to $\alpha = 0.01$. If we run experiments on 36 systems to test the null hypothesis that $\mu = 2.5$ hours, what is the minimum mean time (larger than 2.5 hours) we would need to observe to reject the hypothesis?*

In plain words. Flip the test around. Instead of asking whether a given mean rejects, find the cutoff mean where Z just hits the critical value. A smaller α means you demand stronger evidence, so the cutoff moves further from 2.5.

1. Two-sided at 0.01, so the critical value is $z[0.005] = 2.576$.
2. *Formula.* Reject when $\dfrac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} > z[\alpha/2]$, so $\bar{x} > \mu_0 + z[\alpha/2]\,\sigma/\sqrt{n}$.
3. *Plug in.* $\bar{x} > 2.5 + 2.576 \times 0.8/6$.
4. *Work it out.* $0.8/6 = 0.1333$, $2.576 \times 0.1333 = 0.343$, $2.5 + 0.343$.
5. *Result.* $\bar{x} > 2.843$.
6. A one-sided $H_1 : \mu > 2.5$ would use $z[0.01] = 2.326$, so $2.5 + 2.326 \times 0.1333 = 2.5 + 0.310 = 2.810$.

Answer. About 2.84 h two-sided (2.81 one-sided). Matches main.tex.

Power version (HT p.19). How many runs to detect a shift of $\delta = 0.2$ h with 80% power at 0.05? *Formula.* $n = \left(\dfrac{(z[\alpha/2] + z[\beta])\,\sigma}{\delta}\right)^2$, where δ is the smallest shift you care about, power is $1 - \beta = 0.80$, and $z[0.20] = 0.84$. *Plug in.* $\left(\dfrac{(1.96 + 0.84) \times 0.8}{0.2}\right)^2$. *Work it out.* $2.80 \times 0.8 = 2.24$, $2.24/0.2 = 11.2$, $11.2^2 = 125.4$. *Result.* Round up to 126.

HW5 2.1. Suspect the mean time to first fault is below 740 h. H_0 and H_1?

Question. *Problem 2, Choosing the null hypothesis. A research group develops an autonomous driving system. Because of the specific sensor model adopted, they suspect that the mean time μ to experience the first fault in the sensor subsystem is less than $\mu_0 = 740$ hours. 1. If we use hypothesis testing to confirm this assertion, what should be the null hypothesis H_0? What should be the alternative hypothesis H_1? Carefully justify your choices for H_0 and H_1.*

In plain words. A test can only ever reject H_0, never prove it. So whatever you want to show goes in H_1, here that the mean is below 740. The test is one-sided because only one direction matters.

1. The claim to establish goes in H_1, because a test only supports a claim by rejecting H_0 (HT p.4).
2. $H_0 : \mu = 740$ (or ≥ 740), $H_1 : \mu < 740$, one-sided (HT p.5).

Answer. $H_0 : \mu = 740$ h, $H_1 : \mu < 740$ h, one-sided, because rejecting H_0 is the only way the data can confirm the suspicion.

Usual mistake. Putting the claim in H_0, so a small noisy experiment "supports" it by failing to reject. Picking the side after seeing the data is a disguised two-sided test at double α (HT p.5).

HW5 2.2. $n = 50$, $\bar{x} = 706$, $s = 96.1$, $\alpha = 0.01$. Can we confirm $\mu < 740$?

Question. *2. Suppose we test 50 such systems and obtain a mean time to first fault of 706 hours, with a standard deviation of 96.1 hours. If the target level of significance is $\alpha = 0.01$, can we confirm, based on these experiments, the assertion that the mean time to first fault is less than 740 hours?*

In plain words. Now actually run the test from 2.1. σ isn't known, but with 50 systems the sample SD s is a good stand-in, so the Z formula works. One-sided at 0.01 means you reject only if Z is below -2.326.

1. σ is unknown but $n = 50$, so s stands in and $t \approx Z$ (HT p.11).
2. *Formula.* $Z = \dfrac{\bar{x} - \mu_0}{s/\sqrt{n}}$, where $\bar{x} = 706$, $\mu_0 = 740$, $s = 96.1$ the sample SD, $n = 50$.
3. *Plug in.* $Z = \dfrac{706 - 740}{96.1/\sqrt{50}}$.
4. *Work it out.* SE $= 96.1/7.071 = 13.59$. Top $= -34$. $Z = -34/13.59$.
5. *Result.* $Z = -2.50$.
6. Decision. One-sided, reject if $Z < -z[0.01] = -2.326$. Since $-2.50 < -2.326$, reject.
7. p-value. *Formula.* one-sided lower tail $p = \Phi(Z)$. *Plug in.* $\Phi(-2.50) = 1 - \Phi(2.50)$. *Work it out.* $1 - 0.9938$. *Result.* $p = 0.0062 < 0.01$.

Answer. Reject H_0, so the data confirm $\mu < 740$ at 0.01. With t_{49} (crit 2.405) the verdict is the same.

Small version. $n = 36$, $\bar{x} = 716$, $s = 60$. SE $= 60/6 = 10$. $Z = (716 - 740)/10 = -24/10 = -2.4 < -2.326$, reject. $p = 1 - \Phi(2.4) = 1 - 0.9918 = 0.0082$.

Exam variant, small n and the side matters. $n = 9$, $\bar{x} = 12$, $s = 3$, $\mu_0 = 10$, population roughly normal. SE $= 3/\sqrt{9} = 3/3 = 1$. $t = (12 - 10)/1 = 2.0$ with $9 - 1 = 8$ df. Two-sided at 0.05 needs $t[0.025, 8] = 2.306$, so fail to reject. One-sided $H_1 \colon \mu > 10$ (stated before the data) needs $t[0.05, 8] = 1.860$, so reject. That's why the side has to be fixed in advance (HT p.5, HT p.11).

HW5 Problem 3. Hypothesis testing in the wild, questions 0 to 6

Question. *Problem 3, A paper from the wild (60 points). Work through the homework exercise in Part III of the companion handout Hypothesis Testing in the Wild. Read only the experimental setup and results sections of the paper you choose, and answer all seven questions there. Two or three sentences per question is enough, except for the last question, which asks you to rewrite the paper's key result as an effect size with a confidence interval. Give the full citation at the top. Choosing the paper is part of the exercise. Pick the entry from Part I of the handout that sits closest to your own research area, or any peer-reviewed paper from that area, ideally one you would cite anyway, that makes a claim of the form "A is better than B". Papers that make that claim with no test and no statement of uncertainty are also fair game, and often more revealing. Several entries (1, 3, 4, 7, 9 and 10) are about statistical practice rather than about one system beating another. If you pick one of them, apply the questions to the paper's own central empirical claim.* [Defaults if no area fits are Mytkowicz et al., Producing Wrong Data Without Doing Anything Obviously Wrong!, ASPLOS 2009, or Musgrave et al., A Metric Learning Reality Check, ECCV 2020. The seven handout questions, 0 to 6, are printed in italics at the start of each answer under "His paper" below.]

In plain words. Pick a paper that claims A beats B and audit its testing with 7 fixed questions. Mostly you're checking the things papers skip, like what counts as one independent observation, how many tests they ran, and whether they peeked. The last question rewrites the main result as an effect size, the actual size of the difference, with a CI.

Checklist.

0. Paper and why.
1. H_0, H_1, one or two-sided. Reconstruct if missing and say so.
2. Test and assumptions (normality, independence, pairing, equal variances). The unit, and does n count units or measurements (HT p.20)?
3. α, n, power. With no power analysis, the detectable effect is about 2.8×SE at $\alpha = 0.05$ and 80% power (HT p.19).
4. Count the tests. FWER $= 1 - (1 - \alpha)^m$ (HT p.27). Any correction, and FWER or FDR (HT p.28)?
5. Stopping rule (peeking, HT p.23 to 24), interference (HT p.22), right estimand.
6. Rewrite as an effect size with a CI (HT p.16, p.18). Big enough to matter?

His paper. Spang et al., Unbiased Experiments in Congested Networks, IMC 2021 (entry 8).

0. *Say what you picked and why. Give the full citation and one or two sentences on how the paper relates to your own work.* Closest to video streaming, and his comparisons share a bottleneck too.
1. *Write down H_0 and H_1. For the paper's central claim, state the null and alternative hypotheses, and whether the test is one- or two-sided. If the paper never states them, reconstruct them, and say so.* Not stated. Reconstructed $H_0 \colon \text{TTE} = 0$ against $H_1 \colon \text{TTE} \neq 0$, two-sided since the CIs are two-sided.
2. *Identify the test and its assumptions. Which test was used, if any? What does it assume (normality, independence, pairing, equal variances)? Do the authors check or address those assumptions? What is the experimental unit, and does n count units or measurements (slide 20)?* No named test. Regression on hourly means with Newey-West SEs. The unit is the session but n counts hour cells, since sessions in one hour share congestion. An A/A test and a baseline week check that the links match.
3. *Find α, n, and β. What significance level? How many observations, and what is an observation (a run, a seed, a benchmark, a user)? Is there any power analysis? If not, estimate roughly what effect size the experiment could have detected.* $\alpha = 0.05$ implied. About 14M sessions but only a few dozen hour cells. No power analysis. SE about 2.6, so the detectable effect is about $2.8 \times 2.6 \approx 7\%$.
4. *Count the tests. How many comparisons does the paper make in total? What is the family-wise error rate if none were corrected (slide 27)? Is there a correction, and does the choice between family-wise error and false discovery rate control fit the paper's claims (slide 28)?* 28 intervals in Fig. 5 (7 metrics × 4), plus 24 baseline metrics. FWER $= 1 - 0.95^{28} = 1 - 0.24 = 0.76$. No correction, and FDR fits the claim better.
5. *Check the stopping rule and the estimand. Was n fixed in advance, or could the authors have kept running until the result was significant (slides 23 to 25)? Could units in the experiment have interfered with each other (slide 22)? Does the test measure the quantity the paper's claim is about (pitfall 7)?* 5 days, not said to be fixed in advance. Interference is the paper's point. The naive A/B test measures the effect at 5% allocation, not the effect of deploying to everyone.
6. *Rewrite the result. Rewrite the paper's key sentence as an effect size with a confidence interval, as slides 16 and 18 recommend. Is the effect large enough to matter? Does the claim become stronger, weaker, or different?* Capping raised throughput

12% (95% CI about 7% to 17%), while the naive test said −5% (about −8% to −2%). The claim gets stronger since the sign flips, and the exact size is less certain.

By hand, FWER. *Formula.* FWER $= 1-(1-\alpha)^m$, the chance of at least one false positive when you run m tests at level α and every H_0 is true (HT p.27). *Plug in.* $m=10$, $\alpha=0.05$, so $1-0.95^{10}$. *Work it out.* $0.95^{10}=0.599$, $1-0.599$. *Result.* 0.40. Bonferroni (HT p.28) tests each at $\alpha/m=0.05/10=0.005$.

Where 2.8 comes from. $z[0.025]+z[0.20]=1.96+0.84=2.80$, the same sum as in the HW5 1.3 power version.

Exam scenario. "A new congestion controller ran on 10% of flows sharing links with the old one. We checked p daily and stopped on day 6 at $p=0.04$. Of 8 metrics, throughput was significantly better."

0. The claim is that the new controller gives better throughput, from a live test at 10% allocation. There's no paper to pick, so say what's being claimed.
1. No H_0 stated. Reconstruct H_0 no throughput difference, H_1 better, and ask if the side was fixed in advance (HT p.5).
2. Unit is the flow, but flows on one link aren't independent (HT p.20).
3. No power analysis, n wasn't fixed (HT p.19).
4. 8 metrics give FWER $=1-0.95^8=1-0.66=0.34$ with no correction (HT p.27 to 28). Bonferroni would need $p<0.05/8=0.00625$, and 0.04 fails that.
5. Daily peeking inflates the Type I error well past 5% (HT p.23 to 24). At 10% allocation the new flows interfere with the old, so this isn't the effect of deploying to everyone (HT p.22).
6. Report the throughput difference with a CI, not "significantly better" (HT p.16). Honest headline. "At 10% allocation, flows on the new controller had higher throughput by [difference] (95% CI [low, high]). This was 1 of 8 metrics, checked daily with no correction for peeking or multiple tests, so it's a lead to retest with a fixed sample size, not a finding."

HW6. Comparing two alternatives

HW6 P1. Same workload on routing A and B, 9 rows of delay. Is A better at 95%?

Question. *Problem 1, Two routing algorithms (20 points). Two packet routing algorithms are being evaluated. Experiments were conducted by injecting the same workload in the network and measuring the average end-to-end packet delay. The results of the various experiments are shown in Table 1. Can you say with 95% confidence that routing algorithm A is better than algorithm B?* [Table 1 is the A and B columns below, 9 experiments, rows 1 to 5 on the left half and 6 to 9 on the right.]

In plain words. The same workload ran on both, so each row is a matched pair. Take the difference in each row and build an ordinary one-sample CI on those 9 differences. If the CI includes 0, you can't say either one is better.

A	B	d	d^2	A	B	d	d^2
115.0	125.3	−10.3	106.09	120.3	87.9	32.4	1049.76
89.3	97.6	−8.3	68.89	98.9	104.2	−5.3	28.09
102.1	85.3	16.8	282.24	87.6	96.3	−8.7	75.69
94.3	101.3	−7.0	49.00	104.5	113.2	−8.7	75.69
117.3	112.7	4.6	21.16				

1. The same workload ran under both, so paired. Work on $d_i=A_i-B_i$, the delay of A minus the delay of B in row i. Negative favours A, since lower delay is better (Comp p.5, p.9).
2. Mean difference. *Formula.* $\bar{d}=\frac{1}{n}\sum d_i$, where $n=9$ pairs. *Plug in.* Positives $16.8+4.6+32.4=53.8$, negatives $-10.3-8.3-7.0-5.3-8.7-8.7=-48.3$. *Work it out.* $\sum d=53.8-48.3=5.5$, $\bar{d}=5.5/9$. *Result.* 0.611.
3. SD of the differences. *Formula.* $s_d^2=\dfrac{\sum d_i^2-(\sum d_i)^2/n}{n-1}$. *Plug in.* $\sum d^2=1756.61$ from the table, so $\dfrac{1756.61-5.5^2/9}{8}$. *Work it out.* $5.5^2/9=30.25/9=3.36$, $1756.61-3.36=1753.25$, $1753.25/8=219.16$. *Result.* $s_d=\sqrt{219.16}=14.80$.
4. SE. *Formula.* SE $=s_d/\sqrt{n}$. *Plug in.* $14.80/\sqrt{9}$. *Work it out.* $14.80/3$. *Result.* 4.935.
5. CI. *Formula.* $\bar{d}\pm t[\alpha/2,n-1]\times$SE, with df $=9-1=8$ pairs minus one. *Plug in.* $0.611\pm2.306\times4.935$. *Work it out.* $2.306\times4.935=11.38$. $0.611-11.38=-10.77$ and $0.611+11.38=11.99$. *Result.* CI $=(-10.77, 11.99)$, which contains 0.

Answer. No, you can't say A is better at 95%. Matches main.tex. It isn't evidence they're equal either, since the CI allows A 10.8 better or 12.0 worse.

Check. "Is A better than B?" is a one-sided question by HT p.5, and he used a two-sided interval. The verdict doesn't change. A is better only if $\bar{d}<0$, and $\bar{d}=+0.611$ points the other way, so the one-sided test with $t[0.05,8]=1.860$ can't reject either.

By hand. Four differences 2, 4, 3, 3. $\bar{d}=12/4=3$. Deviations $-1,1,0,0$, so $s_d^2=2/3=0.667$, $s_d=0.816$. SE $=0.816/\sqrt{4}=0.408$. df $=3$, margin $3.182\times0.408=1.30$. CI $3\pm1.30=(1.70,4.30)$ excludes 0.

Sign test on the HW6 data (Comp p.23). *Formula.* Under H_0 each difference is equally likely to be positive or negative, so the count of positives S follows a binomial with $n=9$ and chance 1/2. $P[S\le s]=\sum_{k=0}^{s}\binom{9}{k}/2^9$. *Plug in.* 3 positive out of 9, so $P[S\le3]=\dfrac{\binom{9}{0}+\binom{9}{1}+\binom{9}{2}+\binom{9}{3}}{512}$. *Work it out.* $\dfrac{1+9+36+84}{512}=\dfrac{130}{512}=0.254$. Two-sided doubles it. *Result.* $p=0.51$.

Usual mistake. Analysing paired data as unpaired. Comp p.8 shows the same data giving $(0.65, 3.00)$ paired and $(-22.0, 25.7)$ unpaired. The df come from pairs, $n-1=8$ (Comp p.9).

HW6 P2. CPU times, 30 on A and 24 on B. Is A faster at 90%?

Question. *Problem 2, Two computer systems (20 points). Two computer systems are being evaluated for their performance. Data corresponding to the CPU time for running some transactions on System A and System B are reported in Table 2. Is System A faster than System B at the 90% confidence level?* [Table 2, CPU time in seconds. System A (30 values) 1.3779, 0.0457, 1.4415, 1.3589, 0.6898, 0.7076, 3.9083, 2.2081, 0.9244, 6.5770, 4.4775, 4.6199, 0.1775, 0.3354, 0.3687, 0.0956, 1.2116, 3.8814, 0.2922, 3.0970, 3.2150, 1.2880, 3.0046, 6.8230, 0.7478, 0.3845, 0.2354, 0.9951, 0.3045, 1.6950. System B (24 values) 0.9584, 0.5726, 1.3915, 0.8075, 0.1344, 0.1277, 1.8032, 1.2048, 9.5949, 4.9107, 0.2834, 0.3841, 0.0812, 2.1362, 4.6396, 1.0127, 2.9434, 1.0844, 0.7860, 2.1536, 1.4948, 11.7346, 0.9436, 3.1111.]

In plain words. Different transactions ran on each system, in different counts, so there's nothing to pair. You compare the two means with the Welch method, which doesn't assume both systems have the same variance. It builds a CI on the difference in means and has its own formula for the degrees of freedom, called ν.

1. Different transactions and unequal sizes, so unpaired. Welch by default (Comp p.12 to 13).
2. From the data, A has $n_A=30$, $\bar{x}_A=1.8830$, $s_A=1.9017$. B has $n_B=24$, $\bar{x}_B=2.2623$, $s_B=2.9108$. Difference $\bar{x}_A-\bar{x}_B=1.8830-2.2623=-0.3793$.

3. SE of the difference. *Formula.* $s = \sqrt{s_A^2/n_A + s_B^2/n_B}$, where s_A^2 is A's sample variance and n_A its count, same for B. *Plug in.* $\sqrt{1.9017^2/30 + 2.9108^2/24}$. *Work it out.* $1.9017^2 = 3.6166$, $/30 = 0.12055$. $2.9108^2 = 8.4727$, $/24 = 0.35303$. Sum 0.47358. *Result.* $s = \sqrt{0.47358} = 0.6882$.
4. Welch df. *Formula.* $\nu = \dfrac{(s_A^2/n_A + s_B^2/n_B)^2}{\dfrac{(s_A^2/n_A)^2}{n_A - 1} + \dfrac{(s_B^2/n_B)^2}{n_B - 1}}$. *Plug in.* $\nu = \dfrac{0.47358^2}{0.12055^2/29 + 0.35303^2/23}$. *Work it out.* Top $0.47358^2 = 0.22428$. Bottom $0.014532/29 = 0.000501$ plus $0.124630/23 = 0.005419$, total 0.005920. $0.22428/0.005920$. *Result.* $\nu = 37.89$.
5. CI. *Formula.* $(\bar{x}_A - \bar{x}_B) \pm t[\alpha/2, \nu] \times s$, with 90% so $\alpha/2 = 0.05$. *Plug in.* $-0.3793 \pm 1.686 \times 0.6882$. *Work it out.* $1.686 \times 0.6882 = 1.160$. $-0.3793 - 1.160 = -1.540$ and $-0.3793 + 1.160 = 0.781$. *Result.* CI = $(-1.540, 0.781)$.
6. Test statistic for reference. $t = (\bar{x}_A - \bar{x}_B)/s = -0.3793/0.6882 = -0.55$, $p = 0.58$.

Answer. 0 is inside, so A isn't shown to be faster at 90%. Matches main.tex. Rounding ν down to 37 by hand gives 1.687 and $(-1.540, 0.782)$.

Check. The same one-sided point applies. "Is A faster" is one-sided. At 90% one-sided, $t[0.10, 37.9] = 1.304$ and the upper bound is $-0.379 + 1.304(0.688) = -0.379 + 0.897 = 0.518 > 0$. Still not shown, so the verdict holds, but say the test is one-sided.

By hand. A = 2, 4, 6, 8 and B = 6, 8, 10, 12, 14.

1. Means $20/4 = 5$ and $50/5 = 10$. Variances $s_A^2 = (9 + 1 + 1 + 9)/3 = 6.67$ and $s_B^2 = (16 + 4 + 0 + 4 + 16)/4 = 10$.
2. Pieces $6.67/4 = 1.667$ and $10/5 = 2$. $s = \sqrt{1.667 + 2} = \sqrt{3.667} = 1.915$.
3. $\nu = \dfrac{3.667^2}{1.667^2/3 + 2^2/4} = \dfrac{13.44}{0.926 + 1} = \dfrac{13.44}{1.926} = 6.98$, round down to 6.
4. 90% CI. $t[0.05, 6] = 1.943$, margin $1.943 \times 1.915 = 3.72$. $(5 - 10) \pm 3.72 = (-8.72, -1.28)$, which excludes 0, so A is smaller.

Usual mistake. Using $n - 1$ of one sample as the df, or reading overlap of two separate CIs as "no difference" (Comp p.15 to 16).

Slide twin, paired (Comp p.11). Cache hit ratio on 6 workloads under policies A and B

In plain words. The same paired recipe as HW6 P1, on a tiny example from the slides. The lesson is to check which t value the slide used, since that decides whether it's a 90% or a 95% interval.

1. Differences $d_i = A_i - B_i$ are +0.07, +0.09, −0.05, −0.06, +0.10, −0.03. *Formula.* $\bar{d} = \frac{1}{n}\sum d_i$. *Plug in.* $\sum d = 0.26 - 0.14 = 0.12$, $n = 6$. *Result.* $\bar{d} = 0.12/6 = 0.020$.
2. $\sum d^2 = 0.0049 + 0.0081 + 0.0025 + 0.0036 + 0.0100 + 0.0009 = 0.0300$.
3. *Formula.* $s_d^2 = \dfrac{\sum d^2 - (\sum d)^2/n}{n - 1}$. *Plug in.* $\dfrac{0.0300 - 0.12^2/6}{5}$. *Work it out.* $0.0144/6 = 0.0024$, $(0.0300 - 0.0024)/5 = 0.0276/5 = 0.00552$. *Result.* $s_d = 0.0743$.
4. SE $= 0.0743/\sqrt{6} = 0.0743/2.449 = 0.0303$.
5. The slide multiplies by $t[0.05, 5] = 2.015$, so margin $2.015 \times 0.0303 = 0.061$ and its interval $0.020 \pm 0.061 = (-0.041, +0.081)$ is a 90% interval. For 95% use $t[0.025, 5] = 2.571$, giving $0.020 \pm 0.078 = (-0.058, 0.098)$.

Answer. Contains 0, so no difference shown, and the width is the finding (Comp p.11).

Slide twin, unpaired Welch (Comp p.14). A has $n = 7$, mean 0.2657, SD 0.0934. B has $n = 9$, mean 0.4011, SD 0.1447

In plain words. The same Welch recipe as HW6 P2, with small numbers from the slides. This time the CI misses 0, so this one does show a difference.

1. *Formula.* $s = \sqrt{s_A^2/n_A + s_B^2/n_B}$. *Plug in.* $\sqrt{0.0934^2/7 + 0.1447^2/9}$. *Work it out.* $0.001246 + 0.002326 = 0.003572$. *Result.* $s = 0.0598$.
2. *Formula.* Welch ν as in HW6 P2. *Plug in.* $\nu = \dfrac{0.003572^2}{0.001246^2/6 + 0.002326^2/8}$. *Work it out.* Top 1.276×10^{-5}. Bottom $2.59 \times 10^{-7} + 6.76 \times 10^{-7} = 9.35 \times 10^{-7}$. *Result.* $\nu = 13.6$.
3. Difference $0.2657 - 0.4011 = -0.135$. $t[0.05, 13.6] = 1.765$, margin $1.765 \times 0.0598 = 0.106$.
4. 90% CI $= -0.135 \pm 0.106 = (-0.241, -0.029)$.

Answer. The CI excludes 0, so B has the higher hit ratio, by 3 to 24 points.

HW6 Problem 3. Comparing two systems in the wild, 8 questions

Question. *Problem 3, A paper from the wild (60 points). Work through the homework exercise in Part III of the companion handout Comparing Two Systems in the Wild. Read only the experimental setup and results sections of the paper you choose, and answer all eight questions there. Two or three sentences per question is enough, except for the last question, which asks you to rewrite the paper's headline comparison as an effect with a confidence interval and to say what the experiment could not have detected. Give the full citation at the top. Choosing the paper is part of the exercise. Pick the entry from Part I of the handout that sits closest to your own research area, or any peer-reviewed paper from that area, ideally one you would cite anyway, that compares exactly two systems, models, algorithms or configurations. "Ours" versus a baseline is ideal. Papers that make that comparison with no test and no statement of uncertainty are also fair game, and often more revealing. Several entries (4, 5, 9, 10 and 11) are guides to, or surveys of, statistical practice rather than a comparison of two systems. If you pick one of them, apply the questions to the paper's own central empirical claim.* [Defaults if no area fits are Curtsinger and Berger, STABILIZER, ASPLOS 2013 (unpaired t), Dietterich, Approximate Statistical Tests for Comparing Supervised Classification Learning Algorithms, 1998 (paired t), or Klees et al., Evaluating Fuzz Testing, CCS 2018 (Mann-Whitney U). The eight handout questions are printed in italics at the start of each answer under "His paper" below.]

In plain words. Pick a paper comparing two systems and audit it with 8 fixed questions. The big ones are whether the data should've been paired, whether error-bar overlap was used as a test, and whether the difference is big enough to matter. The last question rewrites the headline as a difference with a CI, plus the smallest difference the experiment could have caught.

Checklist.

1. Paper and why.
2. Design. Paired or unpaired, the unit, units per system (Comp p.5).
3. If unpaired, could it be paired? How much variance is between units (Comp p.6)?
4. Which test, and are df and an exact p reported? If no test, what instead?
5. Is the response normal, or a bounded score, ratio, quantile? Would a rank test or bootstrap be better (Comp p.17 to 18)?

6. Are error bars defined in the caption? Any conclusion from overlap (Comp p.15 to 16)?
7. Effect size in real units with an interval. Big enough, and who decides (Comp p.24)?
8. Rewrite the headline with an interval plus what it couldn't detect (Comp p.25).

His paper. Miller, Adding Error Bars to Evals, arXiv 2411.00640, 2024 (entry 3).

1. *Say what you picked and why. Full citation and a sentence on its relation to your work.* He compares VLMs on the same benchmark questions.
2. *Identify the design. Paired or unpaired? What is the unit, a benchmark, a test-set example, a run, a user? How many units per system?* Paired, with the question as the unit (the cluster for MGSM). Example models on 5000 MATH, 164 HumanEval, 2500 MGSM.
3. *If it is unpaired, could it have been paired? If yes, estimate what that would have bought. How much of the observed variance is between units rather than between systems?* Already paired. In Miller's example a correlation of 0.5 cuts the variance from 1/6 to 1/9, so the SE shrinks by $\sqrt{(1/9)/(1/6)} = \sqrt{2/3} = 0.82$, about 18%.
4. *Find the comparison. Which test, if any? Is the statistic reported with its degrees of freedom and an exact p? If there is no test, what is offered instead?* Paired z on per-question differences, ±1.96 SE. No df, no p. Significant if the CI excludes 0.
5. *Check the assumptions. Is the response plausibly normal, or is it a bounded score, a ratio, or a quantile? Would a rank-based test or a bootstrap have been more defensible (slides 17 to 18)?* Scores are 0 or 1, not normal, so it leans on the CLT. Fine for 5000, but a paired bootstrap or sign test is safer for 164.
6. *Look at the figure. If there are error bars, does the caption say what they are? Is any conclusion drawn from their overlap (slides 15 to 16)?* No figures, only SEs in tables. Single-model CIs on MATH overlap though the paired CI excludes 0.
7. *Find the effect size. What is the difference, in the units of the response, and does the paper give an interval for it? Is the difference large enough to matter, and who decides?* MATH difference +2.5 points, 95% CI (1.2, 3.8). Whether that matters is left to the user.
8. *Rewrite the claim. Restate the paper's headline comparison as an effect with an interval, and add the sentence about what the experiment could not have detected.* Galleon beat Dreadnought by 2.5 points on MATH (1.2 to 3.8). HumanEval −3.1 (−7.2, 1.0) and MGSM −2.7 (−6.1, 0.7) are consistent with no difference. HumanEval's SE is 2.1, so it couldn't reliably detect under about $2.8 \times 2.1 \approx 6$ points.

Exam scenario. "Compiler X and Y were each run on the same 8 benchmarks. An unpaired t test gave $p = 0.4$, and the error bars overlap, so the compilers perform the same."

1. The claim is that two compilers perform the same, based on a non-significant unpaired test and overlapping error bars over 8 benchmarks. There's no paper to pick, so say what's being claimed.
2. Same benchmarks under both, so it's paired with the benchmark as the unit (Comp p.5).
3. Unpaired throws away the correlation. Between-benchmark spread swamps the difference, as in Comp p.8 where the same data give $p = 0.87$ unpaired and $p = 0.008$ paired.
4. They ran an unpaired t and gave only p. Use a paired t on the 8 differences with $8 - 1 = 7$ df and report t, df and the exact p (Comp p.26).
5. Run times are skewed, so analyse log times or use the signed-rank test (Comp p.18, Comp p.21).
6. The bars aren't defined, and overlap isn't a test (Comp p.15).
7. Give the mean difference with a CI. "Perform the same" doesn't follow from a non-significant result.
8. Honest headline. "On the same 8 benchmarks, X's run time minus Y's averaged $\bar{d}$, with paired 95% CI $\bar{d} \pm 2.365\, s_d/\sqrt{8}$, using $t[0.025, 7] = 2.365$. With only 8 benchmarks we couldn't reliably detect a difference smaller than about $3.3\, s_d/\sqrt{8}$." The 3.3 is $t[0.025, 7] + t[0.20, 7] = 2.365 + 0.896$, the small-sample version of the 2.8×SE rule (HT p.19).

One checklist for any experiment on the exam

1. Claim and hypotheses, with the direction chosen before the data (HT p.4 to 5).
2. Unit and n. Count runs, not requests from one run (HT p.20, CI p.66).
3. Paired or unpaired, and could it be paired (Comp p.5)?
4. Test fits the response, whether z, t, Welch, rank or bootstrap (Comp p.17 to 18, CI p.67).
5. Power or detectable effect (HT p.19).
6. Number of tests, and peeking (HT p.23 to 28).
7. Effect size with an interval in real units (HT p.16, Comp p.24).
8. What the interval misses, such as systematic error and interference (CI p.17, HT p.22).
9. Reading. No "equivalent" from non-significance, no verdicts from overlap, no "95% probability" for a CI (HT p.14, Comp p.15, CI p.33).

Practice. "Our scheduler cuts mean latency from 12.1 to 11.4 ms, $p < 0.05$, from one run of each with 10,000 requests, with the best of three workloads shown." The unit is the run, so $n = 1$ per system and the p means nothing. Three workloads are a hidden family of tests. There's no interval on the 0.7 ms. Fix it with several independent runs per system on all three workloads, paired by workload, reporting the difference and its CI.

Homework-style mock problems, no calculator

Mock 1 (HW3 style). Three runs of equal work report 40, 60 and 120 req/s. Mean rate?

In plain words. Three rates from equal amounts of work, so the honest mean rate is total work over total time, which is the harmonic mean. Using fractions with a common denominator keeps it calculator-free.

1. Rates with equal work, so harmonic (Summ p.28). *Formula.* $x_H = \dfrac{n}{\sum 1/x_i}$, with $n = 3$ runs.
2. *Plug in.* $x_H = \dfrac{3}{1/40 + 1/60 + 1/120}$.
3. *Work it out.* Over 120, $3/120 + 2/120 + 1/120 = 6/120 = 0.05$. Then $3/0.05$.
4. *Result.* $x_H = 60$ req/s. The AM would say $(40 + 60 + 120)/3 = 220/3 = 73.3$.

Answer. 60 req/s, harmonic mean.

Mock 2 (HW4 style). $n = 25$, $\bar{x} = 50$, $s = 10$, σ unknown. 90% CI?

In plain words. Small sample and unknown σ means a t interval with $n - 1 = 24$ degrees of freedom. It's the mean plus or minus t times the standard error $s/\sqrt{n}$.

1. $n < 30$ and σ unknown, so t with 24 df, assuming roughly normal data (CI p.37). *Formula.* $\bar{x} \pm t[0.05, 24]\, s/\sqrt{n}$.
2. *Plug in.* SE $= 10/\sqrt{25} = 10/5 = 2$. $t[0.05, 24] = 1.711$.

3. *Work it out.* Margin $1.711 \times 2 = 3.42$. $50 - 3.42 = 46.58$ and $50 + 3.42 = 53.42$.
4. *Result.* CI = (46.58, 53.42).

Answer. 90% CI $(46.58, 53.42)$.

Mock 3 (HW5 style). $\sigma = 4$ known, $n = 64$, $\bar{x} = 51$, $H_0\colon \mu = 50$, two-sided, $\alpha = 0.05$

In plain words. σ is known, so it's a Z test. Count how many standard errors 51 is away from 50 and compare that with 1.96.

1. *Formula.* $Z = \dfrac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$. *Plug in.* $\dfrac{51 - 50}{4/\sqrt{64}}$. *Work it out.* SE $= 4/8 = 0.5$, $Z = 1/0.5$. *Result.* $Z = 2.0$.
2. $|Z| = 2.0 > 1.96$, reject. *Formula.* $p = 2(1 - \Phi(|Z|))$. *Plug in.* $2(1 - \Phi(2.0)) = 2(1 - 0.9772)$. *Work it out.* 2×0.0228. *Result.* $p = 0.0456$.
3. At $\alpha = 0.01$ (2.576) you'd fail to reject. Report p and the CI $51 \pm 1.96 \times 0.5 = 51 \pm 0.98 = (50.02, 51.98)$, not just the verdict (HT p.16).

Answer. $Z = 2.0$, $p = 0.046$, reject H_0 at 0.05.

Mock 4 (HW6 style). Paired differences 1, 3, 2, 4, 0 on 5 benchmarks. 95% CI on the mean difference

In plain words. The differences are already paired, so this is a one-sample t CI on the differences with $n - 1 = 4$ df. Then check whether 0 is inside.

1. Mean. $\bar{d} = (1 + 3 + 2 + 4 + 0)/5 = 10/5 = 2$.
2. SD. *Formula.* $s_d^2 = \dfrac{\sum(d_i - \bar{d})^2}{n - 1}$. *Plug in.* Deviations $-1, 1, 0, 2, -2$, squares $1, 1, 0, 4, 4$. *Work it out.* $10/4 = 2.5$. *Result.* $s_d = \sqrt{2.5} = 1.581$.
3. SE $= 1.581/\sqrt{5} = 1.581/2.236 = 0.707$.
4. CI. $t[0.025, 4] = 2.776$, margin $2.776 \times 0.707 = 1.963$. $2 - 1.963 = 0.04$ and $2 + 1.963 = 3.96$.

Answer. CI = (0.04, 3.96). It just excludes 0, so a difference is shown. The size is anywhere from almost nothing to about 4, which you should say.